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Squigonometry: The Study of Imperfect Circles

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Squigonometry: The Study of Imperfect Circles

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This book is dedicated to our students, whose curiosity and hard work fuel our own.

Preface

Our main goals in writing *Squigonometry: The Study of Imperfect Circles* were to compile results in generalized trigonometry that had been known for years, to introduce new facts and relationships that we had only recently discovered, and to create a book that exposes students to the rewards available from taking ideas from their early experiences in mathematics and pushing on them. In particular, we wanted to write a book that could entice undergraduate students to explore the back alleyways of a familiar corner of mathematics.

Squigonometry brings a modern perspective to some old mathematics, offering many opportunities for students to explore a variety of mathematical niches and maybe discover something new, or see something old in a new way.

Generalization and abstraction have always been part of the fiber of mathematics. To take something we have known to be true for a long time, a canonical equation, for example, and disassemble it, thoroughly examine it, and then reassemble it with some different parts is often the key to discovering structure. Doing this can often act as a gateway to a new mathematical world. This book is an introduction to generalizations of concepts that students learn early in their mathematical careers, namely,

1. In the plane, the distance between two points (x_1, y_1) and (x_2, y_2) is given by the Pythagorean theorem:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

2. The unit circle, the set of all points (x, y) that are a distance of 1 unit from $(0, 0)$, has equation $x^2 + y^2 = 1$.
3. The points on the unit circle can be given as $(x, y) = (\cos(t), \sin(t))$ where t is the measure of the angle subtended along the arc of the circle from $(1, 0)$ to (x, y). (This gives the relationship $\cos(t)^2 + \sin(t)^2 = 1$.)

As students continue through calculus, they learn that $\frac{d}{dt}\cos(t) = -\sin(t)$ and $\frac{d}{dt}\sin(t) = \cos(t)$ and that the parameter t represents not only an angle measure, but is also twice the area of the sector of the unit circle swept out by the angle, as well

as the length of the arc of the unit circle subtended by the angle. Students also learn the importance and ubiquity of π in these contexts.

The sine and cosine functions are so durable and so flexible that we can define them in several different ways and obtain the same functions—the unit circle approach and differential equation relationship above yield the same two functions, as does defining sine and cosine functions via integrals.

An entry into generalized trigonometry comes from the search for variants of sine and cosine functions that can parameterize the closed curve $|x|^p + |y|^p = 1$. Once we select a variant, we need to answer some questions. What does this curve look like? Do all the connections between area, angle, and arclength hold? Can we measure distance in the same way as before? Are there ways of approximating these functions?

The philosophy of the book is to follow these ideas in small steps and build up consequences of our choices in each chapter. While the change of a 2 to a generic p in an equation may seem only cosmetic, the ripples from dropping that stone go a long way, enveloping not only the trigonometric functions but also our conceptions of distance, and the hyperbolic, exponential, and logarithmic functions.

Audience and Structure

Squigonometry is suitable for an undergraduate reading course in mathematics as well as a source for senior capstone research. Students should know the basic trigonometric functions as well as both differential and integral single-variable calculus, initial-value problems, and results on infinite series. Familiarity with multivariable and vector calculus will allow additional insights into Chapters 13 and 18, and some basic tools from linear algebra would also assist in Chapter 18.

One notion that ties much of the book together is that of *parameterization*, the idea that we can set $x = f(t)$ and $y = g(t)$ for some values of t and the ordered pair $(x, y) = (f(t), g(t))$ describes a curve in the plane. A primer (or refresher) on parametric equations can be found in Appendix C.

The book is meant to function as both an introduction to generalized trigonometry and an invitation into further research. The chapters are short by design and focus on one aspect of squigonometry. Fundamental results about our new functions such as relational equations, derivatives, integrals, and series approximations are all developed. Variations on π are introduced through the lens of so-called special functions. From there, readers can explore offshoot topics such as hyperbolic, exponential, and logarithmic functions; metrics and norms; and more advanced topics such as lemniscatic and elliptic functions, and other generalizations of the trigonometric functions that are different from the ones focused on here and that have cropped up in recent years. There are historical reasons for the creation of these generalized functions and these are sprinkled throughout the exposition. Students should also come away with the idea that there is no last word on squigonometric functions.

The introduction in Chapter 1 gives a brief recap of the ways in which we define the usual trigonometric functions as well as the relations among them. We reintroduce familiar formulas we will generalize later in the book, and in Chapter 2, we show how replacing the perfection of the circle with an imperfect shape—one with far less symmetry than the circle—can lead to issues.

Chapter 3 introduces us to the p-trigonometric (or *squigonometric*) functions, the generalized trigonometric functions of Grammel and Shelupsky and the notion of the p-circle, the locus of points related by the equation $|x|^p + |y|^p = 1$. We also define generalizations of the tangent, cotangent, secant, and cosecant.

In Chapter 4, we explore the idea of defining distance differently than we are accustomed to by introducing the p-metric. In particular, we will be able to look at the unit p-circle as the set of all points that are one unit away from the origin *in the p-metric*.

We return in Chapter 5 to the integrals that led mathematicians to this study in the first place, and look at the inverses of the squigonometric functions. We also develop a generalized version of right triangle trigonometry using the p-metric.

Up until this point in the book, the relationships and formulas between the various squigonometric functions are established through purely formal manipulation. In Chapter 6, we introduce a generalization of π for each p by appealing to so-called special functions. We then establish domains and ranges for our generalized trigonometric functions and their inverses.

In Chapters 7 and 8, we note that in generalizing the trigonometric functions in this way, we may preserve knowledge of the area of the p-circle, but we lose information such as the length of an arc and the angle subtended. (The case $p = 2$ has been the "go to" for years because it preserves all of that information.) In particular, in Chapter 8, we develop a squigonometry inspired by the generalizations of π in [2, 30].

Chapter 9 introduces not only the integral formulas listed by Grammel [42], but also some tweaks of known integration techniques for trigonometric functions (such as inverse trigonometric substitution) that will allow us to evaluate some integrals that previously eluded mathematicians. We also look at three applications of this kind of integration in Chapter 10, two of which are via mathematician Piet Hein: finding the volume of the central square in Stockholm, and determining the exact volume of Hein's so-called "superegg." The third involves the trisection of a region between two curves.

Chapter 11 introduces series representations for the squigonometric functions and for the inverse sines and tangents, as well as extensions of our functions to the complex plane. We then use these series in Chapter 12 to find reasonable series approximations for the various π's, as well as examine continued fraction representation of the π's, and how to find exact sums of related series.

Polar coordinates are often introduced to students in precalculus or calculus and in Chapter 13, we introduce *squircular* coordinates. Just as polar coordinates generalize to cylindrical and spherical coordinates, squircular coordinates

generalize to p-cylindrical and p-spherical coordinates. This will allow us to examine other three-dimensional objects such as superellipsoids.

Most of the previous research on generalized trigonometric functions also included generalizations of the hyperbolic functions, and we go there in Chapter 14. We establish relationships between the generalized trigonometric and hyperbolic functions via complex numbers and a special function called the Gudermannian. We also determine the radii of convergence for the series representations of the generalized transcendental functions discovered thus far. In an example of a seemingly small change having perhaps unforeseen circumstances, this leads to generalizations of the exponential and logarithmic functions in Chapter 15.

Lemniscates—curves with polar equation $r^2 = \cos(2\theta)$—provided the initial impetus to generalize trigonometric functions and in Chapter 16, we dive into the relationship between these lemniscatic functions, the related elliptic integrals, and our squigonometric functions in the $p = 3$ and $p = 4$ cases. In particular, we look at addition and double-angle formulas for Dixon elliptic functions. We further explore these relationships in Chapter 17 as well as how so-called elliptic functions—generalizations of trigonometric functions that parameterize certain ellipses—fit in.

We had earlier used the p-metric to ease the discovery of relationships between squigonometric functions and their inverses. In Chapter 18, we introduce the related idea of the p-norm (a metric measures distances between two things; a norm measures the size of a single thing). Norms and the notion of convexity allow us tie up some loose ends in our understanding of the p-metric, most notably the triangle inequality. With these tools in hand, we formally define the dual norm in Chapter 19 with a focus on geometric understanding of the special relationship between conjugate p-metrics (when $\frac{1}{p} + \frac{1}{q} = 1$).

In particular, we look at the relationship between area and arclength, between curves and their so-called pedals, and how to "invert" the curves we have examined thus far. We also get a geometric look at some classical theorems of analysis, including Hölder's inequality.

We wrap things up in Chapter 20 by briefly looking at a few other generalizations of trigonometric functions and some additional directions for students to explore these functions.

The chapters are generally brief. Each chapter begins with a short introduction discussing some of the history behind that chapter's focus and then presents the main results with proofs. Many of the proofs in the book are computational in nature and use known results from calculus and the theory of special functions to discover properties of our squigonometric functions.

Some of these properties are developed further in the exercise sections at the end of each chapter. The exercises vary in difficulty from one-or two-line proofs to extensive applications of the material, but the material necessary to complete them can be found in the text. They are meant to offer practice with newfound skills and reinforce connections. The appendix includes solutions and hints to many exercises.

In addition to exercises, we have included several projects throughout the text to invite deeper dives into the material or serve as a springboard to independent study or research where students can play with their new mathematical tools. They are typically written with the upper level undergraduate in mind to pursue for senior thesis or capstone project. These investigations will fill out parts of the theory, but guided by questions with potential for new discovery. In many cases, the same project may have geometric, analytic, or computational access points that could attract students of diverse interests and backgrounds. Many could be expanded to full research projects or carved into classroom projects.

The appendix compiles several formulas found or referenced in the body of the book, a curve menagerie that illustrate various geometric features in six key cases, hints and solutions for selected exercises, and other miscellany, including some modest proposals for alternate dates for "Pi Day."

We have done our best to untangle this subject's convoluted history via in-text references, endnotes, and a full bibliography. We have taken the assignment of credit very seriously and we apologize for any oversights.

Researching and writing *Squigonometry* has been a thrill for us and we hope that students and faculty will find the subject as inspiring and enjoyable as we have.

Acknowledgements

This book began with two articles the authors separately published on squigonometry in Mathematics Magazine in 2011 and 2016 [83, 105]. We met for the first time at the Joint Mathematics Meetings in 2017. We decided there was more to squigonometry than a collaborative article [107] could cover and the idea of this book was born.

We would like to thank our mathematical forebears—that is, all who wrote the works cited in the bibliography—for providing us with such rich source material. As Bill puts it, "This stuff just never stops giving." The research for this book was a pure joy.

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Rob's Acknowledgements

At the turn of the millennium, Adler and Tanton [2] and, separately, Euler and Sadek [30] defined a generalized π that yielded the correct arclength and circumference for p-circles. This led me to think about generalized trigonometric functions for a p-circle for the first time. (These functions are described in Chapters 7 and 8.) I thank these authors for altering my career trajectory.

A large part of the writing was done in the spring and summer of 2020 while I was on a sabbatical from Norwich University and I thank Lea Williams, Norwich University's then-Faculty Development coordinator, and then-Provost Sandra Affenito for their support. Additionally, the Norwich University Department of Mathematics and College of Science and Mathematics have been very supportive of this endeavor. I also thank my departmental colleagues for their enthusiastic backing. Many Norwich students had to suffer through various iterations of this material as part of their senior capstone. The book is dedicated to them, but I want to acknowledge Ashley Johnson and Yu-Wei Wu, whose work led to breakthroughs on generalized exponential functions. I owe a debt of gratitude to friends and colleagues who read early versions of this material or tried it out with their students for various papers and projects: Vince Ferlini and Karen Stanish of Keene State College, and Joe Fields of Southern Connecticut State University.

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Bill's Acknowledgements

"Squigonometry" began for me in a calculus class I taught at Hendrix College early in my career. The course was modeled on a differential equations approach [17, 18], wherein we defined transcendental functions in terms of the initial-value problems they solved. With students seeming unimpressed with the derivations of familiar results, I started changing 2's into 4's to see what would happen. I could not anticipate the journey that followed as that idea grew to a class project, paper, several undergraduate research projects, a book chapter, and now this book. I am grateful to those who engaged me along the way, especially the myriad students—to whom this book is dedicated—who have enjoyed (or at least endured) the various exercises, talks, and projects on this material that I couldn't help sharing with them.

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And special thanks to my wife Toni and our twins Orson and Tabitha for their love, support, and patience through this process. I promise, kids, it will be my greatest pleasure to show what all of those funny symbols mean, but some of it will have to wait until you finish third grade.

Northfield, VT, USA — Robert D. Poodiack
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Introduction

1

Trigonometry has been studied for millenia, as far back as the Sumerians, the Babylonians, the Hellenic Greeks, and the ancient Egyptians [80]. While trigonometry literally means "triangle measure," the initial work on trigonometric functions related to chords of a circle, the straight lines connecting two points on the curve. Hipparchus, in the second century B.C., was interested in spherical triangles, the imaginary triangle formed by three stars on the celestial sphere. He knew that to compute the distances between stars he needed to know the length of the chord of the circle connecting them. In turn, this meant he had to figure the central angle subtending that chord. The solution is related to finding the sine of an angle since for a circle of radius 1, the length of the chord subtended by an angle of t radians is $2\sin(t/2)$.

It wasn't until about the year 500 A.D. when the Hindus transmuted the ideas of chords into the modern sine function. The term *trigonometry* did not appear for the first time until the end of 16th century even though several familiar addition and multiple-angle formulas had been discovered by then.

The circle comes with a natural parameterization by the classical cosine and sine functions. Those functions have identities that capture the beauty of the perfect object they parameterize. The Pythagorean formula $\cos(t)^2 + \sin(t)^2 = 1$ is well-known to precalculus students and beyond as are some of its corollaries, such as $1 + \tan(t)^2 = \sec(t)^2$.

It has only been in the past 300 or so years that mathematicians have thought about using similar functions to parameterize other curves that lack the circle's perfection. Let's remind ourselves of this perfection by laying out several methods of parameterizing the unit circle, all of which lead to the same two functions. (See Appendix C for a primer on working with parameterizations.)

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We often define the sine and cosine functions in one of three ways:

- via differential equations,
- geometrically, as coordinates of points on the unit circle $x^2 + y^2 = 1$,
- analytically, via the use of definite integrals.

Differential equations approach

Recall from calculus that $\frac{d}{dt}\cos(t) = -\sin(t)$ and $\frac{d}{dt}\sin(t) = \cos(t)$. We view these as defining properties for the cosine and sine functions. By choosing to start the parameterization on the x-axis, we further adopt the initial conditions $\cos(0) = 1$ and $\sin(0) = 0$. Thus we can say that cosine and sine are the functions satisfying

$$\begin{cases} x'(t) = -y(t) \\ y'(t) = x(t) \\ x(0) = 1 \\ y(0) = 0, \end{cases} \tag{1.1}$$

where x corresponds to the cosine and y to the sine. This is an example of a *coupled initial value problem* (or CIVP). It turns out problems like these always have unique solutions [14]. Therefore, we may *define* cosine and sine to be the unique solution to (1.1). We could then use Euler's method to approximate values, and the graph of the phase plane would trace out the unit circle. Further, the equations (1.1) are enough to derive all of the familiar properties of these functions.

For example, suppose $x(t)$ and $y(t)$ are functions which satisfy equation (1.1) and consider the function $f(t) = x(t)^2 + y(t)^2$, whose derivative is given by $f'(t) = 2x(t)x'(t) + 2y(t)y'(t)$. Since $x'(t) = -y(t)$ and $y'(t) = x(t)$ then $f'(t) = 0$ and $f(0) = 1$. A function whose derivative is always zero must be constant, so $f(t) = 1$ for all t. Hence we arrive at $x(t)^2 + y(t)^2 = 1$, which is more familiar in the form $\cos(t)^2 + \sin(t)^2 = 1$. This is an example of a general method for proving identities with functions defined as solutions to (coupled) initial value problems.

As another example, we show how to use these techniques to prove the symmetries of cosine and sine. Let $x(t) = \cos(-t)$ and $y(t) = -\sin(-t)$. Then $x(0) = 1$ and $y(0) = 0$, $x'(t) = \sin(-t) = -y(t)$, and $y'(t) = \cos(-t) = x(t)$. In other words, x and y are solutions to the CIVP (1.1) that defines sine and cosine. But the solution of that CIVP is unique, so it must be that $\cos(-t) = x(t) = \cos(t)$ and $-\sin(-t) = y(t) = \sin(t)$.

We also know the tangent function as $\tan(t) = \frac{\sin(t)}{\cos(t)}$, and its derivative

$$\frac{d}{dt}\tan(t) = \frac{d}{dt}\frac{\sin(t)}{\cos(t)} = \frac{1}{\cos(t)^2} = 1 + \tan(t)^2.$$

We could thus redefine tangent as the solution to the initial value problem

$$\frac{dx}{dt} = 1 + x^2, \quad x(0) = 0,$$

a familiar separable differential equation.

We now discuss this existence/uniqueness result in more generality. The theorems for the uncoupled and coupled cases are as follows:

Theorem 1.1

Let x_0, t_0 be real numbers and let $F(x,t)$ be continuous with continuous partial derivatives. Then there is a unique function $x(t)$ defined on an interval containing t_0 satisfying

$$\begin{cases} x'(t) = F(x(t),t) \\ x(t_0) = x_0. \end{cases}$$

Theorem 1.2

Let x_0, y_0, and t_0 be real numbers and let $F(x,y,t)$ and $G(x,y,t)$ be continuous with continuous partial derivatives. Then there are unique functions $x(t)$ and $y(t)$ defined on an interval containing t_0 satisfying

$$\begin{cases} x'(t) = F(x(t),y(t),t) \\ y'(t) = G(x(t),y(t),t) \\ x(t_0) = x_0 \\ y(t_0) = y_0. \end{cases}$$

These results are really just special cases of a more general theorem that says that IVP's defined on vector-valued functions have unique solutions. Theorems 1.1 and 1.2 are just the one- and two-dimensional cases, respectively.

Unit circle approach

The unit circle is the set of points (x,y) that are exactly 1 unit from the origin. Such points satisfy the equation $x^2 + y^2 = 1$.

Any line through the origin intersects the unit circle at a point t units away from $(1,0)$ as measured along the arc of the circle. We define the *radian* as a measure of angle such that one radian subtends an arc of the unit circle of length 1. Thus the line passing through our point subtends an angle of t radians, and we define the cosine and sine of t to be the x- and y-coordinates, respectively, of our intersection point. (See Figure 1.1.) Thus we have that $\cos(t)^2 + \sin(t)^2 = 1$ for all $t \in \mathbb{R}$.

There seems to be no reason to expect that the sine and cosine functions defined here are the same ones defined in (1.1), so we will prove that these functions fulfill the CIVP (1.1) and, by uniqueness, must be the same functions.

We note first that when $t = 0$, our line from the origin would intersect the unit circle at $(1,0)$, meaning that $\cos(0) = 1$ and $\sin(0) = 0$. We then recall that the line tangent to the unit circle at the point $P(\cos(t), \sin(t))$ is perpendicular to the line from the origin through P. The line from the origin to P has slope $\frac{\sin(t)}{\cos(t)} = \tan(t)$;

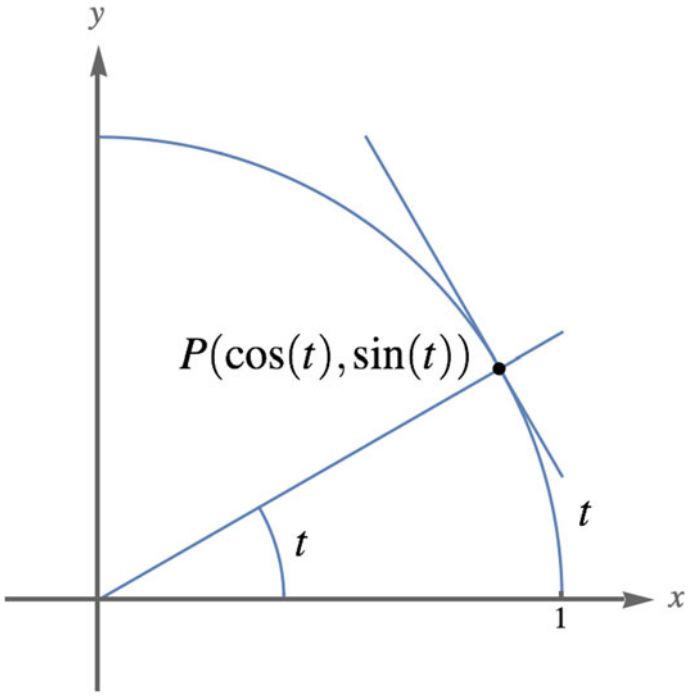


Figure 1.1 The first quadrant of the unit circle

thus the tangent line at P has slope $-\cot(t) = -\frac{\cos(t)}{\sin(t)}$. But all this tells us is that our new cosine and sine fulfill the CIVP

$$\begin{cases} x'(t) = -y(t)\zeta(x(t), y(t), t) \\ y'(t) = x(t)\zeta(x(t), y(t), t) \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{1.2}$$

for some appropriate differentiable function $\zeta(x(t), y(t), t)$, with x corresponding to our new cosine, and y corresponding to our new sine.

To show our cosine and sine are the same ones defined by (1.1), we define radius and angle functions by the following relations:

$$r(t) = \sqrt{x(t)^2 + y(t)^2} \quad \text{and} \quad \tan(\theta(t)) = \frac{y(t)}{x(t)}.$$

(In essence, we are defining polar coordinates.) On the unit circle, $r(t) = 1$ for all t and we have that $\theta(t) = t$ by the definition of a radian. So we need to find the function ζ in (1.2) that makes $\theta'(t) = 1$. We use implicit differentation to obtain

$$\begin{aligned}
\sec(\theta(t))^2\theta'(t) &= \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right)\\
[1+\tan(\theta(t))^2]\theta'(t) &= \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right)\\
\left[1+\left(\frac{y(t)}{x(t)}\right)^2\right]\theta'(t) &= \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right)\\
\Rightarrow \theta'(t) &= \frac{1}{1+\left(\frac{y(t)}{x(t)}\right)^2}\cdot\frac{d}{dt}\left(\frac{y(t)}{x(t)}\right)\\
&= \frac{x(t)^2}{x(t)^2+y(t)^2}\cdot\frac{x(t)\cdot x(t)\zeta(x(t),y(t),t)+y(t)\cdot y(t)\zeta(x(t),y(t),t)}{x(t)^2}\\
&= \zeta(x(t),y(t),t),
\end{aligned}$$

showing that the cosine and sine we defined as coordinates on the unit circle satisfy the CIVP (1.1) and are the same functions we defined via differential equations.

We can use properties of the unit circle to enhance our knowledge of our functions' properties. As a full circuit of the unit circle subtends 2π radians, each point on the unit circle is associated with infinitely many angles, all multiples of 2π radians apart. In particular, this means that the cosine and sine functions are periodic – they repeat their values every 2π radians:

$$\begin{aligned}
\cos(t+2\pi k) &= \cos(t)\\
\sin(t+2\pi k) &= \sin(t)
\end{aligned}$$

for all $k \in \mathbb{Z}$.

The argument t in cosine and sine thus simultaneously specifies both the arclength and the angle. It also tells us (twice) the area of the sector. We see that a sector traced out by t radians has an area that is $\frac{t}{2\pi}$ times the area of the whole circle, or $\frac{t}{2\pi}\times\pi = \frac{t}{2}$ square units. Thus the parameter t provides information on the angle subtended, the arclength traced out, and the area of the corresponding sector. We will come to appreciate this remarkable feature of Euclidean geometry when we lose it in the geometries we will soon explore.

Analytic approach

We can also define the sine and cosine as the solutions to integral equations. We define a sine function $S(t)$ such that

$$\int_0^{S(t)} \frac{1}{\sqrt{1-u^2}}\,du = t. \tag{1.3}$$

In a similar manner, we can define the cosine function to be the function $C(t)$ such that

$$\int_{C(t)}^{1} \frac{1}{\sqrt{1-u^2}}\,du = t. \tag{1.4}$$

Again, there seems to be no reason to expect that the sine and cosine functions defined here are the same ones defined in (1.1). We will prove that $S(t)$ and $C(t)$ fulfill the CIVP (1.1) and, by uniqueness, must be the same functions.

We first claim that $C(t) = \sqrt{1 - S(t)^2}$ (and thus $S(t) = \sqrt{1 - C(t)^2}$ and $C(t)^2 + S(t)^2 = 1$). If we substitute $s = \sqrt{1 - u^2}$ into Equation (1.3), then $u = \sqrt{1 - s^2}$, $du = -\frac{s}{\sqrt{1-s^2}}\, ds$, and

$$t = \int_0^{S(t)} \frac{1}{\sqrt{1-u^2}}\, du = \int_1^{\sqrt{1-S(t)^2}} \frac{1}{s} \cdot \frac{-s}{\sqrt{1-s^2}}\, ds = \int_{\sqrt{1-S(t)^2}}^1 \frac{1}{\sqrt{1-s^2}}\, ds.$$

Comparing this last integral to (1.4), we see that $C(t) = \sqrt{1 - S(t)^2}$. Furthermore, Equations (1.3) and (1.4) show that $S(0) = 0$ and $C(0) = 1$.

Taking the derivatives with respect to t across Equations (1.3) and (1.4) gives

$$1 = \frac{1}{\sqrt{1 - S(t)^2}} \cdot S'(t) = -\frac{1}{\sqrt{1 - C(t)^2}} \cdot C'(t)$$

and so $S'(t) = \sqrt{1 - S(t)^2} = C(t)$ and $C'(t) = -\sqrt{1 - C(t)^2} = -S(t)$. Thus S and C fulfill the CIVP (1.1) and must be the same sine and cosine functions found there.

Once again, we see that these three approaches inevitably yield the same cosine and sine functions.

Defining the other functions

We have already seen the tangent and cotangent functions but in the interest of completeness, we recall the other trigonometric functions, the secant and cosecant.

$$\tan(t) = \frac{\sin(t)}{\cos(t)}, \qquad \cot(t) = \frac{\cos(t)}{\sin(t)},$$
$$\sec(t) = \frac{1}{\cos(t)}, \qquad \csc(t) = \frac{1}{\sin(t)}.$$

Just as cosine and sine are inextricably linked by the CIVP (1.1) (and many other formulas!), tangent and secant can be so linked. If we let $x = \cos(t)$ and $y = \sin(t)$, we can define $z = \tan(t) = \frac{y}{x}$ and $w = \sec(t) = \frac{1}{x}$ and we can derive a new CIVP from (1.1):

$$\begin{cases} z'(t) = w(t)^2 \\ w'(t) = w(t)z(t) \\ z(0) = 0 \\ w(0) = 1 \end{cases} \tag{1.5}$$

Exercises

▶ **Exercise 1.1** *Give a definition of* $u(t) = \cot(t)$ *as the solution of an initial value problem.*

▶ **Exercise 1.2** *Derive a CIVP for the cotangent and cosecant functions. What curve is parameterized by* $(\cot(t), \csc(t))$*?*

▶ **Exercise 1.3** *Prove the trigonometric doubling formulas by finding a CIVP to which the functions* $x(t) = \cos(2t)$, $y(t) = \sin(2t)$ *and* $X(t) = 1 - 2\sin(t)^2$, $Y(t) = 2\sin(t)\cos(t)$ *are both solutions.*

▶ **Exercise 1.4** *Generalize the previous exercise by proving the trigonometric angle sum formulas by finding a CIVP for* $x(t) = \cos(t + a)$, $y(t) = \sin(t + a)$ *are a solution, holding a fixed.*



Imperfection

2

One way to view the perfection of the circle is that almost any attempt to parameterize it – by area, arclength, or angle – will lead to the same functions, cosine and sine. Another way to view it is to note that if we try to parameterize an imperfect curve, we may get some of the story – the area *or* the arclength *or* the angle – but not all of it.

Let's try introducing some imperfection by considering the graph of the equation $|x| + |y| = 1$, whose graph is a square with vertices 1 unit along the coordinate axes (Figure 2.1).

We can think of this formula as that of the classical circle $x^2 + y^2 = 1$ with the twos replaced by ones (the absolute values were secretly always there, but were redundant since two is even). Our usual definition of the unit circle is as the locus of all points that are distance 1 from the origin. The quantity $|x| + |y|$ measures the distance to the origin differently than we are used to. It is sometimes called *taxicab metric* because it is the distance a taxi must travel in a city grid – our usual Euclidean "as the crow flies" distances do not apply since you cannot drive through city blocks. (We will soon call it the 1-metric.) So we can think of the curve $|x| + |y| = 1$ as the unit circle if we measure distance using the taxicab metric.

What are the natural trigonometric functions to parameterize this circle?

Suppose we want the parameterization $(x(t), y(t))$ so that t equals twice the area of the corresponding sector of the curve. Beginning in the first quadrant and referring to Figure 2.2, we see $t = 2 \cdot \frac{1}{2} \cdot 1 \cdot y = y$ and $x = 1 - y$. Thus we get what we call the areal parameterization $x(t) = 1 - t$, $y(t) = t$ for $0 \leq t \leq 1$. Now suppose Q is in the second quadrant. Then x is negative and we must account for the triangle already swept in the first quadrant, giving $t = 2\left(\frac{1}{2} + \frac{1}{2} \cdot 1 \cdot (-x)\right) = 1 - x$ and $y = 1 + x = 2 - t$. Thus $x(t) = 1 - t$, $y(t) = 2 - t$ for $1 \leq t \leq 2$. Continuing this way gives a full definition for cosine and sine in taxicab geometry for $0 \leq t \leq 4$ where t is twice the area of the portion of taxicab circle swept by a ray from the origin. We may then extend this definition by periodicity to all real numbers.

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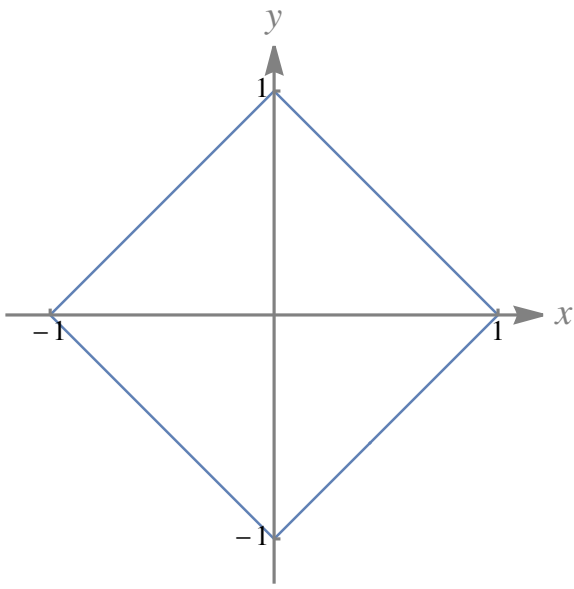


Figure 2.1 The curve $|x| + |y| = 1$

$$x(t) = \begin{cases} 1-t & 0 \le t \le 2 \\ t-3 & 2 < t \le 4 \end{cases} \qquad (2.1)$$

$$y(t) = \begin{cases} t & 0 \le t \le 1 \\ 2-t & 1 < t \le 3 \\ t-4 & 3 < t \le 4 \end{cases}$$

Now let's suppose instead we want the parameter t to represent the arclength traveled along our curve from the point $B(1, 0)$. In Figure 2.2, the length of segment QB is $s = y\sqrt{2} = t\sqrt{2}$. We thus deduce that the arclength parameter s is simply a rescaling of the area parameter t and therefore the corresponding arclength trigonometric functions are the same as the areal functions, with the period stretched from 4 (twice the area of the square) to $4\sqrt{2}$ (the perimeter of the square). Although the functions are closely related, we see that they are both reasonable but distinct options. In particular, it is not clear what "π" should mean in this geometry.

When we parameterize by angle, however, we will see that this is not just a matter of scaling. Referring to Figure 2.3, we see the right triangle with vertex at Q and side lengths x and $1 - x$ is similar to a triangle with side lengths $\cos\theta$ and $\sin\theta$. Thus

$$\frac{\sin\theta}{\cos\theta} = \frac{1-x}{x} = \frac{1}{x} - 1$$

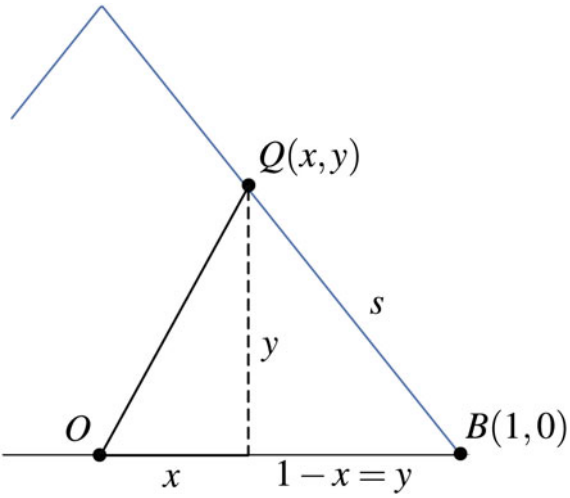


Figure 2.2 Taxicab sine and cosine for the areal and arclength parameterizations

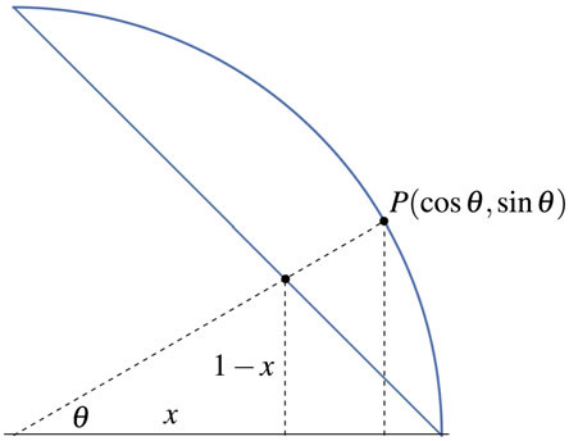
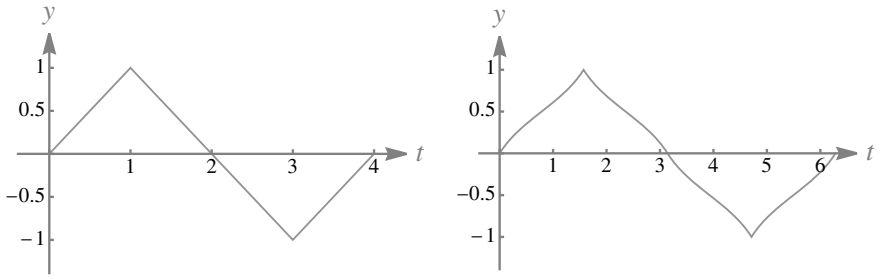


Figure 2.3 Taxicab sine and cosine for the angular parameterization

which we solve to obtain $x(\theta) = \frac{\cos\theta}{\sin\theta+\cos\theta}$. After accounting for the other quadrants by adding absolute values and invoking the relation $|y| = 1 - |x|$, we obtain

$$x(\theta) = \frac{\cos\theta}{|\sin\theta| + |\cos\theta|}$$
$$y(\theta) = \frac{\sin\theta}{|\sin\theta| + |\cos\theta|}.$$

The graphs of the areal and angular sine functions are shown in Figure 2.4.



(a) area (b) angular

Figure 2.4 Graphs of the taxicab sine functions

Exercises

▶ **Exercise 2.1** *Verify the piecewise linear definition (2.1) of the areal taxicab sine and cosine functions in all quadrants.*

▶ **Exercise 2.2** *Define the unit max circle as the set* $\{(x, y) \mid \max\{|x|, |y|\} = 1\}$. *Find formulas for the areal trigonometric functions for this circle.*



A squigonometry introduction

3

The development of the sine and cosine functions to parameterize the points of the unit circle led to mathematicians developing generalized trigonometric functions to parameterize other closed curves. Jakob Bernoulli considered the arclength of the lemniscate, a figure-eight type figure we will see more of in Chapter 16, back in 1691. However, it was Erik Lundberg in 1879 [65] who examined differential equations of the form

$$\frac{dy}{dx} = (1 \pm y^p)^{\frac{m}{p}},$$

where m and p are integers and $p \geq m$. (He charmingly refers to trigonometric functions as "goniometric functions." He calls solutions of the above differential equation "hypergoniometric functions," since the functions are generalizations of either trigonometric or hyperbolic functions. A goniometer, by the way, is a device for measuring angles in crystals or, in physical therapy, the range of motion in joints.)

In particular, Lundberg sought to invert integrals of the form

$$x = \int_0^y \frac{1}{(1-u^p)^{\frac{m}{p}}}\, du,$$

so that y would be a function of x. He notated $y = S_{\frac{m}{p}}(x)$, called the *sinualis of the pth order and of the mth degree*. In particular, $S_{\frac{1}{2}}(x) = \sin x$, the customary and usual sine function.

Lundberg examined a few flavors of his hypergoniometric functions. When $m = p$ and the plus sign is chosen, a generalization of the tangent function is born (see Chapter 20). He also investigated the case $m = 2$ and p is an odd positive integer [61].

Much of Lundberg's work looked at the case when $m = p - 1$. That is, he examined the inverse of integrals of the form

$$x = \int_0^y \frac{1}{(1-u^p)^{\frac{p-1}{p}}}\, du.$$

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By setting $y = S(x) = S_{\frac{p-1}{p}}(x)$, and then defining a corresponding *cosinualis* function C via the change of variable $z = (1-u^p)^{1/p}$, Lundberg established the relationship

$$[S(x)]^p + [C(x)]^p = 1,$$

as well as the derivative formulas $S'(x) = C(x)^{p-1}$ and $C'(x) = -S(x)^{p-1}$. This would mean that S and C parameterize the curve $x^p + y^p = 1$, at least in the first quadrant, which was pointed out by Siegmund Günther [46] in 1881. This particular brand of generalized trigonometric function has tended to fade and reappear at intervals ever since [42,89].

We will develop these functions by approaching them in as similar a method as possible to the way we did the ordinary trigonometric functions in the introduction.

3.1 Parameterizing the p-circle

We will use the differential equations approach to define our new cosine and sine functions. If we are to use a coupled initial value problem to define these functions, the solutions $x(t)$ and $y(t)$ must make the function

$$g(t) = x(t)^p + y(t)^p, \tag{3.1}$$

constant for $p \geq 1$. We design the following CIVP with that property in mind:

$$\begin{cases} x'(t) = -y(t)^{p-1} \\ y'(t) = x(t)^{p-1} \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{3.2}$$

Note that Theorem 1.2 is what allows us to use the CIVP as a device to uniquely define these functions. If $x(t)$ and $y(t)$ are the solutions to (3.2) then

$$\begin{aligned} g'(t) &= p \cdot x(t)^{p-1} x'(t) + p \cdot y(t)^{p-1} y'(t) \\ &= -p \cdot x(t)^{p-1} y(t)^{p-1} + p \cdot y(t)^{p-1} x(t)^{p-1} = 0 \end{aligned}$$

so $g(t)$ must be constant. Since $g(0) = 1$, we then have that $g(t) \equiv 1$, as desired.

We denote $\cos_p(t) = x(t)$ and $\sin_p(t) = y(t)$ to be the unique pair of functions that solve (3.2).

In particular, when $p = 4$, the functions $\cos_4(t)$ and $\sin_4(t)$ parameterize the unit 4-circle, which we call the unit *squircle*. (See Figure 3.1.)

The squircle is so named because of its resemblance to a hybrid of a square and a circle. The larger the exponent p, the more squarish the p-circle. (See Figure 3.2.) For this reason, we call the functions defined by (3.2) *squigonometric* functions.[1]

[1] The curve is also called *Lamé's special quartic* after Gabriel Lamé (1795-1870), who performed much research on superellipses [33].

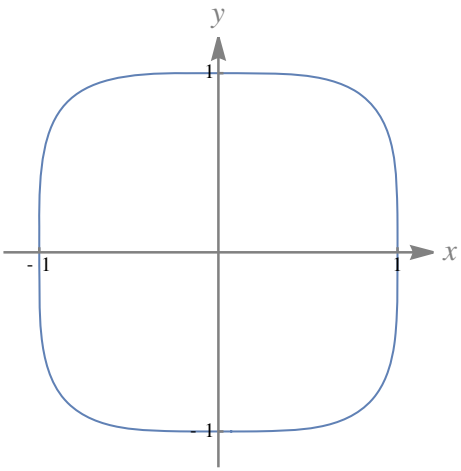


Figure 3.1 The unit squircle $x^4 + y^4 = 1$

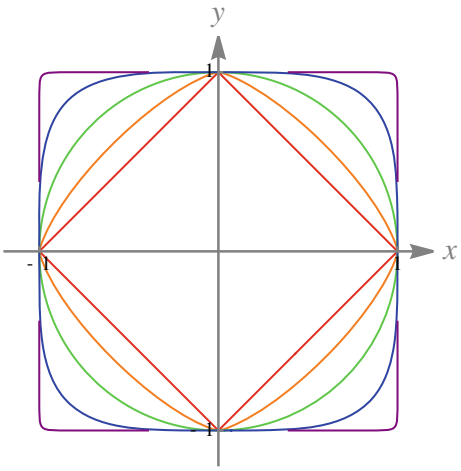


Figure 3.2 Unit p-circles $|x|^p + |y|^p = 1$ for $p = 1$, $\frac{4}{3}$, 2, 4, and 40, respectively (innermost to outermost)

Note that $\cos_p(t)^p + \sin_p(t)^p = 1$. For $p = 2$ or any even positive integer, we get the entire p-circle. In the case where p is not an even integer (or a rational number in reduced form with even numerator), the absolute values in the equation for the p-circle are necessary. (See Figure 3.3.)

To have an equation that describes the entire unit p-circle, we either must use absolute values, as in $|\cos_p(t)|^p + |\sin_p(t)|^p = 1$, or restrict the solutions to (3.2) to a suitable interval and extend the solutions using symmetry. We will see later how to do this.

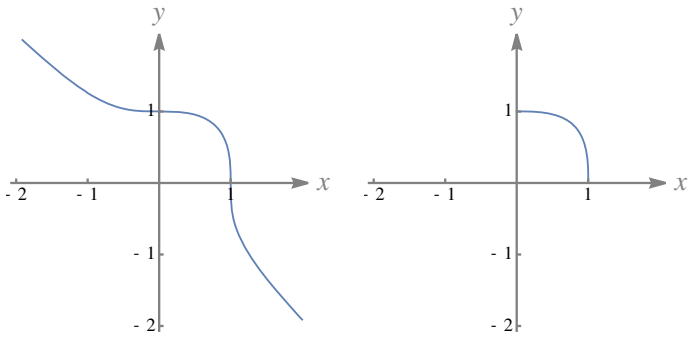


Figure 3.3 The curves $x^3 + y^3 = 1$ and $x^\pi + y^\pi = 1$ illustrate the need for absolute values to define a general p-circle outside of the first quadrant

3.2 Other squigonometric functions

We can define the other squigonometric functions through the usual definitions of ratios of sine and cosine functions:

$$\tan_p(t) = \frac{\sin_p(t)}{\cos_p(t)}, \qquad \sec_p(t) = \frac{1}{\cos_p(t)},$$
$$\cot_p(t) = \frac{\cos_p(t)}{\sin_p(t)} = \frac{1}{\tan_p(t)}, \qquad \csc_p(t) = \frac{1}{\sin_p(t)}.$$

With these in hand, we now have a complete set of functions for parameterizing and analyzing p-circles, which are themselves parameterized by the choice of p.

In the $p = 4$ case, the squigonometric functions have been called *squine*, *cosquine*, *tanquent*, etc. [83,105] We will generally (but not always) favor practicality over whimsy and refer to these functions by referencing p, so for example $\tan_p$ is simply the p-tangent.

We see from their graphs in Figure 3.4 that, when $p > 2$, the squigonometric functions are "flatter" versions of their trigonometric analogs in much the same way as the p-circle is a "flattened" circle. (We should note that we have cheated on this figure, in that we have not yet discussed how to extend p-trigonometric functions beyond their initial region of solution. We will show how to do this in Chapter 5. For now, we just note that the graphs in Figure 3.4 *look* reasonable.)

We also notice from the figure that the periods of these functions seem longer than the usual 2π (in the case of the 4-sine and 4-cosine) or π (in the case of the 4-tangent). We will find out how long these periods are down the road.

We must be careful to remember the broken symmetry of squircles when interpreting these functions. Resuming the approach from Section 3.1, if $x(t)$ and $y(t)$ parameterize the p-circle then $g(t) = x(t)^p + y(t)^p = 1$ and $g'(t) = p \cdot x(t)^{p-1}x'(t) + p \cdot y(t)^{p-1}y'(t) = 0$. Rearranging this expression gives

$$\frac{y'(t)}{x'(t)} = -\left(\frac{x(t)}{y(t)}\right)^{p-1}. \tag{3.3}$$

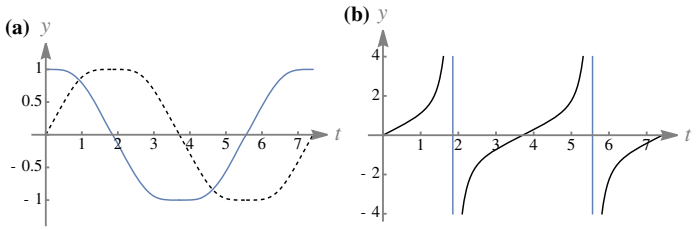


Figure 3.4 Plots of $y = \sin_4(t)$ (3.4a, dashed), $y = \cos_4(t)$ (3.4a, solid), and $y = \tan_4(t)$ (3.4b)

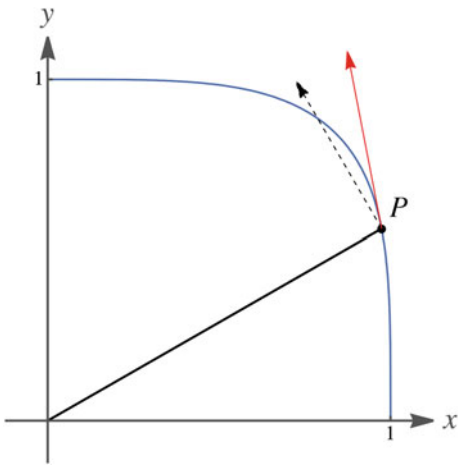


Figure 3.5 P lies at an angle of $\frac{\pi}{6}$ on the 4-circle and so the segment from the origin to P has slope $\frac{1}{\sqrt{3}}$ with perpendicular slope $-\sqrt{3}$ (the dashed arrow). The tangent at P (solid arrow) has slope $-(\sqrt{3})^{4-1} = -3\sqrt{3}$

If we define $x(t)$ and $y(t)$ to be the p-cosine and p-sine, respectively, i.e. the horizontal and vertical coordinate of a point P on the squircle at time t, then the left-hand side of (3.3) is the slope of the tangent line at P. The expression $-\frac{x(t)}{y(t)}$, which would be the negative cotangent, represents the slope of a line perpendicular to a radius connecting the origin to P. If $p = 2$, we see a familiar and essential property of Euclidean circles: their radii are perpendicular to their tangent lines. But this is not the case for $p \neq 2$ as the relationship between the slopes is distorted by a power of $p - 1$. See Figure 3.5. In particular, the connection between the tangent function and the tangent line is less direct.

We emphasize that this discussion applies to any parameterization, not just the choice we made with CIVP (1.1).

3.3 Derivatives

Since we have used a CIVP to define our p-sine and p-cosine functions, we can use the quotient rule to determine the derivatives of the other squigonometric functions.

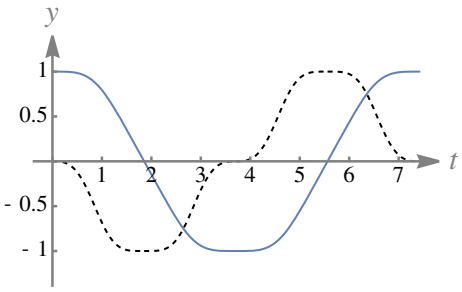


Figure 3.6 Plot of $y = \cos_4(t)$ (solid) and its derivative $y' = -\sin_4(t)^3$ (dashed)

For instance,

$$\frac{d}{dt}\tan_p(t) = \frac{d}{dt}\frac{\sin_p(t)}{\cos_p(t)} = \frac{\cos_p(t)\cdot\cos_p(t)^{p-1} + \sin_p(t)^{p-1}\sin_p(t)}{\cos_p(t)^2}$$
$$= \frac{\sin_p(t)^p + \cos_p(t)^p}{\cos_p(t)^2} = \frac{1}{\cos_p(t)^2} = \sec_p(t)^2.$$

By similar means, we can obtain

$$\frac{d}{dt}\sin_p(t) = \cos_p(t)^{p-1} \qquad \frac{d}{dt}\cos_p(t) = -\sin_p(t)^{p-1}$$
$$\frac{d}{dt}\tan_p(t) = \sec_p(t)^2 \qquad \frac{d}{dt}\cot_p(t) = -\csc_p(t)^2 \tag{3.4}$$
$$\frac{d}{dt}\sec_p(t) = \sec_p(t)^2\sin_p(t)^{p-1} \qquad \frac{d}{dt}\csc_p(t) = -\csc_p(t)^2\cos_p(t)^{p-1} \tag{3.5}$$

A graph of $y = \cos_4(t)$ and its derivative $y' = -\sin_4(t)^3$ can be found in Figure 3.6.

3.4 A differential equation

The customary sine and cosine functions satisfy the linear second-order differential equation $u'' + u = 0$. A glance at the second-order equation satisfied by our p-sine and p-cosine functions gives a glimpse of the complexity to come.

If we take second derivatives of the p-cosine and p-sine functions, we get

$$\frac{d^2}{dt^2}\cos_p(t) = -(p-1)\cos_p(t)^{p-1}\sin_p(t)^{p-2},$$
$$\frac{d^2}{dt^2}\sin_p(t) = -(p-1)\sin_p(t)^{p-1}\cos_p(t)^{p-2}.$$

Thus $\sin_p(t)$ and $\cos_p(t)$ satisfy the second-order nonlinear differential equation

$$u'' + (p-1)u^{p-1}(1-u^p)^{\frac{p-2}{p}} = 0. \tag{3.6}$$

Because the two functions are linearly independent, the general solution for (3.6) is $A\sin_p(t) + B\cos_p(t)$.

For $p = 2$, (3.6) reduces to the linear $u'' + u = 0$. In physics, this differential equation models the *simple harmonic oscillator* and the generalized version is of current interest. See [73] and Project 3.1.

Exercises

▶ **Exercise 3.1** *Use IVP's to show that the p-sine function is odd and the p-cosine is even.*

▶ **Exercise 3.2** *Prove that* $1 + \tan_p(t)^p = \sec_p(t)^p$ *and* $1 + \cot_p(t)^p = \csc_p(t)^p$.

▶ **Exercise 3.3** *Verify that Equations (3.5) are the right generalizations of the well-known derivative formulas for secant and cosecant; that is, show that when* $p = 2$, *we get*

$$\frac{d}{dt}\sec_2(t) = \sec_2(t)\tan_2(t) \quad \text{and} \quad \frac{d}{dt}\csc_2(t) = -\csc_2(t)\cot_2(t).$$

▶ **Exercise 3.4** *Develop a CIVP for the p-tangent and p-secant functions, and for the p-cotangent and p-cosecant functions in the manner of (1.5).*

▶ **Exercise 3.5** *The Wronskian W of two functions* y_1 *and* y_2 *is given as*

$$W(y_1(t), y_2(t)) = y_1(t)y_2'(t) - y_1'(t)y_2(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}$$

If $y_1(t)$ *and* $y_2(t)$ *are solutions to a linear homogeneous second-order differential equation with initial conditions given at* $t = t_0$, *it is known that the general solution of the differential equation can be given as* $y(t) = Ay_1(t) + By_2(t)$ *if and only if* $W(y_1(t), y_2(t)) \neq 0$ *at* $t = t_0$. *[14]*

The solution functions $y_1(t)$ *and* $y_2(t)$ *are called* linearly independent *and form a* fundamental set *of solutions for the differential equation.*

Prove that $y_1(t) = \sin_p(t)$ *and* $y_2(t) = \cos_p(t)$ *form a fundamental set for the differential equation (3.6) and thus the general solution of (3.6) is* $y(t) = A\sin_p(t) + B\cos_p(t)$ *where A and B are real numbers.*

▶ **Exercise 3.6** *This exercise introduces the* graph parameterization, *which models a curve as the graph of a function.*

1. *Show that* $(x, (1 - x^p)^{1/p})$, $0 \le x \le 1$, *parameterizes the unit p-circle in the first quadrant but does not satisfy (3.2).*
2. *Use this parameterization to write an integral representing the area of the p-circle.*

3. *Show that if $x(t)$ is a monotone continuous function with $x(0) = 1$ and $x(1) = 0$ and $y(t) = (1 - x(t)^p)^{1/p}$, then $(x(t), y(t))$ parameterizes the p-circle in the first quadrant.*
4. *What can happen if we don't require x to be continuous?*

▶ **Exercise 3.7** *There are rational points – points where both coordinates are rational numbers – galore on the unit circle ($p = 2$) in the first quadrant; for instance $(3/5, 4/5)$ and $(5/13, 12/13)$, which are derivable from our knowledge of Pythagorean triples. Show that the only rational points on the p-circle for $2 < p < \infty$ (p rational) lie on the coordinate axes. (Equivalently, show that if $p > 2$ then $\sin_p(t)$ and $\cos_p(t)$ are both rational if and only if one of them is 0.)*[2]

Project

Project 3.1 (Generalized harmonic oscillator)
A simple harmonic oscillator is a system in which the force put on a mass m is proportional to its displacement x, i.e. $F = -kx$. The standard example is a mass attached to a spring whose stiffness is represented by the constant k.

Newton's second law gives $F = ma = m\frac{d^2x}{dt^2}$, so we have the second-order differential equation $\frac{d^2x}{dt^2} = -\frac{k}{m}x$ with solution $x(t) = A\cos(\sqrt{\frac{k}{m}}t)$.

Can we use generalized trigonometric functions to model other kinds of periodic motion? For example, what if the force is proportional to a power of x? One can see that the resulting differential equations become quickly unwieldy. Working from the other end, our generalized trigonometric functions get complicated with successive differentiation and so the differential equations they satisfy also get unwieldy. Can these unwieldy calculations be aligned in any way that makes physical sense?

See [73,87,100] for more on how to frame and further investigate this question.

[2] This is about as far as we'll go into number theory, but we note there is much territory to explore after observing that many squircles are algebraic curves. This problem inspired by [108].

Check for updates

p-metrics

4

Having developed these generalized trigonometric functions, we now take a look at the geometry these functions model. Our goal is to define a new distance function called the p-metric and develop a sense of how the geometry compares to classical Euclidean geometry. Our exploration will include some geometric curves and a preview of duality, with a more complete geometric story to be told later after we develop the language of vector spaces.

4.1 Non-Euclidean geometry

We have developed new notions of circles and trigonometry on them. Though analogous to familiar objects from Euclidean geometry, they are not themselves natural Euclidean geometric objects.

We now introduce a family of geometries for which the p-circle is a natural geometric object. Define the *p-distance* between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ to be

$$d_p(P, Q) = \left(|x_1 - x_2|^p + |y_1 - y_2|^p\right)^{1/p}. \tag{4.1}$$

It is easy to verify that this re-definition of distance is consistent with defining a p-circle as the set of points whose p-distance from a center is constant. So our circles really are circles, but is our geometry really a geometry?

We must establish what we mean by this. A *metric* or *distance function* d on a set X is a function $d : X \times X \to \mathbb{R}$ with the following properties:

Positive definite. For all $A, B \in X$, $d(A, B) \geq 0$ with $d(A, B) = 0$ if and only if $A = B$.
Symmetric. $d(A, B) = d(B, A)$ for all $A, B \in X$.
Triangle inequality. $d(A, B) \leq d(A, C) + d(B, C)$ for all $A, B, C \in X$.

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_4

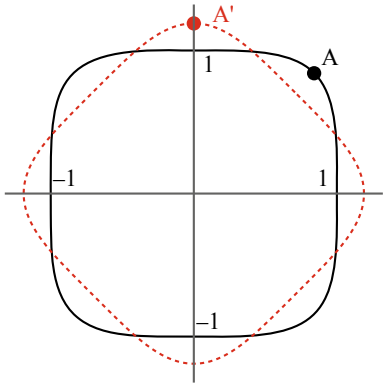


Figure 4.1 The point $A(2^{-1/4}, 2^{-1/4})$ on the unit 4-circle lands at $A'(0, 2^{1/4})$ when rotated by 45° counterclockwise about the origin, increasing the 4-distance to the origin from 1 to $2^{1/4} \approx 1.18921$

Positive definiteness and symmetry are reasonable properties to want from a notion of distance. Distance should not be negative, and a distance of 0 should indicate coinciding points. Symmetry says distance should be directionless,[1] so statements like "the distance between A and B" are well-defined without concern for the order the points are listed. These conditions are usually straightforward to prove.

It is the triangle inequality that makes a geometry "feel" like a geometry. It says that going directly (i.e., along a shortest path in the geometry) from A to B and then from B to C cannot be shorter than going directly from A to C. Without this, the notions of distance and shortest paths no longer correspond.

The p-metrics really are metrics when $p \geq 1$. Positive definiteness and symmetry are easy to verify. The triangle inequality can be proved with an analytic argument (e.g., [21, Theorem 1.2.15(a)]).

We prefer a more geometric approach, but this and some other properties we will discuss are inherited from the fact that this metric is actually a *norm*. This requires some additional vocabulary that we will defer to Chapter 18, but this roughly means the metric behaves properly if we think of points in the plane as vectors.

A useful tool for analyzing a metric is a set of functions that preserve distance, called *isometries*. We can use these to compare the Euclidean $p = 2$ to the non-Euclidean $p \neq 2$ cases. For example, all p-metrics are *translation-invariant*, meaning the distance between two points does not change if we shift them both by the same amount in the same direction. Figure 4.1 shows why rotations are typically *not* isometries if $p \neq 2$.

[1] Sometimes you do want to identify a direction along with a distance, in which case allowing for a signed distance is actually quite useful. Examples you may have seen include the difference between speed and velocity, or signed area in integration.

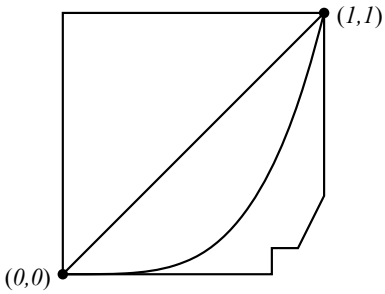


Figure 4.2 The curves shown are all shortest paths from (0, 0) to (1, 1) of length two in taxicab geometry

4.2 Exploring p-metrics

To understand the p-metric geometries, it is useful to explore some recurring examples. Euclidean geometry uses the $p = 2$ metric, but we want to see what changes when p varies.

The case $p = 1$, introduced in Chapter 2 as taxicab geometry, is the smallest p for which the function (4.1) defines a metric. If $p > 1$, the triangle inequality is strict for non-collinear points. This is not the case in taxicab geometry, giving the unusual property that there are infinitely many shortest paths between two points as long as those points are not aligned with the coordinate axes. Any monotone path that stays within the rectangle defined by the points will have length equal to half the perimeter of the rectangle. This is illustrated in Figure 4.2 and will be proved in Section 8.1.

We will also include as a p-metric the limiting case $p = \infty$, called the *max metric*, defined by

$$d_\infty(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\}.$$

The max metric is also called the *Chebyshev metric* or the *chess board metric* because it models the movement of the king in the game of chess. The taxicab and max metrics are useful in part because they are the two extremes and so the behaviors of the other p-metrics generally lie somewhere in between these two. They are also useful for experimentation because many of the drawings and calculations can be done easily by hand. In particular, their circles are Euclidean squares. The downside is that squares have non-differentiable corners and the calculus can get a bit ugly. We will see in the next section that these two metrics have another formal connection.

As an example of the peculiarity of taxicab geometry, consider the *midset* of two points A,B, defined to be the set of points P such that $d(A, P) = d(B, P)$.[2] To construct the midset, we can look at pairs of circles centered at A and B of the same radius. The midset is the collection of intersections of these circles. In Euclidean geometry, this is the perpendicular bisector of the segment connecting A to B.[3]

[2] We strongly recommend [55] for an enjoyable discovery journey through this geometry that requires no calculus background.

[3] This is the classical ruler and compass construction of a perpendicular from Book I, Proposition 12 of Euclid's *Elements* [50].

The circles of radius $\frac{1}{2}|AB|$ intersect at the midpoint and each pair of larger circles intersect at two other points on the midset. For finite $p > 1$, all circles still intersect in at most two points (which we will prove later) and the procedure works the same way, although the midset is no longer a line. See Figure 4.3.

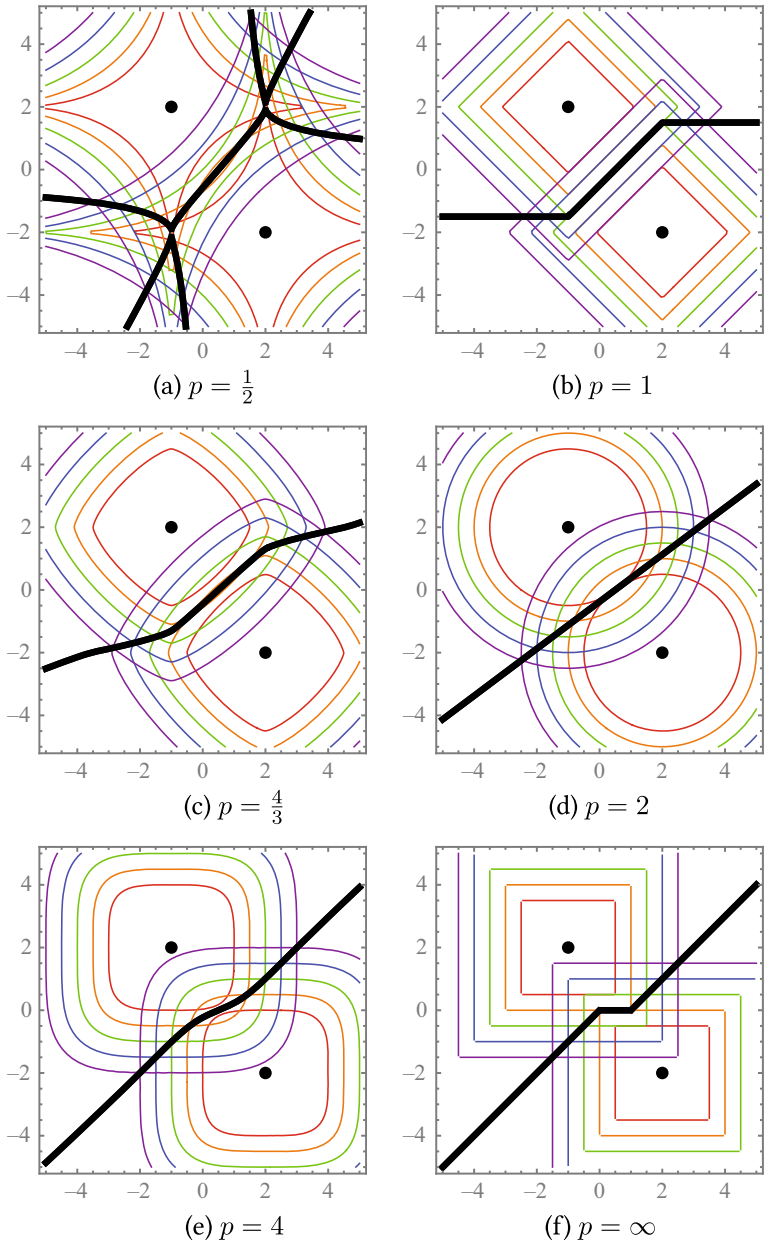


Figure 4.3 Construction of midsets of $(2, -2)$ and $(-1, 2)$ using p-circles

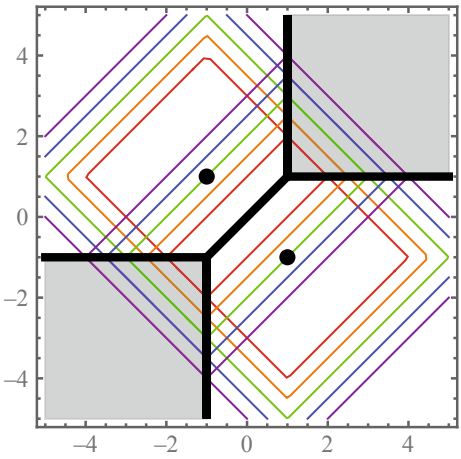


Figure 4.4 Constructing the taxicab midset of the points $(1, -1)$ and $(-1, 1)$

One can think of the midset by starting with the midpoint between the two points, which is clearly in the midset, and then find a direction that moves away from both points at the same rate. There may be a jog inside the rectangle the points determine, but the midset eventually straightens out.

Now consider this question with points that span a square, such as $A(-2, 2)$ and $B(2, -2)$. The taxicab analysis is the same except for what happens after we leave the square. We still can move horizontally to escape A and B at the same rate, but we could also move vertically, or *any direction in between*. We have more options because the triangle inequality is no longer strict for non-collinear points and the midset is two infinite regions connected by a segment. We see in Figure 4.4 that this can happen because taxicab circles can intersect along a line segment; this does not happen for $p > 1$. In that case, it is easy to see that $y = x$ is always a solution to the equation

$$d_p((x, y), A)^p = |x-2|^p+|y+2|^p = |x+2|^p+|y-2|^p = d_p((x, y), B)^p, \quad (4.2)$$

but that solution is not complete if $p = 1$.

Coming back to Euclid, we can *define* the perpendicular bisector as a midset, but our inability to do this for $p \neq 2$ reflects the general problem with working with angles in the non-Euclidean cases.

The definition of midset extends naturally from points to arbitrary sets when we define the distance from a point P to a set $\mathcal{S}$ as $\min\{|PX| \mid X \in \mathcal{S}\}$, the shortest distance from P to a point in $\mathcal{S}$.

Consider the distance from a point to a line, which we know in Euclidean geometry is the perpendicular distance. Observe that the p-distance between two distinct fixed points is strictly increasing in p and thus shortest paths remain shortest as p varies. In particular, the shortest path between two points is still a line segment (but again note that the uniqueness collapses at the extreme case $p = 1$). This argument applies to sets as well, so the distance from a point to a line remains the perpendicular distance.

One can then deduce that the midset of two lines is a pair of lines. If the lines intersect and $p = 2$, these lines bisect the two supplementary angles the intersecting

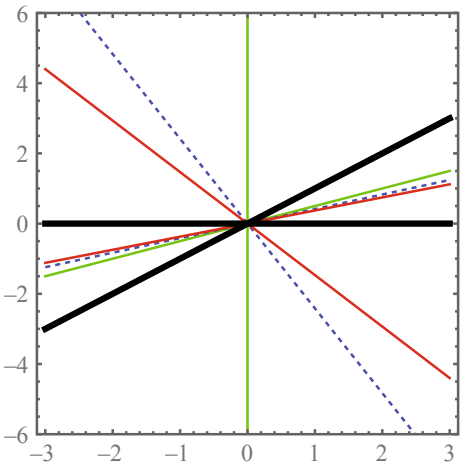


Figure 4.5 Midsets of the lines $y = 0$ and $y = x$ (black) for $p = 1$ (green), $p = 2$ (blue, dashed), and $p = 4$ (red)

lines form. But we know that p-distances are distorted at different angles and so those lines are no longer bisectors when $p \neq 2$. See Figure 4.5 and Exercise 4.8.

4.3 Semimetrics

While he definition requires $p \geq 1$ for the p-metric to be a metric, our analytic definitions do not. For example we can define the $\frac{1}{2}$-circle $\sqrt{|x|} + \sqrt{|y|} = 1$ and parameterize it to get $\frac{1}{2}$-trigonometric functions.

Various complications will arise, the first being that the functions are not differentiable everywhere. The defining CIVP

$$\begin{cases} x'(t) = -y(t)^{-\frac{1}{2}} \\ y'(t) = x(t)^{-\frac{1}{2}} \\ x(0) = 1 \\ y(0) = 0 \end{cases}$$

is ill-defined at $t = 0$ because $y(0) = 0$ but $x'(0) = -\frac{1}{\sqrt{y(0)}}$. We could get around this by dodging the singularity, but we will find a better work-around when we study inverse functions in Chapter 5.

A function that satisfies all conditions of a metric except the triangle inequality is a *semimetric*[4] which we can use to describe the cases $0 < p < 1$. We will use these

[4] In addition to semimetrics, there are also pseudometrics, quasimetrics, metametrics, and other terms that indicate different violations of the definition of metric.

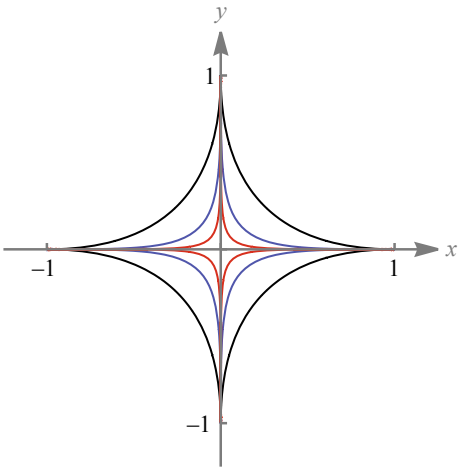


Figure 4.6 The semimetric "circles" $|x|^p + |y|^p = 1$ for $p = \frac{1}{2}$ (outer), $p = \frac{1}{3}$, and $p = \frac{1}{4}$ (inner)

cases as another way to capture our theme of understanding perfection by breaking it. For example, the reader is invited to conjecture a geometric requirement on metric circles by comparing unit circles for $p < 1$, $p = 1$, and $p > 1$, which we will prove in Chapter 18. See Figure 4.6.

We will focus on the metric cases in this book, but we will offer exercises for the reader to explore what goes wrong when $p < 1$ to better understand what goes right otherwise. We also include $p = \frac{1}{2}$ cases in the curve menagerie in Appendix A.

4.4 Conjugate metrics

Two real numbers $p, q > 1$ are *conjugate* if $\frac{1}{p} + \frac{1}{q} = 1$ and we will explore connections between p-metrics and their conjugate q-metrics. The following relationships among conjugate numbers are worth keeping in mind.

Theorem 4.1

If $p, q > 1$ are conjugate then

- $pq = p + q$
- $p = q(p - 1)$
- $p = \frac{q}{q-1}$
- $\frac{p}{q} = p - 1 = \frac{1}{q-1}$
- $p = q$ *if and only if* $p = q = 2$

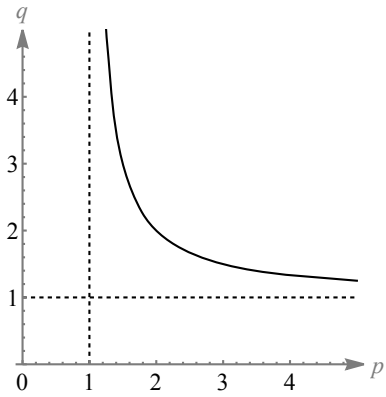


Figure 4.7 The relationship $\frac{1}{p} + \frac{1}{q} = 1$

Note that these expressions are all valid if p and q are exchanged. We also extend the definition to $p = 1$ by defining its conjugate to be $q = \infty$. The relationship is illustrated in Figure 4.7.[5]

The first relationship $pq = p + q$ allows us to think of conjugacy as numbers where the operations of addition and multiplication coincide. For example, conjugate p, q are the points where a logarithmic "freshman's dream" relationship $\log(p+q) = \log p + \log q$ holds.

Getting back to geometry, one elegant property of Euclidean circles is that the motion of a particle moving at constant speed on a circular path is described entirely by the path itself. That is, if a circle is parameterized by $(\cos(t), \sin(t))$ then its derivative $(-\sin(t), \cos(t))$ also parameterizes a circle.

Now consider the p-circle. The analogous parameterization is $u(t) = (\cos_p(t), \sin_p(t))$ with derivative $u'(t) = \left((-\sin_p(t))^{p-1}, (\cos_p(t))^{p-1}\right)$. Let q be conjugate to p. Then the distance from $u'(t)$ to the origin in the q-metric is

$$\left(|-\sin_p(t)|^{q(p-1)} + |\cos_p(t)|^{q(p-1)}\right)^{1/q} = \left(|\sin_p(t)|^p + |\cos_p(t)|^p\right)^{1/q} = 1.$$

Our expression works in the first quadrant, but as usual we must be careful about taking negative numbers to non-integral powers. After checking each quadrant case by case or extending by symmetry, we can confirm that this parameterization will indeed trace out the whole circle and we have proved

Theorem 4.2

The derivative of the parameterization $(\cos_p(t), \sin_p(t))$ *of the* p*-circle parameterizes the* q*-circle, where* p *and* q *are conjugate.*

This theorem offers some insight into the perfection of the Euclidean circle. There is this natural connection between conjugate circles as they parameterize each other's

[5] This is also the picture of what the $p = -1$ circle would be, were we so inclined to define it.

circular motion. This conjugate relationship is obscured when $p = 2$ because the conjugate circles are identical.

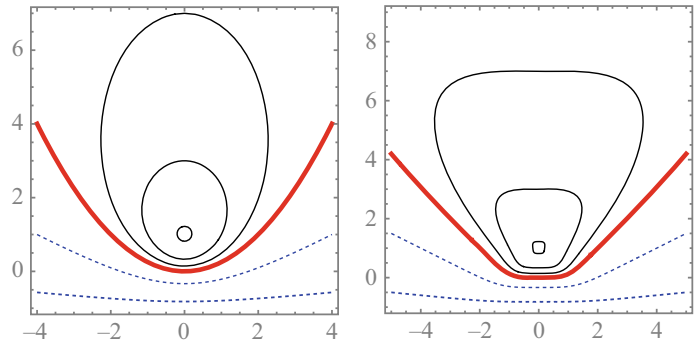
The relationship in play here is *duality*, which we will develop more formally when we view the p-length as the norm of a vector space in Chapter 18. For now we offer a promise that Theorem 4.2 is not the last connection we will find between conjugate p-metrics. The reader might again note the expressions in Theorem 4.1, as their appearance may indicate that something about duality is in play.

4.5 Conics and other curves

The p-metric suggests a new way to study the geometry of algebraic curves. A natural place to start is with conics.

Let F be a fixed point (the *focus*) and ℓ a fixed line (the *directrix*). A *conic* with eccentricity e is defined to be the set of points P such that the distance from P to F is equal to e times the distance from P to ℓ. Of course, we will be taking the distance to be various p-distances. The value of e determines the type of conic obtained: a hyperbola if $e > 1$, a parabola if $e = 1$, and an ellipse if $0 < e < 1$. The case $e = 0$ degenerates to a point, but the ellipses are clearly getting not just smaller but rounder as $e \to 0^+$ so it is natural to define a circle as a conic with eccentricity 0. Notice that the parabolic case $e = 1$ is another example of a midset, in this case between a point and a line, and thus conics can be considered a generalization of such midsets.

Let's set our focus F to be the point $(0, 1)$ and our directrix ℓ to be the horizontal line $y = -1$. The family of Euclidean conics is illustrated in Figure 4.8a. To generalize to other p-metrics, we use the p-distance to $(0, 1)$ given by $(|x|^p + |y - 1|^p)^{1/p}$. Note that the distance to the line $y = -1$ remains $|y + 1|$ regardless of the value of p. Conics for $p = 4$ are illustrated in Figure 4.8b.



(a) $p = 2$ (b) $p = 4$

Figure 4.8 Conics with focus $(0, 1)$, directrix $y = -1$, and eccentricities $\frac{1}{10}$, $\frac{1}{2}$, $\frac{3}{4}$ (ellipses, thin), 1 (parabola, thick), 2, and 10 (hyperbolas, dashed)

The asymmetry of the ellipses for $p = 4$ is striking. We see from Figure 4.10 why this is – the distances to the line and the point do not grow at the same rate as we move vertically.

We see that we will not recover the equation $\frac{x^4}{a^4} + \frac{y^4}{b^4} = 1$ from this definition, which is what we may have guessed for a general ellipse. The dashed curves in Figure 4.9 illustrate how curves of this form also defy yet another candidate for the definition of ellipse: the set of points whose distance to two foci sum to a constant.

The story gets worse when you consider that the p-metric is not rotationally invariant as we see with the parabolas in Figure 4.11.

We see from this excursion that multiple definitions of conics lead to different curves for general p-metrics. The definitions of conic coincide only if $p = 2$ and

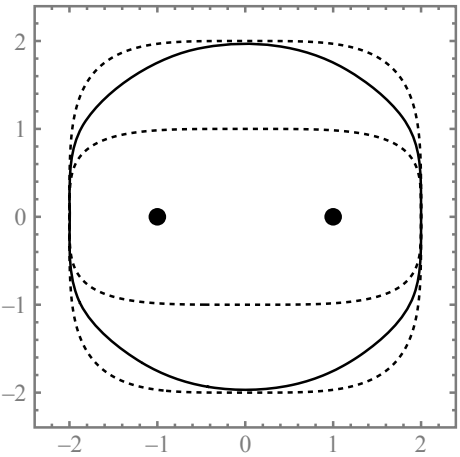


Figure 4.9 The solid curve is the set of points whose 4-distances to $(-1, 0)$ and $(1, 0)$ sum to 4. The dashed curves satisfy $x^4 + y^4 = 16$ and $\frac{x^4}{16} + y^4 = 1$

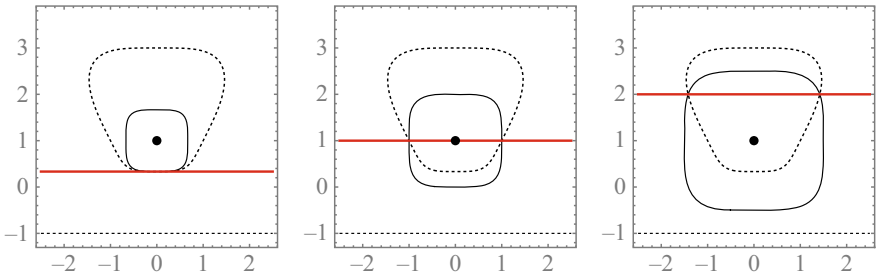


Figure 4.10 4-ellipse with focus $F(0, 1)$, directrix $y = -1$, and eccentricity 1/2. Solid curves are the 4-circle of radius k centered at F and the line of points distance $2k$ from the directrix

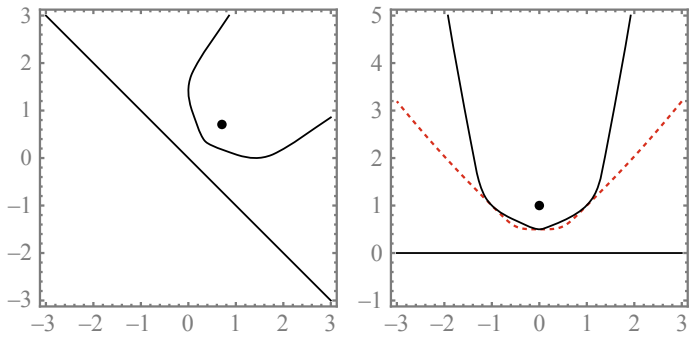


Figure 4.11 On the left is the 4-parabola with focus $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and directrix $y = -x$. On the right is the same set rotated 45° and superimposed atop the 4-parabola with directrix $y = 0$ and focus $(0, 1)$

these (possibly degenerate) conics completely classify polynomials in two variables of degree at most two.

The p-metric still offers a lens through which one can understand the geometry of higher order curves, more of which we defer to the menagerie, but the lens is a little cracked, making the analysis considerably more complicated.

Exercises

▶ **Exercise 4.1** *Find the distance between* $(0, 0)$ *and* $(3, 4)$ *in the* p*-metric for* $p = 4/3, 1, 2, 4, 10, \infty$. *(Later we will call this the* p*-norm of* $(3, 4)$.)

▶ **Exercise 4.2** *We may define the Paris metric* d_P *on the plane so that* $d_P(A, B)$ *is equal to the Euclidean distance between* A *and* B *if and only if they lie on the same ray from the origin. Otherwise it is the sum of their distances to the origin,* $d_P(A, B) = d(O, A) + d(O, B)$. *In vector notation, this is*

$$d_P(\boldsymbol{u}, \boldsymbol{v}) = \begin{cases} \|\boldsymbol{u} - \boldsymbol{v}\| & \boldsymbol{u} = \lambda \boldsymbol{v}, \lambda \geq 0 \\ \|\boldsymbol{u}\| + \|\boldsymbol{v}\| & \text{otherwise.} \end{cases}$$

The Paris metric is so named because it models the streets of Paris radiating from the Arc de Triomphe (Figure 4.12). Show that this is a metric on the plane and that the Paris distance between two points is greater than or equal to their Euclidean distance. When are they equal?[6]

[6] The Paris metric induces a completely different topology on the plane. It is an example of a *hedgehog space*, named for infinitely many spines emanating from a central point. Also called the *French railway metric* or *radial metric* [92].



Figure 4.12 Place Charles de Gaulle, Paris, France. The Paris metric models a distance in which going from one ray (street) to another requires passing through the origin (the Arc de Triomphe). The Champs-Élysées is visible as the negative y-axis. Image courtesy Institut national de l'information géographique et forestière, Licence Ouverte, via Wikimedia Commons

▶ **Exercise 4.3** *Verify Theorem 4.1 and extend it to the extremal cases* $p = 1$ *and* $p = \infty$.

▶ **Exercise 4.4** *Where exactly are those points of intersection in Figure 4.1?*

▶ **Exercise 4.5** *The fact that taxicab circles can intersect along a segment was necessary to create the pathological midset in Figure 4.4. But max norm* ($p = \infty$) *circles also have this property and we do not have this problem, as shown in Figure 4.3f. Explain.*

▶ **Exercise 4.6** *Explicitly calculate the* $p = 4$ *midset of* $(1, -1)$ *and* $(-1, 1)$ *by writing out the expression for and solving for the points* (x, y) *satisfying*

$$d_4((x, y), (1, -1)) = d_4((x, y), (-1, 1)).$$

▶ **Exercise 4.7** *How many points of intersection can two circles in the* $p = \frac{1}{2}$ *metric have? Use this as starting point to study midsets and conics for the semimetric* $p = \frac{1}{2}$. *Also note Figure 4.3a.*

▶ **Exercise 4.8** *Verify that the p-midset of two lines is a pair of lines and find a general formula in terms of $p \geq 1$ for the slopes of the lines that form the midsets of the two lines in Figure 4.5. What more can you say about these and other similar midsets?*

▶ **Exercise 4.9** *Find an expression for the height of the p-ellipse with eccentricity e, focus $(0, 1)$, and directrix $y = -1$.*

▶ **Exercise 4.10** *Explain how the midset of two points can be considered as a branch of a degenerate hyperbola.*

▶ **Exercise 4.11** *In this exercise we introduce ellipse geometry, which will be a running example*[7] *and we will refer to this exercise again as we develop more tools for analyzing and interpreting functions defined like this.*

Let $\mathcal{E}$ be the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for $a, b > 0$.

1. *Find a distance function $d_{\mathcal{E}}$ for which $\mathcal{E}$ is the unit circle of the metric.*
2. *Explicitly solve the CIVP*
$$\begin{cases} x'(t) = -\dfrac{a}{b} y(t) \\ y'(t) = \dfrac{b}{a} x(t) \\ x(0) = a \\ y(0) = 0. \end{cases} \tag{4.3}$$
We use these solutions to define the ellipse trigonometric functions $\cos^{\mathcal{E}}(t) = x(t)$ and $\sin^{\mathcal{E}}(t) = y(t)$.
3. *Verify that $(\cos^{\mathcal{E}}(t), \sin^{\mathcal{E}}(t))$ parameterizes $\mathcal{E}$.*
4. *Show that $\cos^{\mathcal{E}}(t)$ and $\sin^{\mathcal{E}}(t)$ are solutions to the second-order differential equation (3.6) for $p = 2$.*
5. *Let $a, b > 0$. Define $x(t) = \cos_p^{\mathcal{E}}(t)$, $y(t) = \sin_p^{\mathcal{E}}(t)$ for $p > 0$ in the first quadrant as the solutions to the CIVP*
$$\begin{cases} x'(t) = -\dfrac{a}{b} y(t)^{p-1} \\ y'(t) = \dfrac{b}{a} x(t)^{p-1} \\ x(0) = a \\ y(0) = 0 \end{cases}. \tag{4.4}$$
Show that these functions (with suitably expanded domains) parameterize a curve $\mathcal{E}_p$ with equation $\frac{x^p}{a^p} + \frac{y^p}{b^p} = 1$. We will still refer to this as generalized ellipse geometry, but remember that this is just a generalization of the equation and that this curve does not retain the geometric definition of an ellipse.

[7] Other exercises and references invoking this geometry can be found via the index entry for "ellipse geometry." This is not to be confused with *elliptic* integrals, which will play a prominent role later in this book.

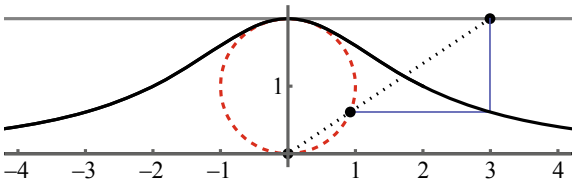


Figure 4.13 Witch of Agnesi

▶ **Exercise 4.12** *The witch of Agnesi*[8] $\mathcal{W}$ *is defined as follows. Let* $\mathcal{C}$ *be the unit circle centered at* $(0, 1)$, ℓ *the line* $y = 2$, *and* O *the origin* $(0, 0)$. *Let* P *be a point on* $\mathcal{C}$ *and* Q *the intersection of the line* $\overleftrightarrow{OP}$ *with* ℓ. *Finally, let* R *be a point that is aligned horizontally with* P *and vertically with* Q. *Then* $\mathcal{W}$ *is the set of all such points* R. *See Figure* 4.13.

We generalize this construction to $\mathcal{W}_p$ *in the obvious way, replacing "circle" with "squircle." Figure* A.3 *shows more of the coven.*

1. *Explain geometrically why* $(0, 2)$ *and* $(2, 1)$ *are in* $\mathcal{W}_p$.
2. *Show that* $\mathcal{W}_p$ *is parameterized by* $\left(\frac{2\cos_p(t)}{1+\sin_p(t)}, 1 + \sin_p(t)\right)$.
3. *Show that* $\mathcal{W}_p$ *satisfies the Cartesion relationship*

$$x^p = 2^p \left(y^{-p} - (1 - y^{-1})^p\right)$$

for $x > 0, 0 < p < \infty$.

Projects

Project 4.1 (Max geometry)
Taxicab geometry is probably the best-studied of the non-Euclidean p-metrics. Besides having numerous applications, it is also highly accessible. Most of the relevant curves and graphs of functions are composed of line segments so one can explore the geometry quite deeply with only some graph paper and a ruler without ever seeing a differential equation. (The downside is that a formal analytic approach devolves into a morass of absolute values and piecewise expressions.) E. F. Krause's *Taxicab Geometry: An Adventure in Non-Euclidean Geometry* [55] offers a beautiful elementary journey of discovery of this geometry.

[8] The witch of Agnesi has an interesting history [56, 72, 104]. It is named for Maria Gaetana Agnesi who studied (but did not discover) this curve in her 1748 treatise *Instituzioni analitiche* [3], the oldest surviving mathematical publication written by a woman. The name "witch" has nothing to do with the curve – it is a mistranslation that stuck. The curve is essentially the probability density function for the Cauchy distribution, which we will investigate in Project 6.2.

Compose a similar exploration for the max geometry $p = \infty$. Like its taxicab dual, its circles are Euclidean squares and many natural curves are polygonal. Develop trigonometric functions, conics, and other curves for the max norm as far as you can without any calculus.

Project 4.2 (Squircular wheels)
Perhaps the founding result of applied geometry is that a round wheel rolls smoothly on flat ground. Prove it.

Now we generalize: if we changed the ground, could a vehicle roll smoothly with square wheels? We can set up a differential equation to model this situation and discover that terrain made of perfectly positioned catenaries will work, as Leon Hall and Stan Wagon demonstrate in [47]. See Figure 4.14. They and others have since constructed physical square-wheeled tricycles that ride smoothly on catenary tracks.

Of course, we would call these the $p = 2$ and $p = \infty$ tricycles. For general p, what kind of terrain would we need to smoothly ride a vehicle with squircular wheels? Do the curves retain any geometric properties of catenaries?

Project 4.3 (Rendezvous numbers)
If you choose any finite set of points on the unit circle, there is guaranteed to be a point somewhere on the circle whose average distance to the points you chose is exactly $\frac{4}{\pi}$. Perhaps more surprisingly, $\frac{4}{\pi}$ is the *only* number with this

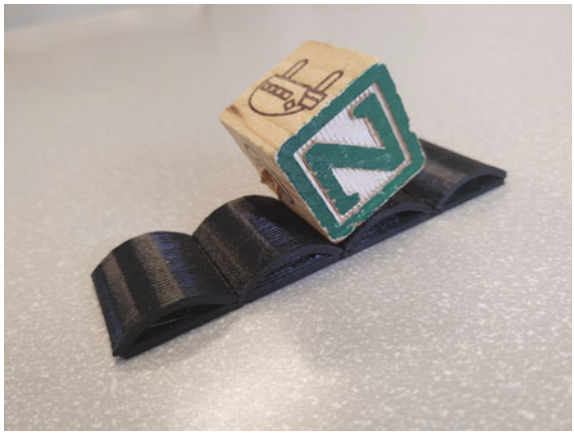


Figure 4.14 A square wheel rolls smoothly on a track of catenary arcs

property. And most surprising: every compact metric space has a unique number like this. The formal theorem, due to O. Gross [45], is as follows:

> **Theorem**. *For any compact metric space X there is a unique number r with the following property: for any finite set $\{x_1, \ldots, x_n\}$ of points in X there is some point $z \in X$ such that the average of the distances from z to each of the x_i is r.*

We call the number r above the *rendezvous number* of the metric space. It is a curious but elusive quantity, and difficult to compute. Cleary, Morris, and Yost [19] calculate the Euclidean rendezvous numbers of several sets, including our leading example of the circle, but their methods exploit the symmetries of these sets in essential ways and do not offer a general method. The rendezvous number of an ellipse, for example, is unknown.

Since the result holds in any metric, p-metrics offer new territory on which to explore the rendezvous number. Can we find the p-rendezvous numbers of any interesting sets? Part of the computational challenge with rendezvous numbers is that average Euclidean distance is expressed as a sum of roots, making the extreme cases $p = 1$ and $p = \infty$ particularly inviting.

Check for updates

Inverse squigonometric functions

5

In Chapter 3, we noted that the period of the 4-sine and 4-cosine is longer than the usual $2\pi \approx 6.28$ and that the first vertical asymptote in the tangent graph is well beyond the classical $\pi/2 \approx 1.57$. The plots in Figure 5.1 bear out that the period of the p-sine and p-cosine seem to increase with p from about 4 when $p = 1$ to nearly 8 when $p = 10$. Recall from Figure 2.4 that our first graph here is the $p = 1$ graph when our parameter t represents twice the area of a sector of a taxicab circle swept out by a ray from the origin, not the angle.

This would indicate that π should be considered as a function of p, rather than as a constant. We can determine π as a function of p by first defining inverse functions for our p-sine and p-cosine functions. In turn, in the next chapter, we will be able to determine intervals of definition for our sine and cosine functions in (3.2) and then extend those functions to be defined on the entire real line.

5.1 Inverting sine and cosine

Let $y = \sin_p(t)$. Then $dy/dt = \cos_p(t)^{p-1} = (1 - y^p)^{\frac{p-1}{p}}$. This is a separable differential equation and, since $\sin_p(0) = 0$, we can solve for t to obtain

$$t = \arcsin_p(y) = \int_0^y \frac{1}{(1 - u^p)^{\frac{p-1}{p}}}\, du, \tag{5.1}$$

where we assume for now that $0 \le y \le 1$. In a similar way, with $x = \cos_p(t)$, since $\cos_p(0) = 1$, we can write

$$t = \arccos_p(x) = \int_x^1 \frac{1}{(1 - u^p)^{\frac{p-1}{p}}}\, du, \tag{5.2}$$

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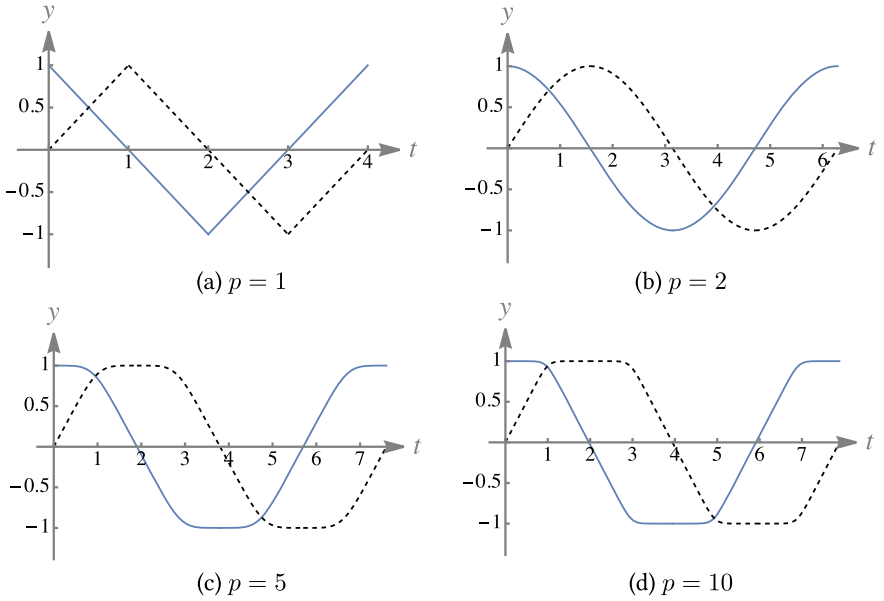


Figure 5.1 Graphs of $y = \cos_p(t)$ (solid) and $y = \sin_p(t)$ (dashed) for various p-values.

with $0 \le x \le 1$ for the time being. This of course leads to two new derivative formulas:

$$\frac{d}{dx}\arcsin_p(x) = \frac{1}{(1-x^p)^{\frac{p-1}{p}}} \quad \text{and} \quad \frac{d}{dx}\arccos_p(x) = -\frac{1}{(1-x^p)^{\frac{p-1}{p}}}. \tag{5.3}$$

The first two exercises of this chapter reinforce the idea that the sine and cosine functions we have defined give an areal parameterization of the unit p-circle.

5.2 Inverting the rest

In Exercise 3.4 of Chapter 3, we proved that $1 + \tan_p(t)^p = \sec_p(t)^p$ while (3.4) told us that $\frac{d}{dt}\tan_p(t) = \sec_p(t)^2$. We can combine the two to rewrite the derivative formula for the p-tangent as

$$\frac{d}{dt}\tan_p(t) = (1 + \tan_p(t)^p)^{\frac{2}{p}}.$$

Separating variables and integrating as before and noting that $\tan_p(0) = 0$ gives us the definition for the p-arctangent:

$$\arctan_p(x) = \int_0^x \frac{1}{(1+u^p)^{\frac{2}{p}}}\,du. \tag{5.4}$$

A similar calculation, and the fact that $\cot_p(t) = \frac{\cos_p(t)}{\sin_p(t)} \to \infty$ as $t \to 0^+$ gives us the definition

$$\operatorname{arccot}_p(x) = \int_x^\infty \frac{1}{(1+u^p)^{\frac{2}{p}}}\,du.$$

Now we can use (3.5) to write the derivative of the p-secant in terms of only itself:

$$\begin{aligned}\frac{d}{dt}\sec_p(t) &= \sec_p(t)^2 \sin_p(t)^{p-1} \\ &= \sec_p(t)^2 \cdot (1-\cos_p(t)^p)^{\frac{p-1}{p}} \\ &= \sec_p(t)^2 \left(1 - \frac{1}{\sec_p(t)^p}\right)^{\frac{p-1}{p}} \\ &= \sec_p(t)^{2-(p-1)} \left(\sec_p(t)^p - 1\right)^{\frac{p-1}{p}}\end{aligned}$$

Thus we can once again separate and integrate to get

$$\operatorname{arcsec}_p(x) = \int_1^x \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du, \quad x \geq 1.$$

In a similar manner, we can obtain

$$\operatorname{arccsc}_p(x) = \int_x^\infty \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du, \quad x \geq 1.$$

Thus we have the following derivatives for the inverse squigonometric functions:

$$\begin{aligned}\frac{d}{dx}\operatorname{arcsin}_p(x) &= \frac{1}{(1-x^p)^{\frac{p-1}{p}}}, & \frac{d}{dx}\operatorname{arccos}_p(x) &= -\frac{1}{(1-x^p)^{\frac{p-1}{p}}}, \\ \frac{d}{dx}\operatorname{arctan}_p(x) &= \frac{1}{(1+x^p)^{\frac{2}{p}}}, & \frac{d}{dx}\operatorname{arccot}_p(x) &= -\frac{1}{(1+x^p)^{\frac{2}{p}}}, \\ \frac{d}{dx}\operatorname{arcsec}_p(x) &= \frac{x^{p-3}}{(x^p-1)^{\frac{p-1}{p}}}, & \frac{d}{dx}\operatorname{arccsc}_p(x) &= -\frac{x^{p-3}}{(x^p-1)^{\frac{p-1}{p}}}\end{aligned}$$

5.3 Composition of squig and inverse squig functions

The inverse squigonometric functions we defined are meant to be true inverse functions in the sense that composing a squigonometric function with its inverse returns the original argument. For example,

$$\sin_p(\arcsin_p(x)) = x \quad \text{and} \quad \arcsin_p(\sin_p(t)) = t$$

for values of x and t to be determined. But we would also like to figure out what happens when we compose a p-trigonometric function with any inverse p-trigonometric function.

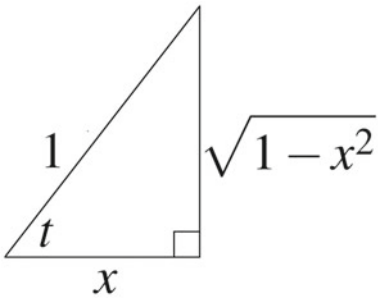


Figure 5.2 Proof that $\sin(\arccos(x)) = \sqrt{1-x^2}$

If we compose a trigonometric with an inverse trigonometric function of a real variable x, we get an algebraic function of x – all the traces of trigonometry are wiped from the expression.

One way to establish such relationships is through differentiating and integrating. For example, if we wish to determine an algebraic equivalent to $\sin(\arccos(x))$, we note that

$$\frac{d}{dx}\sin(\arccos(x)) = \cos(\arccos(x)) \cdot \frac{d}{dx}\arccos(x) = -\frac{x}{\sqrt{1-x^2}}.$$

We can integrate the last expression via substitution, giving us that $\sin(\arccos(x)) = \sqrt{1-x^2}$ (since $\arccos(1) = 0$).

Right triangle trigonometry

These relationships are often illustrated by the use of right triangles. We of course remember the mnemonic SOH CAH TOA, shorthand for

$$\sin(t) = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos(t) = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan(t) = \frac{\text{opposite}}{\text{adjacent}}.$$

A proof that $\sin(\arccos(x)) = \sqrt{1-x^2}$ consists of returning to the definition of trigonometric functions on right triangles. We let $t = \arccos(x)$. Then $x = \cos(t) = \frac{x}{1}$. So we draw a right triangle with angle t, adjacent side x, and hypotenuse 1. By the Pythagorean theorem, the measure of the remaining side of the triangle (the side opposite angle t) must be $\sqrt{1-x^2}$. (See Figure 5.2.)

So to compute $\sin(\arccos(x)) = \sin(t)$, we simply take the ratio of the side opposite t to the hypotenuse and get that $\sin(\arccos(x)) = \frac{\sqrt{1-x^2}}{1}$.

The calculus approach

For general p-trigonometric functions and their inverses, we can use the calculus trick to simplify compositions of those functions.

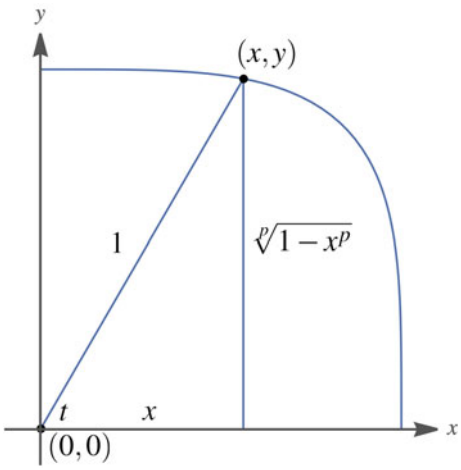


Figure 5.3 A right triangle inscribed in the first quadrant of a p-circle that shows $\sin_p(\arccos_p(x)) = \sqrt[p]{1-x^p}$

If we wish to find the value of $\sin_p(\arccos_p(x))$, for example, we can use (3.2) and (5.3) and the chain rule to write

$$\frac{d}{dx}\sin_p(\arccos_p(x)) = \big(\cos_p(\arccos_p(x))\big)^{p-1} \cdot \frac{d}{dx}\arccos_p(x) = -\frac{x^{p-1}}{(1-x^p)^{\frac{p-1}{p}}}.$$

We can integrate the last expression via the substitution $u = 1 - x^p$ to get that

$$\sin_p(\arccos_p(x)) = \sqrt[p]{1-x^p}.$$

The p-metric approach

As a visual aid, it would be nice to use right triangles in some way. It turns out that we can do this if we recall the p-metrics defined in Chapter 4. We can think of the *unit p-circle* $|x|^p + |y|^p = 1$ as the locus of points that are 1 unit away from the origin *in the p-metric* (Figure 5.3). Right triangles inscribed in the unit p-circle with legs parallel to the coordinate axes then follow a pth-power Pythagorean theorem: If the legs of a right triangle have measures a and b, and the hypotenuse has measure c, then $|a|^p + |b|^p = |c|^p$.

So we can use right triangles to simplify compositions of squigonometric and inverse squigonometric functions in a familiar manner. (We can still use SOH CAH TOA for every p!) For example, suppose we wish to simplify $\cos_4(\arccsc_4(x))$. Let $t = \arccsc_4(x)$. Then $x = \csc_4(t)$. This implies that $\frac{1}{x} = \sin_4(t)$ and we can draw the triangle in Figure 5.4, using our Pythagorean Theorem for $p = 4$.

We can then write

$$\cos_4(\arccsc_4(x)) = \cos_4(t) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\sqrt[4]{x^4-1}}{x}.$$

A table of these compositions can be found in Appendix B.3.

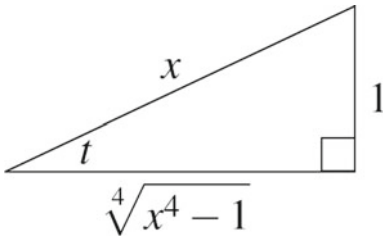


Figure 5.4 Right triangle proof that $\cos_4(\operatorname{arccsc}_4(x)) = \frac{\sqrt[4]{x^4-1}}{x}$

When we draw these right triangles, it is wise to remember that t is simply a parameter value and does not represent an angle as when $p = 2$, though there is a one-to-one correspondence between t and the angle $\theta(t)$: $\tan(\theta(t)) = \frac{y(t)}{x(t)}$. The parameter t actually represents twice the area, but it is only in the Euclidean $p = 2$ case where this happens to be the angle measure as well.

The identities for squigonometric functions are similar enough to their trigonometric analogs that we can find relations among them. For example,

$$\frac{d}{dt}\operatorname{arcsin}_2(\sin_4(t)^2) = \frac{1}{\sqrt{1-\sin_4(t)^4}} \cdot 2\sin_4(t)\cos_4(t)^3 = 2\sin_4(t)\cos_4(t).$$

Exercises

▶ **Exercise 5.1** *Show that the area of the sector of the p-circle between the points $(0, 0)$, $(1, 0)$ and $(x, y) = (\cos_p(t), \sin_p(t))$ is $\arcsin_p(y)/2$ square units.*

▶ **Exercise 5.2** *Show that the area of the sector given in the previous exercise is also $\arccos_p(x)/2$ square units.*

▶ **Exercise 5.3** *Show that for $x \in [0, 1]$, $\arcsin_1(x) = x$ and $\arcsin_\infty(x) = x$.*

▶ **Exercise 5.4** *Evaluate the following:*

1. $\cos_4(\arctan_4(\frac{5}{12}))$
2. $\sin_3(\arctan_3(\frac{7}{4}))$
3. $\cos_5(\arcsin_5(\frac{x}{3}))$ *for* $-3 \le x \le 3$.

▶ **Exercise 5.5** *Find an expression for the p-metric perimeter of the triangle with vertices at $(0, 0)$, $(1, 0)$, and $(0, 1)$. Then rotate the triangle by $45°$ and calculate its perimeter. For which $p \ge 1$ are they the same?*

▶ **Exercise 5.6** *We observed in Section 4.1 that solving the CIVP (1.1) is obstructed by a singular point at 0 in the semimetric case $0 < p < 1$. One way around this is to take the integral (5.1) as a definition.*

1. *Show that* $\arcsin_{\frac{1}{2}}(y) = y - \frac{2}{3}y^{\frac{3}{2}}$ *for* $0 \leq y \leq 1$.
2. *Show that* $\sin_{\frac{1}{2}}(t)$ *is algebraic on some interval. (The precise interval will be articulated in Exercise 6.4. The reader is invited to find the explicit formula for* $\sin_{\frac{1}{2}}(t)$ *but a computer algebra system is strongly recommended. The full expression is in Appendix B.)*
3. *Under what circumstances will* $\sin_p(t)$ *be an algebraic function for* $0 < p < 1$*?*
4. *Explore the other* $\frac{1}{2}$*-trigonometric functions, starting with* $\arctan_{\frac{1}{2}}(x)$.

The many values of Pi

6

By now, we have done a lot in the way of formal manipulation of the p-sine and p-cosine functions, finding derivatives and some integrals, while avoiding dealing with matters like domains or singularities of the squigonometric functions. We now formally determine these.

The definitions (5.1) and (5.2) lead organically to defining π as a function of the parameter p. We note that in the classical $p = 2$ case, we have that $\pi/2 = \arcsin 1 = \arccos 0$. We then analogously define

$$\pi_p = 2\int_0^1 \frac{1}{(1-u^p)^{\frac{p-1}{p}}}\,du = 2\arcsin_p(1) = 2\arccos_p(0). \tag{6.1}$$

This gives us that $\sin_p(\pi_p/2) = 1$ and $\cos_p(\pi_p/2) = 0$.

The p-circle has lines of symmetry wherever the radius function $r(t)$ is at a maximum or minimum. Recall that $r(t) = \sqrt{x(t)^2 + y(t)^2}$. Thus

$$\begin{aligned} r'(t) &= \frac{d}{dt}\sqrt{x(t)^2 + y(t)^2} \\ &= \frac{x(t)x'(t) + y(t)y'(t)}{\sqrt{x(t)^2 + y(t)^2}} \\ &= \frac{-x(t)y(t)^{p-1} + y(t)x(t)^{p-1}}{\sqrt{x(t)^2 + y(t)^2}} \\ &= \frac{x(t)y(t)[x(t)^{p-2} - y(t)^{p-2}]}{r(t)} \end{aligned}$$

So, in the first quadrant, $r'(t) = 0$ whenever $x(t) = 0$, $y(t) = 0$ or $x(t) = y(t)$; that is, when $t = 0$, $t = \pi_p/2$ or $t = \pi_p/4$. In the last case, we obtain $\sin_p(\pi_p/4) = \cos_p(\pi_p/4) = \sqrt[p]{1/2}$.

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p	π_p
1	2.000000000
2	3.141592654
3	3.533277501
4	3.708149355
5	3.800600556
6	3.855242593
7	3.890174738
8	3.913843288
9	3.930614379
10	3.942927898

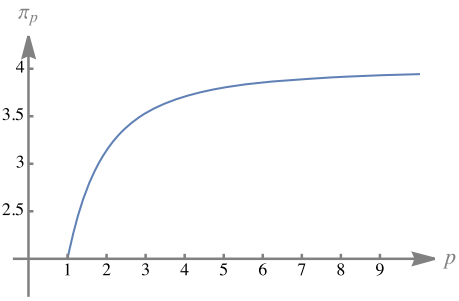


Figure 6.1 π_p values via numerical integration

In Exercise 6.4, we are asked to show that the area of the unit p-circle is π_p square units. Assuming this, we know that $\pi_1 = 2$, $\pi_2 = \pi$, and π_p increases towards 4 as $p \to \infty$. While the integrals for π_p can be computed directly for $p = 1$ and $p = 2$, they are intractable for other $p > 1$ and so we use numerical integration (see Figure 6.1).

The idea of π having nontraditional values goes back as far as Biblical times. The Old Testament tells of Solomon building the altar for the Temple in Jerusalem in about 950 BCE [30]:

> He made the Sea of cast metal, circular in shape, measuring ten cubits from rim to rim and five cubits high. It took a line of thirty cubits to measure around it.
>
> — 1 Kings 7:23 (also 2 Chronicles 4:2)

In this case we would have $\pi = 3$, which is achieved when $p \approx 1.79148$. This was not even accurate for its time as the Babylonians in the 18th century BCE computed π as 3.125, and the Egyptian Rhind Papyrus from 1650 BCE came up with an approximation of 3.16 [82].

An 1897 bill introduced in the Indiana State Legislature dragged specious values of π into the spotlight. The bill sprang from a doctor's claim that he had found a way to "square the circle." That is, the doctor thought he had found a way to construct a square with the same area as a given circle using a finite number of steps using only straightedge and compass. This construction had been proven impossible by Lindemann 15 years earlier when he showed the transcendence of π. The doctor's article on his "proof," published essentially as a letter to the editor in the *American Mathematical Monthly* [41], contained several contradictory values of π, including 4 and 3.2. The doctor had the legislature consider accepting his circle-squaring technique as a royalty-free gift for educational purposes in Indiana schools, along with free use of his various values of π. The Indiana House of Representatives passed the bill unanimously (most representatives likely never read the bill but they liked gifts); the Senate wisely tabled it [26].

6.1 Special functions

Though not expressible in terms of elementary functions, π_p can be written in terms of *special functions* called the beta and gamma functions [4].

The *beta function* is a function of two variables defined in terms of an integral:

$$\mathrm{B}(x, y) = \int_0^1 u^{x-1}(1-u)^{y-1}\, du.$$

The *gamma function* is also defined as an integral:

$$\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\, dt.$$

We easily have that

$$\Gamma(1) = \int_0^\infty e^{-t}\, dt = \Big[-e^{-t}\Big]_0^\infty = 1.$$

We also have

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t}\, dt.$$

We integrate this by parts with $u = t^x$ and $dv = e^{-t}\, dt$. This gives us $du = xt^{x-1}\, dt$, $v = -e^{-t}$ and

$$\Gamma(x+1) = \Big[-t^x e^{-t}\Big]_0^\infty + x\int_0^\infty t^{x-1}e^{-t}\, dt$$

$$\Gamma(x+1) = x\Gamma(x). \tag{6.2}$$

This last equation is the functional relationship for the gamma function. In particular, if n is a positive integer, we write with $n = x + 1$,

$$\begin{aligned}
\Gamma(n) &= (n-1)\Gamma(n-1)\\
&= (n-1)(n-2)\Gamma(n-2)\\
&= (n-1)(n-2)(n-3)\Gamma(n-3)\\
&\vdots\\
&= (n-1)(n-2)(n-3)\cdots 2\cdot 1\cdot \Gamma(1)\\
&= (n-1)(n-2)(n-3)\cdots 2\cdot 1\\
&= (n-1)!
\end{aligned}$$

So the gamma function can be thought of as an extension of the factorial to all positive real numbers. (The functional relationship can be used to extend the gamma function further to all real numbers except for 0, -1, -2, -3, …). The above implies that $0! = (1-1)! = \Gamma(1) = 1$.

There is a relationship between the beta and gamma functions:

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}. \tag{6.3}$$

Another identity that will be used in Chapter 14 is

$$\mathrm{B}(x,\,y) = \int_0^\infty t^{x-1}(1+t)^{-(x+y)}\,dt \tag{6.4}$$

for $x,\,y \geq 0$. Proofs of these facts are found in Appendix D.

6.2 A simple definition

If we make the change of variable $u = t^p$ in equation (6.1), we can use (6.3) to get

$$\pi_p = \frac{2}{p}\int_0^1 u^{\frac{1}{p}-1}(1-u)^{\frac{1}{p}-1}\,du = \frac{2}{p}\mathrm{B}(\tfrac{1}{p},\,\tfrac{1}{p}) = \frac{2}{p}\,\frac{\left[\Gamma(\frac{1}{p})\right]^2}{\Gamma(\frac{2}{p})}. \tag{6.5}$$

We can use the gamma functional relationship (6.2) with the *Euler reflection formula*

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)} \tag{6.6}$$

to derive some facts and some particular values for the gamma functions, and exact values for some π_p's. See [7] for a proof.

Legendre duplication formula

The table and graph in Figure 6.1 hint that $\pi_p \to 4$ as $p \to \infty$. We can formally prove this using the *Legendre duplication formula*, which obtains the value of $\Gamma(2z)$ from the known values of $\Gamma(z)$ and $\Gamma(z+\frac{1}{2})$:

$$\Gamma(2z) = \frac{\Gamma(z)\Gamma(z+\frac{1}{2})}{2^{1-2z}\sqrt{\pi}}. \tag{6.7}$$

A proof of (6.7) that uses the fact that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (see Exercise 6.10) can be found in Appendix D. With the result in hand, we can use (6.5) to write (see [42])

$$\pi_\infty = \lim_{p\to\infty}\pi_p = \lim_{p\to\infty}\frac{2}{p}\,\frac{\left[\Gamma(\frac{1}{p})\right]^2}{\Gamma(\frac{2}{p})} = 2\lim_{p\to\infty}\frac{\Gamma(1+\frac{1}{p})\Gamma(\frac{1}{p})}{\Gamma(\frac{2}{p})},$$

where we have used (6.2) in the last equation. As $p \to \infty$, $\Gamma(1+\frac{1}{p}) \to \Gamma(1) = 1$. Thus

$$\pi_\infty = 2\lim_{p\to\infty}\frac{\Gamma(\frac{1}{p})}{\Gamma(\frac{2}{p})} = 2\lim_{x\to 0^+}\frac{\Gamma(x)}{\Gamma(2x)},$$

where we have set $x = \frac{1}{p}$. We can now use (6.7) on $\Gamma(2x)$ to obtain

$$\pi_\infty = 2\lim_{x\to 0^+}\frac{\Gamma(x)}{\Gamma(2x)} = 2\lim_{x\to 0^+}\frac{\Gamma(x)\cdot 2^{1-2x}\sqrt{\pi}}{\Gamma(x)\Gamma(x+\frac{1}{2})} = 2\cdot\frac{2\sqrt{\pi}}{\Gamma(\frac{1}{2})} = 4,$$

by Exercise 6.10.

6.3 Exact values for the gamma function

We can use the formulas (6.5), (6.2), and (6.6) to get exact values for two particular values of the gamma function. Since

$$\pi_3 = \frac{2\left[\Gamma(\frac{1}{3})\right]^2}{3\Gamma(\frac{2}{3})} = \frac{\left[\Gamma(\frac{1}{3})\right]^3}{\sqrt{3}\,\pi} \quad \text{and} \quad \pi_4 = \frac{2\left[\Gamma(\frac{1}{4})\right]^2}{4\Gamma(\frac{1}{2})} = \frac{\left[\Gamma(\frac{1}{4})\right]^2}{2\sqrt{\pi}},$$

we obtain the exact values

$$\Gamma(\tfrac{1}{3}) = \sqrt[3]{\pi_3 \pi \sqrt{3}} \quad \text{and} \quad \Gamma(\tfrac{1}{4}) = \sqrt{2\pi_4 \sqrt{\pi}}. \tag{6.8}$$

Since these two values of the gamma function have long been considered "unknown" (although the value for $\Gamma(\frac{1}{4})$ appears in [89]), a good source of classroom discussion could encompass whether these values are now solved. (Both values have been proved to be transcendental [13,98].) We will look at additional values of the gamma function and an application in Chapter 10.

6.4 Domains and ranges

With the various π_p's in hand, we can now formally set the domain of solution for (3.2) and establish the periodic nature of the p-sine and p-cosine.

We first declare that the solution functions of the CIVP (3.2), the p-sine and p-cosine, are valid for $0 \le t \le \pi_p/2$, and the same is true for the equation $\cos_p(t)^p + \sin_p(t)^p = 1$. This would parameterize the first quadrant of the p-circle.

We can then extend the function definitions via symmetry, first to $0 \le t \le \pi_p$, then to $0 \le t < 2\pi_p$ and finally, via periodicity, to the entire real line:

$$\sin_p(t) = \begin{cases} \sin_p(\pi_p - t) & \pi_p/2 < t \le \pi_p, \\ -\sin_p(2\pi_p - t) & \pi_p < t < 2\pi_p, \end{cases}$$

$$\cos_p(t) = \begin{cases} -\cos_p(\pi_p - t) & \pi_p/2 < t \le \pi_p, \\ \cos_p(2\pi_p - t) & \pi_p < t < 2\pi_p. \end{cases}$$

We finish by setting $\sin_p(t + 2\pi_p k) = \sin_p(t)$ for $k \in \mathbb{Z}$ and do the same for the p-cosine function.

As the p-sine and p-cosine parameterize the unit p-circle, we have that $-1 \le \sin_p(t) \le 1$ and $-1 \le \cos_p(t) \le 1$ for all $p \ge 1$. (See Figure 5.1.) The period of our p-sine and p-cosine functions is $2\pi_p$.

For the inverse squigonometric functions, we would like the domains to mirror their trigonometric counterparts. It follows from equations (5.1), (5.2), and (6.1) that $\arcsin_p(x) + \arccos_p(x) = \frac{\pi_p}{2}$ for $0 \le x \le 1$, and we would like to extend this to $-1 \le x \le 1$. When $p = 2$, there is no problem in doing this since, in the integral defining the arcsine and arccosine functions, $(-u)^2 = u^2$. Once again, the absolute

value conveniently saves the day for us. We similarly have to add in the absolute values in defining arctangent and arccotangent. So we have

$$\begin{aligned}
\arcsin_p(x) &= \int_0^x \frac{1}{(1-|u|^p)^{\frac{p-1}{p}}}\,du, && |x| \le 1\\
\arccos_p(x) &= \int_x^1 \frac{1}{(1-|u|^p)^{\frac{p-1}{p}}}\,du, && |x| \le 1\\
\arctan_p(x) &= \int_0^x \frac{1}{(1+|u|^p)^{\frac{2}{p}}}\,du, && x \in \mathbb{R}\\
\operatorname{arccot}_p(x) &= \int_x^\infty \frac{1}{(1+|u|^p)^{\frac{2}{p}}}\,du, && x \in \mathbb{R}.
\end{aligned}$$

From our work on right triangles we know that $\arcsin_p(x) = \arctan_p(\frac{x}{\sqrt[p]{1-x^p}})$. As $x \to 1^-$, we have that $\frac{x}{\sqrt[p]{1-x^p}} \to \infty$, thus

$$\frac{\pi_p}{2} = \lim_{x\to 1^-} \arcsin_p(x) = \lim_{x\to 1^-} \arctan_p\left(\frac{x}{\sqrt[p]{1-x^p}}\right) = \arctan_p(\infty).$$

Using the integral definition for p-arctangent, we obtain a new integral formula for π_p:

$$\pi_p = 2\int_0^\infty \frac{1}{(1+u^p)^{\frac{2}{p}}}\,du \tag{6.9}$$

and the formula $\arctan_p(x) + \operatorname{arccot}_p(x) = \frac{\pi_p}{2}$ for all $x \in \mathbb{R}$.

To extend the domain of the p-arcsecant and p-cosecant to $(-\infty,-1]\cup[1,\infty)$ we have to establish some facts first. Since $\arcsin_p(x) + \arccos_p(x) = \frac{\pi_p}{2}$, we can easily affirm the expected cofunction relationships:

$$\begin{aligned}
\sin_p\left(\tfrac{\pi_p}{2} - t\right) &= \cos_p(t), & \cos_p\left(\tfrac{\pi_p}{2} - t\right) &= \sin_p(t),\\
\tan_p\left(\tfrac{\pi_p}{2} - t\right) &= \cot_p(t), & \cot_p\left(\tfrac{\pi_p}{2} - t\right) &= \tan_p(t),\\
\sec_p\left(\tfrac{\pi_p}{2} - t\right) &= \csc_p(t), & \csc_p\left(\tfrac{\pi_p}{2} - t\right) &= \sec_p(t).
\end{aligned}$$

The last four identities in turn imply that

$$\begin{aligned}
\arctan_p(x) + \operatorname{arccot}_p(x) &= \tfrac{\pi_p}{2},\\
\operatorname{arcsec}_p(x) + \operatorname{arccsc}_p(x) &= \tfrac{\pi_p}{2}.
\end{aligned}$$

For the arcsecant, we choose to have its range be in the first and second quadrants so that the function satisfies the formula

$$\operatorname{arcsec}_p(x) = \arccos_p\left(\frac{1}{x}\right)$$

for $|x| \ge 1$. Since $\cos_p(t) = -\cos_p(\pi_p - t)$ for $\pi_p/2 < t \le \pi_p$, we have $\operatorname{arcsec}_p(x) = \pi_p - \operatorname{arcsec}_p(-x)$ for $x \le -1$. It then follows that

$$\begin{aligned}\operatorname{arccsc}_p(x) &= \frac{\pi_p}{2} - \operatorname{arcsec}_p(x)\\ &= \frac{\pi_p}{2} - (\pi_p - \operatorname{arcsec}_p(-x))\\ &= \operatorname{arcsec}_p(-x) - \frac{\pi_p}{2}.\end{aligned}$$

We can now complete the definitions of our inverse squigonometric functions:

$$\begin{aligned}
\operatorname{arcsec}_p(x) &= \int_1^x \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du, && x \geq 1\\
\operatorname{arccsc}_p(x) &= \int_x^\infty \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du, && x \geq 1\\
\operatorname{arcsec}_p(x) &= \pi_p - \operatorname{arcsec}_p(-x) && x \leq -1\\
\operatorname{arccsc}_p(x) &= \operatorname{arcsec}_p(-x) - \frac{\pi_p}{2} && x \leq -1.
\end{aligned}$$

We have the expected ranges:

$$\begin{aligned}
-\pi_p/2 \leq \operatorname{arcsin}_p(x) &\leq \pi_p/2,\\
0 \leq \operatorname{arccos}_p(x) &\leq \pi_p,\\
-\pi_p/2 < \operatorname{arctan}_p(x) &< \pi_p/2,\\
0 < \operatorname{arccot}_p(x) &< \pi_p,\\
0 \leq \operatorname{arcsec}_p(x) &\leq \pi_p, && \operatorname{arcsec}_p(x) \neq \pi_p/2\\
-\pi_p/2 \leq \operatorname{arccsc}_p(x) &\leq \pi_p/2, && \operatorname{arccsc}_p(x) \neq 0.
\end{aligned}$$

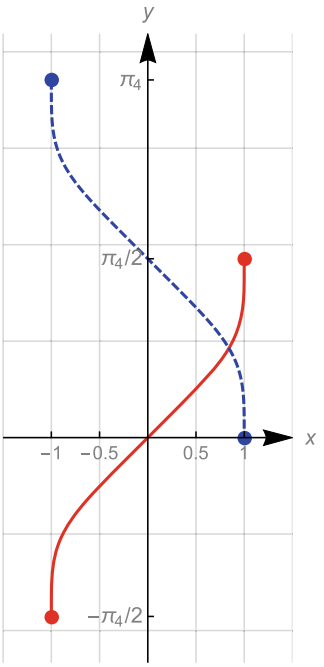
(See Figure 6.2.)

Exercises

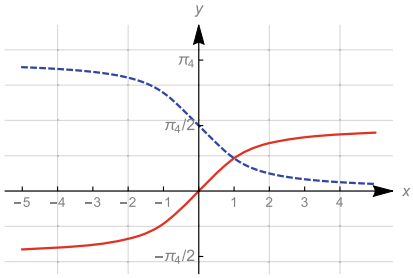
▶ **Exercise 6.1** *The ancient Egyptians approximated the area of a circle with the area of a square whose side length was $\frac{8}{9}$ of diameter of the circle [8,29]. Determine which p-metric the Egyptians were using.*

▶ **Exercise 6.2** *Deduce the following symmetry relationships directly from the assumption that the p-cosine and p-sine functions satisfy CIVP (3.2) for all real numbers and the definition* $\cos_p(\pi_p/2) = 0$.

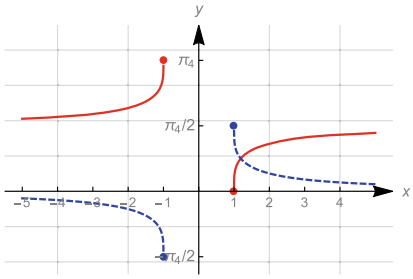
1. $\sin_p(t + \pi_p/2) = \cos_p(t)$ *and* $\cos_p(t + \pi_p/2) = -\sin_p(t)$
2. $\sin_p(t + \pi_p) = -\sin_p(t + \pi_p)$ *and* $\cos_p(t + \pi_p) = -\cos_p(t + \pi_p)$
3. $\sin_p(t + 2\pi_p) = \sin_p(t)$ *and* $\cos_p(t + 2\pi_p) = \cos_p(t)$



(a) $y = \operatorname{arcsin}_4(x)$ (solid) and $y = \operatorname{arccos}_4(x)$ (dashed)



(b) $y = \operatorname{arctan}_4(x)$ (solid) and $y = \operatorname{arccot}_4(x)$ (dashed)



(c) $y = \operatorname{arcsec}_4(x)$ (solid) and $y = \operatorname{arccsc}_4(x)$ (dashed)

Figure 6.2 Graphs of inverse 4-trigonometric functions

▶ **Exercise 6.3** *The integrals that define both* $\arccos_p$ *and* π_p *are improper integrals for* $p > 1$ *since the integrand goes to* ∞ *as* $u \to 1^-$. *Use the geometry of the unit p-circle and the unit circle to show that these integrals converge at 1 for* $p > 1$. *(Handle the* $1 < p < 2$ *and* $p > 2$ *cases separately.)*

▶ **Exercise 6.4** *Show that the area of the unit p-circle is* π_p *square units.*

▶ **Exercise 6.5** *Show that*

$$\pi_p = 2\int_1^\infty \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du.$$

(This result is necessary to complete Exercise 14.3.)

▶ **Exercise 6.6** *Jordan's inequality gives simple bounds on the sine function in the first quadrant: For* $t \in [0, \frac{\pi}{2}]$,

$$\frac{2}{\pi}t \le \sin(t) \le t.$$

The inequality states that the graph of $\sin(t)$ *for* $t \in [0, \frac{\pi}{2}]$ *lies between the tangent line to the graph at 0, and the secant line connecting* $(0, 0)$ *to* $(\frac{\pi}{2}, 1)$.

Draw a graph (a "proof without words") that demonstrates a generalized version of the Jordan inequality: For $t \in [0, \frac{\pi_p}{2}]$,

$$\frac{2}{\pi_p} t \leq \sin_p(t) \leq t.$$

▶ **Exercise 6.7** *Calculate* $\pi_{\frac{1}{2}}$. *(Recall that we briefly investigate the* $p = \frac{1}{2}$ *functions in Exercise 5.3.),*

▶ **Exercise 6.8** *Show that* $\sin_p(t)$ *and* $\cos_p(t)$ *are infinitely differentiable for all* t *if and only if* $p > 1$ *is an integer.*

▶ **Exercise 6.9** *Let* p, q *be conjugate, i.e.* $\frac{1}{p} + \frac{1}{q} = 1$ *(see Section 4.4).*

1. *Show that* $\sin(\frac{\pi}{p}) = \sin(\frac{\pi}{q})$.
2. *Show that*

$$\pi_p \pi_q = \frac{4\pi^2}{pq \sin(\frac{\pi}{p})^2 \Gamma(\frac{2}{p})\Gamma(\frac{2}{q})}.$$

Note the significance of the first part to this result.

▶ **Exercise 6.10** *Prove that*

$$\int_0^\infty e^{-u^2}\, du = \frac{\sqrt{\pi}}{2}.$$

and use this to show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ *(or, in an abuse of notation, show that* $(-\frac{1}{2})! = \sqrt{\pi}$*).*

▶ **Exercise 6.11** *We now generalize the previous exercise.*

1. *Show that*

$$\int_{-\infty}^\infty e^{-|u|^p}\, du = \frac{2}{p}\,\Gamma\left(\frac{1}{p}\right) = 2\,\Gamma\left(\frac{p+1}{p}\right). \tag{6.10}$$

2. *Use Equation (6.7) to prove that*

$$\int_{-\infty}^\infty e^{-|u|^p}\, du = \pi_p \frac{\Gamma(\frac{1}{p} + \frac{1}{2})}{2^{1-\frac{2}{p}}\sqrt{\pi}} \tag{6.11}$$

▶ **Exercise 6.12** *Let*

$$f_p(x) = \frac{p}{2k\Gamma(1/p)} e^{-(\frac{|x-\mu|}{k})^p}$$

for $p, k > 0$, *and any real* μ. *Use the results of Exercise 6.11 to show that this function defines a probability density function with mean* μ. *This is the* generalized normal distribution*; see Figure A.4. Calculate the variance in terms of* p. *What known distributions can you identify for special values of* p*?*

Projects

Project 6.1 (Geometric estimates for π_p)
There is a long history of geometric estimates for π. Try adapting them to estimate π_p. For example, [106] uses the triangles in Figure 6.3 to construct upper and lower bounds on π_p. Also see Project 16.4.

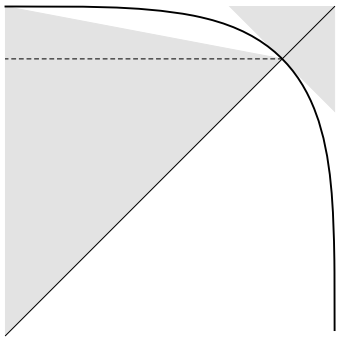


Figure 6.3 A quick estimate of π_p

Project 6.2 (Probability distributions)
We are using p-metrics as a way to classify special functions that relate to these geometries. Can we do something similar for distributions? Our first effort is in Exercise 6.12, wherein we develop a parameterized family of "normal" distributions, and we saw some other familiar distributions appear in that family.

Another example is the Cauchy distribution, which has density of the form

$$g(x) = \frac{1}{\pi\gamma\left(1 + (\frac{x-x_0}{\gamma})^2\right)}.$$

This distribution is notable for not having a well-defined mean because $g(x)$ has a finite integral over the reals but $xg(x)$ does not. Note that up to some parameters and normalization this is really just $f(x) = \frac{1}{1+x^2}$, just as the the normal density essentially amounts to e^{-x^2}, so we will focus on those functions for now.

How should we generalize the Cauchy distribution? An obvious choice would be to spot the 2 and make it a p, so $f_p(x) = \frac{1}{1+|x|^p}$. But what do we mean by generalization? Is it the formula we want to generalize, or some property it holds? This was clearer with squigonometric functions where this question was guided by the interpretation of the parameter. If the main property of the Cauchy distribution is its pathological mean, then notice the distributions f_p all have finite mean for $p > 2$. This is an interesting way to approach the question of what having an undefined mean means, but perhaps if wanted these distributions to share that property we should use $f_p(x) = \frac{|x|^{p-2}}{1+|x|^p}$. We still just replaced 2's with p's, but one of them was tricky to spot.

Or maybe it's the geometry of the curve itself. The graph of the Cauchy density is a witch of Agnesi, a geometric curve we saw in Exercise 4.5 and generalized geometrically. Maybe Figure A.3 should be our class of Cauchy densities.

There are as many probability distributions as there are functions with finite integrals, just as there are as many generalized trigonometric functions as there are parameterizations of the circle, and broad classes of these distributions have been thoroughly studied. Still, we could experiment with defining parameterized families of distributions and calculating their moments and other properties to see if any useful patterns emerge. But beware – many of these calculations will involve some hideous integrals and closed-form expressions may be hard to come by, even with our expanded catalog of squigonometric functions.



Parameterizations

7

Sine and cosine parameterize the unit circle by angle, arclength, and area all at once. We have come to appreciate how miraculous this is now that we have seen this not happen for non-Euclidean squircles.

But still we must proceed. In this chapter we will investigate how to interpret the parameters of generalized trigonometric functions and how to adapt them to access different geometric information. Although we are focused on our original definition, as a practical matter the "right" definition of the squigonometric functions depends on what you want to do with them. Since one definition won't do it all, we must be flexible enough to define the right function for the right task.

7.1 Interpreting the parameter

When we reviewed trigonometric functions, we noted that the argument t in our customary $\cos(t)$ and $\sin(t)$ contains much geometric information: the angle in radians subtended between $(1, 0)$ and the point $(\cos(t), \sin(t))$ on the unit circle, the distance between those two points as we walk along the unit circle, and twice the area of the sector traced out by this angle.

However, we noted that for the taxicab $p = 1$ case we get different trigonometric functions depending on what information we wish to encode in the parameter t. We now investigate the geometric information the argument t for $\cos_p(t)$ and $\sin_p(t)$ divulges for general p.

To see how these measurements will interact differently, let's explore how the relationship between radius and angle changes. As usual, we work in the first quadrant to ease calculation. We recall the radius and angle functions, defined in the Introduction by

$$r(t) = \sqrt{x(t)^2 + y(t)^2}, \qquad \theta(t) = \arctan\left(\frac{y(t)}{x(t)}\right).$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_7

In the case of the unit circle parameterized by cosine and sine, these give the constant unit function and the identity map, respectively. In Chapter 6, we saw that

$$r'(t) = \frac{x(t)y(t)[x(t)^{p-2} - y(t)^{p-2}]}{r(t)}.$$

This illustrates precisely what makes a p-circle different from a circle. The zeros of $r'(t)$ correspond to the lines of symmetry of the p-circle (that is, when $x(t) = \pm y(t)$ or when $x(t) = 0$ or $y(t) = 0$), with the anticipated sign changes in between. For example, for $p \geq 2$, the derivative is positive as we move counterclockwise along the p-circle from the point $(1, 0)$, indicating that the p-circle is breaking outward away from the circle (or inward if $1 \leq p < 2$, when the derivative is negative). This continues until the intersection with the line $y = x$ at $(\sqrt[p]{1/2}, \sqrt[p]{1/2})$, where the radius begins to decrease (or increase for $1 \leq p < 2$) until it meets back with the unit circle at $(0, 1)$.

If we differentiate the angle function, we get

$$\begin{aligned}
\theta'(t) &= \frac{d}{dt} \arctan\left(\frac{y(t)}{x(t)}\right) \\
&= \frac{1}{1 + \frac{y(t)^2}{x(t)^2}} \cdot \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right) \\
&= \frac{\cos_p(t)^2}{\cos_p(t)^2 + \sin_p(t)^2} \cdot \frac{d}{dt} \tan_p(t) \\
&= \frac{\cos_p(t)^2}{\cos_p(t)^2 + \sin_p(t)^2} \cdot \sec_p(t)^2 = \frac{1}{r(t)^2}.
\end{aligned}$$

If $p > 2$, where the p-circle circumscribes the unit circle, we have $r(t) \geq 1$ for all t and so $0 < \theta'(t) \leq 1$, meaning that the parameterization (3.2) progresses counterclockwise around the unit p-circle, but at a slower rate than that of the unit circle. If $1 \leq p < 2$, the p-circle inscribes the unit circle, we have $r(t) \leq 1$ for all t, and the parameterization progresses around the unit p-circle more quickly than the unit circle. Either way, $\theta(t) \neq t$ and t doesn't tell us about the subtended angle.

It turns out the the t in $\cos_p(t)$ and $\sin_p(t)$ does not give us information about the arclength traveled either. We have for the unit circle parameterized as $x = \cos(t)$, $y = \sin(t)$ that the arclength $s(t)$ is given by

$$\begin{aligned}
s(t) &= \int_0^t \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du \\
&= \int_0^t \sqrt{(-\sin u)^2 + (\cos u)^2}\, du \\
&= \int_0^t 1\, du = t.
\end{aligned}$$

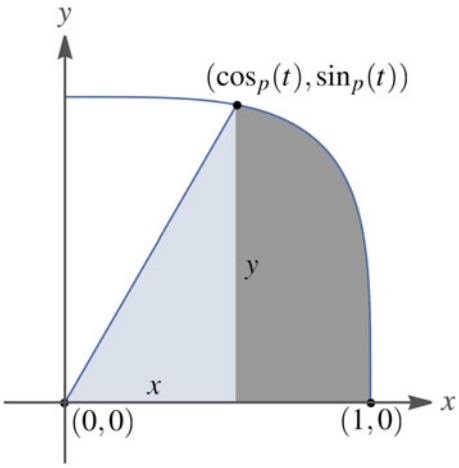


Figure 7.1 Area of a p-circular sector

For general p, with the parameterization $x = \cos_p(t)$, $y = \sin_p(t)$, we have

$$\begin{aligned} s_p(t) &= \int_0^t \sqrt{\left|\frac{dx}{du}\right|^p + \left|\frac{dy}{du}\right|^p}\, du \\ &= \int_0^t \sqrt{|-\sin_p(u)^{p-1}|^p + |\cos_p(u)^{p-1}|^p}\, du \end{aligned}$$

For example, for the squircle (the 4-circle), we have

$$s_4(t) = \int_0^t \sqrt{\sin_4(u)^{12} + \cos_4(u)^{12}}\, du.$$

Since $s_4'(t)$ is not identically 1, we have that $s_4(t) \neq t$ and so the parameter t does not correspond to the arclength, and thus area and arclength yield different parameterizations.

7.2 Areal trigonometric functions

We claim that the area of the p-circular sector between the points $(0, 0)$, $(1, 0)$ and $(\cos_p(t), \sin_p(t))$ is $t/2$ square units, i.e. that our definitions are compatible with the areal interpretation of the parameter. This approach is due to Levin [59].

From Figure 7.1, we have that $x = \cos_p(t)$ and $y = \sin_p(t)$. Let $A(t)$ be the area of the given p-circular sector. Then $A(t)$ is the sum of the area of the light gray triangle in Figure 7.1 and the dark gray area under the curve. That is,

$$\begin{aligned} A(t) &= \frac{xy}{2} + \int_x^1 \sqrt[p]{1-u^p}\, du \\ &= \frac{\cos_p(t)\sin_p(t)}{2} + \int_{\cos_p(t)}^1 \sqrt[p]{1-u^p}\, du \end{aligned}$$

If we differentiate with respect to t, we get

$$\begin{aligned} 2A'(t) &= -\sin_p(t)^{p-1} \cdot \sin_p(t) + \cos_p(t) \cdot \cos_p(t)^{p-1} - 2\sqrt[p]{1-\cos_p(t)^p} \cdot (-\sin_p(t)^{p-1}) \\ &= -\sin_p(t)^p + \cos_p(t)^p + 2\sin_p(t)^p \\ &= \cos_p(t)^p + \sin_p(t)^p = 1. \end{aligned}$$

Since $A(0) = 0$, we have that $A(t) = \frac{t}{2}$ as claimed. For this reason, we will call the CIVP (3.2) the *areal parameterization*.

As noted in the introduction, the squigonometric functions defined by (3.2) are not new and we owe much to the expositors who periodically shed new light on them (see e.g., [42] or [89]).

Getting other information

As we have seen previously, if we want our parameter t to give us information about either angle or arclength, we will have to develop other versions of our squigonometric functions.

Analogously to the CIVP (1.2), we can in fact parameterize the unit p-circle as

$$\begin{cases} x'(t) = -y(t)^{p-1}\zeta(x(t), y(t), t) \\ y'(t) = x(t)^{p-1}\zeta(x(t), y(t), t) \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{7.1}$$

where $\zeta(x(t), y(t), t)$ is any appropriate differentiable function. To see this, suppose x and y satisfy (7.1) for some function ζ and let $g(t) = x(t)^p + y(t)^p$. Then $g(0) = 1$ and

$$\begin{aligned} g'(t) &= p \cdot x(t)^{p-1}x'(t) + p \cdot y(t)^{p-1}y'(t) \\ &= -p \cdot x(t)^{p-1}y(t)^{p-1}\zeta(x(t), y(t), t) + p \cdot x(t)^{p-1}y(t)^{p-1}\zeta(x(t), y(t), t) = 0. \end{aligned}$$

and so $g(t)$ is identically 1, just as we saw earlier for our p-sine and p-cosine.

The idea is that our choice of ζ would depend on what information we're trying to glean from the parameter t. Are there functions ζ that will result in t giving us an angular or an arclength parameterization? There are but we see that when $p \neq 2$, generalized trigonometry can be like a visit to the optometrist. When one aspect such as area comes sharply into focus, others become unseeable.

7.3 Angular trigonometric functions

Having discovered that our parameterization captures information about the area, where does that leave us with angle? (We will postpone discussion on arclength until Chapter 8.)

We now seek a function ζ for (7.1) such that t gives the angle in radians subtended by the arc from $(1, 0)$ to $(x(t), y(t))$. In other words, we want $\theta(t) = t$.

We can get an explicit formula by applying the standard polar substitution $x = r\cos\theta$, $y = r\sin\theta$ to our defining equation:

$$\begin{aligned} |x|^p + |y|^p = 1 &\Rightarrow |r\cos\theta|^p + |r\sin\theta|^p = 1 \\ &\Rightarrow r^p = \frac{1}{|\cos\theta|^p + |\sin\theta|^p}. \end{aligned}$$

Thus

$$\sin_p^{\text{ang}}(\theta) = \frac{\sin\theta}{(|\cos\theta|^p + |\sin\theta|^p)^{1/p}}, \qquad \cos_p^{\text{ang}}(\theta) = \frac{\cos\theta}{(|\cos\theta|^p + |\sin\theta|^p)^{1/p}}. \tag{7.2}$$

While it is convenient to have these functions expressed in terms of familiar, well-implemented functions (and very helpful for obtaining graphs in computer algebra systems), they are not convenient for finding functional relationships.

We thus return to our differential equations approach and work in the first quadrant. Recall that $\theta(t) = \arctan(y(t)/x(t))$ there. We then have

$$\begin{aligned} \theta'(t) &= \frac{1}{1 + \frac{y(t)^2}{x(t)^2}} \cdot \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right) \\ &= \frac{1}{1 + \frac{y(t)^2}{x(t)^2}} \cdot \frac{x(t)^p\zeta(x(t), y(t), t) + y(t)^p\zeta(x(t), y(t), t)}{x(t)^2} \\ &= \frac{\zeta(x(t), y(t), t)[x(t)^p + y(t)^p]}{x(t)^2 + y(t)^2} \\ &= \frac{\zeta(x(t), y(t), t)}{x(t)^2 + y(t)^2}, \end{aligned}$$

since $x(t)^p + y(t)^p = 1$ in the first quadrant. Because we want $\theta(t) = t$, we also want $\theta'(t) = 1$. Setting these two quantities equal to each other, we get that $\zeta(x(t), y(t), t) = x(t)^2 + y(t)^2$.

Thus the coupled initial value problem that leads to our angular parameterization of the unit p-circle is

$$\begin{cases} x'(t) = -y(t)^{p-1}(x(t)^2 + y(t)^2) \\ y'(t) = x(t)^{p-1}(x(t)^2 + y(t)^2) \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{7.3}$$

where as usual $x(t)$ will be the angular cosine function $\cos_p^{\text{ang}}(t)$ and $y(t)$ the angular sine function $\sin_p^{\text{ang}}(t)$.

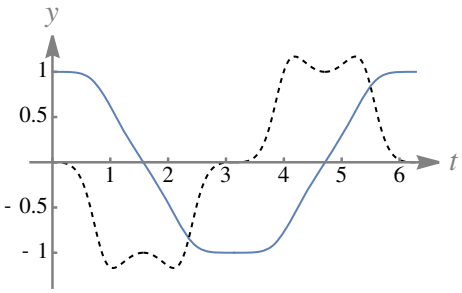


Figure 7.2 Graphs of $y = \cos_4^{\text{ang}}(t)$ (solid) and its derivative (dashed)

The definition of π_p^{ang} is anticlimactic under this parameterization: $\pi_p^{\text{ang}} = \pi$ for all p. So our CIVP (7.3) will hold for $0 \le t \le \pi/2$ and we can extend the angular sine and cosine functions to the real line in the usual manner. A graph of $y = \cos_4^{\text{ang}}(t)$ and its derivative

$$y' = -\sin_4^{\text{ang}}(t)^3(\cos_4^{\text{ang}}(t)^2 + \sin_4^{\text{ang}}(t)^2)$$

can be seen in Figure 7.2 while graphs for various p-values can be seen in Figures 7.3 and A.5. We note that for $p = 1$, the graphs of the sine and cosine are not piecewise linear as they are in the areal parameterization.

The angular sine and cosine functions defined by (7.3) have familiar properties:

$$\begin{gathered}
|\cos_p^{\text{ang}}(t)|^p + |\sin_p^{\text{ang}}(t)|^p = 1, \\
\sin_p^{\text{ang}}\left(\tfrac{\pi}{2} - t\right) = \cos_p^{\text{ang}}(t), \qquad \cos_p^{\text{ang}}\left(\tfrac{\pi}{2} - t\right) = \sin_p^{\text{ang}}(t), \\
\sin_p^{\text{ang}}(0) = 0, \qquad \sin_p^{\text{ang}}\left(\tfrac{\pi}{2}\right) = 1, \\
\sin_p^{\text{ang}}\left(\tfrac{\pi}{4}\right) = \cos_p^{\text{ang}}\left(\tfrac{\pi}{4}\right) = \tfrac{1}{\sqrt[p]{2}}, \\
\cos_p^{\text{ang}}(0) = 1, \qquad \cos_p^{\text{ang}}\left(\tfrac{\pi}{2}\right) = 0.
\end{gathered}$$

We can define angular tangent, secant, and cosecant functions as the usual ratios of the angular sine and cosine. But note from (7.2) that the tangent and cotangent will always be the same as their classical analogues.

$$\tan_p^{\text{ang}}(t) = \frac{\sin_p^{\text{ang}}(t)}{\cos_p^{\text{ang}}(t)} = \tan(t), \qquad \sec_p^{\text{ang}}(t) = \frac{1}{\cos_p^{\text{ang}}(t)},$$

$$\cot_p^{\text{ang}}(t) = \frac{\cos_p^{\text{ang}}(t)}{\sin_p^{\text{ang}}(t)} = \frac{1}{\tan_p^{\text{ang}}(t)} = \cot(t), \qquad \csc_p^{\text{ang}}(t) = \frac{1}{\sin_p^{\text{ang}}(t)}.$$

We also have the Pythagorean relationships

$$\begin{aligned}
1 + |\tan_p^{\text{ang}}(t)|^p &= |\sec_p^{\text{ang}}(t)|^p, \\
|\cot_p^{\text{ang}}(t)|^p + 1 &= |\csc_p^{\text{ang}}(t)|^p.
\end{aligned}$$

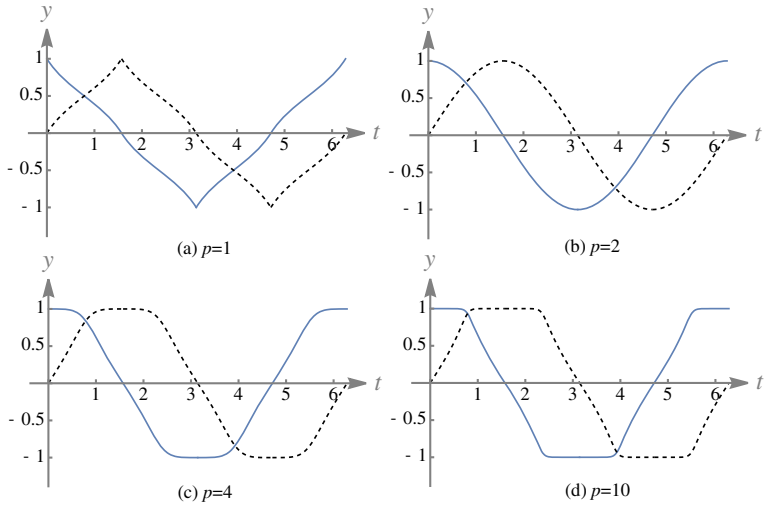


Figure 7.3 Graphs of $y = \cos_p^{\text{ang}}(t)$ (solid) and $y = \sin_p^{\text{ang}}(t)$ (dashed) for various p-values

We can use (7.3) and the quotient rule to compute the derivative formulas. See [67] for some additional identities.

$$\begin{aligned}
\frac{d}{dt}\sin_p^{\text{ang}}(t) &= \cos_p^{\text{ang}}(t)^{p-1}(\cos_p^{\text{ang}}(t)^2 + \sin_p^{\text{ang}}(t)^2),\\
\frac{d}{dt}\cos_p^{\text{ang}}(t) &= -\sin_p^{\text{ang}}(t)^{p-1}(\cos_p^{\text{ang}}(t)^2 + \sin_p^{\text{ang}}(t)^2),\\
\frac{d}{dt}\tan_p^{\text{ang}}(t) &= 1 + \tan_p^{\text{ang}}(t)^2, \qquad (7.4)\\
\frac{d}{dt}\sec_p^{\text{ang}}(t) &= \sec_p^{\text{ang}}(t)^2 \sin_p^{\text{ang}}(t)^{p-1}(\cos_p^{\text{ang}}(t)^2 + \sin_p^{\text{ang}}(t)^2), \qquad (7.5)\\
\frac{d}{dt}\cot_p^{\text{ang}}(t) &= -(1 + \cot_p^{\text{ang}}(t)^2),\\
\frac{d}{dt}\csc_p^{\text{ang}}(t) &= -\csc_p^{\text{ang}}(t)^2 \cos_p^{\text{ang}}(t)^{p-1}(\cos_p^{\text{ang}}(t)^2 + \sin_p^{\text{ang}}(t)^2).
\end{aligned}$$

7.4 Inverse angular trigonometric functions

Let $y = \sin_p^{\text{ang}}(t)$, with $0 \le t \le \pi/2$. (Thus $0 \le y \le 1$.) Then

$$\begin{aligned}
\frac{dy}{dt} &= \cos_p^{\text{ang}}(t)^{p-1}(\cos_p^{\text{ang}}(t)^2 + \sin_p^{\text{ang}}(t)^2)\\
&= (1 - y^p)^{(p-1)/p}((1 - y^p)^{2/p} + y^2)\\
&= (1 - y^p)^{(p+1)/p} + y^2(1 - y^p)^{(p-1)/p}.
\end{aligned}$$

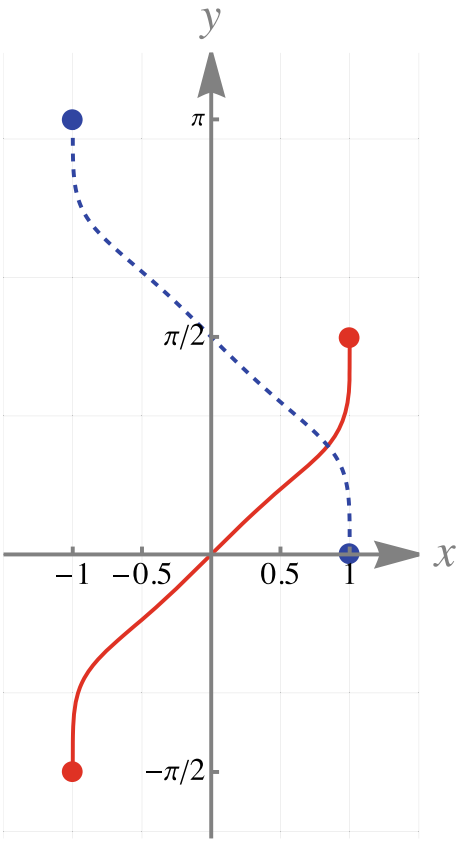


Figure 7.4 Graph of $y = \arcsin_4^{\text{ang}}(x)$ (solid) and $y = \arccos_4^{\text{ang}}(x)$ (dashed)

We can separate and integrate this differential equation to solve for t:

$$t = \arcsin_p^{\text{ang}}(y) = \int_0^y \frac{1}{(1-|u|^p)^{(p-1)/p}((1-|u|^p)^{2/p}+u^2)}\,du = \int_0^y \frac{1}{(1-|u|^p)^{(p+1)/p}+u^2(1-|u|^p)^{(p-1)/p}}\,du. \tag{7.6}$$

In a similar manner, we can define the inverse of the angular cosine function:

$$t = \arccos_p^{\text{ang}}(x) = \int_x^1 \frac{1}{(1-|u|^p)^{(p-1)/p}((1-|u|^p)^{2/p}+u^2)}\,du = \int_x^1 \frac{1}{(1-|u|^p)^{(p+1)/p}+u^2(1-|u|^p)^{(p-1)/p}}\,du. \tag{7.7}$$

Inserting the absolute values gives us a domain for both of these functions of $[-1, 1]$. The range is $[-\pi/2, \pi/2]$ for the inverse sine and $[0, \pi]$ for the inverse

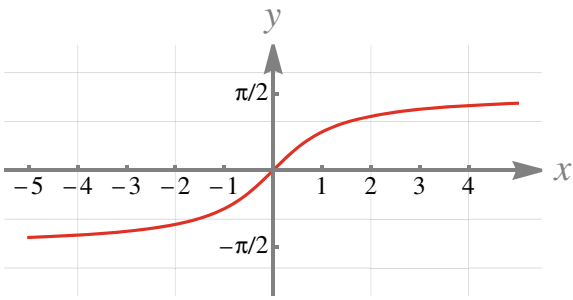


Figure 7.5 Graph of $y = \arctan_4^{\text{ang}}(x) = \arctan(x)$

cosine. (See Figures 7.4 and 7.5.) Because $\pi_p^{\text{ang}} = \pi$ for all p, this gives us an entire family of integrals that evaluate to π.

$$\pi = 2\int_0^1 \frac{1}{(1-u^p)^{(p-1)/p}((1-u^p)^{2/p}+u^2)}\,du, \quad p \geq 1. \tag{7.8}$$

Especially given (7.6) and (7.7), it is surprising that (7.4) leads to a formula for the angular arctangent independent of p:

$$\arctan_p^{\text{ang}}(x) = \int_0^x \frac{1}{1+u^2}\,du = \arctan(x).$$

Exercises

▶ **Exercise 7.1** *Show that the generalized ellipse trigonometric functions* $(\cos_p^{\mathcal{E}}(t), \sin_p^{\mathcal{E}}(t))$ *of Exercise 4.5 give the areal parameterization of the curve* $\mathcal{E}_p$.

▶ **Exercise 7.2** *In CIVP (7.1), what goes wrong if* $\zeta(t) = t$*?*

▶ **Exercise 7.3** *Verify the formula for* $\sin_1^{ang}(t)$ *and* $\cos_1^{ang}(t)$ *directly from the geometry of the circle. Relate these formulas to the area of the squircular arc subtended by* θ*, thus obtaining a relationship between angular and areal parameterizations for this case.*

▶ **Exercise 7.4** *Express* $\sin_\infty^{ang}(t)$ *explicitly as a piecewise function.*

▶ **Exercise 7.5** *Verify the formula (7.8) for* π *for the cases* $p = 1$ *and* $p = 2$ *by direct integration.*



Arclength parameterization

8

Two articles appeared very close to each other in 1999 and 2000 [2,30] that defined new values of π for the p-metric. The new definitions were based not on π being the area of the unit circle, as we have for the most part, but on π being the ratio of the circumference of a circle to its diameter. In particular, double these values of π gave the proper circumference for a p-circle in the p-metric.

In Chapter 7, we discovered that while the parameter t for our p-trigonometric functions gives us information about the area of a p-circular sector, it has no meaningful relationship to the arclength. We investigate in this chapter a different parameterization of the unit p-circle that will be tied to the arclength of the p-circle, and so our generalized versions of π will align with those found in [2,30]. We will not go into the depth that we have for the areal parameterization, but will give the basic facts, leaving others for exercises or research projects.

8.1 p-Arclength

Measuring length in the p-metric works just as in Euclidean geometry. To measure the p-length of a differentiable curve we proceed as in the Euclidean case by approximating it with line segments. As shown in Figure 8.1, each infinitesimal segment has p-length $\sqrt[p]{|dx|^p + |dy|^p}$.

If $f(x)$ is a differentiable function for all $a \le x \le b$ then the arclength of f on this interval is

$$\int_a^b \sqrt[p]{|dx|^p + |dy|^p}\, dx = \int_a^b \sqrt[p]{1 + \left|\frac{dy}{dx}\right|^p}\, dx$$
$$= \int_a^b \sqrt[p]{1 + |f'(x)|^p}\, dx,$$

which is a familiar expression from calculus.

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_8

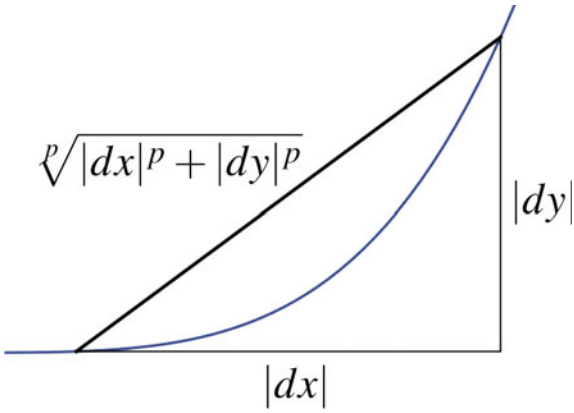


Figure 8.1 Approximating arclength with segments

Example 8.1

Let $f(x)$ be a monotone real-valued function on $[a, b]$. Show that the taxicab arclength of the graph of $f(x)$ depends only on the values of f at the endpoints.

Solution. Suppose $f(x)$ is monotone increasing so that $f'(x) = |f'(x)|$. Then the arclength from a to b in the 1-metric is

$$\begin{aligned}\int_a^b \left(1 + f'(x)\right)\, dx &= x\Big|_a^b + \int_a^b f'(x)\, dx \\ &= (b-a) + (f(b) - f(a))\,.\end{aligned}$$

In particular, the value depends only on the endpoints and not on the path taken. This proves the phenomenon illustrated in Figure 4.2. The monotone decreasing case is handled similarly.

8.2 Pi redux

We can parameterize the first quadrant portion of the unit p-circle $|x|^p + |y|^p = 1$ as $x = u$ and $y = (1 - u^p)^{\frac{1}{p}}$. In this case, the p-arclength element is given by

$$ds = \sqrt[p]{\left|\frac{dx}{du}\right|^p + \left|\frac{dy}{du}\right|^p}\, du = \left(1 + (u^{-p} - 1)^{1-p}\right)^{\frac{1}{p}}\, du.$$

Therefore we can define a new Pi, π_p^{arc}, to represent the arclength traversed along half of a p-circle. By symmetry, we can define this as

$$\pi_p^{\text{arc}} = 4\int_0^{2^{-\frac{1}{p}}} \left(1 + (u^{-p} - 1)^{1-p}\right)^{\frac{1}{p}}\, du. \tag{8.1}$$

p	π_p^{arc}
1	4.000000000
2	3.141592654
3	3.259767993
4	3.396934824
5	3.498456808
6	3.572683184
7	3.628476072
8	3.671684916
9	3.706037566
10	3.733960809

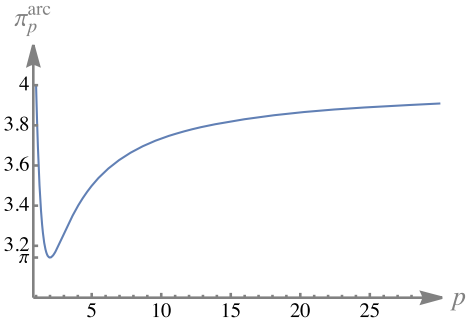


Figure 8.2 π_p^{arc} values via numerical integration

Recall that arclength of a curve is independent of how the curve is parameterized. We can get an alternate definition of π_p^{arc} by parameterizing the first quadrant of the unit p-circle as $x = u^{1/p}$, $y = (1-u)^{1/p}$. Then the p-arclength element becomes

$$ds = \frac{1}{p}\left[u^{1-p} + (1-u)^{1-p}\right]^{\frac{1}{p}}\, du,$$

and we obtain

$$\pi_p^{\text{arc}} = \frac{2}{p}\int_0^1 \left[u^{1-p} + (1-u)^{1-p}\right]^{\frac{1}{p}}\, du \tag{8.2}$$

When $p = 1$, Equation (8.2) gives that $\pi_1^{\text{arc}} = 4$. For $p = 2$, we can use Equation (8.1) to obtain

$$\begin{aligned}
\pi_2^{\text{arc}} &= 4\int_0^{1/\sqrt{2}} \left(1 + (u^{-2}-1)^{-1}\right)^{\frac{1}{2}}\, du \\
&= 4\int_0^{1/\sqrt{2}} \left(1 + \frac{1}{u^{-2}-1}\right)^{\frac{1}{2}}\, du \\
&= 4\int_0^{1/\sqrt{2}} \left(1 + \frac{u^2}{1-u^2}\right)^{\frac{1}{2}}\, du \\
&= 4\int_0^{1/\sqrt{2}} \frac{1}{\sqrt{1-u^2}}\, du \\
&= 4\Big[\arcsin(u)\Big]_0^{1/\sqrt{2}} = \pi.
\end{aligned}$$

When $p = \infty$, the integrand in (8.1) approaches 1, as does the upper limit, and so $\pi_\infty^{\text{arc}} = 4$. For other p, we need to numerically integrate both (8.1) and (8.2). (See Figure 8.2.) We note that the value of π_p^{arc} has a minimum value of π when $p = 2$ [2]. In general, $\pi_p < \pi_p^{\text{arc}}$ when $1 \le p < 2$ but $\pi_p^{\text{arc}} < \pi_p$ when $p > 2$. We will prove this in Section 19.2.

The arclength π's have an interesting relationship. It was suggested in [2] and shown in [54] that if p and q are conjugate, then $\pi_p^{\text{arc}} = \pi_q^{\text{arc}}$. We will study this proof carefully in Section 19.2.

Unlike either of our previous π_p's, π_p^{arc} represents the ratio of circumference to diameter of our p-circles in the p-metric.

8.3 Arclength trigonometric functions

We can use the definition of π_p^{arc} to analytically present an arclength parameterization of the unit p-circle. The goal is to define new sine and cosine functions $\sin_p^{\text{arc}}$ and $\cos_p^{\text{arc}}$ so that the p-arclength traversed in walking counterclockwise along the unit p-circle from $(1, 0)$ to the point $(\cos_p^{\text{arc}}(t), \sin_p^{\text{arc}}(t))$ is t units.

Invoking (8.1), we define our new sine function $S(t)$ to be the function such that

$$t = \int_0^{S(t)} \left(1 + (u^{-p} - 1)^{1-p}\right)^{1/p} du, \tag{8.3}$$

and our new cosine function $C(t)$ to be the function such that

$$t = \int_{C(t)}^1 \left(1 + (u^{-p} - 1)^{1-p}\right)^{1/p} du, \tag{8.4}$$

with $0 \le t \le \pi_p^{\text{arc}}/2$. We first claim that $S(t)^p + C(t)^p = 1$. In Equation (8.3), set $s = (1 - u^p)^{1/p}$. Then $u = (1 - s^p)^{1/p}$ and $du = -\frac{s^{p-1}}{(1-s^p)^{(p-1)/p}}\, ds$. Thus Equation (8.3) becomes

$$\begin{aligned}
t &= \int_1^{(1-S(t)^p)^{1/p}} \left(1 + \left(\frac{1}{1-s^p} - 1\right)^{1-p}\right)^{1/p} \cdot \frac{-s^{p-1}}{(1-s^p)^{(p-1)/p}}\, ds \\
&= \int_{(1-S(t)^p)^{1/p}}^1 \left(1 + \left(\frac{s^p}{1-s^p}\right)^{1-p}\right)^{1/p} \cdot \frac{s^{p-1}}{(1-s^p)^{(p-1)/p}}\, ds \\
&= \int_{(1-S(t)^p)^{1/p}}^1 \left[\left(1 + \left(\frac{s^p}{1-s^p}\right)^{1-p}\right) \cdot \frac{s^{p(p-1)}}{(1-s^p)^{p-1}}\right]^{1/p} ds \\
&= \int_{(1-S(t)^p)^{1/p}}^1 \left[\frac{s^{p(p-1)}}{(1-s^p)^{p-1}} + 1\right]^{1/p} ds \\
&= \int_{(1-S(t)^p)^{1/p}}^1 \left[\left(\frac{1-s^p}{s^p}\right)^{1-p} + 1\right]^{1/p} ds \\
&= \int_{(1-S(t)^p)^{1/p}}^1 \left((s^{-p} - 1)^{1-p} + 1\right)^{1/p} ds,
\end{aligned}$$

which is (8.4) with $C(t) = (1 - S(t)^p)^{1/p}$, and the claim follows.

If we differentiate (8.3) with respect to t, we get

$$1 = (1 + (S(t)^{-p} - 1)^{1-p})^{1/p} \cdot S'(t)$$

$$S'(t) = \frac{1}{\left[1 + \left(\frac{1-S(t)^p}{S(t)^p}\right)^{1-p}\right]^{1/p}}$$

$$S'(t) = \frac{1}{\left[1 + \left(\frac{C(t)^p}{S(t)^p}\right)^{1-p}\right]^{1/p}}$$

$$S'(t) = \frac{1}{\left[1 + \left(\frac{S(t)^p}{C(t)^p}\right)^{p-1}\right]^{1/p}}$$

$$S'(t) = \frac{C(t)^{p-1}}{\left[C(t)^{p(p-1)} + S(t)^{p(p-1)}\right]^{1/p}}$$

Through similar means, we can derive

$$C'(t) = \frac{-S(t)^{p-1}}{\left[C(t)^{p(p-1)} + S(t)^{p(p-1)}\right]^{1/p}}$$

From (8.3) and (8.4), we have that $S(0) = 0$, and $C(0) = 1$. We will call S and C the *arclength sine* and *arclength cosine* functions, respectively, notate them as $\sin_p^{\text{arc}}$ and $\cos_p^{\text{arc}}$, and note that they are the unique solutions on $0 \le t \le \pi_p^{\text{arc}}/2$ of the CIVP

$$\begin{cases} x'(t) = -y(t)^{p-1} \cdot [x(t)^{p(p-1)} + y(t)^{p(p-1)}]^{-1/p} \\ y'(t) = x(t)^{p-1} \cdot [x(t)^{p(p-1)} + y(t)^{p(p-1)}]^{-1/p} \\ x(0) = 1 \\ y(0) = 0, \end{cases} \tag{8.5}$$

with x corresponding to the arclength cosine and y corresponding to the arclength sine. We can then extend the functions to the entire real line in the usual manner via symmetry. A graph of $y = \cos_4^{\text{arc}}(t)$ and its intimidating derivative

$$y' = \frac{-\sin_4^{\text{arc}}(t)^3}{[(\cos_4^{\text{arc}}(t))^{12} + (\sin_4^{\text{arc}}(t))^{12}]^{1/4}}$$

can be seen in Figure 8.3, while graphs of the arclength sine and cosine for various values of p can be seen in Figure 8.4.

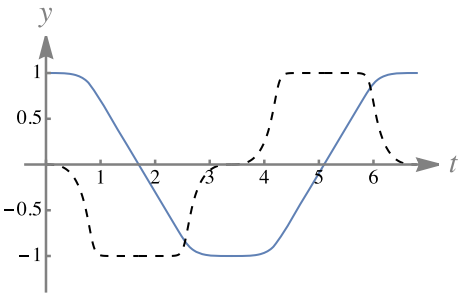


Figure 8.3 Graphs of $y = \cos_4^{\text{arc}}(t)$ (solid) and its derivative (dashed)

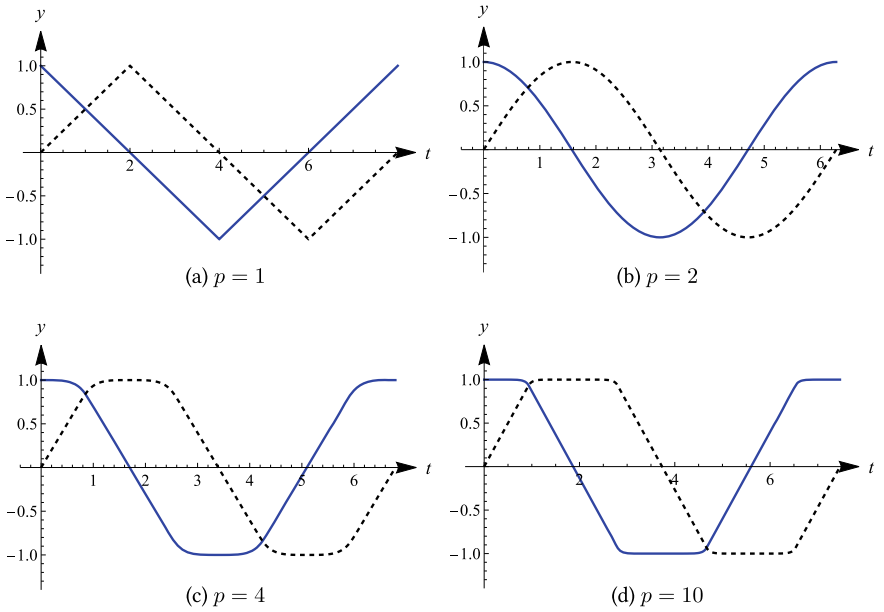


Figure 8.4 Graphs of $y = \cos_p^{\text{arc}}(t)$ (solid) and $y = \sin_p^{\text{arc}}(t)$ (dashed) for various p-values

The arclength sine and cosine functions defined by (8.5) have many of the same characteristics as their areal counterparts:

$$|\cos_p^{\text{arc}}(t)|^p + |\sin_p^{\text{arc}}(t)|^p = 1,$$

$$\sin_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{2} - t\right) = \cos_p^{\text{arc}}(t), \qquad \cos_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{2} - t\right) = \sin_p^{\text{arc}}(t),$$

$$\sin_p^{\text{arc}}(0) = 0, \qquad \sin_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{2}\right) = 1,$$

$$\sin_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{4}\right) = \cos_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{4}\right) = \frac{1}{\sqrt[p]{2}},$$

$$\cos_p^{\text{arc}}(0) = 1, \qquad \cos_p^{\text{arc}}\left(\frac{\pi_p^{\text{arc}}}{2}\right) = 0.$$

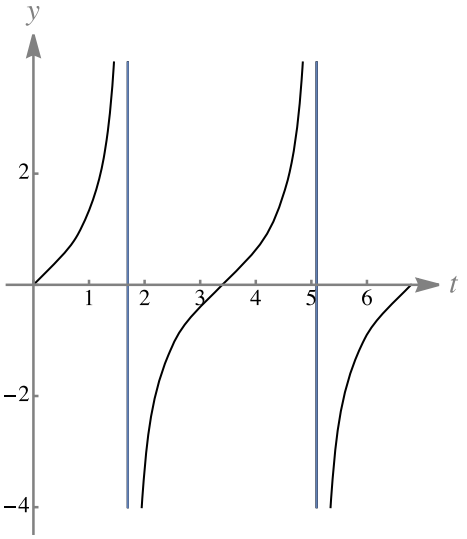


Figure 8.5 Graph of $y = \tan_4^{\text{arc}}(t)$ with vertical asymptotes at $t = \pi_4^{\text{arc}}/2 \approx 1.69847$ and $t = 3\pi_4^{\text{arc}}/2 \approx 5.0954$

We can define new arclength tangent, secant, and cosecant functions as the usual ratios of the arclength sine and cosine.

$$\tan_p^{\text{arc}}(t) = \frac{\sin_p^{\text{arc}}(t)}{\cos_p^{\text{arc}}(t)}, \qquad \sec_p^{\text{arc}}(t) = \frac{1}{\cos_p^{\text{arc}}(t)},$$
$$\cot_p^{\text{arc}}(t) = \frac{\cos_p^{\text{arc}}(t)}{\sin_p^{\text{arc}}(t)} = \frac{1}{\tan_p^{\text{arc}}(t)}, \qquad \csc_p^{\text{arc}}(t) = \frac{1}{\sin_p^{\text{arc}}(t)}.$$

The arclength sine, cosine, secant, and cosecant have period $2\pi_p^{\text{arc}}$ while the tangent and cotangent have period π_p^{arc} (Figure 8.5). We also have the Pythagorean relationships

$$1 + |\tan_p^{\text{arc}}(t)|^p = |\sec_p^{\text{arc}}(t)|^p, \tag{8.6}$$
$$|\cot_p^{\text{arc}}(t)|^p + 1 = |\csc_p^{\text{arc}}(t)|^p.$$

Derivatives

We can use the quotient rule to compute the following derivative formulas.

$$\frac{d}{dt}\sin_p^{\text{arc}}(t) = \frac{\cos_p^{\text{arc}}(t)^{p-1}}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}},$$
$$\frac{d}{dt}\cos_p^{\text{arc}}(t) = \frac{-\sin_p^{\text{arc}}(t)^{p-1}}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}},$$
$$\frac{d}{dt}\tan_p^{\text{arc}}(t) = \frac{\sec_p^{\text{arc}}(t)^2}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}}, \tag{8.7}$$

$$\frac{d}{dt}\sec_p^{\text{arc}}(t) = \frac{\sec_p^{\text{arc}}(t)^2 \sin_p^{\text{arc}}(t)^{p-1}}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}}, \tag{8.8}$$

$$\frac{d}{dt}\csc_p^{\text{arc}}(t) = \frac{-\csc_p^{\text{arc}}(t)^2 \cos_p^{\text{arc}}(t)^{p-1}}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}}, \tag{8.9}$$

$$\frac{d}{dt}\cot_p^{\text{arc}}(t) = \frac{-\csc_p^{\text{arc}}(t)^2}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}}. \tag{8.10}$$

These are similar to the derivative formulas we obtained for the areal parameterization, but each has the extra denominator. See Figure A.6 for graphs of arclength sine functions with their derivatives.

8.4 Inverse arclength trigonometric functions

From (8.1) and (8.2) we can define inverses of the arclength sine and cosine functions:

$$\arcsin_p^{\text{arc}}(x) = \int_0^x \left(1 + (|u|^{-p} - 1)^{1-p}\right)^{1/p} du,$$

$$\arccos_p^{\text{arc}}(x) = \int_x^1 \left(1 + (|u|^{-p} - 1)^{1-p}\right)^{1/p} du,$$

where $-1 \le x \le 1$. To define the inverse of the arclength tangent, we need to tweak the derivative of the tangent a bit to get it entirely in terms of the tangent function. We use (8.6) in the first quadrant to note that

$$1 + \tan_p^{\text{arc}}(t)^p = \sec_p^{\text{arc}}(t)^p = \frac{1}{\cos_p^{\text{arc}}(t)^p} \Longrightarrow \sec_p^{\text{arc}}(t)^2 = \left(1 + \tan_p^{\text{arc}}(t)^p\right)^{2/p}$$

Therefore,

$$\begin{aligned}
\frac{d}{dt}\tan_p^{\text{arc}}(t) &= \frac{\sec_p^{\text{arc}}(t)^2}{[\cos_p^{\text{arc}}(t)^{p(p-1)} + \sin_p^{\text{arc}}(t)^{p(p-1)}]^{1/p}} \\
&= \frac{\left(1 + \tan_p^{\text{arc}}(t)^p\right)^{2/p}}{\left(1 + \tan_p^{\text{arc}}(t)^{p(p-1)}\right)^{1/p} \cdot \cos_p^{\text{arc}}(t)^{p-1}} \\
&= \frac{\left(1 + \tan_p^{\text{arc}}(t)^p\right)^{2/p} \left(1 + \tan_p^{\text{arc}}(t)^p\right)^{(p-1)/p}}{\left(1 + \tan_p^{\text{arc}}(t)^{p(p-1)}\right)^{1/p}} \\
&= \frac{\left(1 + \tan_p^{\text{arc}}(t)^p\right)^{(p+1)/p}}{\left(1 + \tan_p^{\text{arc}}(t)^{p(p-1)}\right)^{1/p}}
\end{aligned}$$

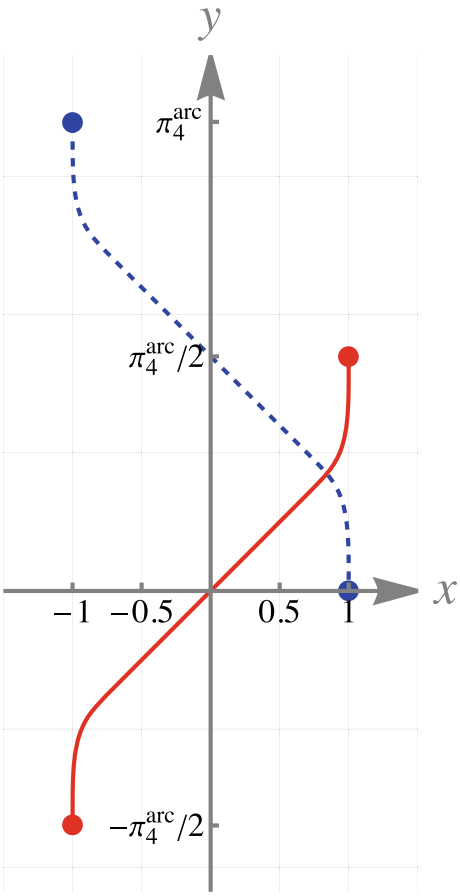


Figure 8.6 Graph of $y = \arcsin_4^{\text{arc}}(x)$ (solid) and $y = \arccos_4^{\text{arc}}(x)$ (dashed)

Because $\tan_p^{\text{arc}}(0) = 0$ we can separate and integrate to obtain

$$\arctan_p^{\text{arc}}(x) = \int_0^x \frac{\left(1 + |u|^{p(p-1)}\right)^{1/p}}{(1 + |u|^p)^{(p+1)/p}}\, du.$$

The domain for both the inverse sine and cosine is $[-1, 1]$ while the domain for the inverse tangent is $(-\infty, \infty)$. The ranges are as expected: $[-\pi_p^{\text{arc}}/2, \pi_p^{\text{arc}}/2]$ for the arcsine, $[0, \pi_p^{\text{arc}}]$ for the cosine, and $(-\pi_p^{\text{arc}}/2, \pi_p^{\text{arc}}/2)$ for the arctangent. Graphs of these functions for $p = 4$ can be found in Figures 8.6 and 8.7.

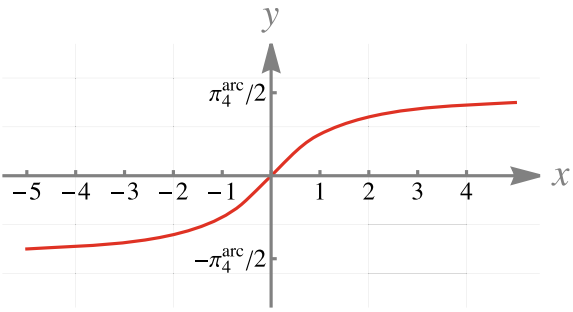


Figure 8.7 Graph of $y = \arctan_4^{\text{arc}}(x)$

8.5 Duality of area and arclength

We can compute the arclength of any curve with respect to any metric. Euclidean and p-arclength are obvious choices for the p-functions, but we have seen that the formulas can get unwieldy. But something interesting happens when we take the arclength in terms of the conjugate q where $\frac{1}{p} + \frac{1}{q} = 1$.

We parameterize the first quadrant of the unit p-circle as $x = \cos_p(u)$, $y = \sin_p(u)$ and calculate the q-arclength element

$$\begin{aligned} ds &= \sqrt[q]{\left|\frac{dx}{du}\right|^q + \left|\frac{dy}{du}\right|^q}\, du \\ &= \sqrt[q]{|\sin_p(u)|^{q(p-1)} + |\cos_p(u)|^{q(p-1)}}\, du \\ &= \sqrt[q]{|\sin_p(u)|^p + |\cos_p(u)|^p}\, du \\ &= du. \end{aligned}$$

But we have already shown in Section 7.2 that this is the areal parameterization. Recall that one of our goals was to resolve the inconvenience that area, arclength, and angle yield different parameterizations. This calculation shows that the properties area and arclength split across two conjugate p-metrics and align in the self-dual case of $p = 2$. To summarize, we have the following[1]

Theorem 8.2

Let p and q be conjugate. Then the q-arclength parameterization of the p-metric is the areal parameterization. In particular, the q-circumference of the p-circle is $2\pi_p$.

[1] The observation that the q-circumference of the p-circle is $2\pi_p$ originates from Lindqvist and Peetre [62]; The proof of Theorem 8.2 is a slight generalization of their argument.

Exercises

▶ **Exercise 8.1** *Use Equation (8.1) to show that* $\pi_1^{arc} = 4$.

▶ **Exercise 8.2** *Use Equation (8.2) to show that* $\pi_2^{arc} = \pi$.

▶ **Exercise 8.3** *Find a CIVP for the arclength tangent and arclength secant functions.*

▶ **Exercise 8.4** *Derive a definition for the inverse arclength secant function.*

▶ **Exercise 8.5** *Repeat the analysis of radius, angle, and arclength from Section 7.1 for the semimetric case of* $p = \frac{1}{2}$.

▶ **Exercise 8.6** *Calculate the* $\frac{4}{3}$*-circumference of the* 4*-circle.*

▶ **Exercise 8.7** *Calculate taxicab circumference of the unit* p*-circle for any* $p > 0$.

▶ **Exercise 8.8** *State and prove a formula for calculating the taxicab arclength of a parameterized curve.*

▶ **Exercise 8.9** *Show that the areal and arclength 1-sine and 1-cosine are piecewise linear, but the angular versions are not.*

▶ **Exercise 8.10** *Show that the* q*-surface area of the* p*-sphere* $|x|^p + |y|^p + |z|^p = 1$ *is* $\frac{4}{3}\pi_p$. *Generalize.*

Integrating squigonometric functions

9

In his 1948 paper, Grammel [42] revealed several integration formulas for the p-trigonometric functions. We will spend some time rediscovering them here while a much fuller integral table can be found in the appendix.

9.1 Antiderivatives

We have already established various derivative formulas for the squigonometric functions, which can be rewritten as integral formulas:

$$\int \cos_p(t)^{p-1}\,dt = \sin_p(t) + C$$

$$\int \sin_p(t)^{p-1}\,dt = -\cos_p(t) + C$$

$$\int \sec_p(t)^2\,dt = \int \frac{1}{\cos_p(t)^2}\,dt = \tan_p(t) + C$$

$$\int \csc_p(t)^2\,dt = \int \frac{1}{\sin_p(t)^2}\,dt = -\cot_p(t) + C$$

$$\int \sec_p(t)^2 \sin_p(t)^{p-1}\,dt = \int \frac{\sin_p(t)^{p-1}}{\cos_p(t)^2}\,dt = \sec_p(t) + C$$

$$\int \csc_p(t)^2 \cos_p(t)^{p-1}\,dt = \int \frac{\cos_p(t)^{p-1}}{\sin_p(t)^2}\,dt = -\csc_p(t) + C$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_9

We have just defined the inverse squigonometric functions as integrals, so we may add them to our growing table:

$$\int \frac{1}{(1-|t|^p)^{\frac{p-1}{p}}}\,dt = \operatorname{arcsin}_p(t) + C$$
$$\int \frac{1}{(1+|t|^p)^{\frac{2}{p}}}\,dt = \operatorname{arctan}_p(t) + C$$
$$\int \frac{t^{p-3}}{(t^p-1)^{\frac{p-1}{p}}}\,dt = \operatorname{arcsec}_p(t) + C$$

An extensive integral table, adapted from [42], can be found in the Appendix.

9.2 Substitution

To develop other formulas, we use tricks we have picked up in our integration experience ([93], for instance) and tweak them a little.

Example 9.1

Evaluate

$$\int \frac{1}{(a^4-x^4)^{\frac{3}{4}}}\,dx, \quad a>0,$$

Solution. We first factor out the a^4 from the expression in the denominator:

$$\int \frac{1}{(a^4-x^4)^{\frac{3}{4}}}\,dx = \int \frac{1}{\left[a^4\left(1-\left(\frac{x}{a}\right)^4\right)\right]^{\frac{3}{4}}}\,dx$$
$$= \frac{1}{a^3}\int \frac{1}{\left(1-\left(\frac{x}{a}\right)^4\right)^{\frac{3}{4}}}\,dx.$$

Now let $u = x/a$ so that $du = dx/a$ and $dx = a\,du$. Then

$$\int \frac{1}{(a^4-x^4)^{\frac{3}{4}}}\,dx = \frac{1}{a^3}\int \frac{a}{(1-u^4)^{\frac{3}{4}}}\,du$$
$$= \frac{1}{a^2}\operatorname{arcsin}_4(u) + C$$
$$= \frac{1}{a^2}\operatorname{arcsin}_4\left(\frac{x}{a}\right) + C$$

By similar techniques, we also have

$$\int \frac{1}{(a^p - x^p)^{\frac{p-1}{p}}}\,dx = \frac{1}{a^{p-2}} \operatorname{arcsin}_p\left(\frac{x}{a}\right) + C$$
$$\int \frac{1}{(a^p + x^p)^{\frac{2}{p}}}\,dx = \frac{1}{a} \arctan_p\left(\frac{x}{a}\right) + C$$

Example 9.2

Find the antiderivative of $\sin_4(t)\cos_4(t)$ *(Exercise 8 in [105]).*

Solution. To integrate a p-sine or p-cosine, we need to have $p-1$ cosines or sines, respectively, in the integral in order to use substitution. We will often have to multiply and divide by powers of sines and cosines to achieve this.

We will try the substitution $u = \sin_4(t)$. We will need $\cos_4(t)^3$ in the integral. We write

$$\int \sin_4(t)\cos_4(t)\,dt = \int \frac{\sin_4(t)\cos_4(t)^3}{\cos_4(t)^2}\,dt = \int \frac{\sin_4(t)\cos_4(t)^3}{\sqrt{1-\sin_4(t)^4}}\,dt.$$

We can now make our substitution $u = \sin_4(t)$, $du = \cos_4(t)^3\,dt$ to yield

$$\begin{aligned}\int \sin_4(t)\cos_4(t)\,dt &= \int \frac{u}{\sqrt{1-u^4}}\,du = \tfrac{1}{2}\arcsin(u^2) + C \\ &= \tfrac{1}{2}\arcsin(\sin_4(t)^2) + C.\end{aligned}$$

□

9.3 Powers of squigonometric functions

When we integrate even powers of sines and cosines, we often use double-angle identities to lower the exponents in the integral:

$$\cos(t)^2 = \tfrac{1}{2}(1+\cos(2t)),$$
$$\sin(t)^2 = \tfrac{1}{2}(1-\cos(2t)).$$

As we've already shown, though, the rate of change of the angle with respect to our parameter t, $\theta'(t) = \frac{1}{r(t)^2}$; that is, it is nonconstant for $p \neq 2$. The parameter t, which we know to represent area, is not so conducive to doubling formulas. We will get some in Chapter 16 but a quick glance will make clear that they will not be useful for simplifying anything.

Usually, if we want to integrate a power of a squigonometric function, we will have to use integration by parts and invoke reduction formulas [42]:

$$\int \cos_p(t)^m\,dt = \frac{1}{m-p+2}\cos_p(t)^{m-p+1}\sin_p(t) + \frac{m-p+1}{m-p+2}\int \cos_p(t)^{m-p}\,dt, \tag{9.1}$$

$$\int \sin_p(t)^m \, dt = -\frac{1}{m-p+2} \sin_p(t)^{m-p+1} \cos_p(t) + \frac{m-p+1}{m-p+2} \int \sin_p(t)^{m-p} \, dt,$$

for $m \neq p-2$.

Example 9.3

Determine $\int \cos_p(t)^p \, dt$.

Solution. We choose our parts so $u = \cos_p(t)$ and $dv = \cos_p(t)^{p-1} \, dt$. Then $du = -\sin_p(t)^{p-1} \, dt$, $v = \sin_p(t)$, and we can write

$$\begin{aligned} \int \cos_p(t)^p \, dt &= \cos_p(t) \sin_p(t) + \int \sin_p(t)^p \, dt \\ &= \cos_p(t) \sin_p(t) + \int \left[1 - \cos_p(t)^p\right] dt \\ &= t + \cos_p(t) \sin_p(t) - \int \cos_p(t)^p \, dt. \end{aligned} \tag{9.2}$$

Solving the equation for our integral yields

$$\int \cos_p(t)^p \, dt = \tfrac{1}{2}\left[t + \cos_p(t) \sin_p(t)\right] + C \tag{9.3}$$

□

We can similarly derive

$$\int \sin_p(t)^p \, dt = \tfrac{1}{2}\left[t - \cos_p(t) \sin_p(t)\right] + C.$$

We can also rewrite (9.2) to obtain

$$\int \left[\cos_p(t)^p - \sin_p(t)^p\right] dt = \cos_p(t) \sin_p(t) + C.$$

Some time in calculus courses is spent on handling antiderivatives of the form $\int \sin(t)^m \cos(t)^n \, dt$. This becomes problematic in the case where $p > 2$ mainly because direct formulas for $\int \sin_p(t)^m \, dt$ or $\int \cos_p(t)^m \, dt$ do not exist for $m < p-1$. (This is not true for related definite integrals; see Exercise 9.4.) We can, however, expand upon Euler's result that

$$\int_0^{\pi/2} \cos(t)^m \sin(t)^n \, dt = \tfrac{1}{2} \mathrm{B}\left(\tfrac{m+1}{2}, \tfrac{n+1}{2}\right) = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{m+n+2}{2}\right)}.$$

Shelupsky [89] introduced the KN_p function to expand this identity to the $p \neq 2$ setting. For $x > -1$, define

$$KN_p(x) = \int_0^\infty u^x e^{-u^p/p} \, du. \tag{9.4}$$

We note that $KN_p(x) = p^{\frac{x-p+1}{p}} \Gamma\left(\frac{x+1}{p}\right)$ and Equation (6.2) implies that $KN_p(x) = (x-p+1)KN_p(x-p)$.

With this in hand, we claim, for any m and n for which the integral converges,

$$\int_0^{\pi_p/2} \cos_p(t)^m \sin_p(t)^n \, dt = \frac{KN_p(m) \cdot KN_p(n)}{KN_p(m+n+1)}. \tag{9.5}$$

We prove this by first writing

$$\begin{aligned} KN_p(m) \cdot KN_p(n) &= \left(\int_0^\infty x^m e^{-x^p/p} \, du \right) \left(\int_0^\infty y^n e^{-y^p/p} \, dy \right) \\ &= \int_0^\infty \int_0^\infty x^m y^n e^{-(x^p+y^p)/p} \, dx \, dy. \end{aligned}$$

This is a double integral over the entire first quadrant. We now make the change of variable

$$x = R \cos_p(t), \quad y = R \sin_p(t), \tag{9.6}$$

where $0 \le t \le \pi_p/2$ and $R^p = x^p + y^p$. These are a form of "squircular" polar coordinates which we will examine further in Chapter 13. As with the usual polar coordinates, these variables with these ranges cover the entire first quadrant exactly once.

We must compute the Jacobian of the transformation (9.6):

$$\begin{aligned} \frac{\partial(x,y)}{\partial(R,t)} &= \begin{vmatrix} \partial x/\partial R & \partial y/\partial R \\ \partial x/\partial t & \partial y/\partial t \end{vmatrix} \\ &= \begin{vmatrix} \cos_p(t) & \sin_p(t) \\ -R \sin_p(t)^{p-1} & R \cos_p(t)^{p-1} \end{vmatrix} \\ &= R \cos_p(t)^p + R \sin_p(t)^p = R. \end{aligned}$$

Thus our area element in the double integral is $dA = R \, dR \, dt$ and our expression becomes

$$\begin{aligned} KN_p(m) \cdot KN_p(n) &= \int_0^{\pi_p/2} \int_0^\infty [R \cos_p(t)]^m [R \sin_p(t)]^n e^{-R^p/p} \, R \, dR \, dt \\ &= \int_0^\infty R^{m+n+1} e^{-R^p/p} \, dR \int_0^{\pi_p/2} \cos_p(t)^m \sin_p(t)^n \, dt \\ &= KN_p(m+n+1) \int_0^{\pi_p/2} \cos_p(t)^m \sin_p(t)^n \, dt \end{aligned}$$

Therefore we have shown (9.5) to be true. Rewriting it in terms of the gamma and beta functions, we obtain

$$\int_0^{\pi_p/2} \cos_p(t)^m \sin_p(t)^n \, dt = \frac{\Gamma(\frac{m+1}{p})\Gamma(\frac{n+1}{p})}{p\,\Gamma(\frac{m+n+2}{p})} = \frac{1}{p} \mathrm{B}(\tfrac{m+1}{p}, \tfrac{n+1}{p}). \tag{9.7}$$

9.4 Squigonometric substitution

Suppose we wish to find the area of the hyperellipse $|\frac{x}{a}|^p + |\frac{y}{b}|^p = 1$ where $a, b > 0$. The hyperellipse is symmetric about both the x- and y-axes, so the total area A will be four times the area in the first quadrant. (See Figure 9.1.) We can solve the hyperellipse equation for y to get the part of the hyperellipse in the first quadrant:

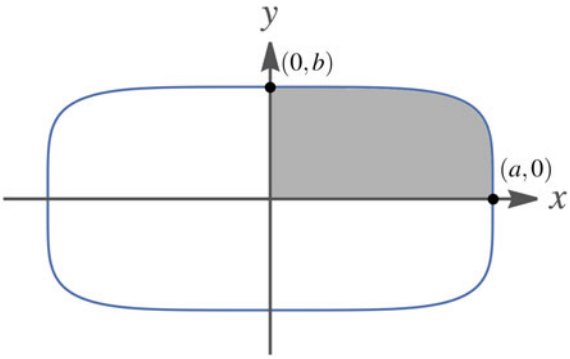


Figure 9.1 The hyperellipse $|x/a|^p + |y/b|^p = 1$

Table 9.1 Table of squigonometric substitutions

Expression	Substitution	Identity
$\sqrt[p]{a^p - x^p}$	$x = a\sin_p(t), -\frac{\pi_p}{2} \le t \le \frac{\pi_p}{2}$	$1 - \sin_p(t)^p = \cos_p(t)^p$
$\sqrt[p]{a^p + x^p}$	$x = a\tan_p(t), -\frac{\pi_p}{2} < t < \frac{\pi_p}{2}$	$1 + \tan_p(t)^p = \sec_p(t)^p$
$\sqrt[p]{x^p - a^p}$	$x = a\sec_p(t), 0 \le t < \frac{\pi_p}{2}$ or $\frac{\pi_p}{2} < t \le \pi_p$	$\sec_p(t)^p - 1 = \tan_p(t)^p$

$$y = \frac{b}{a}\sqrt[p]{a^p - x^p}, \quad 0 \le x \le a.$$

Thus the area A is given by

$$A = 4\int_0^a \frac{b}{a}\sqrt[p]{a^p - x^p}\,dx.$$

In some cases like this, we may be able to determine areas and volumes, or solve some indefinite integrals, using a method akin to trigonometric substitution:

- If the integrand contains $a^p - x^p$, we substitute $x = a\sin_p(t)$ and use the identity $1 - \sin_p(t)^p = \cos_p(t)^p$.
- If the integrand contains $a^p + x^p$, we substitute $x = a\tan_p(t)$ and use the identity $1 + \tan_p(t)^p = \sec_p(t)^p$.
- If the integrand contains $x^p - a^p$, we substitute $x = a\sec_p(t)$ and use the identity $\sec_p(t)^p - 1 = \tan_p(t)^p$.

Keep in mind that these are actually *inverse* substitutions – we must ensure that the substitution we define is invertible. To do this, we restrict t to a suitable domain (Table 9.1).

Example 9.4

Determine the area of the hyperellipse $|x/a|^p + |y/b|^p = 1$.

Solution. As noted earlier, the area is given by

$$A = \frac{4b}{a} \int_0^a \sqrt[p]{a^p - x^p}\, dx.$$

Set $x = a \sin_p(t)$. Then $dx = a \cos_p^{p-1}(t)\, dt$ and

$$\sqrt[p]{a^p - x^p} = \sqrt[p]{a^p - a^p \sin_p(t)^p} = \sqrt[p]{a^p \cos_p(t)^p} = a \cos_p(t),$$

since $0 \le t \le \pi_p/2$. Then Equation (9.3) gives us

$$\begin{aligned} A &= \frac{4b}{a} \int_0^a \sqrt[p]{a^p - x^p}\, dx \\ &= \frac{4b}{a} \int_0^{\pi_p/2} a \cos_p(t) \cdot a \cos_p(t)^{p-1}\, dt \\ &= 4ab \int_0^{\pi_p/2} \cos_p(t)^p\, dt \\ &= 2ab \Big[t + \cos_p(t) \sin_p(t) \Big]_0^{\pi_p/2} \\ &= \pi_p ab. \end{aligned} \tag{9.8}$$

□

We note a few splendid side effects of this example. First, a p-circle with radius a (that is, a hyperellipse with $a = b$) has area $\pi_p a^2$. Second, if the radius of our p-circle is 1 (that is $a = b = 1$), we get yet another integral expression for π_p:

$$\pi_p = 4 \int_0^1 \sqrt[p]{1 - x^p}\, dx. \tag{9.9}$$

Third, if we had back substituted for t in the last equation, using our p-triangle technique (see Figure 9.2), we get the antiderivative formula

$$\begin{aligned} \int \sqrt[p]{a^p - x^p}\, dx &= \frac{1}{2} a^2 \Big[t + \sin_p(t) \cos_p(t) \Big] + C \\ &= \frac{1}{2} a^2 \left[\operatorname{arcsin}_p\left(\frac{x}{a}\right) + \frac{x}{a} \cdot \frac{\sqrt[p]{a^p - x^p}}{a} \right] + C \\ &= \frac{1}{2} \Big[a^2 \operatorname{arcsin}_p\left(\frac{x}{a}\right) + x \sqrt[p]{a^p - x^p} \Big] + C. \end{aligned}$$

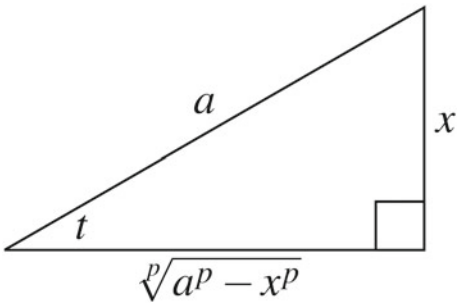


Figure 9.2 p-triangle for Example 9.4

In general, squigonometric substitution is an integration technique to be used sparingly; it can result in some amazing messes. But when it works, it can lead to discovering relationships between our new p-trigonometric functions and the traditional functions we know already. In the end, this sort of substitution can serve as an alternate route to approximations as well as a bridge between integrands with rational coefficients and integrands that are rational functions. The integration can become difficult as p increases.

Example 9.5

Evaluate

$$\int \frac{\sqrt[3]{1-x^3}}{x^3}\,dx.$$

Solution. We can use inverse squigonometric substitution with $x = \sin_3(t), -\pi_3/2 \leq t \leq \pi_3/2$. Then $dx = \cos_3(t)^2\,dt$ and

$$\sqrt[3]{1-x^3} = \sqrt[3]{1-\sin_3(t)^3} = \sqrt[3]{\cos_3(t)^3} = \cos_3(t),$$

since t leads to a Quadrant I or Quadrant IV angle. So we get

$$\begin{aligned}\int \frac{\sqrt[3]{1-x^3}}{x^3}\,dx &= \int \frac{\cos_3(t)}{\sin_3(t)^3}\,\cos_3(t)^2\,dt \\ &= \int \frac{\cos_3(t)^3}{\sin_3(t)^3}\,dt \\ &= \int \frac{1-\sin_3(t)^3}{\sin_3(t)^3}\,dt \\ &= \int \left(\frac{1}{\sin_3(t)^3} - 1\right) dt\end{aligned}$$

We note that (see Equation (34) in [42])

$$\int \frac{1}{\sin_p(t)^p}\,dt = \frac{p-2}{p-1}\,t - \frac{1}{p-1}\,\frac{\cos_p(t)}{\sin_p(t)^{p-1}} + C. \tag{9.10}$$

Using this with $p = 3$ above, we have

$$\begin{aligned}\int \frac{\sqrt[3]{1-x^3}}{x^3}\,dx &= \frac{1}{2}\left(t - \frac{\cos_3(t)}{\sin_3(t)^2}\right) - t + C \\ &= -\frac{1}{2}\left(t + \frac{\cos_3(t)}{\sin_3(t)^2}\right) + C.\end{aligned}$$

Since $x = \sin_3(t)$, we have that $\cos_3(t) = \sqrt[3]{1-\sin_3(t)^3} = \sqrt[3]{1-x^3}$ and thus

$$\int \frac{\sqrt[3]{1-x^3}}{x^3}\,dx = -\frac{1}{2}\left(\arcsin_3(x) + \frac{\sqrt[3]{1-x^3}}{x^2}\right) + C. \tag{9.11}$$

□

Example 9.6

Evaluate $\displaystyle\int \frac{1}{\sqrt[4]{x^4-a^4}}\,dx$ *for* $a > 0$.

Solution. Let $x = a\sec_4(t)$, $t \in [0, \pi_4/2) \cup (\pi_4/2, \pi]$. Then $dx = a\sec_4^2(t)\sin_4^3(t)\,dt$ and

$$\sqrt[4]{x^4-a^4} = \sqrt[4]{a^4(\sec_4(t)^4 - 1)} = \sqrt[4]{a^4\tan_4(t)^4} = a|\tan_4(t)|.$$

It must be the case that $x^4 > a^4$; that is, either $x > a$ or $x < -a$. If $x > a$, then $t \in [0, \pi_4/2)$, then $|\tan_4(t)| = \tan_4(t)$, and we have

$$\begin{aligned}\int \frac{dx}{\sqrt[4]{x^4-a^4}} &= \int \frac{a\sec_4(t)^2\sin_4(t)^3}{a\tan_4(t)}\,dt \\ &= \int \frac{\sin_4(t)^2}{\cos_4(t)}\,dt \\ &= \int \frac{\sin_4(t)^2\cos_4(t)^3}{\cos_4(t)^4}\,dt \\ &= \int \frac{\sin_4(t)^2}{1-\sin_4(t)^4}\cos_4(t)^3\,dt\end{aligned}$$

Now we set $u = \sin_4(t)$. Then $du = \cos_4(t)^3\,dt$ and we get

$$\int \frac{dx}{\sqrt[4]{x^4-a^4}} = \int \frac{u^2}{1-u^4}\,du$$

Employing partial fractions, we obtain

$$\begin{aligned}\int \frac{dx}{\sqrt[4]{x^4-a^4}} &= \int \left[\frac{1}{4(1+u)} + \frac{1}{4(1-u)} - \frac{1}{2(1+u^2)}\right] du \\ &= \frac{1}{4}\ln|1+u| - \frac{1}{4}\ln|1-u| - \frac{1}{2}\arctan_2 u + C \\ &= \frac{1}{4}\ln\left|\frac{1+\sin_4(t)}{1-\sin_4(t)}\right| - \frac{1}{2}\arctan_2(\sin_4(t)) + C\end{aligned}$$

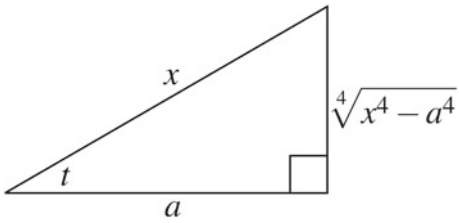


Figure 9.3 4-triangle for Example 9.6

From the triangle in Figure 9.3, we see that if $\sec_4(t) = x/a$, then $\sin_4(t) = \sqrt[4]{x^4-a^4}/x$ and so

$$\begin{aligned}\int \frac{dx}{\sqrt[4]{x^4-a^4}} &= \frac{1}{4}\ln\left|\frac{1+\sqrt[4]{x^4-a^4}/x}{1-\sqrt[4]{x^4-a^4}/x}\right| - \frac{1}{2}\arctan_2\left(\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C\\ &= \frac{1}{4}\ln\left|\frac{x+\sqrt[4]{x^4-a^4}}{x-\sqrt[4]{x^4-a^4}}\right| - \frac{1}{2}\arctan\left(\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C\end{aligned}$$

If $x < -a$ then $t \in (\pi_4/2, \pi_4]$, then $\tan_4(t) < 0$ and $|\tan_4(t)| = -\tan_4(t)$. So we let $w = -x$ so that $dw = -dx$ and

$$\begin{aligned}\int \frac{dx}{\sqrt[4]{x^4-a^4}} &= -\int \frac{dw}{\sqrt[4]{w^4-a^4}}\\ &= -\frac{1}{4}\ln\left|\frac{w+\sqrt[4]{w^4-a^4}}{w-\sqrt[4]{w^4-a^4}}\right| + \frac{1}{2}\arctan\left(\frac{\sqrt[4]{w^4-a^4}}{w}\right)+C\\ &= -\frac{1}{4}\ln\left|\frac{(-x)+\sqrt[4]{(-x)^4-a^4}}{(-x)-\sqrt[4]{(-x)^4-a^4}}\right| + \frac{1}{2}\arctan\left(\frac{\sqrt[4]{(-x)^4-a^4}}{(-x)}\right)+C\\ &= -\frac{1}{4}\ln\left|\frac{-x+\sqrt[4]{x^4-a^4}}{-x-\sqrt[4]{x^4-a^4}}\right| + \frac{1}{2}\arctan\left(-\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C\\ &= \frac{1}{4}\ln\left|\frac{-x-\sqrt[4]{x^4-a^4}}{-x+\sqrt[4]{x^4-a^4}}\right| - \frac{1}{2}\arctan\left(\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C\\ &= \frac{1}{4}\ln\left|\frac{x+\sqrt[4]{x^4-a^4}}{x-\sqrt[4]{x^4-a^4}}\right| - \frac{1}{2}\arctan\left(\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C.\end{aligned}$$

In either case, we get

$$\int \frac{dx}{\sqrt[4]{x^4-a^4}} = \frac{1}{4}\ln\left|\frac{x+\sqrt[4]{x^4-a^4}}{x-\sqrt[4]{x^4-a^4}}\right| - \frac{1}{2}\arctan\left(\frac{\sqrt[4]{x^4-a^4}}{x}\right)+C.$$

Exercises

▶ **Exercise 9.1** *Find* $\displaystyle\int \frac{1}{x^{10}\sqrt[10]{x^{10}+1024}}\,dx.$

▶ **Exercise 9.2** *Calculate* $\displaystyle\int_0^{3/2} \frac{x^7}{(16x^4+81)^{\frac{5}{4}}}\,dx.$

▶ **Exercise 9.3** *Calculate* $\displaystyle\int_0^{1} \frac{1}{(1+x^5)^{\frac{11}{5}}}\,dx.$

▶ **Exercise 9.4** *Find* $\displaystyle\int \operatorname{arctan}_4 x\,dx.$

▶ **Exercise 9.5** *Prove Equation (9.10) by writing the integrand as*

$$\frac{1}{\sin_p(t)^p} = \frac{\sin_p(t)^p + \cos_p(t)^p}{\sin_p(t)^p} = 1 + \frac{\cos_p(t)^p}{\sin_p(t)^p}.$$

Integrate the second term by parts with

$$u = \cos_p(t), \quad dv = \frac{\cos_p(t)^{p-1}}{\sin_p(t)^p}\,dt.$$

▶ **Exercise 9.6** *Show that, for* $k > 0$,

$$\int_0^{\pi_p/2} \sin_p(t)^k\,dt = \int_0^{\pi_p/2} \cos_p(t)^k\,dt = \frac{1}{p}\,\mathrm{B}\left(\frac{k+1}{p}, \frac{1}{p}\right).$$



Three applications

10

As academic an exercise as implementing our squigonometric functions can be, and despite the difficulties in working with these functions, we can use them to extend results in real analysis and to solve real-world problems. We will examine three applications of these new functions to questions of area and volume.

10.1 The area of Sergels Torg

The first two problems come from the poet, philosopher and scientist Piet Hein (1905–1996). He made his indelible mark in mathematics as a designer.

Hein, a Dane, had a long career as a philosopher; many know him for his large number of "grooks," his 7000 short poems that often gave a quick moral. One of Hein's most famous grooks goes

> Taking fun as simply fun
> and earnestness in earnest
> Shows how thoroughly thou
> none of the two discernest

Hein had a lot of earnest fun in the field of recreational mathematics. He invented the Soma cube (additionally marketed these days as the Block by Block puzzle). He and John Nash invented (independently of each other) the game Hex.

He also came up with a great philosophical math thought experiment. Hein noticed that many of the objects of modern life encompassed circles or rectangles, and devoted time to determine which closed curve would be a happy medium, a curve

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_10

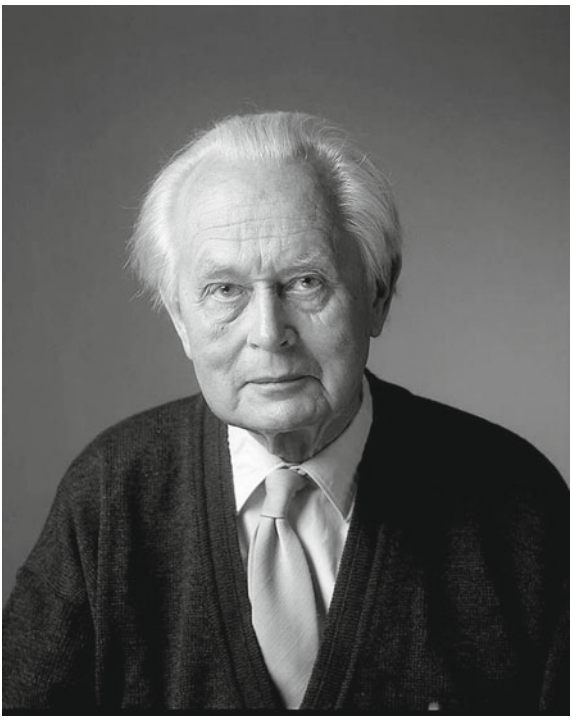


Figure 10.1 Piet Hein (1905–1996)

that would incorporate the most pleasing aspects of both [35][1]. Hein addressed this problem at the request of urban planners in Stockholm, Sweden in 1959. The planners were attempting to flow traffic through a rectangular plot of land, in the center of which was an oval-shaped fountain. Hein's solution was to lay out the street surrounding the fountain in the shape of a hyperellipse (or as Hein called it, a "superellipse"). The resulting central square, Sergels Torg, is considered to be an example of harmonious city design.

This p-hyperellipse, the curve with equation $|x/a|^p + |y/b|^p = 1$, $p > 2$ which we've seen already, and its associated solid of revolution, the *superegg*, became a model for Scandinavian architectural and furniture design throughout the 1960s. (Hein used $p = 2.5$ for his design of Sergels Torg and for his supereggs mainly because he found that value to be the most visually pleasing.)

As we discovered in Equation (9.8), the area of a p-hyperellipse with semi-axes a and b is given by $\pi_p ab$. Using Google Earth for measurement, we find that the

[1] Squarish-circles are not uncommon in art, but not all are squircles as we have defined them. For example, one can replace the corners of a square with circular arcs. The construction of *polycentric ovals* (not ellipses) formed by combining circular arcs that meet with common tangents has a rich history in art and architecture, along with some compelling mathematical questions [71]. It might be interesting to explore what could be done with authentic squircular arcs.



Figure 10.2 Sergels Torg in Stockholm, Sweden. Photograph by Anders Bengtsson, distributed under a CC-BY 2.0 license.

semi-major axis for Sergels Torg is 26.26 m and the semi-minor axis is 24.95 m. This gives us an area for Sergels Torg of approximately

$$\pi_{2.5}(26.26)(24.95) \approx 2215.14 \text{ m}^2,$$

or about 0.55 acres. (The value of $\pi_{2.5}$ is approximately 3.38094.)

10.2 The volume of a superegg

Hein's superegg became a *cause célèbre* in the late 1960s and early 1970s. The superegg has zero curvature on its top and bottom, and can sit on a table without toppling, unlike its $p = 2$ cousin the ellipsoid. At the website named for Hein [22], one can obtain small supereggs to chill a drink, larger hyperelliptical mixing bowls, a superellipse shaped barstool, and huge sculptures to dress up a room. (See Figure 10.3[2]).

Since the superegg is the solid of revolution for the hyperellipse, we can find its volume via the usual formula. We assume the hyperellipse $|x/a|^p + |y/b|^p = 1$ is rotated about the x-axis. Then the volume of the resulting superegg is

$$V_p = 2\int_0^a \pi \left(\frac{b}{a}\sqrt[p]{a^p - x^p}\right)^2 dx.$$

[2] Grook and photos included with the kind permission of Piet Hein A/S - Denmark – http://piethein.com



(a) Twin bowls



(b) Pepper grinder



(c) Lamp

Figure 10.3 Various hyperellipse and superegg housewares

We can use inverse squigonometric substitution here. Let $x = a\sin_p(t)$. Then $dx = a\cos_p(t)^{p-1}\,dt$ and we have

$$\begin{aligned} V_p &= \frac{2\pi b^2}{a^2}\int_0^a (a^p - x^p)^{\frac{2}{p}}\,dx \\ &= \frac{2\pi b^2}{a^2}\int_0^{\pi_p/2} [a\cos_p(t)]^2 \cdot a\cos_p(t)^{p-1}\,dt \\ &= 2\pi ab^2 \int_0^{\pi_p/2} \cos_p(t)^{p+1}\,dt. \end{aligned}$$

We now use the reduction formula (9.1) to obtain

$$\begin{aligned} V_p &= 2\pi ab^2\left[\frac{1}{3}\cos_p(t)^2\sin_p(t)\Big|_0^{\pi_p/2} + \frac{2}{3}\int_0^{\pi_p/2}\cos_p(t)\,dt\right] \\ &= \frac{4\pi ab^2}{3}\int_0^{\pi_p/2}\cos_p(t)\,dt. \end{aligned}$$

We now use formula (9.7) with $m = 1$ and $n = 0$ to get

$$\begin{aligned} V_p &= \frac{4\pi ab^2}{3}\cdot\frac{1}{p}\,\mathrm{B}\left(\tfrac{2}{p},\tfrac{1}{p}\right) \\ &= \frac{4\pi ab^2}{3p}\,\frac{\Gamma(\frac{1}{p})\Gamma(\frac{2}{p})}{\Gamma(\frac{3}{p})} \end{aligned}$$

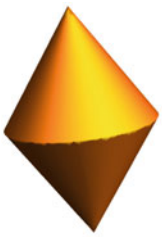
We can now determine exact volumes for some supereggs. When $p = 1$, we have

$$V_1 = \frac{4\pi ab^2}{3\cdot 1}\cdot\frac{\Gamma(1)\Gamma(2)}{\Gamma(3)} = \frac{4\pi ab^2}{3}\cdot\frac{0!1!}{2!} = \frac{2\pi ab^2}{3},$$

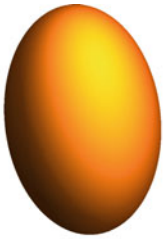
or about $2.0944ab^2$, and we rediscover the formula for the volume of a (double) cone. (See Figure 10.4.)

For $p = 2$, we get via Equation (6.2):

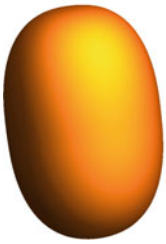
$$V_2 = \frac{4\pi ab^2}{3\cdot 2}\cdot\frac{\Gamma(\frac{1}{2})\Gamma(1)}{\Gamma(\frac{3}{2})} = \frac{2\pi ab^2}{3}\cdot\frac{\Gamma(\frac{1}{2})\cdot 0!}{\frac{1}{2}\Gamma(\frac{1}{2})} = \frac{4\pi ab^2}{3} \approx 4.1888ab^2.$$



(a) $p = 1$



(b) $p = 2$



(c) $p = 2.5$



(d) $p = 4$

Figure 10.4 Supereggs for various p

When $p = 3$, Equation (6.6) gives

$$\begin{aligned} V_3 &= \frac{4\pi ab^2}{3 \cdot 3} \cdot \frac{\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})}{\Gamma(1)} \\ &= \frac{4\pi ab^2}{9} \cdot \Gamma(\tfrac{1}{3})\Gamma(1 - \tfrac{1}{3}) \\ &= \frac{4\pi ab^2}{9} \cdot \frac{\pi}{\sin \frac{\pi}{3}} \\ &= \frac{4\pi ab^2}{9} \cdot \frac{2\pi}{\sqrt{3}} \\ &= \frac{8\pi^2 ab^2}{3^{5/2}} \approx 5.0651 ab^2. \end{aligned}$$

For $p = 4$, we obtain from Equations (6.6) and (6.8) and Exercise 6.10,

$$\begin{aligned} V_4 &= \frac{4\pi ab^2}{3 \cdot 4} \cdot \frac{\Gamma(\frac{1}{4})\Gamma(\frac{2}{4})}{\Gamma(\frac{3}{4})} \\ &= \frac{\pi ab^2}{3} \cdot \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\pi/(\Gamma(\frac{1}{4})\sin(\pi/4))} \\ &= \frac{\pi ab^2}{3} \cdot \frac{\left[\Gamma(\frac{1}{4})\right]^2 \sqrt{\pi}}{\sqrt{2}\,\pi} \end{aligned}$$

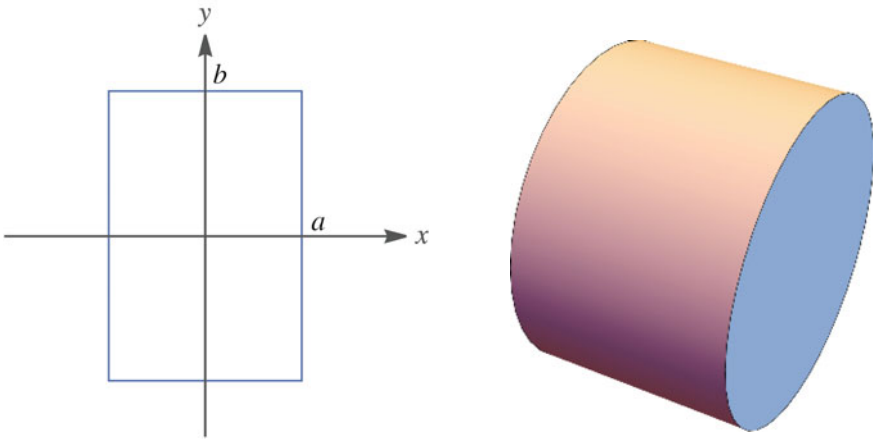


Figure 10.5 Hyperellipse (left) and superegg (right) for $p = \infty$

$$= \frac{\pi ab^2}{3} \cdot \frac{2\pi_4\sqrt{\pi}\cdot\sqrt{\pi}}{\sqrt{2}\,\pi}$$
$$= \frac{\pi_4\,\pi\sqrt{2}\,ab^2}{3} \approx 5.4916ab^2.$$

We notice that the coefficient of ab^2 is increasing along with p. Geometrically, supereggs approach a circular cylinder as $p \to \infty$. For $p = \infty$, a p-hyperellipse would be a rectangle of width $2a$ and height $2b$ and the superegg obtained from rotating the rectangle about the x-axis would have volume

$$V_\infty = (\text{area of base})(\text{height}) = \pi \cdot b^2 \cdot 2a = 2\pi ab^2 \approx 6.28319ab^2.$$

Hein's superegg

While we've met with success in devising exact formulas for volumes of p-supereggs for $p = 1, 2, 3$ and 4 (and ∞) there appears to be no set formula for these volumes solely in terms of various π_p's (as opposed to gamma functions). For instance, though there are for $p = 6$ and $p = 8$, we have not yet found any for $p = 5$ or $p = 7$. (See the exercises at the end of this chapter.)

The volume of Hein's superegg (with $p = 5/2$) is given by

$$V_{5/2} = \frac{4\pi ab^2}{3\left(\frac{5}{2}\right)} \cdot \frac{\Gamma(\frac{1}{5/2})\Gamma(\frac{2}{5/2})}{\Gamma(\frac{3}{5/2})} = \frac{8\pi ab^2}{15} \cdot \frac{\Gamma(\frac{2}{5})\Gamma(\frac{4}{5})}{\Gamma(\frac{6}{5})}. \tag{10.1}$$

We would like to come up with an exact value for this involving various π_p's. We first note that, using Equation (6.6), we have

$$\pi_5 = \frac{2\left[\Gamma(\frac{1}{5})\right]^2}{5\Gamma(\frac{2}{5})} = \frac{2}{5} \cdot \frac{\Gamma(\frac{1}{5})}{\Gamma(\frac{2}{5})} \cdot \frac{\pi}{\Gamma(\frac{4}{5})\sin(\frac{\pi}{5})}. \tag{10.2}$$

There are several ways to evaluate $\sin(\frac{\pi}{5})$. If we let $\theta = \frac{\pi}{5}$, then $5\theta = \pi$. This means that $3\theta = \pi - 2\theta$ and thus $\sin(3\theta) = \sin(\pi - 2\theta)$. Since $\sin(\pi - t) = \sin(t)$ for any acute angle t, we have that $\sin(3\theta) = \sin(2\theta)$. Expanding both of these, we get

$$3\sin(\theta) - 4\sin(\theta)^3 = 2\sin(\theta)\cos(\theta).$$

Since $\theta = \frac{\pi}{5}$, $\sin(\theta) \neq 0$ and we can divide our equation by $\sin(\theta)$, obtaining

$$\begin{aligned} 3 - 4\sin(\theta)^2 &= 2\cos(\theta) \\ 3 - 4(1 - \cos(\theta)^2) &= 2\cos(\theta) \\ 4\cos(\theta)^2 - 2\cos(\theta) - 1 &= 0 \end{aligned}$$

We can solve this quadratic equation in $\cos(\theta)$ to get

$$\cos(\theta) = \frac{1 \pm \sqrt{5}}{4}.$$

But θ is an acute angle and thus $\cos(\theta) > 0$. So it must be that $\cos(\theta) = \frac{1+\sqrt{5}}{4}$. Therefore,

$$\begin{aligned} \sin(\theta) &= \sqrt{1 - \left(\frac{1+\sqrt{5}}{4}\right)^2} \\ &= \sqrt{1 - \left(\frac{6+2\sqrt{5}}{16}\right)} \\ &= \sqrt{\frac{16 - 6 - 2\sqrt{5}}{16}} \\ &= \frac{\sqrt{10 - 2\sqrt{5}}}{4}. \end{aligned}$$

Thus $\sin(\frac{\pi}{5}) = \frac{\sqrt{10-2\sqrt{5}}}{4}$. Plugging this into Equation (10.2) yields

$$\begin{aligned} \pi_5 &= \frac{2}{5} \cdot \frac{\Gamma(\frac{1}{5})}{\Gamma(\frac{2}{5})} \cdot \frac{\pi}{\Gamma(\frac{4}{5})\sin(\frac{\pi}{5})} \\ &= 2 \cdot \frac{\frac{1}{5}\Gamma(\frac{1}{5})}{\Gamma(\frac{2}{5})\Gamma(\frac{4}{5})} \cdot \frac{4\pi}{\sqrt{10 - 2\sqrt{5}}} \\ &= \frac{8\pi}{\sqrt{10 - 2\sqrt{5}}} \cdot \frac{\Gamma(\frac{6}{5})}{\Gamma(\frac{2}{5})\Gamma(\frac{4}{5})}, \end{aligned}$$

which leads to

$$\frac{\Gamma(\frac{2}{5})\Gamma(\frac{4}{5})}{\Gamma(\frac{6}{5})} = \frac{8\pi}{\pi_5\sqrt{10 - 2\sqrt{5}}}.$$

Substituting this into Equation (10.1) gives us an expression for the volume of Hein's superegg:

$$\begin{aligned} V_{5/2} &= \frac{8\pi ab^2}{15} \cdot \frac{8\pi}{\pi_5\sqrt{10-2\sqrt{5}}} \\ &= \frac{64\pi^2\sqrt{10+2\sqrt{5}}}{15\pi_5\sqrt{80}}ab^2 \\ &= \frac{16\pi^2}{15\pi_5}\sqrt{\frac{10+2\sqrt{5}}{5}}\,ab^2 \approx 4.7126ab^2. \end{aligned}$$

In particular, the 10 cm diameter bowls seen in Figure 10.3a (two of which comprise a superegg) each can contain approximately $\frac{1}{2}(4.7126)(0.5\times 10)^3 \approx 294.535$ cubic centimeters, or 0.295 liters of material.

10.3 The trisection of an area

One reason we use absolute values in the defining equation of a p-circle is that, for $p > 1$ an odd integer, the graph of the curve $x^p + y^p = 1$ traces only the first quadrant portion of the p-circle, but then runs off to the northwest and southeast outside of the first quadrant, toward the oblique asymptote $y = -x$. (See Figure 10.6.)

In [23], Ding and Chebolu used a geometric argument to prove that the area between the curve $x^3 + y^3 = 1$ and the slant asymptote is trisected by the coordinate axes. That is, from the figure, $A_1 = A_2 = A_3$. Their proof used the fact that $|\sin_3(t)| > 1$ only if t is complex but not real. (We will explore this idea in Chapter 14.) We will prove this via integration over a real interval using inverse squigonometric substitution. (See Section 9.4.)

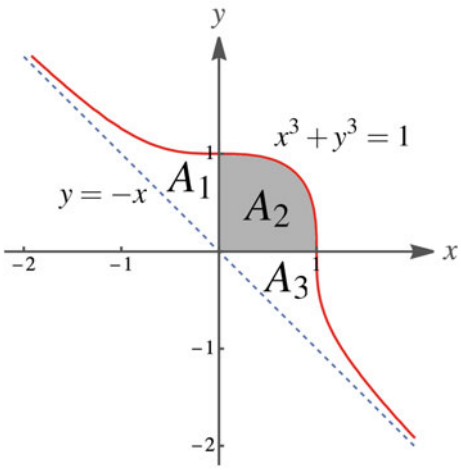


Figure 10.6 The curve $x^3 + y^3 = 1$ (solid) and its oblique asymptote $y = -x$ (dashed).

We first note that

$$A_2 = \int_0^1 \sqrt[3]{1-x^3}\,dx = \frac{\pi_3}{4}$$

by Equation (9.8). We next see from Figure 10.6 that

$$A_1 = \int_{-\infty}^0 \left[\sqrt[3]{1-x^3} - (-x)\right] dx = \int_{-\infty}^0 \left[\sqrt[3]{1-y^3} - (-y)\right] dy = A_3.$$

Therefore, our goal will be to show that $A_1 = A_3 = \frac{\pi_3}{4}$. In A_1, we make the substitution $x \mapsto -x$ to write

$$A_1 = \int_0^\infty \left[\sqrt[3]{1+x^3} - x\right] dx$$

(Geometrically, think of flipping the graph in Figure 10.6 about the y-axis.)

We now set $x = \tan_3(t)$. Then $dx = \sec_3(t)^2\,dt$ and $\sqrt[3]{1+x^3} = \sec_3(t)$. We obtain via Formula 19 in Appendix B.2

$$\begin{aligned}
A_1 &= \int_0^\infty \left[\sqrt[3]{1+x^3} - x\right] dx \\
&= \int_0^{\pi_3/2} \left[\sec_3(t) - \tan_3(t)\right] \sec_3(t)^2\,dt \\
&= \int_0^{\pi_3/2} \sec_3(t)^3\,dt - \int_0^{\pi_3/2} \tan_3(t)\sec_3(t)^2\,dt \\
&= \left[\frac{1}{2}\,t + \frac{1}{2}\,\frac{\sin_3(t)}{\cos_3(t)^2}\right]_0^{\pi_3/2} - \left[\frac{\tan_3(t)^2}{2}\right]_0^{\pi_3/2}
\end{aligned}$$

We know that both expressions are infinite at $t = \frac{\pi_3}{2}$ but we will hold off on taking the limits just yet:

$$\begin{aligned}
A_1 &= \frac{\pi_3}{4} + \lim_{t\to\frac{\pi_3}{2}^-}\left[\frac{1}{2}\,\frac{\sin_3(t)}{\cos_3(t)^2}\right] - \lim_{t\to\frac{\pi_3}{2}^-}\left[\frac{1}{2}\,\frac{\sin_3(t)^2}{\cos_3(t)^2}\right] \\
&= \frac{\pi_3}{4} + \frac{1}{2}\lim_{t\to\frac{\pi_3}{2}^-}\left[\frac{\sin_3(t)}{\cos_3(t)^2} - \frac{\sin_3(t)^2}{\cos_3(t)^2}\right]
\end{aligned}$$

Let ℓ be the remaining limit. We can write

$$\begin{aligned}
\ell &= \lim_{t\to\frac{\pi_3}{2}^-}\left[\frac{\sin_3(t) - \sin_3(t)^2}{\cos_3(t)^2}\right] \\
&= \lim_{t\to\frac{\pi_3}{2}^-}\left[\frac{1-\sin_3(t)}{\cos_3(t)^2}\cdot\sin_3(t)\right] \\
&= \lim_{t\to\frac{\pi_3}{2}^-}\frac{1-\sin_3(t)}{\cos_3(t)^2}
\end{aligned}$$

since $\sin_3(t) \to 1$ as $t \to \frac{\pi_3}{2}^-$. We can now use L'Hôpital's Rule to get

$$\begin{aligned}\ell &= \lim_{t \to \frac{\pi_3}{2}^-} \frac{-\cos_3(t)^2}{-2\cos_3(t)\sin_3(t)^2} \\ &= \lim_{t \to \frac{\pi_3}{2}^-} \frac{\cos_3(t)}{2\sin_3(t)^2} = \frac{0}{2} = 0.\end{aligned}$$

Therefore, $A_1 = \frac{\pi_3}{4}$, which means $A_1 = A_2 = A_3$. The coordinate axes thus trisect the region between the curve $x^3 + y^3 = 1$ and its oblique asymptote $y = -x$.

Exercises

▶ **Exercise 10.1** *Prove that*

$$\Gamma(\tfrac{1}{6}) = \sqrt{3\pi_6 \sqrt[3]{\pi_3 \pi \sqrt{3}}}.$$

and use this result to show that the volume of a $p = 6$ superegg is

$$V_6 = \frac{2\pi\sqrt{\pi_3\pi_6}\,ab^2}{3^{5/4}} \approx 5.87345ab^2.$$

▶ **Exercise 10.2** *Prove that*

$$\Gamma(\tfrac{1}{8}) = 2\sqrt{\pi_8 \sqrt{2\pi_4\sqrt{\pi}}}.$$

and then show that the volume of a $p = 8$ superegg is

$$V_8 = \frac{2^{1/4}\pi_4\pi}{3}\sqrt{1 + \frac{\sqrt{2}}{2}}\,ab^2 \approx 6.03356ab^2.$$

▶ **Exercise 10.3** *Show via limits that the volume of the superegg $V_p \to 2\pi ab^2$ as $p \to \infty$.*

▶ **Exercise 10.4** *Piet Hein invented the Soma cube puzzle in 1933 [22]. The puzzle consists of seven pieces of either three or four unit cubes each, that must be assembled into a $3 \times 3 \times 3$ cube [36].*

1. *Show that there are 240 possible solutions to the Soma cube puzzle.*
2. *Parker Brothers marketed the Soma cube in the United States in the 1970s and claimed on the packaging that there are 1,105,920 solutions to the puzzle. Where did Parker Brothers get this number (or did they just make it up)?*

▶ **Exercise 10.5** *Prove Formula B.2 in Appendix B.2:*

$$\int \sec_p(u)^p\,du = \int \frac{1}{\cos_p(u)^p}\,du = \frac{p-2}{p-1}u + \frac{1}{p-1}\frac{\sin_p(u)}{\cos_p(u)^{p-1}} + C$$

▶ **Exercise 10.6** *Show that the coordinate axes only trisect the region between the curve* $x^p + y^p = 1$ *(*p *an odd integer greater than 1) and its oblique asymptote* $y = -x$ *for* $p = 3$.

Project

Project 10.1 (Values of the gamma function and supereggs)
What other values of the gamma function can be expressed in terms of the π_p's? Can you get any new geometric formulas from them?

Project 10.2 (Computational squigonometry)
A computer can be an invaluable tool in developing intuition, drawing pictures, and forming conjectures, but the functions we have studied are generally not implemented in most computational software. Implementing these functions in a programming language or computer algebra system would provide a robust set of tools to explore them further. Note that this task brings different challenges as you consider these functions symbolically, numerically, and graphically. Consider making use of identities involving functions that may already be implemented, such as the gamma, hypergeometric, and Weierstrass elliptic functions.

Check for updates

Infinite series

11

As with the usual trigonometric functions, we would like to have a method of approximation for squigonometric functions or their inverses. Since the inverse p-trigonometric functions are defined by integrals, we can use methods such as Simpson's rule [93, Section 7.7] to obtain good estimates. For other applications, and for the p-trigonometric functions, it would be good to find representative infinite series.

To ensure differentiability, we restrict our attention in this chapter to positive integers p (see Exercise 6.8).

11.1 Inverse squigonometric functions

It turns out to be easier to obtain first the series for the p-arcsine and p-arctangent functions. We recall that $\frac{d}{dx}(\arcsin_p(x)) = (1-x^p)^{\frac{1-p}{p}}$ for $0 \le x < 1$. We can use the binomial theorem to develop first a series for $(1-x^p)^{\frac{1-p}{p}}$ and then integrate the series term-by-term to obtain a power series for $\arcsin_p(x)$.

$$
\begin{aligned}
&(1-x^p)^{\frac{1-p}{p}} \\
&\quad = \sum_{n=0}^{\infty} \binom{\frac{1-p}{p}}{n} (-x^p)^n \\
&\quad = 1 - \frac{1-p}{p}x^p + \frac{\left(\frac{1-p}{p}\right)\left(\frac{1-p}{p}-1\right)}{2!}x^{2p} - \frac{\left(\frac{1-p}{p}\right)\left(\frac{1-p}{p}-1\right)\left(\frac{1-p}{p}-2\right)}{3!}x^{3p} + \cdots \\
&\quad = 1 - \frac{1-p}{p}x^p + \frac{\left(\frac{1-p}{p}\right)\left(\frac{1-2p}{p}\right)}{2!}x^{2p} - \frac{\left(\frac{1-p}{p}\right)\left(\frac{1-2p}{p}\right)\left(\frac{1-3p}{p}\right)}{3!}x^{3p} + \cdots
\end{aligned}
$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_11

Table 11.1 Power series for p-arcsine functions

$p=1$	$\arcsin_1(x) = x$
$p=2$	$\arcsin_2(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + \frac{35x^9}{1152} + \cdots$
$p=3$	$\arcsin_3(x) = x + \frac{x^4}{6} + \frac{5x^7}{63} + \frac{4x^{10}}{81} + \frac{110x^{13}}{3159} + \cdots$
$p=4$	$\arcsin_4(x) = x + \frac{3x^5}{20} + \frac{7x^9}{96} + \frac{77x^{13}}{1664} + \frac{1155x^{17}}{34,816} + \cdots$
$p=5$	$\arcsin_5(x) = x + \frac{2x^6}{15} + \frac{18x^{11}}{275} + \frac{21x^{16}}{500} + \frac{19x^{21}}{625} + \cdots$
$p=6$	$\arcsin_6(x) = x + \frac{5x^7}{42} + \frac{55x^{13}}{936} + \frac{935x^{19}}{24,624} + \frac{4301x^{25}}{155,520} + \cdots$

$$
\begin{aligned}
&= 1 + \frac{p-1}{p}x^p + \frac{\left(\frac{p-1}{p}\right)\left(\frac{2p-1}{p}\right)}{2!}x^{2p} + \frac{\left(\frac{p-1}{p}\right)\left(\frac{2p-1}{p}\right)\left(\frac{3p-1}{p}\right)}{3!}x^{3p} + \cdots \\
&= 1 + \frac{p-1}{1!p}x^p + \frac{(p-1)(2p-1)}{2!p^2}x^{2p} + \frac{(p-1)(2p-1)(3p-1)}{3!p^3}x^{3p} + \cdots \\
&= 1 + \sum_{n=1}^{\infty} \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{n!p^n}x^{np}
\end{aligned}
$$

If we integrate this series, we get

$$\arcsin_p(x) = x + \sum_{n=1}^{\infty} \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{n!p^n(np+1)}x^{np+1} \tag{11.1}$$

since $\arcsin_p(0) = 0$. Some results for particular p are listed in Table 11.1.

When p is an even integer, the resulting series represents an odd function (which is what we would like our arcsines to be), and thus actually holds for $-1 < x < 1$. When p is an odd integer, the series represents a function with no axis symmetry, so the series converges, but not to $\arcsin_p(x)$ for $-1 < x < 0$. However, for these x's, we define $\arcsin_p(x) = -\arcsin_p(-x)$, and we can use the negative of our series with $-x$ substituted for x.

In a similar manner, we can show that for $0 \le x < 1$,

$$\arctan_p(x) = x + \sum_{n=1}^{\infty}(-1)^n \frac{2(p+2)(2p+2)\cdots((n-1)p+2)}{n!p^n(np+1)}x^{np+1}.$$

(This is left as an exercise at the end of the chapter.) Some results for particular p are shown in Table 11.2. In particular note that $\arctan_1(x) = \frac{x}{1+x}$. Note that for p not an odd integer, we must define $\arctan_p(x) = -\arctan_p(-x)$ for $-1 < x < 0$.

To further examine the arcsine series (11.1), we introduce the *rising factorial* or *Pochhammer symbol*. For n a nonnegative integer, we define

$$(q)_n = \begin{cases} 1 & n = 0 \\ q(q+1)(q+2)\cdots(q+n-1) & n > 0 \end{cases} = \frac{\Gamma(q+n)}{\Gamma(q)}.$$

Table 11.2 Power series for p-arctangent functions

$p=1$	$\arctan_1(x) = x - x^2 + x^3 - x^4 + x^5 - \cdots = \frac{x}{1+x}$
$p=2$	$\arctan_2(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots$
$p=3$	$\arctan_3(x) = x - \frac{x^4}{6} + \frac{5x^7}{63} - \frac{4x^{10}}{81} + \frac{110x^{13}}{3159} - \cdots$
$p=4$	$\arctan_4(x) = x - \frac{x^5}{10} + \frac{x^9}{24} - \frac{5x^{13}}{208} + \frac{35x^{17}}{2176} - \cdots$
$p=5$	$\arctan_5(x) = x - \frac{x^6}{15} + \frac{7x^{11}}{275} - \frac{7x^{16}}{500} + \frac{17x^{21}}{1875} - \cdots$
$p=6$	$\arctan_6(x) = x - \frac{x^7}{21} + \frac{2x^{13}}{117} - \frac{14x^{19}}{1539} + \frac{7x^{25}}{1215} - \cdots$

So, for instance

$$\begin{aligned}\left(\frac{1}{p}\right)_n &= \left(\frac{1}{p}\right)\left(\frac{1}{p}+1\right)\left(\frac{1}{p}+2\right)\cdots\left(\frac{1}{p}+n-1\right)\\ &= \frac{(p+1)(2p+1)\cdots((n-1)p+1)}{p^n}\end{aligned}$$

Similarly,

$$\begin{aligned}\left(\frac{p-1}{p}\right)_n &= \left(\frac{p-1}{p}\right)\left(\frac{p-1}{p}+1\right)\left(\frac{p-1}{p}+2\right)\cdots\left(\frac{p-1}{p}+n-1\right)\\ &= \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{p^n},\end{aligned}$$

and

$$\begin{aligned}\left(\frac{p+1}{p}\right)_n &= \left(\frac{p+1}{p}\right)\left(\frac{p+1}{p}+1\right)\left(\frac{p+1}{p}+2\right)\cdots\left(\frac{p+1}{p}+n-1\right)\\ &= \frac{(p+1)(2p+1)(3p+1)\cdots((n-1)p+1)(np+1)}{p^n},\end{aligned}$$

Therefore,

$$\begin{aligned}\frac{(\frac{1}{p})_n(\frac{p-1}{p})_n}{(\frac{p+1}{p})_n} &= \frac{\frac{(p+1)(2p+1)\cdots((n-1)p+1)}{p^n}\cdot\frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{p^n}}{\frac{(p+1)(2p+1)(3p+1)\cdots((n-1)p+1)(np+1)}{p^n}}\\ &= \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{p^n(np+1)}\end{aligned}$$

This implies that we can write

$$\arcsin_p(x) = x\sum_{n=0}^{\infty}\frac{(\frac{1}{p})_n(\frac{p-1}{p})_n}{(\frac{p+1}{p})_n}\frac{(x^p)^n}{n!} \tag{11.2}$$

The *hypergeometric function* is defined as

$${}_2F_1\left(\begin{matrix}a,\ b\\ c\end{matrix};z\right) = \sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n}\frac{z^n}{n!} = \sum_{n=0}^{\infty}\frac{\Gamma(a+n)\Gamma(b+n)\Gamma(c)}{\Gamma(a)\Gamma(b)\Gamma(c+n)}\frac{z^n}{n!}. \tag{11.3}$$

Comparing this to Equation (11.2), we have

$$\arcsin_p(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{p-1}{p} \\ \frac{p+1}{p} & \end{matrix}; x^p\right) = x\sum_{n=0}^{\infty} \frac{\Gamma(\frac{p-1}{p}+n)}{\Gamma(\frac{p-1}{p})(np+1)} \frac{x^{np}}{n!}. \tag{11.4}$$

for $0 \le x < 1$.

Again, it can be similarly proven that

$$\arctan_p(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{2}{p} \\ \frac{p+1}{p} & \end{matrix}; -x^p\right) = x\sum_{n=0}^{\infty}(-1)^n \frac{\Gamma(\frac{2}{p}+n)}{\Gamma(\frac{2}{p})(np+1)} \frac{x^{np}}{n!}. \tag{11.5}$$

for $0 \le x < 1$.

11.2 Squigonometric functions

The power series for the p-trigonometric functions are, alas, much messier than those for their inverses. To obtain infinite series expansions for the squigonometric functions themselves, we could either employ the usual "two-column" derivations, or use Picard iteration.

Another method is to use series reversion techniques [84,103] to get the necessary coefficients.

Let's try to obtain a series for $x = \sin_p(t)$, having our previous series for $t = \arcsin_p(x)$ in hand.

Series reversion implements the following idea: if $t = f(x)$ where $f(0) = 0$ is represented by the Maclaurin series

$$t = f(x) = a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \cdots$$

then we can write

$$x = f^{-1}(t) = A_1t + A_2t^2 + A_3t^3 + A_4t^4 + \cdots$$

where each A_i can be written in terms of various a_j's for $1 \le j \le i$.

So with

$$t = \arcsin_p(x) = x + \frac{p-1}{p(p+1)}x^{p+1} + \frac{(p-1)(2p-1)}{2!p^2(2p+1)}x^{2p+1} + \frac{(p-1)(2p-1)(3p-1)}{3!p^3(3p+1)}x^{3p+1} + \cdots \tag{11.6}$$

and writing

$$x = \sin_p(t) = A_1t + A_2t^2 + A_3t^3 + \cdots, \tag{11.7}$$

we substitute (11.7) into (11.6) to get

$$\begin{aligned} t = 1(A_1 t + A_2 t^2 + A_3 t^3 + \cdots) \\ + \frac{p-1}{p(p+1)}(A_1 t + A_2 t^2 + A_3 t^3 + \cdots)^{p+1} \\ + \frac{(p-1)(2p-1)}{2!p^2(2p+1)}(A_1 t + A_2 t^2 + A_3 t^3 + \cdots)^{2p+1} \\ + \frac{(p-1)(2p-1)(3p-1)}{3!p^3(3p+1)}(A_1 t + A_2 t^2 + A_3 t^3 + \cdots)^{3p+1} + \cdots \end{aligned} \tag{11.8}$$

We can now solve for $A_1, A_2, \ldots$ by matching coefficients and powers between the left and right sides of (11.8). To start, we know that there is only one t term on each side so $A_1 = 1$. When we look at t^2 terms, there are none on the left side, and only one on the right, $A_2 t^2$, so $A_2 = 0$. By similar argument, $A_3 = A_4 = \cdots = A_p = 0$.

The fun begins when we look at the t^{p+1} terms. There are again none on the left side, but there are a couple of ways to get a t^{p+1} on the right side of (11.8). We have a $A_{p+1}t^{p+1}$ on the first line, but we also have $\frac{p-1}{p(p+1)}A_1^{p+1}t^{p+1}$ from the second line. Thus the total coefficient of $t^{p+1} = 0$, or

$$A_{p+1} + \frac{p-1}{p(p+1)}A_1^{p+1} = 0.$$

Since $A_1 = 1$, we can solve this for A_{p+1} to get

$$A_{p+1} = -\frac{p-1}{p(p+1)}.$$

For A_{p+k}, $1 < k \leq p$, we can get a t^{p+k} term in a couple of different ways. We have the $A_{p+k}t^{p+k}$ term from the first line of (11.8). From the second line, we can get a t^{p+k} in several ways. We can choose $A_1 t$ from p of the terms and a $A_k t^k$ from one, and we can do this in $\binom{p+1}{1} = (p+1)$ ways giving us $\frac{p-1}{p(p+1)}(p+1)[A_1^p A_k t^{p+k}]$. But this is 0 since $A_k = 0$. This leads to $A_{p+k} = 0$.

We look now to determine A_{2p+1} and again we tally the number of ways we can obtain a t^{2p+1} term on the right side of (11.8). We can have the $A_{2p+1}t^{2p+1}$ term from the first line. On the second line, where we must choose from $(p+1)$ factors, we can choose $A_1 t$ from p factors, and $A_{p+1}t^{p+1}$ from one factor. We can do this in $(p+1)$ ways, giving us $\frac{p-1}{p(p+1)}(p+1)A_1^p A_{p+1}t^{2p+1}$. Lastly, from the third line of (11.8), where we must choose a term from each of $(2p+1)$ copies of our series, we can choose $A_1 t$ from each, giving us $\frac{(p-1)(2p-1)}{2!p^2(2p+1)}A_1^{2p+1}t^{2p+1}$. However, there are no t^{2p+1}'s on the left side of (11.8). Matching coefficients gives us the equation

$$A_{2p+1} + \frac{p-1}{p(p+1)}[(p+1)A_1^p A_{p+1}] + \frac{(p-1)(2p-1)}{2!p^2(2p+1)}A_1^{2p+1} = 0.$$

Substituting the already determined values of A_1 and A_{p+1} and solving for A_{2p+1} yields

$$A_{2p+1} = \frac{(p-1)\left(2p^2-3p-1\right)}{2p^2(p+1)(2p+1)}.$$

Table 11.3 Power series for p-sine functions

$p=1$	$\sin_1(t) = t$
$p=2$	$\sin_2(t) = t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040} + \frac{t^9}{362,880} + \cdots$
$p=3$	$\sin_3(t) = t - \frac{t^4}{6} + \frac{2t^7}{63} - \frac{13t^{10}}{2268} + \frac{23t^{13}}{22,113} + \cdots$
$p=4$	$\sin_4(t) = t - \frac{3t^5}{20} + \frac{19t^9}{480} - \frac{469t^{13}}{41,600} + \frac{189,611t^{17}}{56,576,000} + \cdots$
$p=5$	$\sin_5(t) = t - \frac{2t^6}{15} + \frac{34t^{11}}{825} - \frac{719t^{16}}{49,500} + \frac{3034t^{21}}{556,875} + \cdots$
$p=6$	$\sin_6(t) = t - \frac{5t^7}{42} + \frac{265t^{13}}{6552} - \frac{253,595t^{19}}{15,685,488} + \frac{91,657,229t^{25}}{13,175,809,920} + \cdots$

Similar but more involved and painful work leads to

$$A_{2p+2} = A_{2p+3} = \cdots A_{3p} = 0,$$

$$A_{3p+1} = -\frac{(p-1)\left(12p^5-34p^4+p^3+33p^2+11p+1\right)}{3!p^3(p+1)^2(2p+1)(3p+1)},$$

$$A_{3p+2} = A_{3p+3} = \cdots A_{4p} = 0,$$

$$A_{4p+1} = \frac{(p-1)\left(144p^8-644p^7+520p^6+861p^5-565p^4-674p^3-194p^2-23p-1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)(4p+1)}.$$

In particular, $A_k = 0$ for any $k \neq mp+1$, $m \in \{0, 1, 2, \ldots\}$. Thus we have that

$$\begin{aligned}\sin_p(t) = t &- \frac{p-1}{p(p+1)}t^{p+1} + \frac{(p-1)\left(2p^2-3p-1\right)}{2p^2(p+1)(2p+1)}t^{2p+1} \\ &- \frac{(p-1)\left(12p^5-34p^4+p^3+33p^2+11p+1\right)}{3!p^3(p+1)^2(2p+1)(3p+1)}t^{3p+1} \\ &+ \frac{(p-1)\left(144p^8-644p^7+520p^6+\cdots-23p-1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)(4p+1)}t^{4p+1} + \cdots\end{aligned} \tag{11.9}$$

Fortunately, computer algebra systems such as Mathematica have the built-in ability to find the series representing the inverse of a function.[1] A list of some series for p-sine functions can be found in Table 11.3.

Through a similar process, we can show that

$$\begin{aligned}\tan_p(t) = t &+ \frac{2}{p(p+1)}t^{p+1} - \frac{p^2-5p-2}{p^2(p+1)(2p+1)}t^{2p+1} \\ &+ \frac{2\left(2p^5-16p^4+23p^3+44p^2+17p+2\right)}{3p^3(p+1)^2(2p+1)(3p+1)}t^{3p+1} \\ &- \frac{18p^8-193p^7+585p^6+38p^5-1500p^4-1313p^3-443p^2-68p-4}{3p^4(p+1)^3(2p+1)(3p+1)(4p+1)}t^{4p+1} + \cdots\end{aligned} \tag{11.10}$$

[1] The Mathematica command `InverseSeries[Series[f[x],{x,0,10}]]` for instance, will produce the series for $f^{-1}(x)$ up to the degree-10 term.

Table 11.4 Power series for p-tangent functions

$p=1$	$\tan_1(t) = t + t^2 + t^3 + t^4 + t^5 + \cdots = \frac{t}{1-t}$
$p=2$	$\tan_2(t) = t + \frac{t^3}{3} + \frac{2t^5}{15} + \frac{17t^7}{315} + \frac{62t^9}{2835} + \cdots$
$p=3$	$\tan_3(t) = t + \frac{t^4}{6} + \frac{2t^7}{63} + \frac{13t^{10}}{2268} + \frac{23t^{13}}{22,113} + \cdots$
$p=4$	$\tan_4(t) = t + \frac{t^5}{10} + \frac{t^9}{120} + \frac{11t^{13}}{15,600} + \frac{211t^{17}}{3,536,000} + \cdots$
$p=5$	$\tan_5(t) = t + \frac{t^6}{15} + \frac{t^{11}}{825} + \frac{13t^{16}}{49,500} - \frac{37t^{21}}{1,113,750} + \cdots$
$p=6$	$\tan_6(t) = t + \frac{t^7}{21} - \frac{t^{13}}{819} + \frac{368t^{19}}{980,343} - \frac{9692t^{25}}{102,936,015} + \cdots$

Table 11.5 Power series for p-cosine functions

$p=1$	$\cos_1(t) = 1 - t$
$p=2$	$\cos_2(t) = 1 - \frac{t^2}{2} + \frac{t^4}{24} - \frac{t^6}{720} + \frac{t^8}{40,320} + \cdots$
$p=3$	$\cos_3(t) = 1 - \frac{t^3}{3} + \frac{t^6}{18} - \frac{23t^9}{2268} + \frac{25t^{12}}{13,608} + \cdots$
$p=4$	$\cos_4(t) = 1 - \frac{t^4}{4} + \frac{9t^8}{160} - \frac{149t^{12}}{9600} + \frac{15,147t^{16}}{3,328,000} + \cdots$
$p=5$	$\cos_5(t) = 1 - \frac{t^5}{5} + \frac{4t^{10}}{75} - \frac{224t^{15}}{12,375} + \frac{4957t^{20}}{742,500} + \cdots$
$p=6$	$\cos_6(t) = 1 - \frac{t^6}{6} + \frac{25t^{12}}{504} - \frac{15,775t^{18}}{825,552} + \frac{21,301,825t^{24}}{2,635,161,984} + \cdots$

Results for some p-tangent functions are in Table 11.4. Since $\cos_p(t) = \frac{\sin_p(t)}{\tan_p(t)}$, we can simply divide (11.9) by (11.10) to obtain

$$\begin{aligned}\cos_p(t) = 1 &- \frac{1}{p}t^p + \frac{(p-1)^2}{2!p^2(p+1)}t^{2p} \\ &- \frac{(p-1)^2\left(4p^3 - 6p^2 - 3p + 1\right)}{3!p^3(p+1)^2(2p+1)}t^{3p} \\ &+ \frac{(p-1)^2\left(36p^6 - 116p^5 + 41p^4 + 100p^3 - 6p^2 - 8p + 1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)}t^{4p} + \cdots\end{aligned} \tag{11.11}$$

A few particular results for $p = 1$ to 6 are in Table 11.5.

11.3 Convergence and complex numbers

We have dealt with these series as formal power series but now must pay some attention to matters of convergence. Since the series for $\arcsin_p(x)$ and $\arctan_p(x)$ emanate from binomial series, these series converge on $-1 < x < 1$. (We will deal

with the endpoints of 1 and -1 in the next chapter.) As stated earlier, the problem is that the series do not converge to either function for $-1 < x < 0$, p an odd natural number. (Note that the series represents neither an even nor an odd function for p an odd natural number.) So we must define

$$\arcsin_p(x) = \begin{cases} x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, \frac{p-1}{p} \\ \frac{p+1}{p}\end{matrix}; x^p\right) & 0 \le x < 1, \\ -\arcsin_p(-x) & -1 < x < 0 \end{cases}$$

and

$$\arctan_p(x) = \begin{cases} x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, \frac{2}{p} \\ \frac{p+1}{p}\end{matrix}; -x^p\right) & 0 \le x < 1, \\ -\arctan_p(-x) & -1 < x < 0 \end{cases}$$

We would eventually like to construct an analog of the hyperbolic functions and generalize identities such as

$$\begin{aligned} \sinh z &= -i \sin(iz), \\ \cosh z &= \cos(iz), \\ \frac{d}{dz} \sinh z &= \cosh z, \\ \frac{d}{dz} \cosh z &= \sinh z \end{aligned}$$

for $z \in \mathbb{C}$. (See Section 14.1.) To do this, we need to extend further the domain of our p-trigonometric functions into the complex numbers. (For the present, we will assume $p > 2$ is a natural number.)

While the hypergeometric series (11.3) is defined for $|z|<1$, the function ${}_2F_1\left(\begin{matrix}a, b \\ c\end{matrix}; z\right)$ can be analytically continued into the complex realm via the integral representation

$${}_2F_1\left(\begin{matrix}a, b \\ c\end{matrix}; z\right) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\, dt,$$

valid for when $\Re(c) > \Re(b) > 0$, with a branch cut on the real line from 1 to ∞. This means that the p-arcsine and p-arctangent functions can be continued into the complex plane as

$$\begin{aligned} \arcsin_p(z) &= z \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, \frac{p-1}{p} \\ \frac{p+1}{p}\end{matrix}; z^p\right), \\ \arctan_p(z) &= z \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, \frac{2}{p} \\ \frac{p+1}{p}\end{matrix}; -z^p\right), \end{aligned}$$

which have branch cuts from every pth root of unity to ∞.

We might expect the series for $\sin_p(t)$ and $\cos_p(t)$ to have an infinite radius of convergence, but for $p > 2$, the radius of convergence appears to be finite (see Figure 11.1). The tangent series appears to converge on $(-\pi_p/2, \pi_p/2)$. The series for

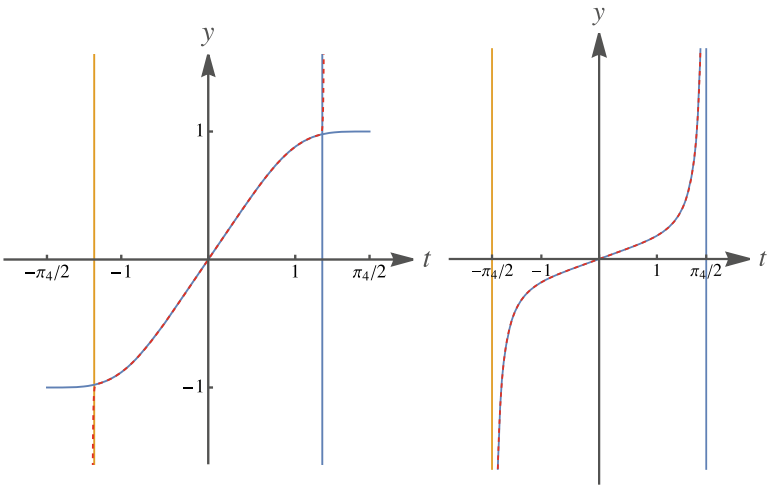


Figure 11.1 Graphs of $y = \sin_4(t)$ (solid, left) and $y = \tan_4(t)$ (solid, right) and their respective degree-201 Maclaurin polynomials (dashed). The vertical lines represent the apparent interval of convergence for each series.

$p = 2$ for sine and cosine, of course converge on $(-\infty, \infty)$ while the tangent series converges on $(-\pi/2, \pi/2)$. This would seem to indicate that a complex singularity exists for the sine and cosine functions for $p > 2$. We will investigate this further in Chapter 14.

When p is an even natural number, the formal Maclaurin series for $\sin_p(t)$, $\cos_p(t)$, and $\tan_p(t)$ converge to those functions within their respective intervals of convergence, positive *and* negative values. For these p, we can extend the domain of the sine, cosine, and tangent functions into the complex realm within a disk D of some radius, say Υ_p, greater than 1. (In general, $\sin_p(\cdot)$ and $\cos_p(\cdot)$ are unable to be analytically continued to meromorphic functions on all of $\mathbb{C}$. It has been shown, though, that $\sin_4(\cdot)^2$, $\cos_4(\cdot)^2$ and $\sin_4(\cdot)\cos_4(\cdot)$ can be [59].)

For p an odd natural number, these Maclaurin series converge within the real interval $(-\Upsilon_p, \Upsilon_p)$ but do not converge to the sine, cosine, or tangent functions on $(-\Upsilon_p, 0)$. (See Figure 11.2.)

So while we can define our functions to equal their respective Maclaurin series on D, we have to make a special definition for $z = t \in (-\Upsilon_p, 0)$ [40]. For example, we can define for $\sin_p(t) = -\sin_p(-t)$ for these t to preserve the odd symmetry, and similarly define $\tan_p(t) = -\tan_p(-t)$ in this domain. To preserve the even symmetry of the p-cosine, we would need to define $\cos_p(t) = \cos_p(-t)$ for $t \in (-\Upsilon_p, 0)$.

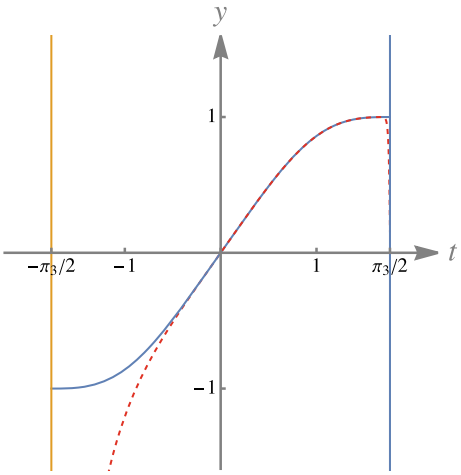


Figure 11.2 Graph of $y = \sin_3(t)$ (solid) and its degree-60 Maclaurin polynomial (dashed). The Maclaurin series converges to $\sin_3(t)$ for $t \geq 0$ but not for $t < 0$. The vertical lines represent the apparent interval of convergence for the series.

Exercises

▶ **Exercise 11.1** *Calculate* $\sin_3'''(0)$. *Deduce that* $\sin_3(t)$ *cannot equal its power series expansion in a neighborhood of 0.*

▶ **Exercise 11.2** *Show that for* $0 \leq x < 1$,

$$\arctan_p(x) = x + \sum_{n=1}^{\infty} (-1)^n \frac{2(p+2)(2p+2)\cdots((n-1)p+2)}{n!p^n(np+1)} x^{np+1}.$$

▶ **Exercise 11.3** *Prove Formula (11.5):*

$$\arctan_p(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{2}{p} \\ \frac{p+1}{p} & \end{matrix}; -x^p\right)$$

for $0 \leq x < 1$.

▶ **Exercise 11.4** *Emulate the steps in Section 11.2 to prove (11.10).*

Project

Project 11.1 (Constructing power series)
The usual form of a Maclaurin series for an analytic function $f(t)$ is

$$f(t) = f(0) + f'(0)t + f''(0)\frac{t^2}{2!} + f'''(0)\frac{t^3}{3!} + \cdots$$

1. If $f(t) = \cos_p(t)$, as we saw in Equation (11.11), the Maclaurin series has the form

$$f(t) = 1 + f^{(p)}(0)\frac{t^p}{p!} + f^{(2p)}(0)\frac{t^{2p}}{(2p)!} + \cdots$$

Let $B_p = f^{(p)}(0)$ and $C_p = f^{(2p)}(0)$.

a) Show that $B_p = -(p-1)!$.

b) Show that the sequence $\{C_k\}$ has the recursive relationship

$$\begin{aligned} C_1 &= 0 \\ C_{k+1} &= [2(p-k)C_k + (2k-1)(p-1)_{2k-1}(p-1)](2p-2k-1) \\ &\quad + 2k(p-1)_{2k}(p-1) \end{aligned}$$

You may wish to first consider the form of $f^{(k)}(t)$ for $k > 0$. Convince yourself that

$$f^{(k)}(t) = \sum_{i=0}^{M_k}\sum_{j=0}^{N_k} a_{ij}^k \cos_p^i(t)\sin_p^j(t)$$

for all k and where M_k and N_k are constants that depend on k. Differentiate the above equation with respect to t and find a recursion relation for the a_{ij}^k. You may wish to make a table of a_{ij}^k values for small values of k to get a sense of pattern.

c) Use a computer algebra system to compute several values for C_p for different p. Convince yourself that $C_p/(2p!)$ gives the same coefficients for t^{2p} as Equation (11.11).

2. If $g(t) = \sin_p(t)$, we can similarly look at Equation (11.9) as having the form

$$g(t) = t + D_p\frac{t^{p+1}}{(p+1)!} + E_p\frac{t^{2p+1}}{(2p+1)!} + \cdots$$

where $D_p = g^{(p+1)}(0)$ and $E_p = g^{(2p+1)}(0)$.

a) Show that $D_p = -(p-1)(p-1)!$.

b) Show that the sequence $\{E_k\}$ fulfills the recursion relation

$$\begin{aligned} E_1 &= 0 \\ E_{k+1} &= [2(p-k)E_k + (2kp-2k-1)(p-1)_{2k-1}](2p-2k-1) \\ &\quad + [(2k+1)p-2k-2](p-1)(p-1)_{2k} \end{aligned}$$

Work similarly to the cosine problem above and use a computer algebra system to convince yourself that $E_p/(2p+1)!$ gives the same coefficient for t^{2p+1} as Equation (11.9).

Consult [66] for more details. A different recursion relation amongst the coefficients in the p-sine and p-cosine series is found in [42, p. 193].

Check for updates

Series and rational approximations

12

Since π is irrational, we have to approximate its value for applications. Two methods of doing this are through the use of the series we developed in the previous chapter, or through finding "nice" fractions (that is, fractions with relatively small denominators, like $\frac{22}{7}$ or $\frac{355}{113}$). We will expand this approach to all π_p, $p > 1$. We will also investigate a series for Catalan's constant G.

12.1 Series for Pi

Since the series for the p-arcsine (11.1) and p-arctangent functions are hypergeometric, they converge for $|x| < 1$, but the Ratio Test gives no information for $x = 1$ and $x = -1$.

To check convergence here, we can use Gauss' test for hypergeometric functions (Theorem 4.15 in [15]). We consider first the series for p-arcsine. Once again, we denote the coefficient of the x^{np+1} term as a_n, and with p fixed, we get

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(p-1)(2p-1)\cdots(np-1)((n+1)p-1)}{(n+1)!p^{n+1}((n+1)p+1)} \cdot \frac{n!p^n(np+1)}{(p-1)(2p-1)\cdots(np-1)} \\ &= \frac{((n+1)p-1)(np+1)}{(n+1)p((n+1)p+1)} \\ &= \frac{p^2n^2+p^2n+(p-1)}{p^2n^2+(2p^2+p)n+(p^2+p)} \\ &= \frac{n^2+n+(p-1)/p^2}{n^2+(2+1/p)n+(1+1/p)},\end{aligned}$$

a ratio of two monic, quadratic polynomials in n.

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_12

Examine the polynomials in the numerator and denominators, and look particularly at the coefficients of the second terms – if the degree of both polynomials is k, look at the n^{k-1} terms. Call the coefficient in the numerator B_{k-1} and that in the denominator b_{k-1}. Gauss' test says the following:

- If $B_{k-1} \geq b_{k-1}$, then the series does not converge.
- If $b_{k-1} > B_{k-1} \geq b_{k-1} - 1$, then the series converges if and only if it is an alternating series.
- If $b_{k-1} - 1 > B_{k-1}$, then the series is absolutely convergent.

In this case, we have that $k = 2$, so we look at the coefficients of n. We have $B_1 = 1$ and $b_1 = 2 + 1/p$. Because $b_1 - 1 = 1 + 1/p > B_1$, the series is absolutely convergent at 1 and -1.

Since $\arcsin_p(1) = \frac{\pi_p}{2}$, this leads to a series converging to π_p:

$$\begin{aligned}\pi_p &= 2\arcsin_p(1)\\ &= 2 + 2\sum_{n=1}^{\infty} \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{n!p^n(np+1)}\\ &= 2 + \frac{2(p-1)}{p(p+1)} + \frac{2(p-1)(2p-1)}{2!p^2(2p+1)} + \frac{2(p-1)(2p-1)(3p-1)}{3!p^3(3p+1)} + \cdots\end{aligned} \tag{12.1}$$

Some particular examples for this series are

$$\pi = 2 + \frac{1}{3} + \frac{3}{20} + \frac{5}{56} + \frac{35}{576} + \cdots$$

$$\pi_4 = 2 + \frac{3}{10} + \frac{7}{48} + \frac{77}{832} + \frac{1155}{17,408} + \cdots$$

$$\pi_{10} = 2 + \frac{9}{55} + \frac{57}{700} + \frac{1653}{31,000} + \frac{64,467}{1,640,000} + \cdots$$

However, as with its $p = 2$ counterpart, the series (12.1) converges very slowly at 1. (See Table 12.1.) The rate of convergence does seem to worsen as p increases.

We can get faster convergence, though, by picking a different evaluation point. For $p \neq 2$, the only other well-known nonzero value for the p-sine function in the first quadrant is $\sin_p(\pi_p/4) = 1/\sqrt[p]{2}$. So we can evaluate our partial sums at $x = 1/\sqrt[p]{2}$ and multiply the result by 4. This leads to the series

$$\begin{aligned}\pi_p &= 4\arcsin_p(1/\sqrt[p]{2})\\ &= \frac{4}{\sqrt[p]{2}} + \frac{4(p-1)}{p(p+1)}\left(\frac{1}{\sqrt[p]{2}}\right)^{p+1} + \frac{4(p-1)(2p-1)}{2!p^2(2p+1)}\left(\frac{1}{\sqrt[p]{2}}\right)^{2p+1} + \cdots\\ &= \frac{4}{\sqrt[p]{2}}\left(1 + \sum_{n=1}^{\infty} \frac{(p-1)(2p-1)(3p-1)\cdots(np-1)}{n!(2p)^n(np+1)}\right)\end{aligned} \tag{12.2}$$

This series converges much more quickly as Table 12.2 bears out.

Table 12.1 Partial sums of arcsine series for π_p at $x = 1$

Number of terms	$p = 2$	$p = 4$	$p = 10$
5	2.63338293650794	2.60473021540347	2.33769666205156
10	2.78338929188658	2.78525095222801	2.45078208245647
20	2.88877269065657	2.93421946287952	2.55328879302673
100	3.02870805687419	3.19174468784359	2.76161986283374
200	3.06178763420401	3.27402838558264	2.84093159972021
500	3.09112580392386	3.36296385482535	2.93752960285587
1000	3.10590868555622	3.41790042557163	3.00489266272935
2000	3.11636080290427	3.46408693360831	3.06772618955630
5000	3.12563482941193	3.51405693022108	3.14436457685457
10,000	3.13030881490639	3.54493864873786	3.19784471684383
$\vdots$	$\vdots$	$\vdots$	$\vdots$
π_p	3.14159265358979	3.70814935460274	3.94292789781003

Table 12.2 Partial sums of arcsine series for π_p at $x = 1/\sqrt[p]{2}$.

Number of terms	$p = 2$	$p = 4$	$p = 10$
5	3.13831684697490	3.70359984478460	3.93982038686803
10	3.14155131496131	3.70808331412542	3.94287900363664
20	3.14159263812515	3.70814932580081	3.94292787446900
100	3.14159265358979	3.70814935460274	3.94292789781003
$\vdots$	$\vdots$	$\vdots$	$\vdots$
π_p	3.14159265358979	3.70814935460274	3.94292789781003

We look now to the series for the p-arctangent, we let c_n be the absolute value of the coefficient of the x^{np+1} term and examine

$$\begin{aligned}\frac{c_{n+1}}{c_n} &= \frac{2(p+2)(2p+2)\cdots((n-1)p+2)(np+2)}{(n+1)!p^{n+1}((n+1)p+1)} \cdot \frac{n!p^n(np+1)}{2(p+2)(2p+2)\cdots((n-1)p+2)} \\ &= \frac{(np+2)(np+1)}{(n+1)p((n+1)p+1)} = \frac{p^2n^2+3pn+2}{p^2n^2+(2p^2+p)n+(p^2+p)} \\ &= \frac{n^2+(3/p)n+(2/p^2)}{n^2+(2+1/p)n+(1+1/p)}.\end{aligned}$$

We use Gauss' test once more. Here, we have $B_1 = 3/p$ and $b_1 = 2 + 1/p$. When $p = 1, b_1 = B_1$ and the series diverges. (This is not a surprise since $\arctan_1(x) = \frac{x}{1+x}$ is represented by the series $x - x^2 + x^3 + \cdots$, which diverges at $x = 1$.)

Table 12.3 Partial sums of arctangent series for π_p at $x = 1$.

Number of terms	$p = 2$	$p = 4$	$p = 10$
5	3.33968253968254	3.73485105580694	3.94564332410312
10	3.04183961892940	3.69892986294449	3.94218986803704
20	3.09162380666784	3.70493933966621	3.94272261445123
100	3.13159290355855	3.70786621242194	3.94291687626253
200	3.13659268483882	3.70804943294881	3.94292474395648
500	3.13959265558978	3.70812410439117	3.94292729300416
1000	3.14059265383979	3.70814043064021	3.94292772424990
2000	3.14109265362104	3.70814620009602	3.94292784798590
⋮	⋮	⋮	⋮
π_p	3.14159265358979	3.70814935460274	3.94292789781003

When $p = 2$, we have that $b_1 > B_1 \geq b_1 - 1$ and so the series converges conditionally at 1. For $p > 2$, $b_1 - 1 > B_1$ and the series is absolutely convergent at 1.

Since $\tan_p(\frac{\pi_p}{4}) = 1$, we have for $p > 1$,

$$\begin{aligned}\pi_p &= 4 \arctan_p(1) \\ &= 4\left(1 + \sum_{n=1}^{\infty} (-1)^n \frac{2(p+2)(2p+2)\cdots((n-1)p+2)}{n!p^n(np+1)}\right) \\ &= 4\left(1 - \frac{2}{p(p+1)} + \frac{2(p+2)}{2!p^2(2p+1)} - \frac{2(p+2)(2p+2)}{3!p^3(3p+1)} + \cdots\right). \quad (12.3)\end{aligned}$$

We recall that when $p = 2$, we get

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right),$$

a most elegant series that is nearly useless for approximations, due to the slow rate of convergence. Other series we get from (12.3) include

$$\pi_4 = 4\left(1 - \frac{1}{10} + \frac{1}{24} - \frac{5}{208} + \cdots\right)$$

$$\pi_{10} = 4\left(1 - \frac{1}{55} + \frac{1}{175} - \frac{11}{3875} + \cdots\right)$$

Unlike the series (12.1), the convergence results for the arctangent series improve as p increases, as can be seen in Table 12.3, though the convergence is not as quick as that for the arcsine series at $1/\sqrt[p]{2}$.

Yet another series formula for π_p comes from the integral formula (9.9)

$$\pi_p = 4\int_0^1 \sqrt[p]{1 - x^p}\, dx.$$

Table 12.4 Partial sums of series (12.4).

Number of terms	$p = 2$	$p = 4$	$p = 10$
5	3.18025793650794	3.73265990290347	3.94937166205156
10	3.15433045399595	3.71787201402992	3.94577783385334
20	3.14594333192751	3.71212073109584	3.94422261349266
100	3.14197133126466	3.70866843673796	3.94314373597066
200	3.14172608425672	3.70836697681194	3.94302832492377
500	3.14162634081748	3.70821846309116	3.94296449357063
1000	3.14160455578304	3.70817839449711	3.94294496134913
2000	3.14159686023027	3.70816156087682	3.94293585613727
⋮	⋮	⋮	⋮
π_p	3.14159265358979	3.70814935460274	3.94292789781003

We can use the binomial series once again to derive a series formula:

$$\begin{aligned}
\pi_p &= 4\int_0^1 (1-x^p)^{\frac{1}{p}}\,dx \\
&= 4\int_0^1 \sum_{n=0}^{\infty} \binom{1/p}{n} (-x^p)^n\,dx \\
&= 4\left[\int_0^1 \left(1 - \frac{1}{p}x^p + \frac{(\frac{1}{p})(\frac{1}{p}-1)}{2!}x^{2p} - \frac{(\frac{1}{p})(\frac{1}{p}-1)(\frac{1}{p}-2)}{3!}x^{3p} + \cdots\right) dx\right] \\
&= 4\left[\int_0^1 \left(1 - \frac{1}{p}x^p + \frac{1-p}{2!p^2}x^{2p} - \frac{(1-p)(1-2p)}{3!p^3}x^{3p} + \cdots\right) dx\right] \\
&= 4\left[x - \frac{1}{p(p+1)}x^{p+1} + \frac{1-p}{2!p^2(2p+1)}x^{2p+1} - \frac{(1-p)(1-2p)}{3!p^3(3p+1)}x^{3p+1} + \cdots\right]_0^1 \\
&= 4\left[1 - \frac{1}{p(p+1)} + \frac{1-p}{2!p^2(2p+1)} - \frac{(1-p)(1-2p)}{3!p^3(3p+1)} + \cdots\right]
\end{aligned}$$

Thus

$$\pi_p = 4\left[1 - \sum_{n=1}^{\infty} \frac{(p-1)(2p-1)\cdots((n-1)p-1)}{n!p^n(np+1)}\right] \tag{12.4}$$

Some partial sums are computed in Table 12.4. The rate of convergence appears to be faster than that of the p-arctangent series, but slower than that of the p-arcsine series at $x = 1/\sqrt[p]{2}$.

12.2 Rational approximations to Pi

By now, everyone knows that "Pi Day" is March 14th, or 3/14 in the US date format. (But see Appendix E for some modestly proposed alternatives to that date, corresponding to other $p \neq 2$.) Many know that "Casual Pi Day" falls on July 22nd, or 22/7 in the date format used in much of the world outside of the US, since $22/7 \approx 3.14286$ is a reasonably good rational approximation of π. Fewer know the better rational approximations of $333/106$ or $355/113$ or $5,419,351/1,725,033$.

One way of getting rational approximants to π_p would be to calculate partial sums for the series in Section 12.1. For example, if we compute the first 10 partial sums for the series (12.4) for π, we get

$$4, \frac{10}{3}, \frac{97}{30}, \frac{1343}{420}, \frac{32,057}{10,080}, \frac{703,049}{221,760}, \frac{36,485,783}{11,531,520}, \frac{72,872,467}{23,063,040}, \frac{19,801,986,719}{6,273,146,880}, \frac{751,927,973,347}{238,379,581,440},$$

with successive differences from π of

$$0.858407, 0.191741, 0.0917407, 0.0560264, 0.0386653, 0.0287221, 0.022412, 0.0181151, 0.0150346, 0.0127378.$$

We[1] could also search for the fraction of lowest denominator within a given error of π_p. Table 12.5 has results for π_2, π_4, and π_{10} to within 10^{-10}. It reveals, among other things, that we have a choice for "Casual π_4 Day": either March 11th (11/3) or July 26th (26/7).

While all of these fractions are valid approximations to π, they are highly unwieldy in that the numerator and denominator become very large well before we are within even 0.01 of π. (The first partial sum of (12.4) that is within 0.01 of π is the 12th one: $\frac{19,745,168,090,239}{6,265,977,569,280}$.)

To find rational approximations of π (or any irrational constant), we usually turn to *continued fractions*. For example, we know that $\pi = 3.14159265358979....$ As a first approximation, we start with 3, the integer part. So $\pi = 3 + 0.14159265358979....$ The reciprocal of the fractional part of π is $1/0.14159265358979... = 7.06251330\,5931$. Use the integer part as an approximation to the reciprocal; this gives us a second approximation of $\pi \approx 3 + \frac{1}{7}$ or $\frac{22}{7}$.

Now write $7.062513305931\cdots = 7 + 0.062513305931\ldots$. The reciprocal of the fractional part is $1/0.062513305931\ldots = 15.9965944067\ldots$. If we use 15 as the approximation to this reciprocal, we can write

$$\pi \approx 3 + \frac{1}{7 + \frac{1}{15}} = \frac{333}{106}.$$

[1] By "we," we mean a computer algebra system. Mathematica has the `Rationalize` command, which is how Table 12.5 was generated.

Table 12.5 Fractions of lowest denominators within 10^{-n} of π_p

n	π_2	π_4	π_{10}
1	$\frac{22}{7}$	$\frac{11}{3}$	4
2	$\frac{22}{7}$	$\frac{26}{7}$	$\frac{59}{15}$
3	$\frac{201}{64}$	$\frac{89}{24}$	$\frac{138}{35}$
4	$\frac{333}{106}$	$\frac{597}{161}$	$\frac{138}{35}$
5	$\frac{355}{113}$	$\frac{864}{233}$	$\frac{1451}{368}$
6	$\frac{355}{113}$	$\frac{3367}{908}$	$\frac{1589}{403}$
7	$\frac{75,948}{24,175}$	$\frac{9237}{2491}$	$\frac{41,452}{10,513}$
8	$\frac{100,798}{32,085}$	$\frac{58,789}{15,854}$	$\frac{65,287}{16,558}$
9	$\frac{103,993}{33,102}$	$\frac{86,500}{23,327}$	$\frac{138,519}{35,131}$
10	$\frac{312,689}{99,532}$	$\frac{528,237}{142,453}$	$\frac{206,984}{52,495}$

We continue one more iteration, writing $15.9965944067\ldots = 15 + 0.9965944067\ldots$ and taking the reciprocal of the fractional part, $1/0.9965944067\ldots = 1.0034172310\ldots$. Taking 1 as the approximation to the reciprocal, we obtain

$$\pi \approx 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1}}} = \frac{355}{113}$$

We can iterate this process as long as we like. Carrying on a few more repetitions leads to

$$\pi = 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1 + \cfrac{1}{292 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{1 + \cdots}}}}}}}}}$$

Note that all of the numerators are 1. This representation is often rewritten as a sequence

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, \dots],$$

generated by reading down the "diagonal" of the continued fraction.

Continued fractions have many remarkable properties. Rational numbers have finite continued fractions, while irrational numbers have infinite ones. Quadratic irrationals (like $\sqrt{2}$ or $\sqrt{19}$) have a repeating pattern in their sequence:

- $\sqrt{2} = [1; 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, \dots]$ has a repeating pattern of length 1.
- $\sqrt{19} = [4; 2, 1, 3, 1, 2, 8, 2, 1, 3, 1, 2, 8, \dots]$ has a repeating pattern of length 6.

All other irrational numbers have infinite continued fractions with no discernible pattern in the sequence.

A *convergent* for a continued fraction is a rational number obtained by stopping the continued fraction process after a finite number of steps. We have already seen that the first four convergents for π are 3, $\frac{22}{7}$, $\frac{333}{106}$, and $\frac{355}{113}$. The first ten are

$$3, \frac{22}{7}, \frac{333}{106}, \frac{355}{113}, \frac{103,993}{33,102}, \frac{104,348}{33,215}, \frac{208,341}{66,317}, \frac{312,689}{99,532}, \frac{833,719}{265,381}, \frac{1,146,408}{364,913}.$$

with respective distances from π of

$$-0.141593, 0.00126449, -0.0000832196, 2.66764 \times 10^{-7}, -5.77891 \times 10^{-10},$$
$$3.31628 \times 10^{-10}, -1.22356 \times 10^{-10}, 2.91434 \times 10^{-11},$$
$$-8.71525 \times 10^{-12}, 1.61071 \times 10^{-12}.$$

Why is this list of convergents any better than the list of partial sums? Without getting too deep into continued fractions, we mention three properties without proof [49]:

1. Each convergent is automatically in lowest terms.
2. If $\frac{p_n}{q_n}$ are convergents that approximate an irrational number x and if $q \le q_n$ and if $\frac{p}{q} \neq \frac{p_n}{q_n}$, then
$$\left|\frac{p_n}{q_n} - x\right| < \left|\frac{p}{q} - x\right|.$$
This means that each convergent of a continued fraction is the best possible rational approximation to x with a denominator of its size or less.
3. Successive convergents alternate in being "too large" or "too small."

In general, the convergents for continued fractions are considered the "best" rational approximations to irrational numbers though, again, there appears to be no pattern in the sequence. We will mention briefly *generalized continued fractions*, which allow numerators other than 1. For these cases, we can get some nice patterns. For instance, if we use Gregory's arctangent series

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

the related generalized continued fraction can be written as

$$\pi = \cfrac{4}{1+\cfrac{1^2}{2+\cfrac{3^2}{2+\cfrac{5^2}{2+\cfrac{7^2}{2+\cdots}}}}} \tag{12.5}$$

(See Project 12.3 below.) Naturally, we can develop continued fractions for all of our π_p. For example,

$$\pi_4 = 3 + \cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{1+\cfrac{1}{8+\cfrac{1}{1+\cfrac{1}{2+\cdots}}}}}}}},$$

or $[3; 1, 2, 2, 2, 1, 8, 1, 2, 1, 2, 9, 8, 6, \ldots]$ with initial convergents

$$3, 4, \frac{11}{3}, \frac{26}{7}, \frac{63}{17}, \frac{89}{24}, \frac{775}{209}, \frac{864}{233}, \frac{2503}{675}, \frac{3367}{908}, \frac{9237}{2491}, \frac{86,500}{23,327}, \frac{701,237}{189,107}, \frac{4,293,922}{1,157,969}, \ldots$$

and respective distances from π_4

$$-0.708149, 0.291851, -0.0414827, 0.00613636, -0.002267, 0.000183979,$$
$$-0.0000153833, 5.15184 \times 10^{-6}, -1.20645 \times 10^{-6}, 4.25133 \times 10^{-7},$$
$$-1.69873 \times 10^{-8}, 2.22138 \times 10^{-10}, -4.5528 \times 10^{-12}, 1.37668 \times 10^{-14}.$$

Continued fractions for π_p's include

- $\pi_1 = [2]$
- $\pi_{4/3} = [2; 1, 1, 5, 1, 1, 62, 1, 3, 1, 3, 1, 185, 1, \ldots]$
- $\pi_{5/2} = [3; 2, 1, 1, 1, 2, 132, 1, 6, 4, 1, 1, 1, 5, 4, 2, 1, \ldots]$
- $\pi_3 = [3; 1, 1, 7, 79, 2, 7, 1, 2, 4, 1, 2, 10, 1, 2, \ldots]$
- $\pi_{10} = [3; 1, 16, 1, 1, 11, 43, 3, 3, 1, 15, 6, \ldots]$.

12.3 Generating new series and sums

We can often obtain new series representations for known quantities by manipulating known series, which happen to be marvelously resilient. We can do this with the series for the p-arcsine and the p-arctangent we developed in the past two chapters to obtain new series whose sums can be computed at particular values of p and x.

For example, we can use the series for the p-arcsine (11.4),

$$\arcsin_p(x) = x \sum_{n=0}^{\infty} \frac{\Gamma(\frac{p-1}{p} + n)}{\Gamma(\frac{p-1}{p})(np+1)} \frac{x^{np}}{n!}.$$

We saw at the start of this chapter that this series converges at $x = 1$ by Gauss' test. Since $\arcsin_p(1) = \frac{\pi_p}{2}$, we can write

$$\sum_{n=0}^{\infty} \frac{\Gamma(\frac{p-1}{p} + n)}{(np+1)n!} = \frac{\pi_p \Gamma(\frac{p-1}{p})}{2}.$$

When $p = 2$, this gives

$$\sum_{n=0}^{\infty} \frac{\Gamma(n + \frac{1}{2})}{(2n+1)n!} = \frac{\pi^{3/2}}{2}.$$

If we use Legendre's duplication formula (6.7) for the gamma function to rewrite the above summand[2] and recall that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, we get that

$$\sum_{n=1}^{\infty} \binom{2n-1}{n} \frac{2^{1-2n}}{2n+1} = \frac{\pi}{2} - 1.$$

When $p = 3$, we obtain

$$\sum_{n=0}^{\infty} \frac{\Gamma(n + \frac{2}{3})}{(3n+1)n!} = \left(\frac{\pi_3 \pi}{3}\right)^{2/3},$$

and for $p = 4$, we get

$$\sum_{n=0}^{\infty} \frac{\Gamma(n + \frac{3}{4})}{(4n+1)n!} = \frac{\pi^{3/4}\sqrt{\pi_4}}{2}.$$

A different series can be derived from the fact that

$$\frac{\pi_p^2}{8} = \frac{1}{2}(\arcsin_p(1))^2 = \int_0^1 \frac{\arcsin_p(x)}{(1-x^p)^{\frac{p-1}{p}}}\, dx. \tag{12.6}$$

[2] The first term of the series has to be taken separately since it involves the quotient $\Gamma(2n)/\Gamma(n)$. The Legendre duplication formula can be used to rewrite the quotient and show that $\lim_{n\to 0^+} \Gamma(2n)/\Gamma(n) = \frac{1}{2}$.

(Substitute $u = \arcsin_p(x)$ in the integral.) We can get a new series by expanding the denominator in the integral and then integrating. (We can either use the series from the beginning of Chapter 11 or differentiate the p-arcsine series (11.4).) We obtain

$$\Gamma\left(\tfrac{1}{p}\right)^2 \sum_{n=0}^{\infty} \frac{\Gamma(n+\frac{2}{p})\Gamma(n+\frac{p-1}{p})}{\Gamma(n+\frac{3}{p})(np+1)n!} = \frac{p\,\pi_p{}^2\,\pi}{8\sin(\frac{\pi}{p})}. \tag{12.7}$$

(The proof is left to the enthusiastic reader.) When $p = 2$, this reduces to

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8},$$

which was known to Euler. When we set $p = 3$ in (12.7), we get

$$\sum_{n=0}^{\infty} \frac{\Gamma(n+\frac{2}{3})^2}{(3n+1)(n!)^2} = \frac{\sqrt[6]{3}\,\sqrt[3]{\pi}\,\pi_3^{4/3}}{4}.$$

For $p = 4$ we get

$$\sum_{n=0}^{\infty} \frac{\Gamma(n+\frac{1}{2})}{(4n+1)n!} = \frac{\pi_4\sqrt{2\pi}}{4}.$$

We can perform similar tricks with the series for the p-arctangent. (See Exercise 12.8.)

Series for Catalan's constant

If we take the series for $\arctan_p(x)$ (11.5), divide it by x, and integrate it from 0 to 1, we obtain

$$\int_0^1 \frac{\arctan_p(x)}{x}\,dx = \sum_{n=0}^{\infty}(-1)^n \frac{\Gamma(n+\frac{2}{p})}{\Gamma(\frac{2}{p})(np+1)^2 n!}.$$

When we set $p = 2$, we have

$$\int_0^1 \frac{\arctan(x)}{x}\,dx = \sum_{n=0}^{\infty}(-1)^n \frac{\Gamma(n+1)}{(2n+1)^2 n!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}, \tag{12.8}$$

since $\Gamma(n+1) = n!$ and $\Gamma(1) = 1$.

The last series is the definition of Catalan's constant G:

$$G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots = 0.915965594177219\ldots \tag{12.9}$$

Named for the French and Belgian mathematician Eugène Charles Catalan (1814–1894), who found quickly converging series for G and gave expressions for it in terms of integrals, Catalan's constant has many formulas for it, yet no one knows if G is rational or irrational, algebraic or transcendental.

We can get a new series representation for G as a consequence of our series (11.2) for $\arcsin_p(x)$:

$$\arcsin_p(x) = x \sum_{n=0}^{\infty} \frac{\Gamma(\frac{p-1}{p} + n)}{\Gamma(\frac{p-1}{p})(np+1)} \frac{x^{np}}{n!}, \qquad 0 \le x \le 1.$$

Substituting $t = \arcsin_p(x)$, we obtain

$$\frac{t}{\sin_p(t)} = \sum_{n=0}^{\infty} \frac{\Gamma(\frac{p-1}{p} + n)}{\Gamma(\frac{p-1}{p})(np+1)} \frac{\sin_p(t)^{np}}{n!}, \qquad 0 \le t \le \frac{\pi_p}{2}.$$

We now integrate both sides from 0 to $\pi_p/2$ and invoke the result of Exercise 9.6:

$$\int_0^{\pi_p/2} \frac{t}{\sin_p(t)}\, dt = \sum_{n=0}^{\infty} \frac{\Gamma(1 - \frac{1}{p} + n)}{\Gamma(1 - \frac{1}{p})n!} \frac{\Gamma(\frac{1}{p})\Gamma(n + \frac{1}{p})}{p\,\Gamma(n + \frac{2}{p})} \frac{1}{np+1}. \tag{12.10}$$

If we substitute $x = \tan(u)$ in (12.8), we get [52]

$$\begin{aligned} \int_0^1 \frac{\arctan(x)}{x}\, dx &= \int_0^{\pi/4} \frac{u}{\tan(u)} \sec(u)^2\, du \\ &= \int_0^{\pi/4} \frac{u}{\sin(u)\cos(u)}\, du \\ &= 2\int_0^{\pi/4} \frac{u}{\sin(2u)}\, du. \end{aligned}$$

Setting $t = 2u$, we obtain

$$\int_0^{\pi/2} \frac{t}{\sin(t)}\, dt = 2G.$$

So setting $p = 2$ in (12.10) yields

$$G = \frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{\Gamma(n + \frac{1}{2})}{n!} \right)^2 \frac{1}{2n+1}. \tag{12.11}$$

If we now use the Legendre duplication formula (6.7), solve for $\Gamma(n + \frac{1}{2})$, and substitute into (12.11), we get

$$G = \frac{\pi}{4} \sum_{n=0}^{\infty} \left(\frac{(2n)!}{(n!)^2\, 2^{2n}} \right)^2 \frac{1}{2n+1}, \tag{12.12}$$

a formula first derived in [16].

Exercises

▶ **Exercise 12.1** *Use the series (12.3) to estimate π_{12} to five decimal places. (Note that that series is an alternating series.)*

▶ **Exercise 12.2** *Do the same for (12.4).*

▶ **Exercise 12.3**

1. *Show that*
$$ {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{p-1}{p} \\ & \frac{p+1}{p}\end{matrix}; 1\right) = \frac{\pi_p}{2}. $$
2. *Show that*
$$ {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{p-1}{p} \\ & \frac{p+1}{p}\end{matrix}; \frac{1}{2}\right) = \frac{\pi_p}{2^{2-1/p}}. $$
3. *Prove that*
$$ {}_2F_1\left(\begin{matrix}\frac{1}{p}, & \frac{2}{p} \\ & \frac{p+1}{p}\end{matrix}; -1\right) = \frac{\pi_p}{4}. $$

▶ **Exercise 12.4** *Compute the first five nonzero terms of the series (12.2) for each of $p = 2, 4,$ and 10. Convince yourself of the correctness of your answer by comparing your series against the first row of Table 12.2.*

▶ **Exercise 12.5** *Compute the first five nonzero terms of the series (12.4) for each of $p = 2, 4$ and 10. Convince yourself of the correctness of your answer by comparing your series against the first row of Table 12.4.*

▶ **Exercise 12.6** *Find the first five convergents of the continued fractions for the π_p's listed above. Which are within .01 of the given π_p?*

▶ **Exercise 12.7** *Calculate directly the values of π_p^{arc} and π_p^{euc} for $p = 1$ and $p = \infty$.*

▶ **Exercise 12.8** *Use the series for the p-arctangent (11.5) to show that*
$$ \sum_{n=0}^{\infty} (-1)^n \frac{\Gamma(\frac{2}{p} + n)}{(np+1)n!} = \frac{\pi_p \Gamma(\frac{2}{p})}{4}. $$
Find the exact sums of this series for $p = 2, 3, 4, 6,$ and 8.

▶ **Exercise 12.9** *Compare the rates of convergence for the series (12.9) and (12.12) for Catalan's constant G.*

Projects

Project 12.1 (Surprising integrals)
A surprising and beautiful integral result is that

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi,$$

which can be proven via polynomial division and term-by-term integration. This integral also provides proof that $\frac{22}{7} > \pi$. A more recent result [64] is that

$$\int_0^1 \frac{x^8(1-x)^8(25+816x^2)}{3164(1+x^2)}\,dx = \frac{355}{113} - \pi.$$

Explore whether there are integral formulas that give, for instance, the difference between π_4 and a continued fraction convergent. For example, is there a reasonable integral whose value is $\frac{26}{7} - \pi_4$ or $\pi_4 - \frac{63}{17}$? You will likely need a computer algebra system for your explorations.

Project 12.2 (Generalized continued fractions)
We implied in Equation (12.5) that there was a connection between Gregory's series and a generalized continued fraction for π. In fact, Euler discovered a connection between alternating series and generalized continued fractions. In particular, the convergents for the continued fraction correspond exactly to the partial sums of the series. Consult [53] and [86] and answer the following questions.

1. Show that the generalized continued fraction (12.5) is the correct one corresponding to Gregory's series.
2. Find a similar continued fraction for π_4 using the series (12.3) for π_p. Convince yourself that your continued fraction is correct by demonstrating that the convergents are exactly the partial sums of (12.3).
3. Find a generalized continued fraction for π_p using (12.3) as a jumping-off point. When you set $p = 2$ in your answer, do you get the same continued fraction? If not, can you adjust your continued fraction so that it does match?

Project 12.3 (Pi as a function and isoperimetric ratios)
However we choose to define π, we can choose to see it as a function of p. That means we can analyze it formally, seeking and interpreting features familiar from calculus. For example, we have already discussed the minimum of π_p^{arc} at $p = 2$ [2]. Does the inflection point in Figure 8.2 have any geometric meaning?

Arclength offers a two-parameter function to analyze. In addition to varying the p for the squircle, we can also vary the metric for calculating the arclength. We might thus define $\pi_{p,q}$ to be half the circumference of the p-circle with respect to the q-arclength. We defined π_p^{arc} so that the arclength is taken with respect to the p-arclength.

For example, we could use Euclidean arclength

$$\pi_p^{\text{euc}} = \pi_{p,2}^{\text{arc}} = 2\int_0^1 \sqrt{1 + \left(t^{p-1}(1-t^p)^{\frac{1}{p}-1}\right)^2}\, dt.$$

Although this does not consider arclength in the circle's natural geometry unless $p = 2$, it offers some insight into the geometry of the p-circle.

The *isoperimetric ratio* of a simple closed curve is the square of the length of the curve divided by its area. Note that this is a dimensionless quantity and thus scale-independent. For the unit p-circle, this is $I_p = \frac{\text{length}^2}{\text{area}} = \frac{(2\pi_p^{\text{euc}})^2}{\pi_p} = \frac{4(\pi_p^{\text{euc}})^2}{\pi_p}$. For $p = 2$, where all of our definitions of π are equivalent, this is 4π. The circle is the solution to the *isoperimetric problem*, meaning that 4π is the minimum possible value of the isoperimetric ratio. That is, the circle is the most efficient shape to enclose the most area with the smallest perimeter [12].

Of course, this is not true for general p. What can be said about the isoperimetric ratio if we use p-arclength instead of Euclidean? We may define

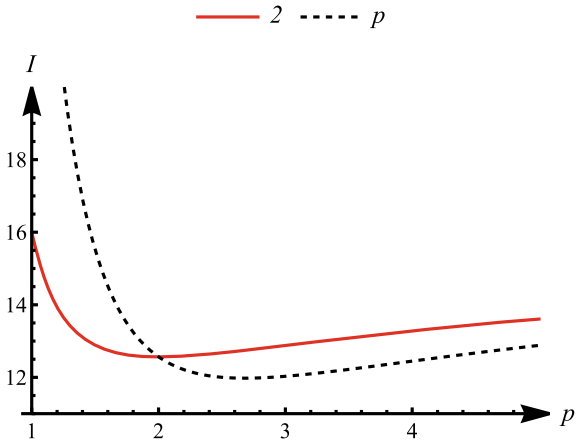


Figure 12.1 Isoperimetric ratios of unit p-circle with respect to the 2 (Euclidean) and p metrics.

$\hat{I}_p = \frac{4(\pi_p^{\text{arc}})^2}{\pi_p}$ and we see that the minimum is not attained at $p = 2$. See Figure 12.1. And we again have the two-parameter option for this function as we separately vary the metric for the circumference and squircle (or three, if we also want to independently vary the area).

Thus we have new functions to explore. What are the extremal and inflection points of the various definitions of π and the isoperimetric ratio? What squircle-arclength combinations give interesting results? We have already seen something curious happen when we take the q-length of a p-circle when p and q are conjugate in Theorem 8.2.



Alternate coordinate systems

13

Up until now we have generally worked in Cartesian coordinates to locate points in the plane. Our past experience with calculus has shown us, though, that Newton's polar coordinate system can be more convenient for describing some curves or performing some integrations. This is all well and good if we're dealing with circles or circular regions. When we have a general p-circle, or we wish to integrate over a p-circular region, we need to make an adjustment.

13.1 Squircular coordinate system

As we have seen in (9.6), we can set up squircular coordinates using the p-metric, via the conversion

$$x = r\cos_p(t), \qquad y = r\sin_p(t),$$

where $|x|^p + |y|^p = r^p$. (That is, the distance between the origin and the point (x, y) on the p-circle is r units in the p-metric. See Figure 13.1.)

In this case we will assign the point P with Cartesian coordinates (x, y) the *squircular* coordinates (r, t). To find t when we know x and y, we can use the relationship

$$\tan_p(t) = \frac{y}{x}.$$

Once more, it is important to distinguish between the value of the parameter t and the actual angle subtended (in radians):

$$\tan(\theta(t)) = \frac{y}{x};$$

that is, $\tan(\theta(t)) = \tan_p(t)$.

As with the classical polar coordinates, each point in the plane has an infinitude of representations in squircular coordinates. The point (r, t) can be represented as well by $(r, t + 2n\pi_p)$ or $(-r, t + (2n + 1)\pi_p)$ for any $n \in \mathbb{Z}$.

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_13

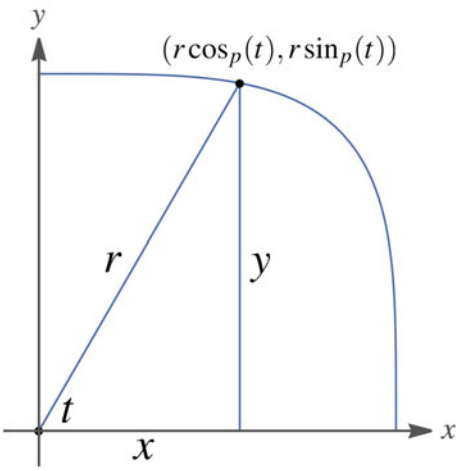


Figure 13.1 Writing point (x, y) in terms of radius r and parameter t

13.2 Squircular curves

We can graph some squircular equations of the form $r = f(t)$ or $F(r, t) = 0$ but as we might guess, the difficulty level is higher for $p \neq 2$. (The examples in the rest of the chapter are adaptations of problems in [93].)

Example 13.1

What curve is represented by the squircular equation $r = 2$?

Solution. The curve consists of all points (r, t) with $r = 2$. Since r represents the distance from the point to the origin in the p-metric, the curve $r = 2$ represents the p-circle centered at the origin and radius 2. In general the equation $r = a$ represents a p-circle centered at the origin of radius $|a|$. (See Figure 13.2 for various examples when $p = 4$.) □

Example 13.2

When $p = 4$, what curve is represented by the equation $t = \pi_4/3$?

Solution. Since $t = \pi_4/3 = \arctan_4(y/x)$, we have that $y = (\tan_4(\pi_4/3))x \approx 1.59x$, a line. In fact, any equation of the form $t = a$ represents a straight line through the origin of slope $\tan_p(a)$. □

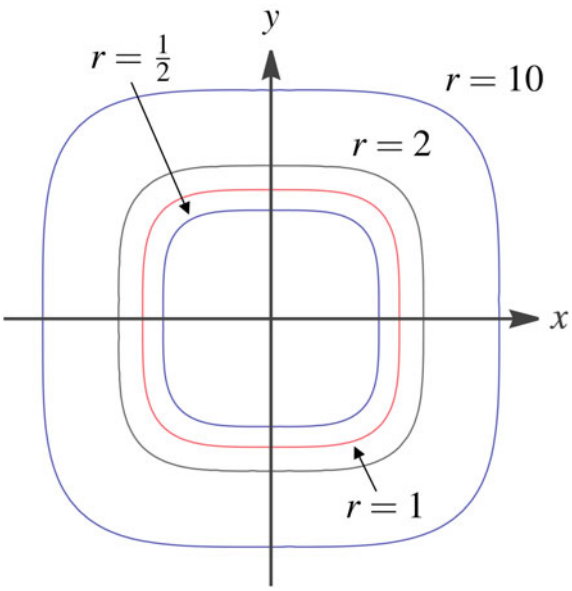


Figure 13.2 The squircular ($p = 4$) curves $r = \frac{1}{2}$, $r = 1$, $r = 2$, and $r = 10$

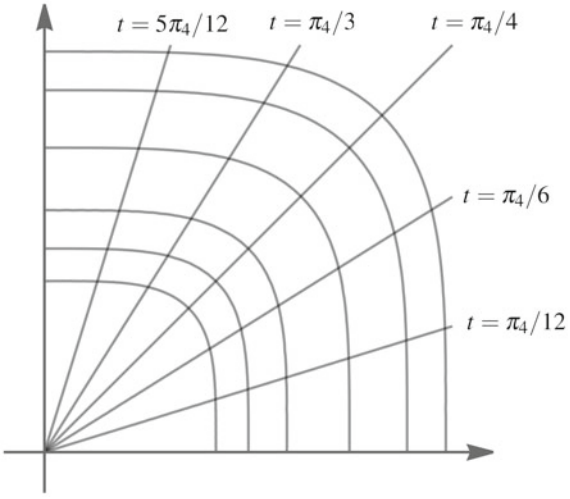


Figure 13.3 Curves of the form $t =$ constant for $p = 4$

As Figure 13.3 bears out, unlike the usual polar coordinates, our squircular coordinate system is not an orthogonal one; that is, the curves $r =$ constant and $t =$ constant generally do not intersect at right angles for $p \neq 2$.

Example 13.3

Sketch the curve with squircular equation $r = 2\cos_p(t)$.

Solution. When $p = 2$, we can easily convert the polar equation to Cartesian:

$$\begin{aligned} r = 2\cos t &\implies r^2 = 2r\cos t \\ &\implies x^2 + y^2 = 2x \\ &\implies x^2 - 2x + y^2 = 0 \\ &\implies (x-1)^2 + y^2 = 1, \end{aligned}$$

where the last equation follows from completing the square in x. This is the equation of a circle of radius 1 with center $(x, y) = (1, 0)$.

Matters are not so simple for $p \neq 2$. If we perform a similar trick to restore Cartesian coordinates, we get the equation $(x^p + y^p)^{2/p} - 2x = 0$, which is not helpful. We notice, on the other hand, for large p, as t increases, the radius r does not change much before $t = \pi_p/4$. So we end up with a much flatter shape than before as Figure 13.4 bears out. This tends to be the general theme for graphs when $p > 2$. □

Example 13.4

Sketch the curve $r = 1 + \sin_4(t)$.

Solution. As with polar coordinates, it is sometimes easier to first sketch the graph in Cartesian coordinates and then use that graph as a way to read values, keeping in mind that the distance r is in the 4-metric. So we get a flattened cardioid. See Figure 13.5.

□

Example 13.5

Sketch the curve $r = \cos_5(t)^5 - \sin_5(t)^5$.

Solution. Recall that in the absence of double-angle formulas for the p-trigonometric functions, we can use $\cos_p(t)^p - \sin_p(t)^p$ as a generalization for $\cos(2t) = \cos(t)^2 - \sin(t)^2$. Happily, we get the expected flattened version of the usual four-leaved rose. See Figure 13.6.

See the menagerie in Appendix A for additional examples of polar curves. □

13.3 Area

In Chapter 6, we established that the area of the p-circle $|x|^p + |y|^p = r^p$ is $\pi_p r^2$ square units. This implies that the area of a sector of this circle is $A = \frac{1}{2}r^2 t$. (Note that this holds if we use the parameter t, but not the angle $\theta(t)$, for $p \neq 2$.) We

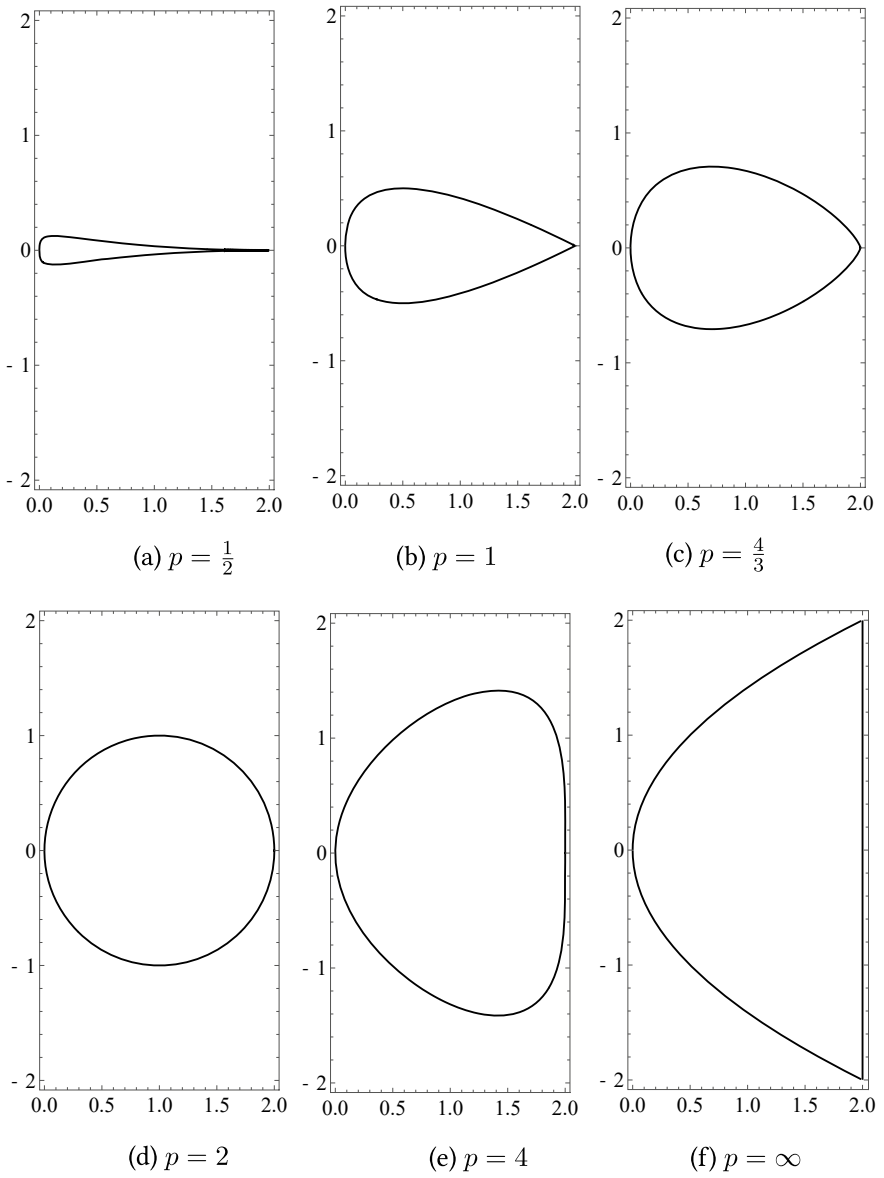
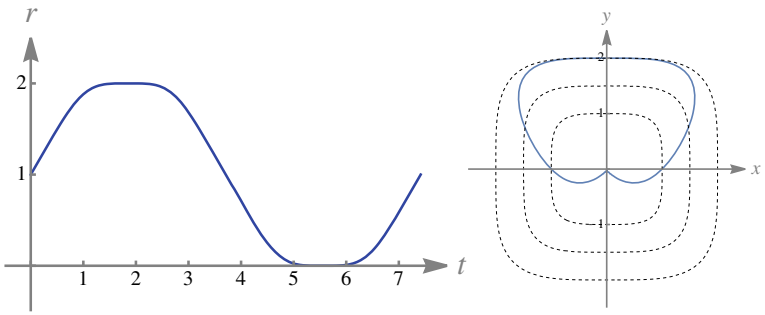


Figure 13.4 Squircular graphs of $r = 2\cos_p(t)$ for various p-values



(a) Cartesian coordinates (b) Squircular coordinates

Figure 13.5 $r = 1 + \sin_4(t), 0 \le t \le 2\pi_4$

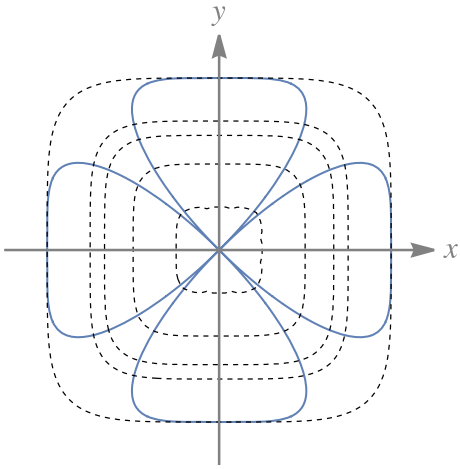


Figure 13.6 Four-leaved rose $r = \cos_5(t)^5 - \sin_5(t)^5$

can use the usual Riemann sum argument then to establish that if $\mathscr{R}$ is the region bounded by the squircular curve $r = f(t)$ and the rays $t = a$ and $t = b$, then

$$A(\mathscr{R}) = \int_a^b \tfrac{1}{2}[f(t)]^2\,dt = \int_a^b \tfrac{1}{2}r^2\,dt.$$

Example 13.6

Find the area enclosed by one loop of the four-leaved rose $r = \cos_5(t)^5 - \sin_5(t)^5$.

Solution. This curve was sketched out in Example 13.5. If we choose the rightmost leaf, a good choice since it encompasses the x-axis where $t = 0$, we note that the

region is swept out by a ray that rotates from $t = -\pi_5/4$ to $t = \pi_5/4$. Therefore,

$$\begin{aligned} A &= \int_{-\pi_5/4}^{\pi_5/4} \tfrac{1}{2} r^2 \, dt \\ &= \tfrac{1}{2} \int_{-\pi_5/4}^{\pi_5/4} (\cos_5(t)^5 - \sin_5(t)^5)^2 \, dt \\ &= \int_0^{\pi_5/4} (\cos_5(t)^5 - \sin_5(t)^5)^2 \, dt \\ &= \int_0^{\pi_5/4} (\cos_5(t)^{10} - 2\cos_5(t)^5 \sin_5(t)^5 + \sin_5(t)^{10}) \, dt. \end{aligned}$$

We first evaluate $\int_0^{\pi_5/4} \cos_5(t)^{10} \, dt$. We can apply the reduction formula (9.1) with $m = 10$ and $p = 5$ to get

$$\begin{aligned} \int_0^{\pi_5/4} \cos_5(t)^{10} \, dt &= \frac{1}{7} \cos_5(t)^6 \sin_5(t) \Big|_0^{\pi_5/4} + \frac{6}{7} \int_0^{\pi_5/4} \cos_5(t)^5 \, dt \\ &= \frac{1}{7} \left(\frac{1}{\sqrt[5]{2}} \right)^7 + \frac{6}{7} \int_0^{\pi_5/4} \cos_5(t)^5 \, dt. \end{aligned}$$

Similarly, formula (9.1) also gives us

$$\int_0^{\pi_5/4} \sin_5(t)^{10} \, dt = -\frac{1}{7} \left(\frac{1}{\sqrt[5]{2}} \right)^7 + \frac{6}{7} \int_0^{\pi_5/4} \sin_5(t)^5 \, dt.$$

Adding these two results yields

$$\int_0^{\pi_5/4} (\cos_5(t)^{10} + \sin_5(t)^{10}) \, dt = \frac{6}{7} \int_0^{\pi_5/4} (\cos_5(t)^5 + \sin_5(t)^5) \, dt = \frac{6}{7} \int_0^{\pi_5/4} 1 \, dt = \frac{3\pi_5}{14}.$$

This leaves us to evaluate $-2 \int_0^{\pi_5/4} \cos_5(t)^5 \sin_5(t)^5 \, dt$. If we let $u = \sin_5(t)^5$, then $du = 5 \sin_5(t)^4 \cos_5(t)^4 \, dt$ and

$$\begin{aligned} -2 \int_0^{\pi_5/4} \cos_5(t)^5 \sin_5(t)^5 \, dt &= -\frac{2}{5} \int_0^{\pi_5/4} \cos_5(t) \sin_5(t) \cdot 5 \sin_5(t)^4 \cos_5(t)^4 \, dt \\ &= -\frac{2}{5} \int_0^{1/2} u^{\frac{1}{5}} (1-u)^{\frac{1}{5}} \, du. \end{aligned}$$

This last integral resembles the integral for the beta function, but the upper limit is wrong. It's an example of an *incomplete beta function*. Denote this integral by I. Let $s = 1 - u$. Then $ds = -du$ and we have

$$I = \int_0^{1/2} u^{\frac{1}{5}} (1-u)^{\frac{1}{5}} \, du = \int_1^{1/2} (1-s)^{\frac{1}{5}} s^{\frac{1}{5}} \, (-ds) = \int_{1/2}^1 s^{\frac{1}{5}} (1-s)^{\frac{1}{5}} \, ds.$$

Therefore,

$$2I = \int_0^1 u^{\frac{1}{5}} (1-u)^{\frac{1}{5}} \, du = \mathrm{B}(\tfrac{6}{5}, \tfrac{6}{5}) = \frac{[\Gamma(\frac{6}{5})]^2}{\Gamma(\frac{12}{5})},$$

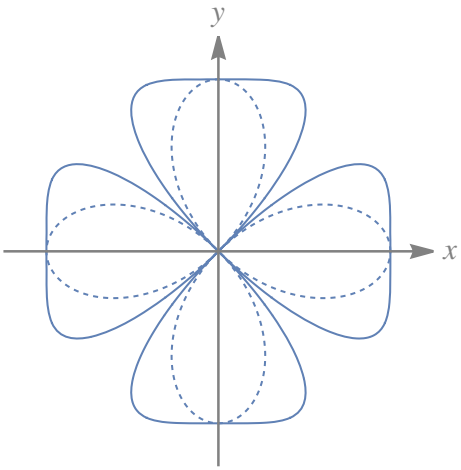


Figure 13.7 Plots of $r = \cos(2t) = \cos(t)^2 - \sin(t)^2$ (dashed) and $r = \cos_5(t)^5 - \sin_5(t)^5$ (solid)

and

$$\begin{aligned}
-2\int_0^{\pi_5/4} \cos_5(t)^5 \sin_5(t)^5\, dt &= -\frac{1}{5}\,\frac{[\Gamma(\frac{6}{5})]^2}{\Gamma(\frac{12}{5})} \\
&= -\frac{1}{5}\,\frac{[\frac{1}{5}\Gamma(\frac{1}{5})]^2}{\frac{7}{5}\Gamma(\frac{7}{5})} \\
&= -\frac{1}{5}\,\frac{\frac{1}{25}\Gamma(\frac{1}{5})^2}{\frac{7}{5}\cdot\frac{2}{5}\Gamma(\frac{2}{5})} \\
&= -\frac{1}{125}\cdot\frac{25}{14}\,\frac{\Gamma(\frac{1}{5})^2}{\Gamma(\frac{2}{5})} \\
&= -\frac{1}{28}\cdot\frac{2}{5}\,\frac{\Gamma(\frac{1}{5})^2}{\Gamma(\frac{2}{5})} = -\frac{\pi_5}{28},
\end{aligned}$$

where we have used (6.2) in the second and third lines. At last this gives us that the area of one loop of $r = \cos_5(t)^5 - \sin_5(t)^5$ is

$$A = \frac{3\pi_5}{14} - \frac{\pi_5}{28} = \frac{5\pi_5}{28} \approx 0.678679 \text{ square units.}$$

This number seems plausible given the plot in Figure 13.7 and the knowledge that the area of one loop of the four-leaved rose $r = \cos(2t)$ is $\pi/8 \approx 0.392699$ square units.

□

13.4 Double integrals

We can use double integrals in squircular coordinates to evaluate an integral like $\iint_{\mathscr{R}} f(x, y)\, dA$, where $\mathscr{R}$ is a region best described using squircular coordinates.

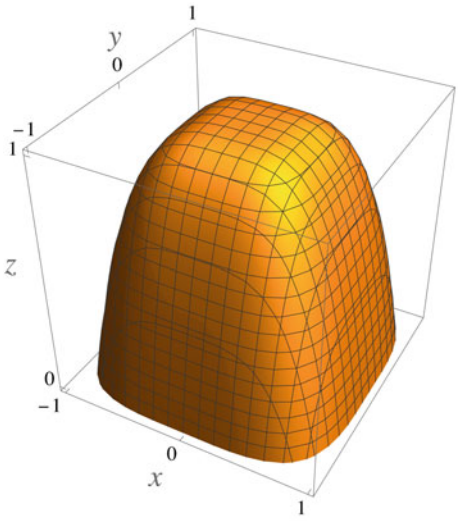


Figure 13.8 The solid bounded by the plane $z = 0$ and the 4-paraboloid $z = 1 - x^4 - y^4$

We computed in Chapter 9 that the area element in computing a double integral in squircular coordinates is $dA = r\,dr\,dt$. (See Equation (9.6).) This means that if we have a "squircular rectangle" $\mathscr{R}$ given by $0 \le a \le r \le b$, $\alpha \le t \le \beta$, where $0 \le \beta - \alpha \le 2\pi_p$, then

$$\iint_{\mathscr{R}} f(x, y)\,dA = \int_{\alpha}^{\beta}\int_{a}^{b} f(r\cos_p(t), r\sin_p(t))\,r\,dr\,dt.$$

We have a similar formula for when the squircular region is bounded by curves of the form $r = g(t)$.

Example 13.7

Find the volume of the solid bounded by the plane $z = 0$ and the 4-paraboloid $z = 1 - x^4 - y^4$. (See Figure 13.8.)

Solution. The solid intersects the xy-plane in the squircle $x^4 + y^4 = 1$, so the planar region D for integration is the squircular disk $x^4 + y^4 \le 1$. In squircular coordinates, this is given by $0 \le r \le 1, 0 \le t \le 2\pi_4$. Because $1 - x^4 - y^4 = 1 - r^4$, the volume is

$$\begin{aligned} V &= \iint_D (1 - x^4 - y^4)\,dA = \int_0^{2\pi_4}\int_0^1 (1 - r^4)\,r\,dr\,dt \\ &= \int_0^{2\pi_4} dt \int_0^1 (r - r^5)\,dr \\ &= 2\pi_4 \left[\frac{r^2}{2} - \frac{r^6}{6}\right]_0^1 \\ &= \frac{2\pi_4}{3} \approx 2.4721 \text{ cubic units.} \end{aligned}$$

□

13.5 Orthogonal trajectories

As noted earlier, the curves $r = \text{constant}$ generally do not meet the curves $t = \text{constant}$ at right angles. This can make some vector operations more difficult as the basis vectors are not orthogonal.

We can determine an orthogonal coordinate system by keeping the concentric p-circles $r = \text{constant}$ and determining the set of orthogonal trajectories for these curves. Orthogonal trajectories are useful in analyzing differential equations and in conformal geometry.

To ease calculations, we will work entirely in the first quadrant. Given the p-circle $x^p + y^p = r^p$, we differentiate implicitly with respect to x and obtain

$$\begin{aligned} x^p + y^p = r^p &\Longrightarrow px^{p-1} + py^{p-1}\frac{dy}{dx} = 0 \\ &\Longrightarrow \frac{dy}{dx} = -\frac{x^{p-1}}{y^{p-1}}. \end{aligned}$$

This means that our orthogonal trajectories have slope $\frac{dy}{dx} = \frac{y^{p-1}}{x^{p-1}}$, the negative reciprocal of the previous slope. This differential equation is separable:

$$\frac{dy}{dx} = \frac{y^{p-1}}{x^{p-1}} \Longrightarrow \int \frac{dy}{y^{p-1}} = \int \frac{dx}{x^{p-1}} \tag{13.1}$$

If $p = 2$, integrating the previous equation leads to

$$\ln y = \ln x + C \Longrightarrow y = e^{\ln x + C} = e^{\ln x} e^C = C_1 x$$

where $C_1 = e^C$. Since $y = 0$ is a solution to the differential equation as well, the solution can be written as $y = C_1 x$ for any $C_1 \geq 0$. These are the lines of constant angle $t = \theta$ in the usual polar coordinates.

If $p \neq 2$, matters turn out differently. We integrate (13.1) to get

$$\begin{aligned} \int y^{1-p}\,dy = \int x^{1-p}\,dx &\Longrightarrow \frac{y^{2-p}}{2-p} = \frac{x^{2-p}}{2-p} + C \\ &\Longrightarrow y = \left(x^{2-p} + C_1\right)^{1/(2-p)}, \end{aligned}$$

where $C_1 = C(2-p)$. A plot of some orthogonal trajectories for $p = 4$ can be found in Figure 13.9.

13.6 Expanding to three dimensions

p-cylindrical coordinates

We can expand the idea of squircular coordinates to three dimensions by using the idea of p-cylindrical coordinates. We simply add the z-coordinate:

$$x = r\cos_p(t), \qquad y = r\sin_p(t), \qquad z = z$$

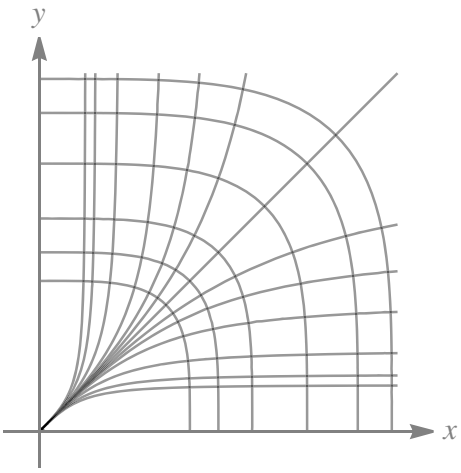


Figure 13.9 The 4-circles $r =$ constant and their orthogonal trajectories

and the conversion from Cartesian to p-cylindrical coordinates is

$$r^p = |x|^p + |y|^p, \qquad \tan_p(t) = \frac{y}{x}, \qquad z = z.$$

Example 13.8

A solid E lies within the cylinder $x^4 + y^4 = 1$, below the plane $z = 4$, and above the 4-paraboloid $z = 1 - x^4 - y^4$. (See Figure 13.10.) The density at any point is proportional to its distance (in the 4-metric) from the axis of the cylinder. Find the mass of E.

Solution. In 4-cylindrical coordinates the cylinder is $r = 1$ and the 4-paraboloid is $z = 1 - r^4$. Therefore we can write

$$E = \{(r, t, z) \mid 0 \le t \le 2\pi_4,\ 0 \le r \le 1,\ 1 - r^4 \le z \le 4\}$$

Since the density $\rho(x, y, z)$ is proportional to the 4-distance from the z-axis, the density function is

$$\rho(x, y, z) = K\sqrt[4]{x^4 + y^4} = Kr,$$

where K is a constant of proportionality. Therefore,

$$\begin{aligned} m &= \iiint_E K\sqrt[4]{x^4 + y^4}\, dV \\ &= \int_0^{2\pi_4} \int_0^1 \int_{1-r^4}^4 (Kr)\, r\, dz\, dr\, dt \\ &= K \int_0^{2\pi_4} dt \int_0^1 r^2[4 - (1 - r^4)]\, dr \end{aligned}$$

$$= 2\pi_4 K \int_0^1 (3r^2 + r^6)\, dr$$
$$= 2\pi_4 K \left[r^3 + \frac{r^7}{7} \right]_0^1$$
$$= \frac{16\pi_4 K}{7}$$

□

p-spherical coordinates

Another way to extend the polar coordinate system into three dimensions is spherical coordinates, which involves a three-dimensional radius ρ measured using the p-metric. Thus a p-sphere would have equation $|x|^p + |y|^p + |z|^p = \rho^p$, where $\rho \geq 0$. (See Figure 13.11.)

We have two coordinates representing angles. We let t, as in squircular coordinates, represent an angle starting from the positive x-axis, and we let u represent an angle beginning from the positive z-axis. The quantities t and u are not angles themselves but are related to the usual spherical coordinate angles through our previous relationships:

$$\tan(\theta(t)) = \tan_p(t), \qquad \tan(\phi(u)) = \tan_p(u).$$

In particular, we have

$$0 \leq t \leq 2\pi_p, \qquad 0 \leq u \leq \pi_p.$$

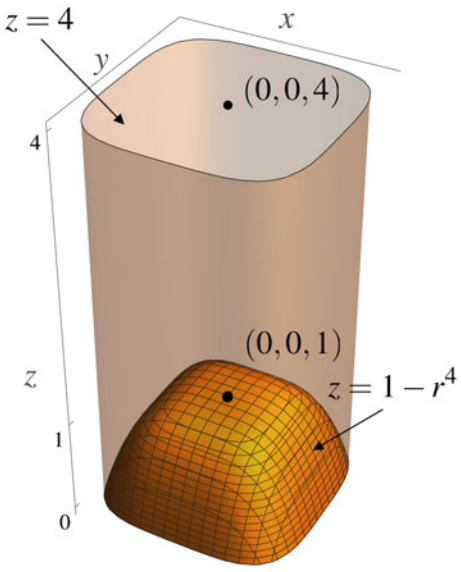
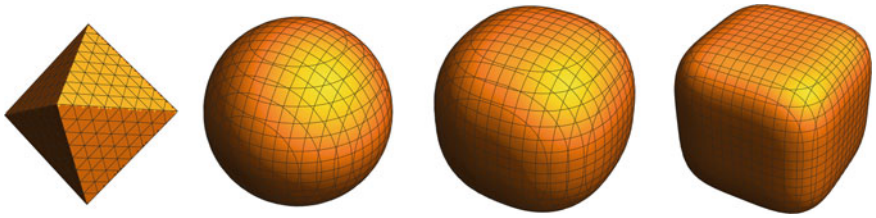


Figure 13.10 The solid bounded on the sides by the 4-cylinder $x^4 + y^4 = 1$, above by the plane $z = 4$, and below by the 4-paraboloid $z = 1 - x^4 - y^4$



(a) $p = 1$ (b) $p = 2$ (c) $p = 2.5$ (d) $p = 4$

Figure 13.11 Some p-spheres for various p

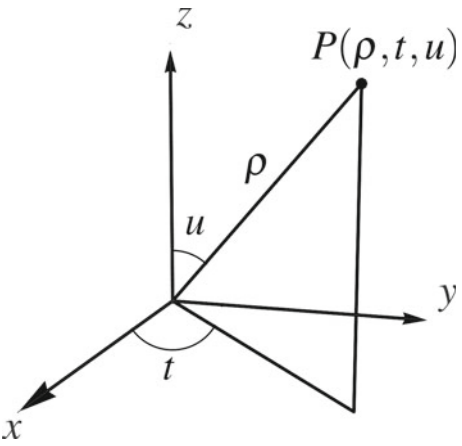


Figure 13.12 The p-spherical coordinates of a point

The relationship between Cartesian coordinates and p-spherical coordinates can be seen in Figure 13.12 and should seem familiar to us from our previous work with spherical coordinates. To convert from p-spherical coordinates to rectangular ones, we use

$$x = \rho \sin_p(u) \cos_p(t), \qquad y = \rho \sin_p(u) \sin_p(t), \qquad z = \rho \cos_p(u).$$

where

$$\rho^p = |x|^p + |y|^p + |z|^p.$$

We use this last equation in converting from rectangular to spherical coordinates. It can be shown that the volume element in p-spherical coordinates is

$$dV = dx\, dy\, dz = \rho^2 \sin_p(u)\, d\rho\, dt\, du.$$

(See Exercise 13.5.) Thus we can perform triple integration in p-spherical coordinates:

$$\iiint_E f(x, y, z)\, dV$$
$$= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin_p(u) \cos_p(t), \rho \sin_p(u) \sin_p(t), \rho \cos_p(u))\, \rho^2 \sin_p(u)\, d\rho\, dt\, du \tag{13.2}$$

where E is a "p-spherical wedge" given by

$$E = \{(\rho, t, u) \mid a \le \rho \le b,\ \alpha \le t \le \beta,\ c \le u \le d\}.$$

We will finish this chapter by finding expressions for volumes of p-spheres and for objects called *superellipsoids*.

Example 13.9

Find the volume of the unit p-sphere $|x|^p + |y|^p + |z|^p = 1$.

Solution. If we let E be the unit p-sphere and let V_p be its volume, we can use (13.2) to write

$$\begin{aligned} V_p &= \iiint_E dV \\ &= \int_0^{\pi_p} \int_0^{2\pi_p} \int_0^1 \rho^2 \sin_p(u)\, d\rho\, dt\, du \\ &= \left(\int_0^{2\pi_p} dt \right) \left(\int_0^1 \rho^2\, d\rho \right) \left(\int_0^{\pi_p} \sin_p(u)\, du \right) \\ &= 2\pi_p \left[\frac{\rho^3}{3} \right]_0^1 \int_0^{\pi_p} \sin_p(u)\, du \\ &= \frac{2\pi_p}{3} \int_0^{\pi_p} \sin_p(u)\, du \\ &= \frac{4\pi_p}{3} \int_0^{\pi_p/2} \sin_p(u)\, du, \end{aligned}$$

by symmetry. Again, we appeal to formula (9.7) with $m = 0$ and $n = 1$ to obtain

$$\begin{aligned} V_p &= \frac{4\pi_p}{3} \cdot \frac{1}{p} B(\tfrac{1}{p}, \tfrac{2}{p}) \\ &= \frac{4\pi_p}{3p} \cdot \frac{\Gamma(\frac{1}{p})\Gamma(\frac{2}{p})}{\Gamma(\frac{3}{p})}. \end{aligned} \tag{13.3}$$

□

This expression is similar to (but not the same as) the formula for the volume of a superegg we discovered in Chapter 10. Employing the same techniques we used there to simplify for various fixed p, we get that for $p = 1$, $V_1 = \frac{4}{3} \approx 1.33333$. For $p = 2$, we obtain the expected $V_2 = \frac{4\pi}{3} \approx 4.18879$. Others can be seen in Table 13.1.

Table 13.1 Volumes of various unit p-spheres

p	Volume of unit p-sphere
1	$V_1 = \frac{4}{3} \approx 1.33333$
4/3	$V_{4/3} = \frac{6\pi_{4/3}\pi}{5\sqrt{2}\,\pi_4} \approx 1.82714$
2	$V_2 = \frac{4\pi}{3} \approx 4.18879$
5/2	$V_{5/2} = \frac{64\pi_{5/2}\pi}{15\pi_5\sqrt{10-2\sqrt{5}}} \approx 5.07159$
3	$V_3 = \frac{8\pi_3\pi}{3^{5/2}} \approx 5.69658$
4	$V_4 = \frac{\sqrt{2}\,\pi_4^2}{3} \approx 6.48199$
6	$V_6 = \frac{2\sqrt{\pi_3}\,\pi_6^{3/2}}{3^{5/4}} \approx 7.20767$
8	$V_8 = \frac{\pi_4\pi_8\sqrt{1+\sqrt{2}}}{3} \approx 7.51670$

Superellipsoids

The three-dimensional analogue of a hyperellipse is a *superellipsoid*, a surface with equation $|x/a|^p + |y/b|^p + |z/c|^p = 1$. (See Figure 13.13.) Note that each cross section of a superellipsoid is a hyperellipse.

Through scaling, and Equation (13.3), we can show that the volume of a superellipsoid is

$$V = V_p abc = \frac{4\pi_p}{3p} \cdot \frac{\Gamma(\frac{1}{p})\Gamma(\frac{2}{p})}{\Gamma(\frac{3}{p})} abc.$$

So for instance, using Table 13.1, the volume of the solid shown in Figure 13.13 would be approximately $6.48199(1)(3/2)(2) \approx 19.446$ cubic units. We note that for $p > 2$ and with $b = c$, the volume of a superellipsoid is generally larger than the volume of a superegg. (For $1 < p < 2$, the volume of the superellipsoid will be smaller.)

Exercises

▶ **Exercise 13.1** *Explore the class of polar curves of the form $r^2 = \cos_p(\theta)$. For $p = 2$, this curve is a lemniscate of Bernoulli which will feature prominently in Section 16.1.*

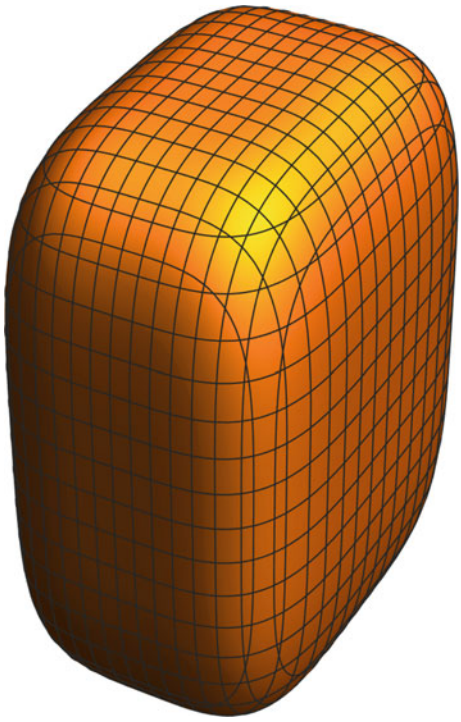


Figure 13.13 The superellipsoid $x^4 + \frac{y^4}{(3/2)^4} + \frac{z^4}{2^4} = 1$

▶ **Exercise 13.2** *Show explicitly that the squircular graph of* $r = 2\cos_\infty(t)$ *in Figure 13.4 really is the capped parabola that it appears to be.*

▶ **Exercise 13.3** *The* osculating circle *of a curve at a point is the circle that best approximates the curve at that point, much like how the tangent is the line that best approximates it at the point.*

If the osculating circle at a point has radius r*, then the* curvature *at that point is* $\frac{1}{r^2}$*. It is loosely a measure of how unlike a line it is at that point. The smaller the circle, the more it bends. As a curve straightens to a line, the radius of the osculating circle grows to infinity and thus the curvature goes to zero.*

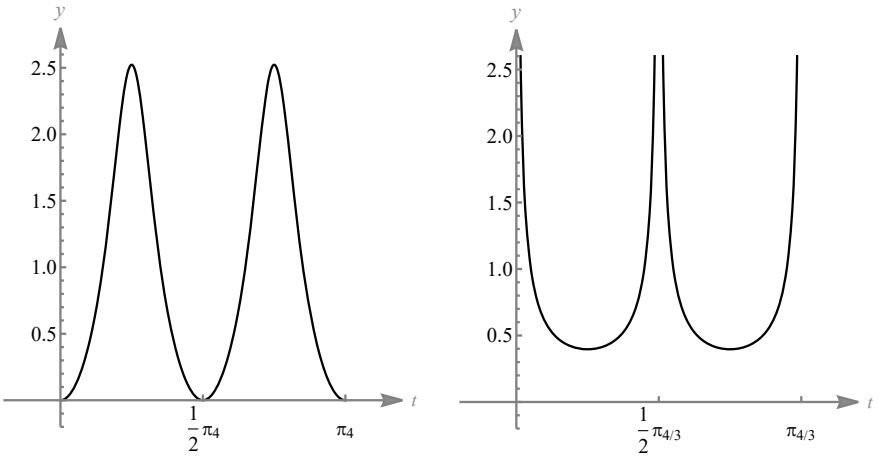
The formula for the curvature of parameterized curve $(x(t), y(t))$ *is (suppressing the t's for clarity)*

$$\kappa = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}}.$$

Show that the curvature for the unit squircle parameterized by $(\cos_p(t), \sin_p(t))$ *is*

$$\kappa_p(t) = |p-1| \frac{|\sin_p(t)\cos_p(t)|^{p-2}}{(|\sin_p(t)|^{2(p-1)} + |\cos_p(t)|^{2(p-1)})^{3/2}}.$$

Confirm that $\kappa_1(t) = 0$ *and* $\kappa_2(t) = 1$*, and explain the behaviors of* $\kappa_p(t)$ *for other* p*. See Figure 13.14.*



(a) $p = 4/3$ (b) $p = 4$

Figure 13.14 Graph of the curvature function $\kappa_p(t)$ of Exercise 13.3

▶ **Exercise 13.4** *Find the area of the region that lies inside the 3-circle* $r = 3\sin_3(t)$ *and the flattened cardioid* $r = 1 + \sin_3(t)$.

▶ **Exercise 13.5** *Prove that the Jacobian for the transformation to p-spherical coordinates is*

$$\frac{\partial(x, y, z)}{\partial(\rho, t, u)} = -\rho^2 \sin_p(u),$$

and thus the volume element in p-spherical coordinates is

$$dV = \rho^2 \sin_p(u)\, d\rho\, dt\, du.$$

▶ **Exercise 13.6** *Show* $V_1 = \frac{4}{3}$ *geometrically.*

▶ **Exercise 13.7** *Derive the formulas in Table 13.1.*

▶ **Exercise 13.8** *Find* $\lim_{p\to\infty} V_p$.

▶ **Exercise 13.9** *Show that the volume of the superellipsoid* $|x/a|^p + |y/b|^p + |z/c|^p = 1$ *is* $V_p abc$, *where* V_p *is the volume of the unit p-sphere.*

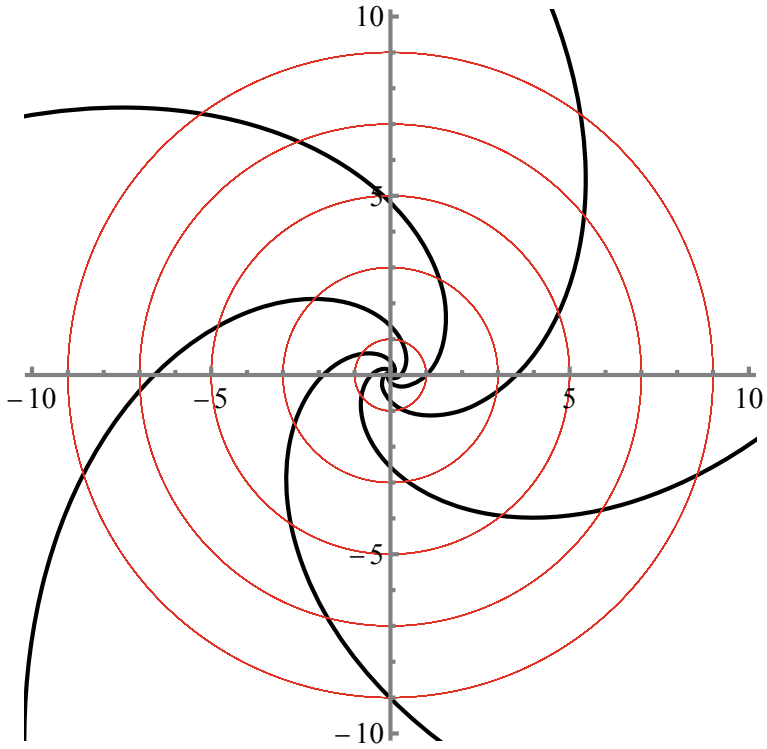


Figure 13.15 Logarithmic spirals are isogonal trajectories of concentric circles

Project

Project 13.1 (Isogonal trajectories)
The orthogonal trajectory problem can be generalized to find families of curves that meet a given set of curves at some specified angle α. Then the orthogonal trajectories are the special case $\alpha = \pi/2$.

Begin the exploration of isogonal trajectories with concentric Euclidean circles and try to recover the *logarithmic spirals* [6] of Figure 13.15. Try $\alpha = \pi/4$, but see what happens as you vary the angle.

Then move on to other p. The resulting differential equations are homogeneous and can be transformed into a separable equation by a substitution, but the resulting integrals are unwieldy. However, they are accessible for some numerical or graphical analysis.



Hyperbolic functions

14

As the pair of functions $(\cos(t), \sin(t))$ parameterize the unit circle $x^2 + y^2 = 1$, the ordered pair $(\cosh(t), \sinh(t))$ parameterizes the right half of the unit hyperbola $x^2 - y^2 = 1$. From the trigonometric functions, it is natural to proceed to generalize the hyperbolic functions (as we do in this chapter), and the exponential and logarithmic functions as well (which we do in the next chapter). As with the trigonometric functions, we will see that these generalizations satisfy variants of the relationships familiar from their classical analogues.

14.1 Definitions

The tools we have developed to generalize trigonometric functions may be applied to other functions as well. The first natural candidates are the hyperbolic trigonometric functions.

Just as we did with our founding trigonometric CIVP (1.1), we define the hyperbolic sine and cosine functions as the unique solution to the CIVP

$$\begin{cases} x'(t) = y(t) \\ y'(t) = x(t) \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{14.1}$$

with $x(t) = \cosh(t)$ and $y(t) = \sinh(t)$, which is constructed so they satisfy the hyperbolic identity. As with the trigonometric functions, the hyperbolic cosine and sine are defined in the first quadrant and then extended via symmetry so that they satisfy the hyperbolic identity $\cosh(t)^2 - \sinh(t)^2 = 1$.

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_14

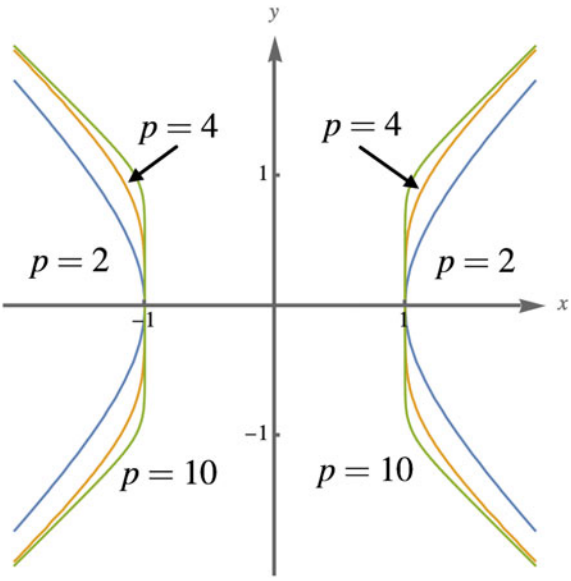


Figure 14.1 The generalized hyperbolas $|x|^p - |y|^p = 1$ for $p = 2$, 4, and 10

We generalize it in the same way. The solutions to the CIVP

$$\begin{cases} x'(t) = y(t)^{p-1} \\ y'(t) = x(t)^{p-1} \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{14.2}$$

define the general hyperbolic functions $\cosh_p(t) = x(t)$ and $\sinh_p(t) = y(t)$. As before, we take this definition in the first quadrant and extend by symmetry to the others so to avoid a glut of absolute values. The curves they parameterize are generalized hyperbolas shown in Figure 14.1.

Moreover, we can adapt the methods of Section 7.2 to show that the parameter is half the area of the hyperbolic sector shown in Figure 14.2 (see Exercise 14.2).

We define $\tanh_p(t) = \frac{\sinh_p(t)}{\cosh_p(t)}$ and the hyperbolic cosecant, secant, and cotangent functions as the reciprocals of the hyperbolic sine, cosine, and tangent functions, respectively.

We can obtain integral expressions for their inverse functions as in Chapter 5.

$$\begin{aligned} \operatorname{arcsinh}_p(x) &= \int_0^x \frac{du}{(1+u^p)^{1-\frac{1}{p}}} \\ \operatorname{arctanh}_p(x) &= \int_0^x \frac{du}{(1-u^p)^{\frac{2}{p}}} \end{aligned} \tag{14.3}$$

The function $\operatorname{arctanh}_4(x) = \int_0^x \frac{du}{\sqrt{1-u^4}}$ was (by another name) a key function in the development of elliptic integrals and will feature prominently in Chapter 16.

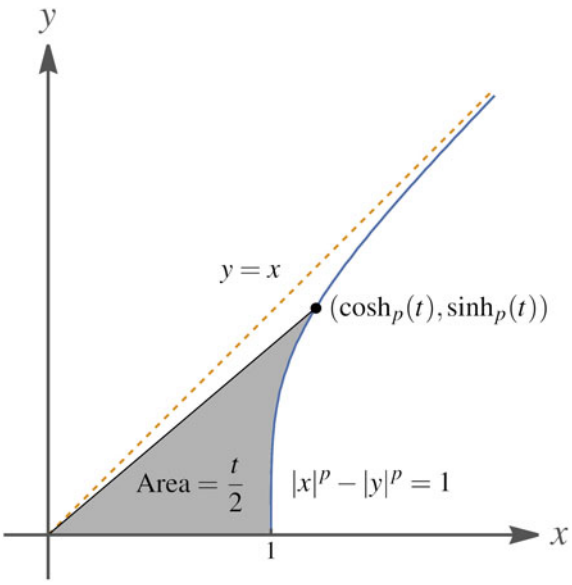


Figure 14.2 The area of the shaded region bounded by the generalized hyperbola $|x|^p - |y|^p = 1$ and the line $y = x$ is $\frac{t}{2}$

The graphs of the hyperbolic functions exhibit three distinct behaviors. Consider the end behavior of the integral expression (14.3):

$$\int_0^\infty \frac{du}{(1+u^p)^{1-\frac{1}{p}}} \approx \int_\epsilon^\infty \frac{du}{u^{p-1}}. \tag{14.4}$$

If $p = 2$, the integrand is approximately $\frac{1}{u}$ and we get logarithmic growth for the classical arcsinh function and thus the expected exponential growth for its inverse, the hyperbolic sine. If $1 < p < 2$, (14.4) grows as a power function of degree less than one, so the p-hyperbolic sine is asymptotically a power function. If $p > 2$, (14.4) converges to what must be a vertical asymptote of the p-hyperbolic sine (Figure 14.3). Let's find those asymptotes.

Theorem 14.1

$$\lim_{t \to \Upsilon_p^-} \sinh_p(t) = \lim_{t \to \Upsilon_p^-} \cosh_p(t) = \infty$$

where $\Upsilon_p = \frac{\Gamma(\frac{1}{p})\Gamma(1-\frac{2}{p})}{p\Gamma(1-\frac{1}{p})}$.

Proof. We will prove this for hyperbolic sine and leave the reader to verify the hyperbolic cosine. We may equivalently find the horizontal asymptotes for the inverse of the p-hyperbolic sine. Thus we need the exact value of

$$\Upsilon_p = \lim_{x \to \infty} \operatorname{arcsinh}_p(x) = \int_0^\infty (1+u^p)^{\frac{1}{p}-1}\, du.$$

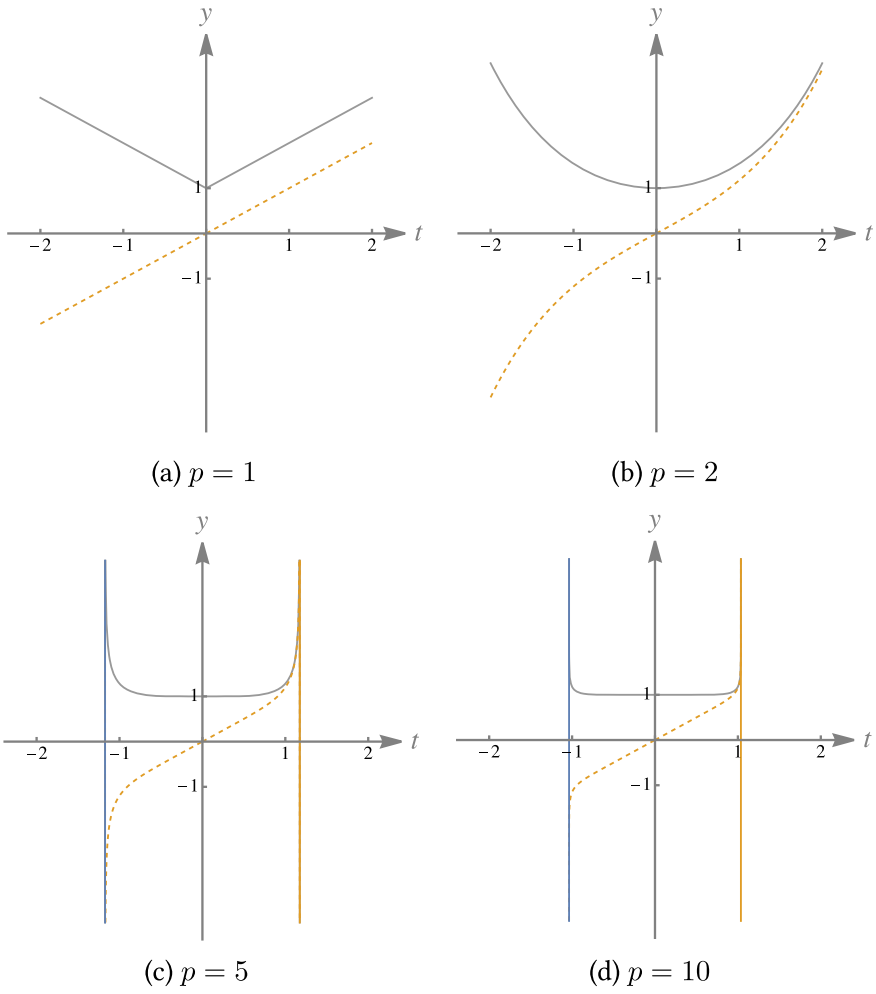


Figure 14.3 Graphs of $y = \cosh_p(t)$ (solid) and $y = \sinh_p(t)$ (dashed) for various p-values, as well as the vertical asymptotes at $\pm\Upsilon_p$ for $p > 2$

If we substitute $t = u^p$, we can write

$$\begin{aligned}\Upsilon_p &= \frac{1}{p}\int_0^\infty t^{\frac{1}{p}-1}(1+t)^{\frac{1}{p}-1}\,dt \\ &= \frac{1}{p}\,\mathrm{B}\left(\tfrac{1}{p}, 1-\tfrac{2}{p}\right) \quad \text{by (6.4)} \\ &= \frac{\Gamma\left(\frac{1}{p}\right)\Gamma\left(1-\frac{2}{p}\right)}{p\,\Gamma\left(1-\frac{1}{p}\right)}\end{aligned}$$

□

p	Υ_p
2	∞
3	1.7666387503
4	1.3110287771
5	1.1744501606
6	1.1129126745
7	1.0794420529
8	1.0590783620
9	1.0457181135
10	1.0364599344

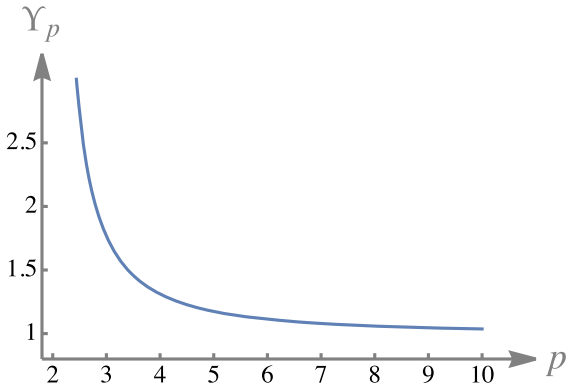


Figure 14.4 Υ_p values

The formula for Υ_p in Theorem 14.1 can be expressed in numerous ways via different identities but has a particularly nice expression in terms of conjugates. If $\frac{1}{p} + \frac{1}{q} = 1$, then

$$\Upsilon_p = \frac{\Gamma(\frac{1}{p})\Gamma(1-\frac{2}{p})}{p\Gamma(1-\frac{1}{p})} = \frac{\Gamma(\frac{1}{p})\Gamma(\frac{1}{q}-\frac{1}{p})}{p\Gamma(\frac{1}{q})}.$$

Υ_p is also related to π_p by

$$\Upsilon_p = \frac{\pi_p}{4}\sec\left(\frac{\pi}{p}\right) \tag{14.5}$$

where $\pi = \pi_2$ as usual, and $p > 2$. The proof is an exercise.

Some complications emerge when we try to extend the definitions of the hyperbolic functions to $p = \infty$. We can see that the graphs of $y = \sinh_p(t)$ and $y = \cosh_p(t)$ are converging pointwise to curves with vertical sides, which is not the graph of a function. The effect is also visible from CIVP (14.1): if x and y are greater than 1, then the derivatives will tend to infinity with p.

The problem with the areal parameterization illustrated in Figure 14.2 is illustrated for $p = \infty$ in Figure 14.5. The definitions $\cosh_\infty(t) = 1$, $\sinh_\infty(t) = t$ give the area $\frac{t}{2}$ of the shaded triangle if $0 \le t \le 1$, as required. But for $t > 1$ the region expands along a line and thus does not increase in area. Since the parameter is supposed to be half the area, there is no way to advance the parameter. Note that we can still parameterize the hyperbola by setting $\sinh_\infty(t) = \cosh_\infty(t) = t$ (or any other increasing function $f(t)$ with $f(1) = 1$) for $t > 1$.

See the appendix and the exercises for additional properties and formulas comparable to those developed in Chapter 5.

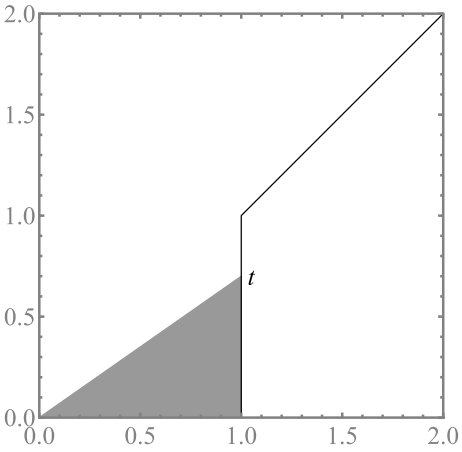


Figure 14.5 Failure of the areal paramaterization of the ∞-hyperbola

14.2 Complex symmetry

If we extend to the complex plane, we see that the hyperbolic and classical trigonometric functions are more than just analogous – they are really just different slices of the same function. The hyperbolic functions are related to their circular counterparts by the complex symmetry relationship

$$\sin(it) = i\sinh(t)$$
$$\cos(it) = \cosh(t).$$

In other words, the hyperbolic trigonometric functions are obtained by evaluating the circular trigonometric functions on the imaginary axis. We will generalize this relationship.

Let $\omega^p = -1$, $p \geq 1$. Note that $\omega = i$ if $p = 2$. Also note that $\omega^{p-1} = \frac{\omega^p}{\omega} = -\frac{1}{\omega}$. Consider the functions

$$f_1(t) = \cos_p(\omega t)$$
$$g_1(t) = \sin_p(\omega t).$$

As usual, we will temporarily restrict to the first quadrant to simplify the calculations. Then

$$\begin{aligned} f_1'(t) &= -\omega \sin_p(\omega t)^{p-1} \\ &= \frac{1}{\omega^{p-1}} \sin_p(\omega t)^{p-1} \\ &= \left(\frac{1}{\omega} g_1(t)\right)^{p-1} \end{aligned}$$

and similarly

$$\begin{aligned} g_1'(t) &= \omega \cos_p(\omega t)^{p-1} \\ &= -\frac{1}{\omega^{p-1}} \sin_p(\omega t)^{p-1} \\ &= -\left(\frac{1}{\omega} f_1(t)\right)^{p-1}. \end{aligned}$$

After confirming the values at $t = 0$, we have shown that these functions satisfy the CIVP

$$\begin{cases} x'(t) = \left(\frac{1}{\omega} y(t)\right)^{p-1} \\ y'(t) = -\left(\frac{1}{\omega} x(t)\right)^{p-1} \\ x(0) = 1 \\ y(0) = 0 \end{cases} \tag{14.6}$$

with $x(t) = f_1(t)$, $y(t) = f_2(t)$.

Now consider $f_2(t) = \cosh_p(t)$, $g_2(t) = \omega \sinh_p(t)$. Then

$$f_2'(t) = \sinh_p(t)^{p-1} = \left(\frac{1}{\omega} g_2(t)\right)^{p-1}$$

and similarly $g_2'(t) = -\left(\frac{1}{\omega} f_2(t)\right)^{p-1}$. Thus f_2 and g_2 also satisfy CIVP (14.6) and by uniqueness of solutions must equal f_1 and g_1, respectively. We have proved

Theorem 14.2

If $\omega^p = -1$ then $\cos_p(\omega t) = \cosh_p(t)$ and $\sin_p(\omega t) = \omega \sinh_p(t)$ for all t for which their CIVPs are satisfied.

Theorem 14.2 allows us develop series expansions for the hyperbolic trigonometric functions without repeating the work of Chapter 11. For example, we may apply (11.11) to deduce

$$\begin{aligned} \cosh_p(t) &= \cos_p(\omega t) \\ &= 1 - \frac{1}{p}(\omega t)^p + \frac{(p-1)^2}{2!p^2(p+1)}(\omega t)^{2p} \\ &\quad - \frac{(p-1)^2\left(4p^3 - 6p^2 - 3p + 1\right)}{3!p^3(p+1)^2(2p+1)}(\omega t)^{3p} \\ &\quad + \frac{(p-1)^2\left(36p^6 - 116p^5 + 41p^4 + 100p^3 - 6p^2 - 8p + 1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)}(\omega t)^{4p} + \cdots \\ &= 1 + \frac{1}{p}t^p + \frac{(p-1)^2}{2!p^2(p+1)}t^{2p} \end{aligned}$$

$$+\frac{(p-1)^2\left(4p^3-6p^2-3p+1\right)}{3!p^3(p+1)^2(2p+1)}t^{3p}$$
$$+\frac{(p-1)^2\left(36p^6-116p^5+41p^4+100p^3-6p^2-8p+1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)}t^{4p}+\cdots.$$

In other words, the coefficients all become positive just as in the classical setting.

The appendix lists additional formulas for the hyperbolic functions, deduced analogously to their counterparts.

Radius of convergence

We can make use of Theorems 14.1 and 14.2 to establish the radius of convergence of the Maclaurin series for the p-trigonometric functions ($p > 2$) we formulated in Chapter 11 as well as for our new p-hyperbolic functions.

We will use the method of the proof of Theorem 14.1 as our starting point in that we will examine what happens to the inverses of our p-trigonometric functions as their arguments grow large. For the p-tangent, we've already seen in (6.9) that $\lim_{x\to\infty}\arctan_p(x)=\pi_p/2$ and so the p-tangent series has radius of convergence $\pi_p/2$.

As we have just seen, however, the p-hyperbolic sine and cosine functions have vertical asymptotes at $t=\pm\Upsilon_p$ and are continuous and differentiable on $(-\Upsilon_p,\Upsilon_p)$, and thus their respective Maclaurin series have radius of convergence Υ_p, with the caveat that these series converge to the proper functions on $[0,\Upsilon_p)$ but only do so on $(-\Upsilon_p,0)$ for p an even natural number.

Because of Theorem 14.2, the Maclaurin series for the p-sine and cosine functions have radius of convergence no bigger than Υ_p. A legitimate question is whether there is another singularity for either function at some point $z\in\mathbb{C}$ with $1<|z|<\Upsilon_p$, which would lower the radius further. The following theorem answers this in the negative.

Theorem 14.3

The Maclaurin series for $\sin_p(t)$ *and* $\cos_p(t)$, $p\geq 2$, *have radius of convergence* Υ_p.

Proof. We examine

$$\lim_{z\to\infty}|\arcsin_p(z)|=\lim_{z\to\infty}|z|\cdot\left|{}_2F_1\left(\begin{matrix}\frac{1}{p},\ \frac{p-1}{p}\\ \frac{p+1}{p}\end{matrix};z^p\right)\right|,$$

where $z\in\mathbb{C}$. (We leave the proof for the cosine function to the reader.) In particular, we assume that $|z|>1$. Because the hypergeometric function has a branch cut along $(1,\infty)$, $\arcsin_p(z)$ has branch cuts from the pth roots of unity to ∞.

There are many ways to rewrite and transform the hypergeometric function. We will use [25, Eq. 15.8.2]:

$$ {}_2F_1\left(\begin{matrix}a,\ b\\ c\end{matrix};z\right)=\frac{\Gamma(c)\Gamma(b-a)}{\Gamma(b)\Gamma(c-a)}(-z)^{-a}{}_2F_1\left(\begin{matrix}a,\ a-c+1\\ a-b+1\end{matrix};\frac{1}{z}\right) + \frac{\Gamma(c)\Gamma(a-b)}{\Gamma(a)\Gamma(c-b)}(-z)^{-b}{}_2F_1\left(\begin{matrix}b,\ b-c+1\\ b-a+1\end{matrix};\frac{1}{z}\right) $$

With $a=1/p$, $b=(p-1)/p$, and $c=(p+1)/p$, this yields

$$ {}_2F_1\left(\begin{matrix}\frac{1}{p},\ \frac{p-1}{p}\\ \frac{p+1}{p}\end{matrix};z^p\right)=\frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{p-2}{p}\right)}{\Gamma\left(\frac{p-1}{p}\right)\Gamma(1)}(-z^p)^{-1/p}{}_2F_1\left(\begin{matrix}\frac{1}{p},\ 0\\ \frac{2}{p}\end{matrix};\frac{1}{z^p}\right) + \frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{2-p}{p}\right)}{\Gamma\left(\frac{1}{p}\right)\Gamma\left(\frac{2}{p}\right)}(-z^p)^{\frac{1}{p}-1}{}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{p-2}{p}\\ \frac{2p-2}{p}\end{matrix};\frac{1}{z^p}\right). \tag{14.7} $$

Write the right-hand side as $(I)+(II)$. We are looking to find $\lim_{z\to\infty}|z|(|(I)+(II)|)$. By the triangle inequality, $|z|(|(I)+(II)|)\le|z||(I)|+|z||(II)|$, so we will examine the last sum.

The advantage of the representation (14.7) is that since $|1/z^p|<1$, we can write the hypergeometric functions on the right-hand side as power series in $1/z^p$. We first examine (I). We note that

$$ {}_2F_1\left(\begin{matrix}\frac{1}{p},\ 0\\ \frac{2}{p}\end{matrix};\frac{1}{z^p}\right)=\sum_{n=0}^{\infty}\frac{\left(\frac{1}{p}\right)_n(0)_n}{\left(\frac{2}{p}\right)_n}\frac{\left(\frac{1}{z^p}\right)^n}{n!}=1 $$

since $(0)_0=\lim_{x\to0^+}(x)_0=1$ and $(0)_n=0$ for $n\ge1$. Therefore

$$ (I)=\frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{p-2}{p}\right)}{\Gamma\left(\frac{p-1}{p}\right)\Gamma(1)}(-z^p)^{-1/p}=\frac{\Gamma\left(1-\frac{2}{p}\right)\Gamma\left(\frac{1}{p}\right)}{p\,\Gamma\left(1-\frac{1}{p}\right)}\left(-\frac{1}{z^p}\right)^{1/p}=\Upsilon_p\cdot\frac{1}{z}\omega_p, $$

where ω_p is a pth root of -1. This gives us that $|z||(I)|=\Upsilon_p$.

We now show that $\lim_{z\to\infty}|z||(II)|=0$. By Gauss' summation formula, we have that

$$ {}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{p-2}{p}\\ \frac{2p-2}{p}\end{matrix};1\right)=\frac{\Gamma\left(2-\frac{2}{p}\right)\Gamma\left(\frac{1}{p}\right)}{\Gamma\left(1-\frac{1}{p}\right)\Gamma(1)} $$

We can also invoke (6.2) to write

$$ \frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{2-p}{p}\right)}{\Gamma\left(\frac{1}{p}\right)\Gamma\left(\frac{2}{p}\right)}=\frac{\Gamma\left(1+\frac{1}{p}\right)}{\Gamma\left(\frac{1}{p}\right)}\frac{\Gamma\left(\frac{2}{p}-1\right)}{\Gamma\left(\frac{2}{p}\right)} =\frac{1}{p}\cdot\frac{1}{\frac{2}{p}-1}=\frac{1}{2-p} $$

Therefore,

$$\begin{aligned}\frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{2-p}{p}\right)}{\Gamma\left(\frac{1}{p}\right)\Gamma\left(\frac{2}{p}\right)}\cdot {}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{p-2}{p}\\ \frac{2p-2}{p}\end{matrix};1\right) &= \frac{1}{2-p}\cdot\frac{\Gamma\left(2-\frac{2}{p}\right)\Gamma\left(\frac{1}{p}\right)}{\Gamma\left(1-\frac{1}{p}\right)\Gamma(1)}\\ &= \frac{1}{\frac{2}{p}-1}\cdot\frac{\Gamma\left(2-\frac{2}{p}\right)\Gamma\left(\frac{1}{p}\right)}{p\,\Gamma\left(1-\frac{1}{p}\right)}\\ &= \frac{1-\frac{2}{p}}{\frac{2}{p}-1}\cdot\frac{\Gamma\left(1-\frac{2}{p}\right)\Gamma\left(\frac{1}{p}\right)}{p\,\Gamma\left(1-\frac{1}{p}\right)}\\ &= -\Upsilon_p.\end{aligned}$$

This leads to

$$\begin{aligned}|z||(II)| &= |z|\left|(-z^p)^{\frac{1}{p}-1}\right|\left|\frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{2-p}{p}\right)}{\Gamma\left(\frac{1}{p}\right)\Gamma\left(\frac{2}{p}\right)}\cdot {}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{p-2}{p}\\ \frac{2p-2}{p}\end{matrix};\frac{1}{z^p}\right)\right|\\ &\le \frac{1}{|z|^{p-2}}\cdot\left|\frac{\Gamma\left(\frac{p+1}{p}\right)\Gamma\left(\frac{2-p}{p}\right)}{\Gamma\left(\frac{1}{p}\right)\Gamma\left(\frac{2}{p}\right)}\cdot {}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{p-2}{p}\\ \frac{2p-2}{p}\end{matrix};1\right)\right|\\ &= \frac{\Upsilon_p}{|z|^{p-2}}\to 0\end{aligned}$$

as $|z|\to\infty$. At last, we have

$$\lim_{z\to\infty}|\arcsin_p(z)| = \lim_{z\to\infty}|z||(I)| = \Upsilon_p.$$

□

Our extension to the complex plane requires some additional attention, as our functions are not complex analytic because there is no power series for the general p-sine in an interval containing 0. Our complex symmetry results indicate why: we forced odd symmetry on a function whose natural symmetries are about pth roots of unity.

We obtain analytic functions if we drop the absolute values and study the curves $x^p + y^p = 1$. In [23], Ding and Chebolu showed that the complex p-sine function maps a region between two rays from the origin conformally onto a polygonal region[1] . These techniques also lead to a geometric proof of the area trisection theorem for $p = 3$ discussed in Section 10.3.

[1] Conformal geometry is beyond this book, but briefly this refers to the Riemann Mapping Theorem which says that any open simply-connected proper subset of the plane can be mapped to the open unit disk – and thus to any other such set – in a way that is bijective and complex analytic, meaning it is infinitely differentiable and preserves angles. See [99] for a historical survey of this theorem.

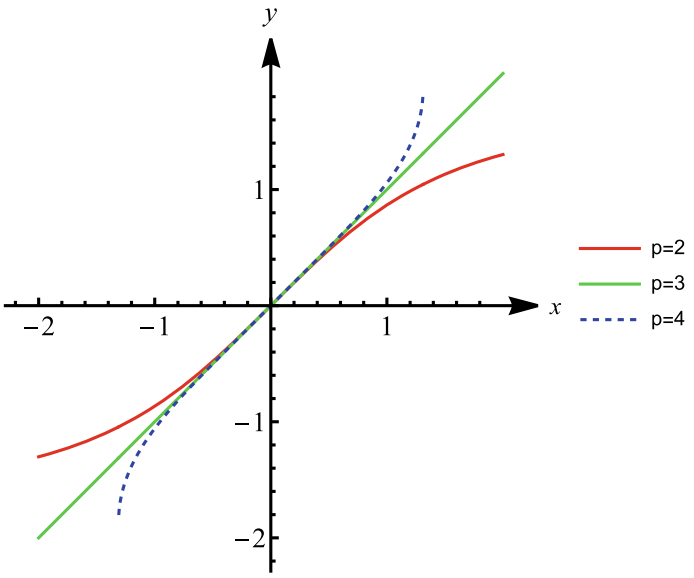


Figure 14.6 Graphs of $y = \mathrm{gd}_p(x)$

Although it comes at the expense of our goal of modeling the p-circles $|x|^p + |y|^p = 1$, complexification can alleviate and explain some of the complications we incurred from adding those absolute values and insisting on real periodicity. We will maintain our real-valued perspective, but we invite the reader to pursue the complex view further.

14.3 The Gudermannian function

Another way to relate the classical and hyperbolic trigonometric functions without using complex numbers is via the *Gudermannian* function $\mathrm{gd}(t) = \int_0^t \frac{1}{\cosh u}\, du$. We obtain the hyperbolic functions by composing trigonometric functions with the Gudermannian, leading to a number of identities such as $\sin(\mathrm{gd}(t)) = \tanh t$. We will see similar results as we generalize the Gudermannian definition to $\mathrm{gd}_p(t) = \int_0^t (\cosh_p(u))^{p-3}\, du$, graphed in Figure 14.6. The domain of the p-Gudermannian is $(-\Upsilon_p, \Upsilon_p)$, the same as the corresponding hyperbolic cosine.

Theorem 14.4

The function $\mathrm{gd}_p(t)$ *satisfies the following identities:*

1. $\mathrm{gd}_p(t) = \arccos_p(\mathrm{sech}_p(t)) \cdot \mathrm{sgn}(t) = \arcsin_p(\tanh_p(t)) = \arccos_p(\mathrm{sech}_p(t)) \cdot \mathrm{sgn}(t) = \arctan_p(\sinh_p(t)) = \mathrm{arcsec}_p(\cosh_p(t)) \cdot \mathrm{sgn}(t)$
2. $\mathrm{gd}_p^{-1}(t) = \int_0^t \cos_p(u)^{p-3}\,du$
3. $\mathrm{gd}_p^{-1}(t) = -\omega\ \mathrm{gd}_p(\omega^{p-1}t)$, *where* $\omega^p = -1$.

Proof. We prove the first equality in (1) by differentiating.

$$\begin{aligned}\frac{d}{dt}\arccos_p(\mathrm{sech}_p(t)) \cdot \mathrm{sgn}(t) &= \frac{-\mathrm{sgn}(t)}{(1-\mathrm{sech}_p(t)^p)^{\frac{p-1}{p}}} \cdot (\mathrm{sech}_p(t))^2 \cdot (\sinh_p(t))^{p-1} \\ &= \mathrm{sgn}(t)^2 \cdot |\tanh_p(t)|^{1-p} \cdot (-\mathrm{sech}_p(t)^2) \cdot (\sinh_p(t))^{p-1} \\ &= \cosh_p(t)^{p-3}.\end{aligned}$$

Note the absolute values that appear because we take a pth root when we apply the Pythagorean identity. The signum function

$$\mathrm{sgn}(x) = \begin{cases} \frac{|x|}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is necessary to resolve this complication. The proof is completed by confirming the values at $t = 0$. The remaining equalities work similarly and are left as exercises.

Part (2) follows from (1). Let $\Theta(t) = \int_0^t \cos_p(u)^{p-3}\,du$. Then

$$\begin{aligned}\frac{d}{dt}\Theta(\mathrm{gd}_p(t)) &= \cos_p(\mathrm{gd}_p(t))^{p-3} \cdot \cosh_p(t)^{p-3} \\ &= \cos_p(\arccos_p(\mathrm{sech}_p(t)))^{p-3} \cdot \cosh_p(t)^{p-3} \\ &= (\mathrm{sech}_p(t) \cdot \cosh_p(t))^{p-3} = 1,\end{aligned}$$

showing $\Theta = \mathrm{gd}_p^{-1}$, again after confirming the values at $t = 0$.

The complex symmetry of Theorem 14.2 along with (2) essentially give us (3):

$$\begin{aligned}\mathrm{gd}_p^{-1}(t) &= \int_0^t \cos_p(u)^{p-3}\,du \\ &= \omega \int_0^{t/\omega} \cos_p(\omega z)^{p-3}\,dz \quad (u = \omega z) \\ &= \omega \int_0^{t/\omega} \cosh_p(z)^{p-3}\,dz \\ &= \omega\ \mathrm{gd}_p\left(\frac{t}{\omega}\right) = \omega\ \mathrm{gd}_p(-\omega^{p-1}t) = -\omega\ \mathrm{gd}_p(\omega^{p-1}t),\end{aligned}$$

the last step following from the fact that the Gudermannian function is odd (since cosh is even). □

The case $p = 3$ is striking, with the 3-Gudermannian being simply the identity function. This is due to the curious relation $\operatorname{arctanh}_3(t) = \operatorname{arcsin}_3(t)$.

Now let's tie this back to our original constructions. Consider the hyperbolic Pythagorean identity $\cosh_p(t)^p - \sinh_p(t)^p = 1$. By dividing by $\cosh_p(t)^p$ and reorganizing, we obtain

$$\operatorname{sech}_p(t)^p + \tanh_p(t)^p = 1$$

so $(\operatorname{sech}_p(t), \tanh_p(p))$ parameterizes the unit p-circle. Per (1.2), we should be able to find a function $\zeta = \zeta(x(t), y(t), t)$ for which $x(t) = \operatorname{sech}_p(t)$, $y(t) = \tanh_p(t)$ is a solution to the CIVP

$$\begin{cases} x'(t) = y(t)^{p-1}\zeta \\ y'(t) = x(t)^{p-1}\zeta \\ x(0) = 1 \\ y(0) = 0. \end{cases} \tag{14.8}$$

We then calculate the ζ that gives our desired solution by

$$\zeta = \frac{y'(t)}{x(t)^{p-1}} = \frac{\operatorname{sech}_p(t)^2}{\operatorname{sech}_p(t)^{p-1}} = \cosh_p(t)^{p-3} = \operatorname{gd}'_p(t).$$

Thus the Gudermannian again appears as a mechanism to transition from a classical to a hyperbolic parameterization of a circle.

Exercises

▶ **Exercise 14.1** *Evaluate the following.*

1. $\displaystyle\int \frac{1}{\sqrt[3]{(1-x^{3/2})^4}}\,dx$
2. $\displaystyle\int \frac{1}{x^{\frac{4}{3}}(1+x^2)^{\frac{5}{6}}}\,dx$

▶ **Exercise 14.2** *Verify that our definition of the p-hyperbolic functions is the areal parameterization as claimed in the caption of Figure 14.2. What can be said about this area generally for $1 \le p < 2$, $p = 2$, and $p > 2$?*

▶ **Exercise 14.3** *Explain the three possible end behaviors of the p-hyperbolic sine by directly looking at CIVP (14.2).*

▶ **Exercise 14.4** *Let $\omega^3 = -1$. Deduce from the assumption $\sin_3(\omega t) = \omega \sinh_3(t)$ that the coefficient of t^n in the power series is 0 if n is not 1 plus a multiple of 3.*

▶ **Exercise 14.5** *Verify Theorem 14.1 for the hyperbolic cosine.*

▶ **Exercise 14.6** *Prove the outstanding identities in Theorem 14.4.*

▶ **Exercise 14.7** *Prove Equation (14.5).*

▶ **Exercise 14.8** *Show* $\pi_p = 4\,\mathrm{gd}_p(\mathrm{arccosh}_p(2^{\frac{p}{p-3}}))$.

▶ **Exercise 14.9** *Verify that* $\zeta = \frac{u'(t)}{v(t)^{p-1}}$ *also gives the derivative of the Gudermannian in (14.8).*

▶ **Exercise 14.10** *Prove that for* $p > 3$, $\lim_{t\to\Upsilon_p} \mathrm{gd}_p(t) = \frac{\pi_p}{2}$. *(Use the result of Exercise 6.5 and substitute* $u = \cosh_p(t)$*.)*

▶ **Exercise 14.11** *Show that* $\lim_{p\to\infty} \Upsilon_p = 1$.

▶ **Exercise 14.12** *Find series expansions for the hyperbolic sine and tangent functions.*

▶ **Exercise 14.13** *Deduce Theorem 14.2 directly from the series expansions of the hyperbolic and classical sine and cosine functions.*

▶ **Exercise 14.14** *Prove that*

$$\sum_{n=0}^{\infty} \frac{\binom{n-\frac{1}{p}}{n}}{((n+1)p-2)} = \sum_{n=0}^{\infty} \frac{\Gamma(n+1-\frac{1}{p})}{\Gamma(n+1)\Gamma(\frac{p-1}{p})((n+1)p-2)} = \Upsilon_p.$$

Project

Project 14.1 (Generalized Gudermannians)
We are unaware of any study of the generalized Gudermannian functions, at least not in this context. There may be additional properties of these functions to explore.

For example, are there any identities for these functions relating Gudermannians for different p? Figure 14.6 suggests possible inverse relationships pivoting around $p = 3$ (but note the domains and Exercise 14.10). Perhaps duality will play a role.

A view from the complex plane may also bear fruit. We may use the complex symmetry relationships of Theorem 14.4 to express gd_p in terms of the same integrand as gd_p^{-1}, but the integration is along a path to a pth root of unity.

Check for updates

Exponentials and logarithms

15

Euler's formula $e^{it} = \cos(t) + i\sin(t)$ points to the intimate link between the exponential and the trigonometric functions. (Physicist Richard Feynman called the formula "the most remarkable formula in mathematics." [32]) The generalized trigonometric and hyperbolic functions we have explored thus far lead to generalizations of the exponential and logarithmic functions as well. We engage here in a short exploration of these transcendental functions.

15.1 Generalized exponentials

There are well-known connections between exponentials and the hyperbolic functions:

$$e^t = \exp(t) = \cosh(t) + \sinh(t), \quad \cosh(t) = \frac{e^t + e^{-t}}{2}, \quad \sinh(t) = \frac{e^t - e^{-t}}{2}.$$

We begin with a definition for a generalized exponential

$$\exp_p(t) = \cosh_p(t) + \sinh_p(t).$$

These p-exponentials will have the same behaviors as the p-hyperbolic functions we developed in the previous chapter: polynomial growth for $1 \le p < 2$, a vertical asymptote at Υ_p for $p > 2$, and only true exponential behavior if $p = 2$. The domain for the p-exponential function is $(-\infty, \infty)$ for $1 \le p \le 2$ and $(-\Upsilon_p, \Upsilon_p)$ for $p > 2$. (Recall, though, that $\lim_{p\to 2+} \Upsilon_p = \infty$.)

The symmetry relations from Theorem 14.2 provide a generalized Euler formula. Let $\omega^p = -1$. Then

$$\begin{aligned}\exp_p(\omega^{p-1}t) &= \cosh_p(\omega^{p-1}t) + \sinh_p(\omega^{p-1}t) \\ &= \cos_p(\omega \cdot \omega^{p-1}t) + \frac{1}{\omega}\sin_p(\omega \cdot \omega^{p-1}t)\end{aligned}$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_15

$$= \cos_p(-t) - \omega^{p-1} \sin_p(-t)$$
$$= \cos_p(t) + \omega^{p-1} \sin_p(t).$$

As with the squigonometric functions, our generalized exponential will not exhibit the elegant properties of its classical version but will satisfy some related identities that simplify when $p = 2$. We summarize them, leaving verification to the reader.

Theorem 15.1

The generalized exponential function $\exp_p(t)$ *satisfies the following:*

1. $\exp_p(\omega^{p-1}t) = \cos_p(t) + \omega^{p-1} \sin_p(t)$
2. $\exp_p(t) \exp_p(-t) = \cosh_p(t)^2 - \sinh_p(t)^2$
3. $\frac{d}{dt} \exp_p(t) = \cosh_p(t)^{p-1} + \sinh_p(t)^{p-1}$
4. $\exp(0) = \exp'(0) = 1$

The generalized Euler formula implies a formula for sines and cosines as linear combinations of exponentials:

$$\cos_p(t) = \frac{\exp(\omega^{p-1}t) + \exp(-\omega^{p-1}t)}{2}, \quad \sin_p(t) = \frac{\exp(\omega^{p-1}t) - \exp(-\omega^{p-1}t)}{2\omega^{p-1}}.$$

When $p = 2$, these identities reduce to more familiar formulas:

$$\cos(t) = \frac{\exp(it) + \exp(-it)}{2} = \frac{e^{it} + e^{-it}}{2}$$
$$\sin(t) = \frac{\exp(it) - \exp(-it)}{2i} = \frac{e^{it} - e^{-it}}{2i}.$$

The last item in Theorem 15.1 reflects a generalization of the definition of e^x as the exponential curve whose tangent at the origin has slope 1.

15.2 Series for exponential functions

We can use our definition to find an infinite series expansion for the p-exponential:

$$\begin{aligned}
\exp_p(t) &= \cosh_p(t) + \sinh_p(t) \\
&= 1 + t + \frac{1}{p}t^p + \frac{p-1}{p(p+1)}t^{p+1} \\
&\quad + \frac{(p-1)^2}{2!p^2(p+1)}t^{2p} + \frac{(p-1)\left(2p^2-3p-1\right)}{2!p^2(p+1)(2p+1)}t^{2p+1} \\
&\quad + \frac{(p-1)^2\left(4p^3-6p^2-3p+1\right)}{3!p^3(p+1)^2(2p+1)}t^{3p} \\
&\quad + \frac{(p-1)\left(12p^5-34p^4+p^3+33p^2+11p+1\right)}{3!p^3(p+1)^2(2p+1)(3p+1)}t^{3p+1}
\end{aligned}$$

$$+\frac{(p-1)^2\left(36p^6-116p^5+41p^4+100p^3-6p^2-8p+1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)}t^{4p}$$
$$+\frac{(p-1)\left(144p^8-644p^7+520p^6+\cdots-23p-1\right)}{4!p^4(p+1)^3(2p+1)(3p+1)(4p+1)}t^{4p+1}+\cdots$$

Some results for particular p are in Table 15.1.

We note that this series representation holds for $0 \le t < \Upsilon_p$ since we know that $\cosh_p(t)$ and $\sinh_p(t)$ – and thus $\exp_p(t)$ – have vertical asymptotes Υ_p for $p > 2$ and $\Upsilon_p \to 1$ as $p \to \infty$. (See Figure 14.4.)

In particular, $\exp_1(t) = 1 + 2t$, exhibiting the linear growth we expect for $p = 1$, and

$$\exp_2(t) = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \cdots = e^t,$$

which actually converges for all $t \in \mathbb{R}$. (We note that $\Upsilon_2 = \infty$.) For $p > 2$, the exponential acts like $1 + t$ for $0 < t < 1$, and then rockets upward very quickly, blowing up in finite time at $t = \Upsilon_p$.

Because our series represent neither even nor odd functions, we will have to separately define the exponential function for $t < 0$. We can use the result of Theorem 15.1 and define

$$\exp_p(t) = \frac{\cosh_p(-t)^2 - \sinh_p(-t)^2}{\exp(-t)}$$

for that region. In the $p = 1$ case we get that

$$\exp_1(t) = \begin{cases} 1 + 2t & t \ge 0, \\ 1 & t < 0. \end{cases}$$

When $p = 2$, we have for $t < 0$,

$$\exp_2(t) = \frac{\cosh(-t)^2 - \sinh(-t)^2}{\exp_2(-t)} = \frac{1}{e^{-t}} = e^t$$

as we would expect.

Table 15.1 Power series for p-exponential functions

$p = 1$	$\exp_1(t) = 1 + 2t$
$p = 2$	$\exp_2(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} + \cdots$
$p = 3$	$\exp_3(t) = 1 + t + \frac{t^3}{3} + \frac{t^4}{6} + \frac{t^6}{18} + \cdots$
$p = 4$	$\exp_4(t) = 1 + t + \frac{t^4}{4} + \frac{3t^5}{20} + \frac{9t^8}{160} + \cdots$
$p = 5$	$\exp_5(t) = 1 + t + \frac{t^5}{5} + \frac{2t^6}{15} + \frac{4t^{10}}{75} + \cdots$
$p = 6$	$\exp_6(t) = 1 + t + \frac{t^6}{6} + \frac{5t^7}{42} + \frac{25t^{12}}{504} + \cdots$

15.3 Generalized logarithms

As our exponential functions are increasing functions, they have inverses, which we will notate by $\ln_p(\cdot)$ and call p-logarithms.[1]

When $p = 1$, we have that $\ln_1(x) = \frac{1}{2}x - \frac{1}{2}$, a line. For $p = 2$, $\ln_2(x) = \ln x$, the usual natural logarithm. Both of these increase to ∞ as x increases, and when $p = 2$, the increase is logarithmically slow.

However, for $p > 2$, our new logarithms increase toward a horizontal asymptote at Υ_p, which approaches 1 from above as $p \to \infty$.

In Figure 15.1, we see the exponential and logarithmic graphs for $p = 4$, where the two functions have respective vertical and horizontal asymptotes at

$$\Upsilon_4 = \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{4\Gamma(\frac{3}{4})} = \frac{\pi_4}{2\sqrt{2}} \approx 1.31103,$$

a number with which we will become more familiar in the next chapter in its guise as $L/2$.

We can now state the following properties of the generalized exponential and logarithmic functions for $p > 2$:

$$\exp_p(0) = 1, \qquad \ln_p(1) = 0,$$
$$\exp_p(-\Upsilon_p) = 0, \qquad \ln_p(0) = -\Upsilon_p,$$

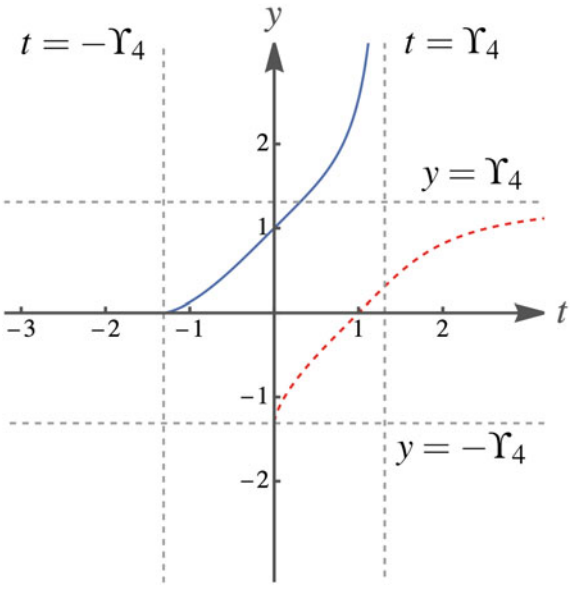


Figure 15.1 Graphs of $y = \exp_4(t)$ (solid) and $y = \ln_4(t)$ (dashed)

[1] We distinguish this from the polylogarithm function $\mathrm{Li}_n(\cdot)$.

$$\lim_{t\to\Upsilon_p^-} \exp_p(t) = \infty, \qquad \lim_{x\to\infty} \ln_p(x) = \Upsilon_p.$$

These can be thought to hold as well for $p = 2$ since $\lim_{p\to 2^+} \Upsilon_p = \infty$.

We can use the result of Theorem 15.1 and implicit differentiation (if $y = \ln_p(x)$ then $\exp_p(y) = x$) to prove

Theorem 15.2

The generalized logarithm functions $\ln_p(x)$ *satisfy*

$$\frac{d}{dx}\ln_p(x) = \frac{1}{\cosh_p(\ln_p(x))^{p-1} + \sinh_p(\ln_p(x))^{p-1}}.$$

Amazingly, when $p = 2$, this formula reduces to a familiar result:

$$\frac{d}{dx}\ln(x) = \frac{1}{\cosh(\ln(x)) + \sinh(\ln(x))} = \frac{1}{\exp(\ln(x))} = \frac{1}{e^{\ln(x)}} = \frac{1}{x}.$$

15.4 Logarithms and inverse hyperbolic functions

As with the usual hyperbolic functions, we can express the inverses of the generalized hyperbolic functions in terms of our new logarithmic functions.

We defined the general exponential function as $\exp_p(t) = \cosh_p(t) + \sinh_p(t)$. Since $|\cosh_p(t)|^p - |\sinh_p(t)|^p = 1$, we have $\cosh_p(t) = \sqrt[p]{1 + |\sinh_p(t)|^p}$. (Note that $\cosh_p(t) \geq 1$ for all p and t.) Thus, $\exp_p(t) = \sinh_p(t) + \sqrt[p]{1 + |\sinh_p(t)|^p}$. We set $x = \sinh_p(t)$ and take the p-logarithm of both sides to get

$$\operatorname{arcsinh}_p(x) = \ln_p\left(x + \sqrt[p]{1 + |x|^p}\right).$$

Through similar means, we can show the following:

Theorem 15.3

The inverse p-hyperbolic functions satisfy

1. $\operatorname{arcsinh}_p(x) = \ln_p\left(x + \sqrt[p]{1 + |x|^p}\right) \quad x \in \mathbb{R}$
2. $\operatorname{arccosh}_p(x) = \ln_p\left(x + \sqrt[p]{x^p - 1}\right) \quad x \geq 1$
3. $\operatorname{arctanh}_p(x) = \ln_p\left(\dfrac{1+x}{\sqrt[p]{1 - |x|^p}}\right) \quad -1 < x < 1$

15.5 Series for logarithms

When $p = 1$, we can simply invert the increasing function $\exp_1(t) = 1+2t$ on $(0, \infty)$ to $\ln_1(t) = \frac{t-1}{2}$. When $p > 1$, we can use series inversion as we did in Chapter 11

Table 15.2 Power series for p-logarithm functions

$p = 1$	$\ln_1(1+x) = \frac{1}{2}x$
$p = 2$	$\ln_2(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \cdots$
$p = 3$	$\ln_3(1+x) = x - \frac{1}{3}x^3 - \frac{1}{6}x^4 + \frac{1}{3}x^5 + \frac{1}{3}x^6 - \cdots$
$p = 4$	$\ln_4(1+x) = x - \frac{1}{4}x^4 - \frac{3}{20}x^5 + \frac{1}{4}x^7 + \frac{9}{32}x^8 + \cdots$
$p = 5$	$\ln_5(1+x) = x - \frac{1}{5}x^5 - \frac{2}{15}x^6 + \frac{1}{5}x^9 + \frac{6}{25}x^{10} + \cdots$
$p = 6$	$\ln_6(1+x) = x - \frac{1}{6}x^6 - \frac{5}{42}x^7 + \frac{1}{6}x^{11} + \frac{5}{24}x^{12} + \cdots$

to find a general series for the p-logarithm. However, since $\exp_p(0) = 1$, we must expand the logarithm series about $t = 1$. We will follow the usual convention for logarithm series, though, by setting $x = t-1$ and writing the series as a generalization of the Mercator series for $\ln(1+x)$:

$$\begin{aligned}\ln_p(1+x) = x &- \frac{1}{p}x^p - \frac{p-1}{p(p+1)}x^{p+1} \\ &+ \frac{1}{p}x^{2p-1} + \frac{3(p-1)}{2p^2}x^{2p} + \frac{(p-1)(2p-1)}{2p^2(2p+1)}x^{2p+1} \\ &- \frac{3p-1}{2p^2}x^{3p-2} - \frac{3(p-1)}{p^2}x^{3p-1} - \frac{(p-1)(11p-7)}{6p^3}x^{3p} \\ &- \frac{(p-1)(2p-1)(3p-1)}{6p^3(3p+1)}x^{3p+1} + \frac{(2p-1)(4p-1)}{3p^3}x^{4p-3} \\ &+ \frac{5(p-1)(4p-1)}{3p^3}x^{4p-2} + \frac{5(p-1)(7p-5)}{6p^3}x^{4p-1} \\ &+ \frac{5(p-1)(2p-1)(5p-3)}{24p^4}x^{4p} \\ &+ \frac{(p-1)(2p-1)(3p-1)(4p-1)}{24p^4(4p+1)}x^{4p+1} + \cdots \end{aligned} \tag{15.1}$$

Particular examples of this series can be found in Table 15.2.

A notable aspect of (15.1) is that for a fixed positive integer n, the series contains all the powers of x from $np - (n-1)$ to $np + 1$. As n approaches p, some overlap occurs. For example, when $p = 2$, we have that $5 = 2p + 1$ but also $5 = 3p - 1$ and $5 = 4p - 3$. Therefore, the coefficient of x^5 in the series for $\ln_2(1+x) = \ln(1+x)$ is

$$\begin{aligned}\frac{(p-1)(2p-1)}{2p^2(2p+1)} - \frac{3(p-1)}{p^2} + \frac{(2p-1)(4p-1)}{3p^3} \\ = \frac{1\cdot 3}{2\cdot 4\cdot 5} - \frac{3(1)}{4} + \frac{3\cdot 7}{3\cdot 8} = \frac{3}{40} - \frac{3}{4} + \frac{7}{8} = \frac{1}{5}.\end{aligned}$$

Exercises

▶ **Exercise 15.1** *Verify the identities in Theorem 15.1.*

▶ **Exercise 15.2** *Prove Theorem 15.2.*

▶ **Exercise 15.3** *Prove parts 2 and 3 of Theorem 15.3.*

▶ **Exercise 15.4** *Prove directly that, for $x > 0$,*

$$\frac{d}{dx} \ln_4 \left(x + \sqrt[4]{1 + x^4} \right) = \frac{1}{(1 + x^4)^{3/4}}.$$

(Use the fact that by Theorem 15.3, $\sinh_4(\ln_4 \left(x + \sqrt[4]{1 + x^4} \right)) = x$ *and* $\cosh_4(\ln_4 \left(x + \sqrt[4]{1 + x^4} \right)) = \sqrt[4]{1 + x^4}$.*)*

Elliptic integrals

16

While Lundberg and others found ways to parameterize the p-circle in the late 19th century, mathematicians have thought about using trigonometric-type functions to parameterize other curves that lack the circle's perfection for the past 300 years, and have linked these functions to so-called elliptic integrals. Jakob Bernoulli considered the arclength of the lemniscate, a figure-eight type curve, back in 1691. His research led to defining lemniscatic versions of the sine and cosine functions, functions that when given a parameter value would return the coordinates of a point on the unit lemniscate. He even found a lemniscatic version of π.

16.1 Lemniscates

A *lemniscate* is the locus of points P such that the product of the distances from P to two fixed foci F_1, F_2 is constant, i.e. $\{P \mid PF_1 \cdot PF_2 = K^2\}$. The calculation of the arclength of this curve led immediately to elliptic integrals and will provide some interesting results for some of our functions.

We will choose our foci to be $(\pm\frac{1}{\sqrt{2}}, 0)$ and constant $K = \frac{1}{2}$. Note that the points $(\pm 1, 0)$ lie on this curve. See Figure 16.1. This curve has equation

$$\sqrt{\left((x-\frac{1}{\sqrt{2}})^2+y^2\right)\cdot\left((x+\frac{1}{\sqrt{2}})^2+y^2\right)} = \frac{1}{2}$$

which simplifies to

$$(x^2+y^2)^2 = x^2 - y^2. \tag{16.1}$$

Equation (16.1) has a nice polar form. With the substitutions $x = r\cos\theta$, $y = r\sin\theta$, and $r^2 = x^2 + y^2$ we get $r^4 = r^2(\cos\theta)^2 - r^2(\sin\theta)^2$, or

$$r^2 = \cos 2\theta. \tag{16.2}$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_16

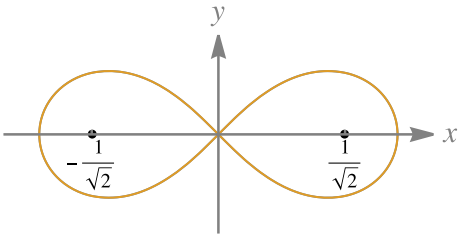


Figure 16.1 The lemniscate $r^2 = \cos(2\theta)$ with foci $(\pm\frac{1}{\sqrt{2}}, 0)$

Notice that we would have an ellipse if we replaced "product" with "sum" in this definition, but the lemniscate is a fourth order curve and thus not a conic section. It does emerge in the study of conics, however.

We now investigate the arclength of the lemniscate, following a similar path as Euler that led to the theory of elliptic integrals.[1]

We begin by differentiating (16.2) to obtain

$$2r = -\sin(2\theta) \cdot 2\frac{d\theta}{dr}$$
$$\frac{d\theta}{dr} = -\frac{r}{\sin(2\theta)}$$

and invoke the polar line element

$$\begin{aligned} ds &= \sqrt{dr^2 + r^2\, d\theta^2} \\ &= \sqrt{1 + r^2\left(\frac{d\theta}{dr}\right)^2}\, dr = \sqrt{1 + r^2\left(\frac{r}{\sin(2\theta)}\right)^2}\, dr \\ &= \sqrt{1 + \frac{r^4}{(\sin(2\theta))^2}}\, dr = \sqrt{1 + \frac{r^4}{1 - (\cos(2\theta))^2}}\, dr \\ &= \sqrt{1 + \frac{r^4}{1 - r^4}}\, dr = \frac{dr}{\sqrt{1 - r^4}}. \end{aligned}$$

The function $\mathrm{sl}^{-1}(x) = \int_0^x \frac{dr}{\sqrt{1-r^4}}$ is the *inverse lemniscatic sine* function , and the *lemniscate constant* $L = 2\int_0^1 \frac{dr}{\sqrt{1-r^4}} = 2\,\mathrm{sl}^{-1}(1)$ is the arclength of one bulb of the lemniscate. (The quantity $L_1 = \frac{1}{2}L$ is known as the *first lemniscatic constant*.) Of course, we have another name for this function: it is the inverse hyperbolic tangent for $p = 4$. We will thus be able to interpret properties of this and related functions as generalized trigonometric identities.

[1] Elliptic integrals were first motivated by calculation of the circumference of an ellipse. We investigated "ellipse geometry" in Exercises 4.5 and 6.4, and an investigation of the corresponding arclength parameterization would takes us down Euler's path.

16.2 The lemniscate constant, Gauss and Pi

We evaluate the integral defining L by substituting $u = r^4$ and obtain a surprising connection to our work with 4-circles:

$$\begin{aligned} L &= 2\int_0^1 \frac{dr}{\sqrt{1-r^4}} \\ &= 2\int_0^1 (1-u)^{-1/2}\cdot \tfrac{1}{4}u^{-3/4}\,du \\ &= \tfrac{1}{2}\int_0^1 u^{\frac{1}{4}-1}(1-u)^{\frac{1}{2}-1}\,du \\ &= \tfrac{1}{2}\,\mathrm{B}(\tfrac{1}{4},\tfrac{1}{2}) = \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{2\Gamma(\frac{3}{4})} \\ &= \frac{\sqrt{2\pi_4\sqrt{\pi}}\,\sqrt{\pi}}{2\,\frac{\pi^{3/4}}{\sqrt{\pi_4}}} \\ &= \frac{\pi_4}{\sqrt{2}} \approx 2.622057554. \end{aligned}$$

Thus the arclength of the lemniscate defined by (16.2) is $2L = \pi_4\sqrt{2}$.

Surprisingly, this quantity is related to *Gauss's constant* G, defined as the reciprocal of the arithmetic-geometric mean of 1 and $\sqrt{2}$ [101]. (We note that Gauss's constant G is *not* the same as Catalan's constant G discussed in Chapter 12.) Because $G = L/\pi$, this means that $G = \frac{\pi_4}{\pi\sqrt{2}} \approx 0.8346268417$. This leads to astonishing answers to some challenging definite integrals, such as

$$\int_0^{\pi/2} \sqrt{\sin(x)}\,dx = \int_0^{\pi/2} \sqrt{\cos(x)}\,dx = \frac{1}{G} = \frac{\pi\sqrt{2}}{\pi_4}. \tag{16.3}$$

We also have that the arithmetic-geometric mean of 1 and $\sqrt{2}$ is exactly $\frac{\pi\sqrt{2}}{\pi_4} = \frac{\pi}{L} = 1.19814023473559\ldots$. We additionally can use (9.7) and the above arclength calculation to compute

$$\int_0^{\pi/2} \frac{1}{\sqrt{\sin(x)}}\,dx = \int_0^{\pi/2} \frac{1}{\sqrt{\cos(x)}}\,dx = \frac{1}{2}\,\frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})} = \frac{\pi_4}{\sqrt{2}} = L.$$

We have already defined the first lemniscatic constant $L_1 = \frac{1}{2}L = \frac{\pi_4}{2\sqrt{2}}$. The *second lemniscatic constant* L_2 is defined as $\frac{1}{2G}$ and it was shown by Euler that $L_1 L_2 = \frac{\pi}{4}$. In particular, this implies [96]

$$L_2 = \int_0^1 \frac{r^2}{\sqrt{1-r^4}}\,dr = \frac{\pi}{\pi_4\sqrt{2}} \approx 0.5990701174.$$

Lemniscatic functions

We note that L plays a similar role for the lemniscate to that of π_p for the p-circle [59]. (Gauss notated the lemniscate constant as ϖ, a variation of π.) We earlier defined the inverse lemniscatic sine (or *lemniscatic arcsine*)

$$\mathrm{sl}^{-1}(x) = \int_0^x \frac{1}{\sqrt{1-r^4}}\,dr, \quad -1 \le x \le 1.$$

We can similarly define an *inverse lemniscatic cosine (or lemniscatic arccosine)* function:

$$\mathrm{cl}^{-1}(x) = \int_x^1 \frac{1}{\sqrt{1-r^4}}\,dr, \quad -1 \le x \le 1.$$

The value $\mathrm{sl}^{-1}(x)$ gives the signed arclength of the lemniscate from the point corresponding to $r = 0$ to the point corresponding to $r = x$. (If $x < 0$, $\mathrm{sl}^{-1}(x) < 0$.)

Because $(\mathrm{sl}^{-1})'(x) = \frac{1}{\sqrt{1-x^4}} > 0$ and $(\mathrm{cl}^{-1})'(x) = -\frac{1}{\sqrt{1-x^4}} < 0$ for all $-1 < x < 1$, both functions have inverses. We can thus define the lemniscatic sine and cosine functions, $\mathrm{sl}(t)$ and $\mathrm{cl}(t)$ respectively, to be the functions such that

$$t = \int_0^{\mathrm{sl}(t)} \frac{1}{\sqrt{1-r^4}}\,dr = \int_{\mathrm{cl}(t)}^1 \frac{1}{\sqrt{1-r^4}}\,dr.$$

We immediately have several familiar-looking relationships:

$$\begin{aligned}
\mathrm{sl}^{-1}(x) + \mathrm{cl}^{-1}(x) = \tfrac{L}{2} = \tfrac{\pi_4}{2\sqrt{2}}, \quad -1 \le x \le 1 \\
\mathrm{sl}(t) = \mathrm{cl}\left(\tfrac{L}{2} - t\right), \quad \mathrm{cl}(t) = \mathrm{sl}\left(\tfrac{L}{2} - t\right) \\
\mathrm{sl}(0) = 0, \quad \mathrm{cl}(0) = 1 \\
\mathrm{cl}\left(\tfrac{L}{2}\right) = 0, \quad \mathrm{sl}\left(\tfrac{L}{2}\right) = 1 \\
\mathrm{sl}(-t) = -\mathrm{sl}(t), \quad \mathrm{cl}(-t) = \mathrm{cl}(t)
\end{aligned}$$

The lemniscatic sine $\mathrm{sl}(t)$ gives the radial distance of the point on the lemniscate $r^2 = \cos 2\theta$ of the point t units in arclength from $(0, 0)$. The lemniscatic cosine $\mathrm{cl}(t)$ gives the distance of the point from the origin as a function of the arclength from $(1, 0)$. Both the lemniscatic sine and cosine have period $2L = \pi_4\sqrt{2} \approx 5.24412$. (See Figure 16.2.) We will discuss this and other non-squircular trigonometric functions further in the next chapter.

16.3 Addition formulas

One familiar trigonometric identity remains to be generalized: the angle sum formula. The non-Euclidean p-metrics do not have a natural notion of angle so it may not be surprising that we will run into some trouble, but we will have some limited success. We have chosen an approach favoring accessibility at the expense of some elegance, but refer to the reader to the notes and references for other developments and some details. [68]

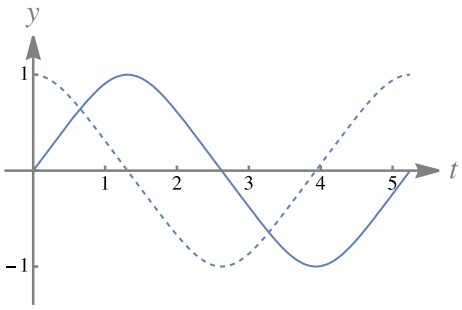


Figure 16.2 Graphs of the lemniscatic functions $y = \mathrm{sl}(t)$ (solid) and $y = \mathrm{cl}(t)$

For any function $f(z)$, we are looking for a rational function R in three variables such that $R(f(a), f(b), f(a+b)) = 0$. Such a function satisfies an *addition theorem*. We will make use of a particular form:

Theorem 16.1

(Euler Addition Theorem). *Let* $g(z) = (1 - z^2)(1 - k^2z^2)$ *for some* $k \in \mathbb{C}$. *Then*

$$\int_0^a \frac{dz}{\sqrt{g(z)}} + \int_0^b \frac{dz}{\sqrt{g(z)}} = \int_0^{c(a,b)} \frac{dz}{\sqrt{g(z)}}$$

where $c(a, b) = \frac{b\sqrt{g(a)}+a\sqrt{g(b)}}{1-k^2a^2b^2}$.

The classical angle addition formulas are the case where $k = 0$ and $g(z) = 1 - z^2$:

$$\begin{aligned}\theta + \phi &= \int_0^{\sin\theta} \frac{dz}{\sqrt{1-z^2}} + \int_0^{\sin\phi} \frac{dz}{\sqrt{1-z^2}} \\ &= \int_0^{c(\sin\theta,\sin\phi)} \frac{dz}{\sqrt{1-z^2}} \\ &= \arcsin(\sin\theta\sqrt{1-\sin^2\phi} + \sin\phi\sqrt{1-\sin^2\theta})\end{aligned}$$

Taking the sine of both sides and applying the Pythagorean identity yields the familiar $\sin(\theta + \phi) = \sin\theta\cos\phi + \sin\phi\cos\theta$.

The Euler Addition Theorem concerns a particular form of *elliptic integral*. The classical arcsine above is an example and the lemniscatic arcsine is another. Indeed, elliptic integrals generalize periodic functions to the complex plane. Whereas real-valued periodic functions have a real constant b such that $f(x + b) = f(x)$ and thus are naturally defined on a circle, elliptic functions have two complex constants ω_1 and ω_2 such that $f(x + \omega_1) = f(x + \omega_2) = f(x)$. These functions are naturally defined on a torus. (For example, the sine and cosine functions have periods 2π and $2\pi i$ while the lemniscatic functions have periods $2L$ and $2Li$.)

Recalling that $\operatorname{arctanh}_4(x) = \int_0^x \frac{du}{\sqrt{1-u^4}}$, the $p = 4$ hyperbolic arctangent fits the form of Theorem 16.1 with $k = i$. This leads to the formula

$$\begin{aligned}\tanh_4(x+y) &= \frac{\tanh_4(x)\sqrt{1-\tanh_4(y)^4} + \tanh_4(y)\sqrt{1-\tanh_4(x)^4}}{1+\tanh_4(x)^2\tanh_4(y)^2} \\ &= \frac{\sinh_4(x)\cosh_4(x) + \sinh_4(y)\cosh_4(y)}{(\cosh_4(x)\cosh_4(y))^2 + (\sinh_4(x)\sinh_4(y))^2}\end{aligned} \tag{16.4}$$

We could push the elliptic integral approach further to get more addition theorems, but it is not necessary (although we will touch on this at the end of the chapter). We will get a formula for the (non-hyperbolic) 4-tangent by the symmetry identities we found in Section 14.2. Let $\omega = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$, so that $\omega^4 = -1$. Recall the identities from Theorem 14.2, which can be rewritten as

$$\begin{aligned}\cosh_4\left(\frac{x}{\omega}\right) &= \cos_4(x) \\ \sinh_4\left(\frac{x}{\omega}\right) &= \frac{1}{\omega}\sin_4(x) \\ \tanh_4\left(\frac{x}{\omega}\right) &= \frac{1}{\omega}\tan_4(x).\end{aligned}$$

Applying these identities to (16.4), we have

$$\begin{aligned}\tan_4(a+b) &= \omega\tanh\left(\frac{a+b}{\omega}\right) \\ &= \omega\,\frac{\sinh_4(\frac{a}{\omega})\cosh_4(\frac{a}{\omega}) + \sinh_4(\frac{b}{\omega})\cosh_4(\frac{b}{\omega})}{(\cosh_4(\frac{a}{\omega})\cosh_4(\frac{b}{\omega}))^2 + (\sinh_4(\frac{a}{\omega})\sinh_4(\frac{b}{\omega}))^2} \\ &= \omega\,\frac{\frac{1}{\omega}\sin_4(a)\cos_4(a) + \frac{1}{\omega}\sin_4(b)\cos_4(b)}{(\cos_4(a)\cos_4(b))^2 + (\frac{1}{\omega^2}\sin_4(a)\sin_4(b))^2} \\ &= \frac{\sin_4(a)\cos_4(a) + \sin_4(b)\cos_4(b)}{(\cos_4(a)\cos_4(b))^2 - (\sin_4(a)\sin_4(b))^2}.\end{aligned}$$

The other trigonometric addition theorems can be obtained from the Pythagorean identities. For example, we may use $\cos_4(t) = \frac{1}{(1+\tan_4(t)^4)^{1/4}}$ to obtain

$$\cos_4(a+b) = \frac{(\cos_4(a)\cos_4(b))^2 - (\sin_4(a)\sin_4(b))^2}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2 - (\sin_4(a)\sin_4(b))^2\right)^4 + (\sin_4(a)\cos_4(a) + \sin_4(b)\cos_4(b))^4}}$$

and similarly, using $\sin_4(t)^4 = 1 - \cos_4(t)^4$,

$$\sin_4(a+b) = \frac{\sin_4(a)\cos_4(a) + \sin_4(b)\cos_4(b)}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2 - (\sin_4(a)\sin_4(b))^2\right)^4 + (\sin_4(a)\cos_4(a) + \sin_4(b)\cos_4(b))^4}}$$

Table 16.1 Doubling formulas for $p = 2$ and $p = 4$ trigonometric functions

	$p = 2$	$p = 4$
$\sin_p(2a)$	$2\sin(a)\cos(a)$	$2\sin_4(a)\cos_4(a) \cdot \frac{1}{\sqrt[4]{(\cos_4(a)^4 - \sin_4(a)^4)^4 + (2\sin_4(a)\cos_4(a))^4}}$
$\cos_p(2a)$	$\cos(a)^2 - \sin(a)^2$	$(\cos_4(a)^2 - \sin_4(a)^2) \cdot \frac{1}{\sqrt[4]{(\cos_4(a)^4 - \sin_4(a)^4)^4 + (2\sin_4(a)\cos_4(a))^4}}$
$\tan_p(2a)$	$\frac{2\tan(a)}{1-\tan(a)^2}$	$\frac{2\tan_4(a)}{1-\tan_4(a)^2} \cdot \frac{1}{\cos_4(a)^2}$

Similarities to the classical case are more easily seen in the doubling formulas in Table 16.1. The hyperbolic formulas work similarly, with the minus signs changed to plus signs. See Appendix B for the complete list.

Rational functions also have addition theorems. For example, if $P(x)$ is a degree one polynomial then $P(a + b) = P(a) + P(b) - P(0)$. Indeed, we could accommodate the $p = 1$ and $p = \infty$ functions if we were to relax our addition theorem requirement to allow R to be piecewise rational.

16.4 Dixon and Weierstrass elliptic functions

We have discussed $p = 2$ and $p = 4$, but what of $p = 3$? A. C. Dixon investigated what we would call the 3-trigonometric functions in an 1890 paper [24] and found the doubling formulas

$$\sin_3(2t) = \frac{\sin_3(t)(1 + \cos_3(t)^3)}{\cos_3(t)(1 + \sin_3(t)^3)} \tag{16.5}$$

$$\cos_3(2t) = \frac{\cos_3(t)^3 - \sin_3(t)^3}{\cos_3(t)(1 + \sin_3(t)^3)} \tag{16.6}$$

These expressions are simple enough that they can be verified by looking at a coupled initial value problem and using the same techniques as in Exercises 1.3 and 1.4. Since by the chain rule, we have $\frac{d}{dt}\cos_3(2t) = -2\sin_3(2t)^2$ and $\frac{d}{dt}\sin_3(2t) = 2\cos_3(2t)^2$, our functions $x(t) = \cos_3(2t)$, $y(t) = \sin_3(2t)$ satisfy

$$\begin{cases} x'(t) = -2y(t)^2 \\ y'(t) = 2x(t)^2 \\ x(0) = 1 \\ y(0) = 0, \end{cases} \tag{16.7}$$

One can less easily verify that $X(t) = \frac{\cos_3(t)^3 - \sin_3(t)^3}{\cos_3(t)(1+\sin_3(t)^3)}$ and $Y(t) = \frac{\sin_3(t)(1+\cos_3(t)^3)}{\cos_3(t)(1+\sin_3(t)^3)}$ also satisfy (16.7), and thus they must be the same functions. (We leave this as an exercise.)

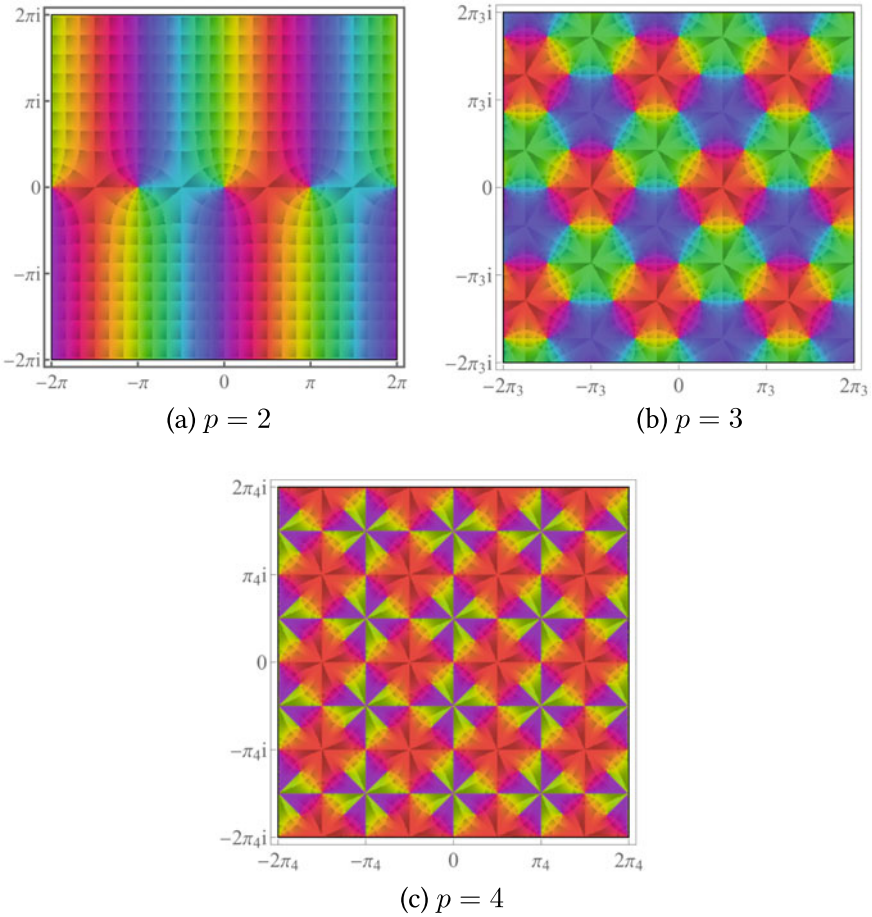


Figure 16.3 Plots of $\sin_p(z)$ for $p = 2, 3, 4$ in the complex plane. We clearly see the real periodicity of the classical sine. Elliptic functions generalize this periodicity to parallelograms, visible in $p = 4$. We also see the parallelograms align when $p = 3$ to provide additional hexagonal symmetry

One can push further to obtain a general sum formula,

$$\cos_3(a+b) = \frac{\cos_3(a)\cos_3(b)^2 - \sin_3(a)^2\sin_3(b)}{\sin_3(a)\cos_3(a)\sin_3(b)^2 + \cos(b)}$$
$$\sin_3(a+b) = \frac{\sin_3(a) + \sin_3(b)\cos_3(b)\cos_3(a)^2}{\sin_3(a)\cos_3(a)\sin_3(b)^2 + \cos(b)}$$

again using a similar IVP approach. See [85] for additional details.

The elementary strategies we have taken in developing these formulas are accessible but limited. Elliptic integrals are better viewed in the complex plane. Whereas a periodic function such as sine or cosine satisfies $f(x+c) = f(x)$ for some real con-

stant c, elliptic integrals are doubly-periodic, satisfying $f(z + w_1) = f(z + w_2) = f(z)$ for some complex constants w_1, w_2. Just as we can understand a periodic function entirely by studying its values on a finite interval (one period), elliptic integrals are determined by parallelograms. Equivalently, where periodic functions can be viewed as functions whose domain is a circle by identifying the ends of the interval, elliptic integrals have tori as a natural domain.

The Weierstrass elliptic functions $\wp(z)$ are defined as the solution to a differential equation of the form $(y')^2 = 4y^3 - g_1 y - g_2$, where g_1 and g_2 are complex constants. We can find identities such as the sum formulas for elliptic functions by first expressing them in terms of $\wp$. Thus identities for certain squigonometric functions can be derived from known properties of $\wp$. For example, $\cot_4^2(t)$ is the Weierstrass elliptic function with $g_1 = -4$ and $g_2 = 0$ [59]. If we take $g_1 = 0, g_2 = 1/27$, we get $\sin_3(t) = \frac{6\wp(t)}{1-3\wp'(t)}$ [85]. The symmetries of the elliptic case are shown in Figure 16.3.

We have fewer tools available for the non-elliptic squigonometric functions, but some progress can be made. We will point toward some of those results in Project 20.2 after we introduce a two-parameter definition of the squigonometric functions that will allow us to broaden the question a little.

Exercises

▶ **Exercise 16.1** *The lemniscate of Bernoulli is the set of points such that the* product *of the distances to two fixed points is a constant (as opposed to an ellipse which considers the sum or the hyperbola which considers the difference). Sketch these curves using different p-distances with foci* $(\pm 1, 0)$ *and product constant 1. Compare with Exercise 13.1.*

▶ **Exercise 16.2** *Continue the pattern in Exercise 16.1 and examine the set of points for which the* ratio *of the p-distances to two fixed points is constant k. Determine what well-known curves you obtain in the Euclidean case* $p = 2$. *What curves do you get for general p when* $k = 1$*?*

▶ **Exercise 16.3** *Prove that* $\cos_4(it) = \cos_4(t)$ *and* $\sin_4(it) = i \sin_4(t)$.

▶ **Exercise 16.4** *Use equation (9.7) and our calculation of the value of L to prove equation (16.3).*

▶ **Exercise 16.5** *Fill in the details for proving the* 3*-squigonometric doubling formulas (16.5) by confirming that both sides of the equations satisfy (16.7). (This can be done with elementary differentiation, but it is a little tedious. This may be a good opportunity to practice getting a computer to complete this sort of symbolic calculation.)*

Project

Project 16.1 (Viète products)
There is a long history of representations of π. One nice expression is Viète's product:

$$\frac{2}{\pi} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \cdot \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \cdots \tag{16.8}$$

This has a geometric derivation that depends heavily on angle sum formulas. Levin [59] was able to adapt this approach to the less symmetric 4-squircle and found a similar expression for π_4:

$$\frac{2\sqrt{2}}{\pi_4} = \frac{2}{L} = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2} + \frac{\frac{1}{2}}{\sqrt{\frac{1}{2}}}} \cdot \sqrt{\frac{1}{2} + \frac{\frac{1}{2}}{\sqrt{\frac{1}{2} + \frac{\frac{1}{2}}{\sqrt{\frac{1}{2}}}}}} \cdots, \tag{16.9}$$

The identities necessary to produce this formula depend on representing the 4-squigonometric functions in terms of the Weierstrass elliptic function.

Could something similar be done to derive a Viète-like product for π_3 using the Dixon elliptic functions? (Levin advises in his paper that this is less straightforward than for π_4.) Could other squigonometric functions with appropriate identities lead to similar formulas? See [34,58,77] for some techniques for producing products like this. Project 20.2 may be related.

Project 16.2 (Geometric curves)
There are lots of geometric curves. We have seen conics, lemniscates, and roulettes – and we will see more – but the study of loci of points that satisfy certain geometric properties has been a featured topic throughout the history of mathematics from its origins to today. p-metrics bring a specific perspective to this topic. But that's a lot of territory to cover, so this isn't a project so much as an invitation to find other projects. The squircular wheels Project 4.2 falls into this category.

One place to start would be to try graphing p-metric versions of these curves, as computers have provided new ways to view some old mathematics. You can find examples of generalized Lissajous and conchoid curves in the menagerie (Figures A.13 and A.14). From there, you may spot some interesting features, make some conjectures, and maybe prove something new, or something old in a new way. Exercises 13.6, 16.1, and 16.2 offer a path with this idea.

The bibliography on curves runs wide and deep. Gibson's trilogy [37–39] offers accessible surveys from the Euclidean, algebraic, differential perspectives. Apostol and Mamikon's development of "visual calculus" also offers a unique look at such curves. [5]



More on lemniscates and ellipses

17

One goal of generalized trigonometric functions is to parameterize imperfect circles, the closed curves that are not quite circles. In Chapter 16, we introduced the lemniscatic constant L and its associated functions $\mathrm{sl}(t)$ and $\mathrm{cl}(t)$, which functioned as a sine and cosine function for the lemniscate. Here we will take a closer look at these functions and their relationship to another set of generalized trigonometric functions, the *Jacobi elliptic functions*, and to our 4-trigonometric functions, $\sin_4(t)$ and $\cos_4(t)$.

17.1 Lemniscatic functions

We previously defined the lemniscatic sine $\mathrm{sl}(t)$ as the straight-line distance from the origin to the point on the lemniscate $r^2 = \cos 2\theta$ that is t units away in arclength, while the lemniscatic cosine $\mathrm{cl}(t)$ is the distance from the point $(1, 0)$ to the point t units in arclength from the origin. (See Figure 17.1.)

The lemniscatic sine and cosine were defined as inverses of elliptic integrals:

$$\mathrm{sl}^{-1}(x) = \int_0^x \frac{1}{\sqrt{1-u^4}}\,du, \qquad \mathrm{cl}^{-1}(x) = \int_x^1 \frac{1}{\sqrt{1-u^4}}\,du,$$

for $-1 \le x \le 1$. In particular, since $\mathrm{sl}^{-1}(x) = \mathrm{arctanh}_4(x)$, then

$$\mathrm{sl}^{-1}(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{4}, \frac{1}{2} \\ \frac{5}{4}\end{matrix}; x^4\right),$$

a hypergeometric function, via Chapter 11. Because $\mathrm{sl}^{-1}(x) = \mathrm{arctanh}_4(x)$, we can obtain several facts about the lemniscatic sine quickly. For instance, the series for

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_17

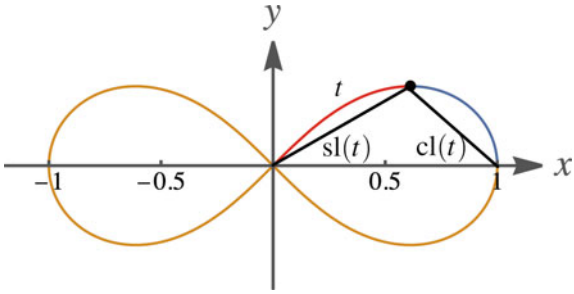


Figure 17.1 Definition of $\mathrm{sl}(t)$ and $\mathrm{cl}(t)$

the inverse lemniscatic sine is just the series for $\arctan_4(x)$ without the alternation:

$$\begin{aligned}\mathrm{sl}^{-1}(x) &= x + \frac{x^5}{10} + \frac{x^9}{24} + \frac{5x^{13}}{208} + \frac{35x^{17}}{2176} + \cdots \\ &= x + \sum_{n=1}^{\infty} \frac{2 \cdot 6 \cdot 10 \cdot \cdots \cdot (4(n-1)+2)}{n!4^n(4n+1)} x^{4n+1}.\end{aligned}$$

As this series represents an odd function, it is valid for $-1 \le x \le 1$. (The series is absolutely convergent on this interval by Gauss' test for hypergeometric series – see Chapter 12.) Setting $x = 1$ yields a series for the lemniscate constant L:

$$\begin{aligned}L &= 2\left(1 + \sum_{n=1}^{\infty} \frac{2 \cdot 6 \cdot 10 \cdot \cdots \cdot (4(n-1)+2)}{n!4^n(4n+1)}\right) \\ &= 2\left(1 + \frac{1}{10} + \frac{1}{24} + \frac{5}{208} + \frac{35}{2176} + \cdots\right). \qquad (17.1)\end{aligned}$$

Similarly, we can obtain the series for the lemniscatic sine from the series for the 4-hyperbolic tangent, which itself is the series for the 4-tangent with alternating terms.

$$\mathrm{sl}(t) = t - \frac{t^5}{10} + \frac{t^9}{120} - \frac{11t^{13}}{15,600} + \frac{211t^{17}}{3,536,000} + \cdots \qquad (17.2)$$

A graph of $\mathrm{sl}(t)$ and its degree-45 Maclaurin polynomial can be found in Figure 17.2. We note that, from our work in Section 14.2, the radius of convergence for this series is $\pi_4/2 = L/\sqrt{2} \approx 1.854074677$.

The lemniscatic sine and cosine are not p-trigonometric functions for any $p \ge 1$ but exhibit some similar characteristics. To find other properties of the lemniscatic sine and cosine, we need to explore the elliptic integrals from which they sprung more closely.

Connection to elliptic integrals

Recall that the lemniscatic constant L was defined as

$$L = 2\,\mathrm{sl}^{-1}(1) = 2\,\mathrm{cl}^{-1}(0) = 2\int_0^1 \frac{1}{\sqrt{1-u^4}}\,du.$$

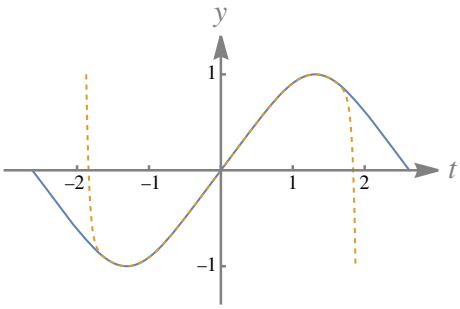


Figure 17.2 Graphs of $y = \mathrm{sl}(t)$ (solid) and its degree-45 Maclaurin polynomial (dashed)

This is an example of a *complete elliptic integral of the first kind*:

$$K(k) = \int_0^1 \frac{1}{\sqrt{1-u^2}\sqrt{1-k^2u^2}}\,du,$$

where k is the *elliptic modulus*. In this case, we have that $K(i) = \frac{L}{2}$. If we set $u = \sin(\theta)$, then $\theta = \arcsin(u)$ and $d\theta = \frac{1}{\sqrt{1-u^2}}\,du$, giving us

$$K(k) = \int_0^{\pi/2} \frac{1}{\sqrt{1-k^2\sin(\theta)^2}}\,d\theta$$

and the integral formula

$$\pi_4 = 2\sqrt{2}\int_0^{\pi/2} \frac{1}{\sqrt{1+\sin(\theta)^2}}\,d\theta$$

from setting $k = i$ once again. In Section 16.3, we saw that elliptic integrals satisfy addition formulas. Other k-values lead to familiar results. For instance,

$$\begin{aligned} K\left(\tfrac{1}{\sqrt{2}}\right) &= \int_0^1 \frac{1}{\sqrt{1-u^2}\sqrt{1-\frac{1}{2}u^2}}\,du \\ &= \sqrt{2}\int_0^1 \frac{1}{\sqrt{1-u^2}\sqrt{2-u^2}}\,du \end{aligned}$$

Let $v = \sqrt{1-u^2}$. Then $u = \sqrt{1-v^2}$, $2-u^2 = 1+v^2$, $du = -\frac{v}{\sqrt{1-v^2}}\,dv$. This gives us

$$\begin{aligned} K\left(\tfrac{1}{\sqrt{2}}\right) &= -\sqrt{2}\int_1^0 \frac{1}{\sqrt{1-v^2}\sqrt{1+v^2}}\,dv \\ &= \sqrt{2}\int_0^1 \frac{1}{\sqrt{1-v^4}}\,dv \\ &= \sqrt{2}\cdot\frac{L}{2} = \frac{\pi_4}{2}. \end{aligned} \tag{17.3}$$

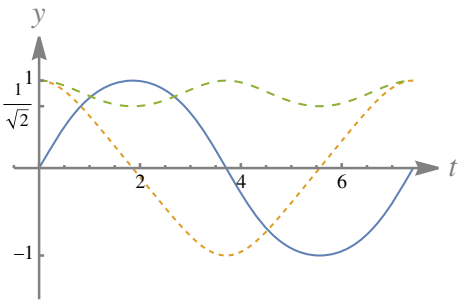


Figure 17.3 Graphs of the Jacobi elliptic functions $y = \mathrm{sn}(t)$ (solid), $y = \mathrm{cn}(t)$ (small dash), and $y = \mathrm{dn}(t)$ (large dash) for $k = 1/\sqrt{2}$. The period for sn and cn is $4K(1/\sqrt{2}) = 2\pi_4$; for dn, the period is π_4

In particular, this means that $K(\frac{1}{\sqrt{2}}) = \sqrt{2}\,K(i)$. (We note in passing that some sources and computer algebra systems, such as Mathematica, refer to elliptic integrals in terms of *the parameter* m, which equals k^2.)

The integral that defines the inverse lemniscatic sine is an example of an *incomplete elliptic integral of the first kind*, defined as

$$F(x;k) = \int_0^x \frac{1}{\sqrt{1-u^2}\sqrt{1-k^2u^2}}\,du,$$

or

$$F(\varphi,k) = \int_0^\varphi \frac{1}{\sqrt{1-k^2\sin(\theta)^2}}\,d\theta.$$

Note the semicolon in the first notation versus the comma in the second. In particular:

- $K(k) = F(\frac{\pi}{2}, k)$,
- $\mathrm{sl}^{-1}(x) = F(x;i)$, and
- $F(\frac{\pi}{2}, i) = \frac{L}{2} = \frac{\pi_4}{2\sqrt{2}}$.

If we set $t = F(\varphi, k)$ for some fixed k, then we define the *elliptic sine* of t, notated $\mathrm{sn}(t)$, as $\mathrm{sn}(t) = \sin(\varphi)$. We similarly define the *elliptic cosine* of t, notated $\mathrm{cn}(t)$, as $\mathrm{cn}(t) = \cos(\varphi)$. We also define a rather strange function, the *delta amplitude* $\mathrm{dn}(t) = \sqrt{1-k^2\sin(\varphi)^2}$. (The value φ is called the *Jacobi amplitude*.) The value $K(k)$ is called the *quarter-period* for the elliptic sine and cosine; thus the pair has period $4K(k)$ and oscillate between -1 and 1. The delta amplitude, on the other hand, has period $2K(k)$ and oscillates between $\sqrt{1-k^2}$ and 1. (See Figure 17.3.)

These three functions are the basic *Jacobi elliptic functions*, which satisfy the following CIVP:

$$\begin{cases} x'(t) = -y(t)z(t), \\ y'(t) = x(t)z(t), \\ z'(t) = -k^2x(t)y(t), \\ x(0) = 1, \\ y(0) = 0, \\ z(0) = 1 \end{cases} \tag{17.4}$$

where $x(t) = \mathrm{cn}(t)$, $y(t) = \mathrm{sn}(t)$, and $z(t) = \mathrm{dn}(t)$. The elliptic sine function is odd, while the elliptic cosine and delta amplitude functions are even.

Because we have three, rather than two, basic elliptic functions, we have a total of *12* Jacobi elliptic functions via ratios of the basic functions and their multiplicative inverses. The naming convention is that a ratio of two Jacobi elliptic functions takes its name from the first letter of each of the functions and its multiplicative inverse simply reverses the letters. For instance,

$$\mathrm{sd}(t) = \frac{\mathrm{sn}(t)}{\mathrm{dn}(t)}, \qquad \mathrm{ds}(t) = \frac{1}{\mathrm{sd}(t)} = \frac{\mathrm{dn}(t)}{\mathrm{sn}(t)}.$$

We would assume that the elliptic sine and cosine would parameterize an ellipse of some sort. We note, though, that since $\mathrm{sn}(t) = \sin(\varphi)$ and $\mathrm{cn}(t) = \cos(\varphi)$, we have $\mathrm{sn}(t)^2 + \mathrm{cn}(t)^2 = 1$ for all t, and the elliptic cosine and sine parameterize the unit circle. (We also have the relation $k^2\,\mathrm{sn}(t)^2 + \mathrm{dn}(t)^2 = 1$.)

This is where that strange delta amplitude function $\mathrm{dn}(t)$ comes in. Suppose we wish to parameterize an ellipse with equation $x^2 + \frac{y^2}{b^2} = 1$, $b > 1$, where as is customary, $x = r\cos(\varphi)$ and $y = r\sin(\varphi)$. We can use the ellipse equation to find an expression for r, the distance from the origin to the point (x, y) on the ellipse:

$$\begin{aligned}
x^2 + \frac{y^2}{b^2} &= 1 \\
r^2\cos(\varphi)^2 + \frac{r^2\sin(\varphi)^2}{b^2} &= 1 \\
r^2\left(\cos(\varphi)^2 + \frac{\sin(\varphi)^2}{b^2}\right) &= 1 \\
r^2 = \frac{1}{\cos(\varphi)^2 + \frac{\sin(\varphi)^2}{b^2}} &= \frac{1}{(1-\sin(\varphi)^2) + \frac{\sin(\varphi)^2}{b^2}} \\
r^2 &= \frac{1}{1 - \sin(\varphi)^2\left(1 - \frac{1}{b^2}\right)}
\end{aligned}$$

If we set $k = \sqrt{1 - \frac{1}{b^2}}$, we then have that

$$r \equiv r(\varphi, k) = \frac{1}{\sqrt{1 - k^2\sin(\varphi)^2}}.$$

Recalling the relation

$$t = F(\varphi, k) = \int_0^{\varphi} \frac{1}{\sqrt{1 - k^2\sin(\theta)^2}}\, d\theta,$$

we see that t represents the total arclength traversed along the ellipse, and thus $r(\varphi, k) = \frac{1}{\mathrm{dn}(t)}$. This leads to the correct parameterization of our ellipse:

$$x = \frac{\mathrm{cn}(t)}{\mathrm{dn}(t)} = \mathrm{cd}(t), \qquad y = \frac{\mathrm{sn}(t)}{\mathrm{dn}(t)} = \mathrm{sd}(t).$$

(See Figures 17.4 and 17.5.)

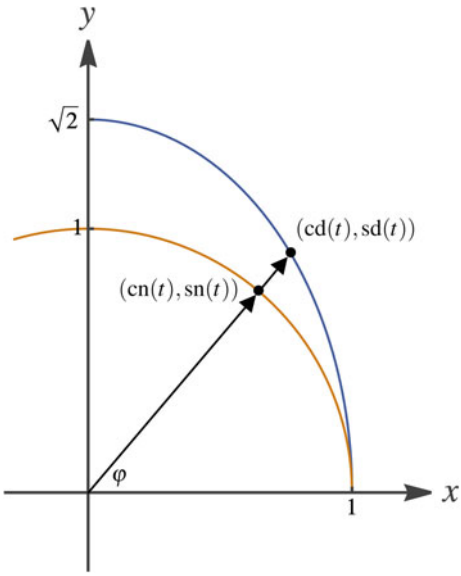


Figure 17.4 The Jacobi ellipse for $k = 1/\sqrt{2}$ ($b = \sqrt{2}$) and the unit circle

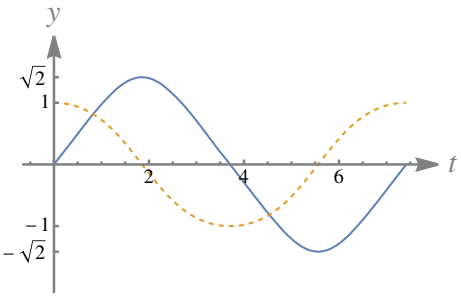


Figure 17.5 Graphs of $y = \mathrm{sd}(t)$ (solid) and $y = \mathrm{cd}(t)$ (dashed) for $k = 1/\sqrt{2}$. Note that the amplitude of $\mathrm{sd}(t)$ in this case is $\sqrt{2}$ rather than 1

The major elliptic functions satisfy several familiar looking relationships:

$$\mathrm{sn}(t)^2 + \mathrm{cn}(t)^2 = 1, \qquad k^2\,\mathrm{sn}(t)^2 + \mathrm{dn}(t)^2 = 1, \tag{17.5}$$

$$\mathrm{sn}(0) = 0, \quad \mathrm{cn}(0) = 1, \quad \mathrm{dn}(0) = 1,$$

$$\mathrm{sn}(K(k)) = 1, \quad \mathrm{cn}(K(k)) = 0, \quad \mathrm{dn}(K(k)) = \sqrt{1-k^2},$$

$$\mathrm{sn}(t + 4K(k)) = \mathrm{sn}(t), \quad \mathrm{cn}(t + 4K(k)) = \mathrm{cn}(t), \quad \mathrm{dn}(t + 2K(k)) = \mathrm{dn}(t),$$

$$\mathrm{sn}(-t) = -\,\mathrm{sn}(t), \quad \mathrm{cn}(-t) = \mathrm{cn}(t), \quad \mathrm{dn}(-t) = \mathrm{dn}(t),$$

$$\frac{d}{dt}\,\mathrm{sn}(t) = \mathrm{cn}(t)\,\mathrm{dn}(t),$$

$$\frac{d}{dt}\,\mathrm{cn}(t) = -\,\mathrm{sn}(t)\,\mathrm{dn}(t),$$

$$\frac{d}{dt}\,\mathrm{dn}(t) = -k^2\,\mathrm{sn}(t)\,\mathrm{cn}(t).$$

We can also define inverse functions for the major elliptic functions. Let $t = \mathrm{sn}^{-1}(x)$. Then $x = \mathrm{sn}(t)$ and $\frac{dx}{dt} = \mathrm{cn}(t)\,\mathrm{dn}(t)$. Using (17.5), we can write this as

$$\frac{dx}{dt} = \sqrt{1 - \mathrm{sn}(t)^2}\sqrt{1 - k^2\,\mathrm{sn}(t)^2}$$

This then leads to

$$\frac{dt}{dx} = \frac{1}{\sqrt{1 - x^2}\sqrt{1 - k^2x^2}}$$

and, since $\mathrm{sn}(0) = 0$, we obtain

$$t = \mathrm{sn}^{-1}(x) = \int_0^x \frac{1}{\sqrt{1 - u^2}\sqrt{1 - k^2u^2}}\,du = F(x; k). \tag{17.6}$$

We can similarly derive

$$\mathrm{cn}^{-1}(x) = \int_x^1 \frac{1}{\sqrt{1 - u^2}\sqrt{1 - k^2 + k^2u^2}}\,du,$$
$$\mathrm{dn}^{-1}(x) = \int_x^1 \frac{1}{\sqrt{1 - u^2}\sqrt{u^2 + k^2 - 1}}\,du.$$

17.2 From elliptic to lemniscatic functions

We can exploit known properties of elliptic integrals to derive properties of elliptic and lemniscatic functions. (See [44] for further details.)

We fix $k = \frac{1}{\sqrt{2}}$ in all that follows.

Theorem 17.1

For $-1 \le x \le 1$,

$$\mathrm{sl}^{-1}(x) = k\,\mathrm{sgn}(x)\left(K(k) - F(\sqrt{1 - x^2}; k)\right)$$

where $\mathrm{sgn}(x) = 1$ *if* $x > 0$, $\mathrm{sgn}(x) = -1$ *if* $x < 0$ *and* $\mathrm{sgn}(x) = 0$ *if* $x = 0$.

Proof. We have from (17.3) that $k \cdot K(k) = K(i) = \frac{L}{2}$. For the present, we consider x in $[0, 1)$. We have

$$\begin{aligned}\mathrm{sl}^{-1}(x) &= \int_0^x \frac{1}{\sqrt{1 - u^4}}\,du \\ &= \int_0^1 \frac{1}{\sqrt{1 - u^4}}\,du - \int_x^1 \frac{1}{\sqrt{1 - u^4}}\,du \\ &= k \cdot K(k) - \int_x^1 \frac{1}{\sqrt{1 - u^2}\sqrt{1 + u^2}}\,du\end{aligned}$$

via (17.3). As in the derivation of (17.3), we set $v = \sqrt{1-u^2}$. Then $u = \sqrt{1-v^2}$, $du = -\frac{v}{\sqrt{1-v^2}}\,dv$ and we get

$$\begin{aligned}
\mathrm{sl}^{-1}(x) &= k \cdot K(k) + \int_{\sqrt{1-x^2}}^{0} \frac{1}{\sqrt{1-v^2}\sqrt{2-v^2}}\,dv \\
&= k \cdot K(k) - \frac{1}{\sqrt{2}} \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{1-v^2}\sqrt{1-\frac{1}{2}v^2}}\,dv \\
&= k\left(K(k) - F(\sqrt{1-x^2}; k)\right).
\end{aligned}$$

The equality also holds for $x = 1$ since $\mathrm{sl}^{-1}(1) = \frac{L}{2} = k \cdot K(k) - F(0; k) = k \cdot K(k)$. Because sl^{-1} is an odd function and $F(1; k) = K(k)$, we get the desired formula for $-1 \le x \le 1$. □

Since $\mathrm{sn}^{-1}(x)$ is defined by an elliptic integral, we can apply Euler's addition theorem (Theorem 16.1) as well as Equations (17.5) to the elliptic sine to obtain (the proof is left as an exercise):

$$\mathrm{sn}(s+t) = \frac{\mathrm{sn}(t)\,\mathrm{cn}(s)\,\mathrm{dn}(s) + \mathrm{sn}(s)\,\mathrm{cn}(t)\,\mathrm{dn}(t)}{1 - k^2\,\mathrm{sn}(s)^2\,\mathrm{sn}(t)^2}.$$

If we then use the relationships $\mathrm{cn}(s+t)^2 = 1 - \mathrm{sn}(s+t)^2$ and $\mathrm{dn}(s+t)^2 = 1 - k^2\,\mathrm{sn}(s+t)^2$ (along with plenty of algebra), we can obtain the similar formulas:

$$\begin{aligned}
\mathrm{cn}(s+t) &= \frac{\mathrm{cn}(s)\,\mathrm{cn}(t) - \mathrm{sn}(s)\,\mathrm{dn}(s)\,\mathrm{sn}(t)\,\mathrm{dn}(t)}{1 - k^2\,\mathrm{sn}(s)^2\,\mathrm{sn}(t)^2}, \qquad (17.7)\\
\mathrm{dn}(s+t) &= \frac{\mathrm{dn}(s)\,\mathrm{dn}(t) - k^2\,\mathrm{sn}(s)\,\mathrm{cn}(s)\,\mathrm{sn}(t)\,\mathrm{cn}(t)}{1 - k^2\,\mathrm{sn}(s)^2\,\mathrm{sn}(t)^2}.
\end{aligned}$$

We can use these addition formulas to develop some formulas for the elliptic functions based on symmetry. (These actually hold for any value of k.)

$$\begin{aligned}
\mathrm{sn}(t + K(k)) &= \frac{\mathrm{sn}(K(k))\,\mathrm{cn}(t)\,\mathrm{dn}(t) + \mathrm{sn}(t)\,\mathrm{cn}(K(k))\,\mathrm{dn}(K(k))}{1 - k^2\,\mathrm{sn}(t)^2\,\mathrm{sn}(K(k))^2} \\
&= \frac{1 \cdot \mathrm{cn}(t)\,\mathrm{dn}(t) + \mathrm{sn}(t) \cdot 0 \cdot \sqrt{1-k^2}}{1 - k^2\,\mathrm{sn}(t)^2 \cdot 1} \\
&= \frac{\mathrm{cn}(t)\,\mathrm{dn}(t)}{\mathrm{dn}(t)^2} \\
&= \frac{\mathrm{cn}(t)}{\mathrm{dn}(t)} = \mathrm{cd}(t). \qquad (17.8)\\
\mathrm{cn}(t + K(k)) &= \frac{\mathrm{cn}(t)\,\mathrm{cn}(K(k)) - \mathrm{sn}(t)\,\mathrm{dn}(t)\,\mathrm{sn}(K(k))\,\mathrm{dn}(K(k))}{1 - k^2\,\mathrm{sn}(t)^2\,\mathrm{sn}(K(k))^2} \\
&= \frac{\mathrm{cn}(t) \cdot 0 - \mathrm{sn}(t) \cdot 1 \cdot \mathrm{dn}(t)\sqrt{1-k^2}}{1 - k^2\,\mathrm{sn}(t)^2 \cdot 1} \\
&= -\frac{\mathrm{sn}(t)\,\mathrm{dn}(t)\sqrt{1-k^2}}{\mathrm{dn}(t)^2}
\end{aligned}$$

$$= -\sqrt{1-k^2}\,\frac{\mathrm{sn}(t)}{\mathrm{dn}(t)} = -\sqrt{1-k^2}\,\mathrm{sd}(t). \tag{17.9}$$

Corollary 17.2

$$\mathrm{sl}(t) = \mathrm{cn}\left(K(k) - \frac{t}{k}\right) = k\,\mathrm{sd}\left(\frac{t}{k}\right)$$

for $-\frac{L}{2} \le t \le \frac{L}{2}$.

Proof. By Theorem 17.1, we have

$$\mathrm{sl}(\mathrm{sl}^{-1}(x)) = \mathrm{sl}\left(k\left(K(k) - F(\sqrt{1-x^2};k)\right)\right) = x,$$

for $-1 \le x \le 1$. Set $t = \mathrm{sl}^{-1}(x) = k\left(K(k) - F(\sqrt{1-x^2};k)\right)$. Because sl and sl^{-1} are inverse functions, we have that $\mathrm{sl}(-\frac{L}{2}) = -1$, $\mathrm{sl}(\frac{L}{2}) = 1$ and $-\frac{L}{2} \le t \le \frac{L}{2}$ since $-1 \le x \le 1$. Rearranging, we obtain

$$F(\sqrt{1-x^2};k) = K(k) - \frac{t}{k}.$$

Thus

$$\mathrm{sn}\left(K(k) - \frac{t}{k}\right) = \mathrm{sn}\left(F(\sqrt{1-x^2};k)\right) = \mathrm{sn}(\mathrm{sn}^{-1}(\sqrt{1-x^2})) = \sqrt{1-x^2},$$

by (17.6). This leads to

$$x^2 = 1 - \mathrm{sn}\left(K(k) - \frac{t}{k}\right)^2 = \mathrm{cn}\left(K(k) - \frac{t}{k}\right)^2.$$

If $0 \le x \le 1$, then $0 \le t \le \frac{L}{2} \Longrightarrow 0 \le \frac{t}{k} \le \frac{\pi_4}{2} = K(k)$. Therefore,

$$-K(k) \le K(k) - \frac{t}{k} \le K(k)$$

and $\mathrm{cn}\left(K(k) - \frac{t}{k}\right) > 0$. In this case, $x = \mathrm{cn}\left(K(k) - \frac{t}{k}\right)$.

If $-1 \le x \le 0$, then $-\frac{L}{2} \le t \le 0 \Longrightarrow -K(k) = -\frac{\pi_4}{2} \le \frac{t}{k} \le 0$ and thus

$$K(k) \le K(k) - \frac{t}{k} \le 2K(k).$$

So $\mathrm{cn}\left(K(k) - \frac{t}{k}\right) < 0$ and once again $x = \mathrm{cn}\left(K(k) - \frac{t}{k}\right)$. Therefore $x = \mathrm{sl}(t) = \mathrm{cn}\left(K(k) - \frac{t}{k}\right)$, giving us the first equality.

The second equation is an immediate consequence of the addition formula (17.9), the oddness of the sn function, the evenness of the dn function, and the fact that, since $k = \frac{1}{\sqrt{2}}$, $\sqrt{1-k^2} = k$:

$$\mathrm{cn}\left(K(k) - \frac{t}{k}\right) = \mathrm{cn}\left(-\frac{t}{k} + K(k)\right)$$

$$= -\sqrt{1-k^2}\,\mathrm{sd}\left(-\frac{t}{k}\right)$$
$$= k\,\mathrm{sd}\left(\frac{t}{k}\right).$$

□

Corollary 17.3

$$\mathrm{cl}(t) = \mathrm{cn}\left(\frac{t}{k}\right).$$

Proof. Since $k \cdot K(k) = \frac{L}{2}$, we have by Corollary 17.2,

$$\mathrm{cl}(t) = \mathrm{sl}\,(k \cdot K(k) - t) = \mathrm{cn}\left(K(k) - \frac{k \cdot K(k) - t}{k}\right) = \mathrm{cn}\left(\frac{t}{k}\right).$$

□

The theorem and corollaries, as well as the addition formulas for the elliptic functions, lead to addition formulas for the lemniscatic functions. For example, using Corollary 17.2, we obtain

$$\mathrm{sl}(s+t) = k\,\mathrm{sd}\left(\frac{s}{k}+\frac{t}{k}\right) = k\,\frac{\mathrm{sn}\left(\frac{s}{k}+\frac{t}{k}\right)}{\mathrm{dn}\left(\frac{s}{k}+\frac{t}{k}\right)}$$
$$= \frac{k\,\mathrm{sn}(\frac{t}{k})\,\mathrm{cn}(\frac{s}{k})\,\mathrm{dn}(\frac{s}{k}) + k\,\mathrm{sn}(\frac{s}{k})\,\mathrm{cn}(\frac{t}{k})\,\mathrm{dn}(\frac{t}{k})}{\mathrm{dn}(\frac{s}{k})\,\mathrm{dn}(\frac{t}{k}) - k^2\,\mathrm{sn}(\frac{s}{k})\,\mathrm{cn}(\frac{s}{k})\,\mathrm{sn}(\frac{t}{k})\,\mathrm{cn}(\frac{t}{k})}$$

Dividing all terms by $\mathrm{dn}(\frac{s}{k})\,\mathrm{dn}(\frac{t}{k})$, we get

$$\mathrm{sl}(s+t) = \frac{\frac{k\,\mathrm{sn}(\frac{t}{k})\,\mathrm{cn}(\frac{s}{k})\,\mathrm{dn}(\frac{s}{k})}{\mathrm{dn}(\frac{s}{k})\,\mathrm{dn}(\frac{t}{k})} + \frac{k\,\mathrm{sn}(\frac{s}{k})\,\mathrm{cn}(\frac{t}{k})\,\mathrm{dn}(\frac{t}{k})}{\mathrm{dn}(\frac{s}{k})\,\mathrm{dn}(\frac{t}{k})}}{1 - \frac{k^2\,\mathrm{sn}(\frac{s}{k})\,\mathrm{cn}(\frac{s}{k})\,\mathrm{sn}(\frac{t}{k})\,\mathrm{cn}(\frac{t}{k})}{\mathrm{dn}(\frac{s}{k})\,\mathrm{dn}(\frac{t}{k})}}$$
$$= \frac{k\,\mathrm{sd}(\frac{t}{k})\,\mathrm{cn}(\frac{s}{k}) + k\,\mathrm{sd}(\frac{s}{k})\,\mathrm{cn}(\frac{t}{k})}{1 - k\,\mathrm{sd}(\frac{s}{k}) \cdot k\,\mathrm{sd}(\frac{t}{k})\,\mathrm{cn}(\frac{s}{k})\,\mathrm{cn}(\frac{t}{k})}$$
$$= \frac{\mathrm{sl}(s)\,\mathrm{cl}(t) + \mathrm{sl}(t)\,\mathrm{cl}(s)}{1 - \mathrm{sl}(s)\,\mathrm{sl}(t)\,\mathrm{cl}(s)\,\mathrm{cl}(t)}, \tag{17.10}$$

by Corollaries 17.2 and 17.3.

We can directly compute as a special case that $\mathrm{sl}(t + \frac{L}{2}) = \mathrm{cl}(t)$ and this leads in turn to $\mathrm{cl}(t + \frac{L}{2}) = \mathrm{sl}(t + L) = -\mathrm{sl}(t)$. We can use these formulas and the addition formula for the lemniscatic sine we just proved to derive an addition formula for the lemniscatic cosine:

$$\mathrm{cl}(s+t) = \frac{\mathrm{cl}(s)\,\mathrm{cl}(t) - \mathrm{sl}(s)\,\mathrm{sl}(t)}{1 + \mathrm{sl}(s)\,\mathrm{sl}(t)\,\mathrm{cl}(s)\,\mathrm{cl}(t)}. \tag{17.11}$$

If we set $s = t$ in these formulas, we get "double-angle" formulas.

$$\operatorname{sl}(2t) = \frac{2\operatorname{sl}(t)\operatorname{cl}(t)}{1 - \operatorname{sl}(t)^2\operatorname{cl}(t)^2}, \tag{17.12}$$

$$\operatorname{cl}(2t) = \frac{\operatorname{cl}(t)^2 - \operatorname{sl}(t)^2}{1 + \operatorname{sl}(t)^2\operatorname{cl}(t)^2}. \tag{17.13}$$

17.3 Derivatives of lemniscatic functions

We seek derivative formulas (and a CIVP) for the lemniscatic functions. We look first at the lemniscatic sine. By Corollary 17.2, we have

$$\begin{aligned}
\frac{d}{dt}\operatorname{sl}(t) &= \frac{d}{dt}\left[k\operatorname{sd}\left(\frac{t}{k}\right)\right] \\
&= \frac{d}{dt}\left[\frac{k\operatorname{sn}(\frac{t}{k})}{\operatorname{dn}(\frac{t}{k})}\right] \\
&= \frac{k\operatorname{cn}(\frac{t}{k})\operatorname{dn}(\frac{t}{k})\left(\frac{1}{k}\right)(\operatorname{dn}(\frac{t}{k})) - k\operatorname{sn}(\frac{t}{k})(-k^2)\operatorname{sn}(\frac{t}{k})\operatorname{cn}(\frac{t}{k})(\frac{1}{k})}{\operatorname{dn}(\frac{t}{k})^2} \\
&= \frac{\operatorname{cn}(\frac{t}{k})\operatorname{dn}(\frac{t}{k})^2 + k^2\operatorname{sn}(\frac{t}{k})^2\operatorname{cn}(\frac{t}{k})}{\operatorname{dn}(\frac{t}{k})^2} \\
&= \operatorname{cl}(t) + k^2\operatorname{sd}(\tfrac{t}{k})^2\operatorname{cn}(\tfrac{t}{k}) \\
&= \operatorname{cl}(t) + \operatorname{sl}(t)^2\operatorname{cl}(t),
\end{aligned}$$

where in the last line, we have used Corollary 17.3 and the fact that since $k = \frac{1}{\sqrt{2}}$, then $\sqrt{1-k^2} = k$. Thus we have

Theorem 17.4

$$\frac{d}{dt}\operatorname{sl}(t) = \operatorname{cl}(t)(1 + \operatorname{sl}(t)^2).$$

We can use this formula to also prove

Theorem 17.5

$$\frac{d}{dt}\operatorname{cl}(t) = -\operatorname{sl}(t)(1 + \operatorname{cl}(t)^2).$$

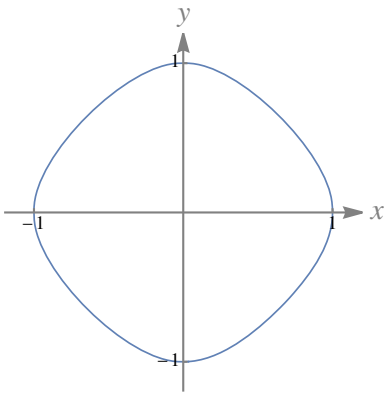


Figure 17.6 The curve parameterized by $(\mathrm{cl}(t), \mathrm{sl}(t))$: $x^2 + x^2y^2 + y^2 = 1$

The lemniscatic functions thus fulfill the CIVP

$$\begin{cases} x'(t) = -y(t)(1 + x(t)^2), \\ y'(t) = x(t)(1 + y(t)^2), \\ x(0) = 1, \\ y(0) = 0, \end{cases} \tag{17.14}$$

where as expected, x is the lemniscatic cosine and y is the lemniscatic sine.

As $\frac{d}{dx}\,\mathrm{sl}^{-1}(x) = \frac{1}{\sqrt{1-x^4}}$, the formula for the derivative of an inverse function gives us that

$$(\mathrm{sl}'(t))^2 = 1 - \mathrm{sl}(t)^4 = (1 - \mathrm{sl}(t)^2)(1 + \mathrm{sl}(t)^2).$$

By Theorem 17.4, however, we have

$$(\mathrm{sl}'(t))^2 = \mathrm{cl}(t)^2(1 + \mathrm{sl}(t)^2)^2.$$

Therefore, it follows that $1 - \mathrm{sl}(t)^2 = \mathrm{cl}(t)^2(1 + \mathrm{sl}(t)^2)$ which means that

$$\mathrm{sl}(t)^2 + \mathrm{sl}(t)^2\,\mathrm{cl}(t)^2 + \mathrm{cl}(t)^2 = 1. \tag{17.15}$$

That is, the lemniscatic cosine and sine parameterize the curve $x^2 + x^2y^2 + y^2 = 1$. (See Figure 17.6.) Although this curve resembles a p-circle for some $1 < p < 2$, the two functions do not fulfill the CIVP (3.2) for any p.

The relationship (17.15) between the lemniscatic sine and cosine is often written as

$$(1 + \mathrm{sl}(t)^2)(1 + \mathrm{cl}(t)^2) = 2,$$

a variation on the Pythagorean identity $\sin(t)^2 + \cos(t)^2 = 1$ which was known in the 18th century [61].

17.4 From lemniscatic to squigonometric functions

As is perhaps to be expected by now, especially given the series (17.2), there are connections between the lemniscatic functions and the 4-trigonometric functions. Here is but one example. (See [81] for others.)

We note first that both the lemniscatic inverse sine and the inverse 4-sine functions are hypergeometric:

$$\mathrm{sl}^{-1}(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{4}, & \frac{1}{2} \\ & \frac{5}{4}\end{matrix}; x^4\right),$$

$$\arcsin_4(x) = x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{4}, & \frac{3}{4} \\ & \frac{5}{4}\end{matrix}; x^4\right).$$

The two listed hypergeometric functions are connected via Euler's hypergeometric transformation:

$${}_2F_1\left(\begin{matrix}a, & b \\ & c\end{matrix}; z\right) = (1-z)^{-a}\, {}_2F_1\left(\begin{matrix}a, & c-b \\ & c\end{matrix}; \frac{z}{z-1}\right).$$

We thus have

$$\omega \cdot x \cdot {}_2F_1\left(\begin{matrix}\frac{1}{4}, & \frac{3}{4} \\ & \frac{5}{4}\end{matrix}; x^4\right) = \left(\frac{x^4}{x^4-1}\right)^{1/4} {}_2F_1\left(\begin{matrix}\frac{1}{4}, & \frac{1}{2} \\ & \frac{5}{4}\end{matrix}; \frac{x^4}{x^4-1}\right),$$

where $\omega^4 = -1$. That is,

$$\omega \arcsin_4(x) = \mathrm{sl}^{-1}\left(\left(\frac{x^4}{x^4-1}\right)^{1/4}\right).$$

If we set $t = \arcsin_4(x)$, then $x = \sin_4(t)$ and

$$\mathrm{sl}(\omega t)^4 = \frac{\sin_4(t)^4}{\sin_4(t)^4 - 1} = -\frac{\sin_4(t)^4}{\cos_4(t)^4} = -\tan_4(t)^4.$$

From the series (17.2), we see that $\mathrm{sl}(iz) = i\,\mathrm{sl}(z)$ and $\mathrm{sl}'(iz) = \mathrm{sl}(z)$ for all $z \in \mathbb{C}$. Therefore,

$$\mathrm{sl}\left(\frac{\sqrt{2}}{1+i}\,t\right)^4 = \mathrm{sl}(i^k \omega t)^4 = \mathrm{sl}(\omega t)^4 = -\tan_4(t)^4,$$

where $k = 0, 1, 2$ or 3. Then

$$\left(\mathrm{sl}'\left(\frac{\sqrt{2}}{1+i}\,t\right)^2\right) = 1 - \mathrm{sl}\left(\frac{\sqrt{2}}{1+i}\,t\right)^4 = 1 + \tan_4(t)^4 = \sec_4(t)^4.$$

The last equation implies that $\mathrm{sl}\left(\frac{\sqrt{2}}{1+i}t\right) = \tan_4(t)$ and $\mathrm{sl}'\left(\frac{\sqrt{2}}{1+i}t\right) = 2\cos_4\left(\frac{t}{1+i}\right)^4 - 1$. There also exist infinite product identities for the lemniscatic sine and cosine derived similarly.

Exercises

▶ **Exercise 17.1** *Show that* $\mathrm{sl}(t)$ *and* $\mathrm{cl}(t)$ *satisfy an Emden-Fowler differential equation:*

$$y'' = -2y^3$$

with appropriate initial conditions for each function.

▶ **Exercise 17.2**
Make a table of partial sums for the series (17.1). Does the series converge quickly or slowly to L*?*

▶ **Exercise 17.3**
Prove the addition formula for the elliptic sine function sn.

▶ **Exercise 17.4**
Use the result of Exercise 17.3 and the formulas (17.5) to prove the addition formulas for the elliptic cosine cn *and the delta amplitude* dn.

▶ **Exercise 17.5**
Use a computer algebra system to plot the functions that fulfill the CIVP (17.14).

▶ **Exercise 17.6**
Use the addition formula (17.10) for the lemniscatic sine sl *to prove the addition and double-angle formulas for the lemniscatic cosine* cl.

▶ **Exercise 17.7**
Use the addition formulas (17.10) and (17.11) to prove the following:

- $\mathrm{sl}(\frac{L}{6}) = \frac{1+\sqrt{3}-\sqrt{2}\sqrt[4]{3}}{2}$,
- $\mathrm{sl}(\frac{L}{3}) = \kappa\,\mathrm{sl}(\frac{L}{6})$,
- $\mathrm{cl}(\frac{L}{3}) = \frac{1+\sqrt{3}-\sqrt{2}\sqrt[4]{3}}{2}$,
- $\mathrm{cl}(\frac{L}{6}) = \kappa\,\mathrm{cl}(\frac{L}{3})$,

where $\kappa = \sqrt{\sqrt{3}+\sqrt{2}\sqrt[4]{3}}$.

▶ **Exercise 17.8**
Use the double-angle formulas (17.12) and (17.13) to prove that

$$\mathrm{sl}\left(\frac{L}{4}\right) = \sqrt{\sqrt{2}-1} = \mathrm{cl}\left(\frac{L}{4}\right).$$



Geometry in the p-norm

18

We have seen how the generalized trigonometric functions naturally describe the p-metric but a few additional tools are useful to complete the picture. If we think of the plane as a vector space, the p-metric is induced naturally from an alternate vector norm. We will also formalize the idea of convexity which will allow us to properly resolve some outstanding geometric properties, most notably the triangle inequality.

18.1 Normed vector spaces

We have introduced the p-metric circles as imperfect variants of Euclidean circles, but they are similar enough that we can still adapt familiar tools to study them.

Most of the niceness of the p-metric comes from being induced by a *normed vector space*. Recall that a vector space is a set of objects that can be added to one another and multiplied by scalars. These operations of addition and scalar multiplication satisfy the properties in Table 18.1, which roughly say that the operations work as you would expect and play well together. Our vector space of concern is $\mathbb{R}^2$, in which vectors add and multiply as usual with

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$
$$r \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ra \\ rb \end{pmatrix}.$$

A *norm* on a vector space V is a function $\|\cdot\| : V \to \mathbb{R}$ for which the following hold for any $\boldsymbol{v}, \boldsymbol{w} \in V$ and $r \in \mathbb{R}$:

Positive definite. $\|\boldsymbol{v}\| = 0$ if and only if $\boldsymbol{v} = \boldsymbol{0}$
Absolute homogeneity. $\|r\boldsymbol{v}\| = |r| \cdot \|\boldsymbol{v}\|$
Triangle inequality. $\|\boldsymbol{v} + \boldsymbol{w}\| \leq \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_18

Table 18.1 The axioms of vector spaces over $\mathbb{R}$, with some elementary properties. These must hold for $\boldsymbol{u}, \boldsymbol{v} \in V$ and $r, s \in \mathbb{R}$.

associativity	$(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$
commutativity	$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$
closure	$\boldsymbol{u} + \boldsymbol{v} \in V, r\boldsymbol{v} \in V$
additive identity	There exists a $\mathbf{0} \in V$ such that $\boldsymbol{v} + \mathbf{0} = \boldsymbol{v}$ for all $v \in V$
additive inverses	For every $\boldsymbol{v} \in V$ there is a $-\boldsymbol{v} \in V$ such that $\boldsymbol{v} + (-\boldsymbol{v}) = \mathbf{0}$
compatibility	$r(s\boldsymbol{v}) = (rs)\boldsymbol{v}$
multiplicative identity	$1\boldsymbol{v} = \boldsymbol{v}$
distributivity	$r(\boldsymbol{u} + \boldsymbol{v}) = r\boldsymbol{u} + r\boldsymbol{v}$ $(r + s)\boldsymbol{v} = r\boldsymbol{v} + s\boldsymbol{v}$
uniqueness of $\mathbf{0}$	If $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{u}$ and $\boldsymbol{u} + \boldsymbol{w} = \boldsymbol{u}$ for all $\boldsymbol{u} \in V$ then $\boldsymbol{v} = \boldsymbol{w}$
uniqueness of additive identity	If $\boldsymbol{u} + \boldsymbol{v} = \mathbf{0}$ and $\boldsymbol{u} + \boldsymbol{w} = \mathbf{0}$ then $\boldsymbol{v} = \boldsymbol{w}$
multiplication by -1	$-1(\boldsymbol{v}) = -\boldsymbol{v}$

The p-norm is thus defined as $\mathbb{R}^2$ with the norm

$$\left\| \begin{pmatrix} a \\ b \end{pmatrix} \right\| = \sqrt[p]{|a|^p + |b|^p}. \tag{18.1}$$

It is easy to verify that this positive definite and absolutely homogeneous. We will show in Section 18.2 that the triangle inequality is satisfied if and only if $p \geq 1$.

Normed vector spaces are metric spaces with the distance function $d(\boldsymbol{u}, \boldsymbol{v}) = \|\boldsymbol{u} - \boldsymbol{v}\|$, and the reader may verify that $\|\cdot\|_p$ really does induce the p-metric we have been using. See Exercise 4.2 for an example of a metric on the plane that is not induced by a norm.

One key property of norms is translation invariance, meaning $d_p(\boldsymbol{u}, \boldsymbol{v}) = d(\boldsymbol{u} - \boldsymbol{w}, \boldsymbol{u} - \boldsymbol{w})$ for any vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$. This means the geometry is the same everywhere, and in particular we are free to center everything at the origin. Moreover, we can infer everything we need about the geometry from the shape of the unit circle.

Note that we can now better justify our definition of the infinity norm $\|(a, b)\|_\infty = \max\{|a|, |b|\}$: this is easily seen to be a norm whose unit circle is the square with vertices at $\pm(1, 1)$ and $\pm(-1, 1)$, the same square to which the unit p-norm circles are clearly converging. Since the circles are the same, the norms must be the same.

We also get that the shortest path between two points is a straight line in a metric induced by a norm. To see this, consider the line segment from the origin (which we just observed is sufficient) to any vector $\boldsymbol{v}$. The length of this segment is $\|\boldsymbol{v}\|_p$ and if we deviate from a straight line in any way, the triangle inequality says that this path can only get longer (unless $p = 1$, in which case there are multiple options in addition to moving straight). The similarities between Euclidean geometry and the other p-metrics are formally captured by the following definition.

Definition 1

Two norms $\|\cdot\|_A$ *and* $\|\cdot\|_B$ *on a vector space* V *are* equivalent *if there is a constant* C *such that* $\frac{1}{C}\|\boldsymbol{v}\|_A \leq \|\boldsymbol{v}\|_B \leq C\|\boldsymbol{v}\|_A$ *for all* $\boldsymbol{v} \in V$.

Figure 18.1 illustrates why all p-norms are equivalent. The p-norms all agree on the coordinate axes and deviate most along the line $y = x$. The worst-case scenario is thus represented by the length of the vector $\boldsymbol{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Since $\|\boldsymbol{u}\|_1 = 2$ and $\|\boldsymbol{u}\|_\infty = 1$, we have that they are equivalent with $C = 2$. Thus all of the p-norms are not only equivalent, they are equivalent with a modest constant.

This shows that p-norm geometry is not so different from Euclidean geometry in a broad sense. So what is the source of all of this imperfection?

The problem is that there is no scalar product. Recall that the Euclidean norm has a convenient operation $\boldsymbol{v} \cdot \boldsymbol{w}$ that returns a scalar with a nice geometric meaning, $\boldsymbol{v} \cdot \boldsymbol{w} = \|\boldsymbol{v}\|\|\boldsymbol{w}\| \cos\theta$. This is a handy way to recover the angle between vectors, but we can instead turn this around and view $\frac{\boldsymbol{v}\cdot\boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}$ as the *definition* of the cosine, and by implication the definition of angle measure.

A consequence of the reduced symmetry is that no such operation exists for $p \neq 2$. This is why our new trigonometric functions can no longer capture everything, why there is no natural notion of angle, and why we must work harder to tease out geometric quantities. See Exercise 18.5 for another way to compare and contrast p-norms with symmetry.

We will mostly work with the plane, but we note that many of the significant applications of the p-norm come in other vector spaces. Perhaps most important are the infinite-dimensional vector spaces. We define the space ℓ^p to be the vector space of infinite sequences with norm $\|(x_1, x_2, \cdots)\|_p = (|x_1|^p + |x_2|^p + \cdots)^{1/p}$. Thus ℓ^1 is the set of absolutely convergent series and ℓ^∞ is the set of bounded sequences. Indeed, observe that ℓ^p contains a copy of every d-dimensional p-norm by restricting

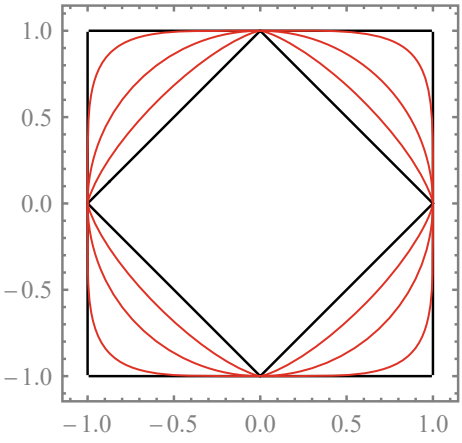


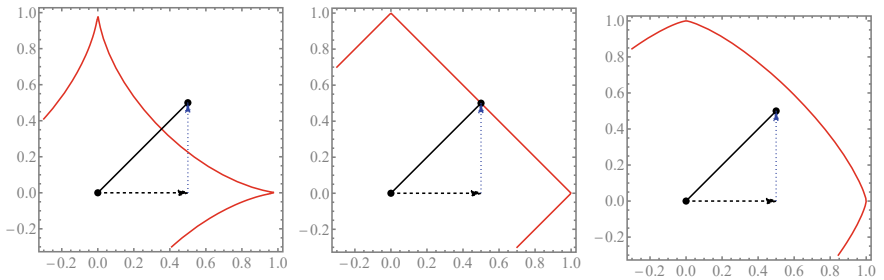
Figure 18.1 All unit p-circles lie between the 1-circle and the ∞-circle

to sequences that are 0 after d terms. Generalizing further, $L^p(S)$ is the vector space of functions defined on a measurable set S with norm $\|f\|_p = \left(\int_S |f|^p \, d\mu\right)^{1/p}$, defined for functions for which this integral is finite. These spaces are standard objects of study in functional analysis.

18.2 Convexity and Minkowski Geometry

The triangle inequality is the property that allows a distance function to capture the idea of measurement. Equation 18.1 gives a norm if and only if $p \geq 1$ (although we will sometimes look at the $p < 1$ case and abuse terminology by still referring to it as a norm, even though *seminorm* is more correct). To see what goes wrong if $p < 1$, consider Figure 18.2. The vectors $\boldsymbol{u} = (0, \frac{1}{2})$ and $\boldsymbol{v} = (\frac{1}{2}, 0)$ are shown with their sum $\boldsymbol{u} + \boldsymbol{v}$ along with unit circles for $p < 1$, $p = 1$, and $p > 1$. Note that $\boldsymbol{u}$ and $\boldsymbol{v}$ have p-length $\frac{1}{2}$ for any p, so the sum of their norms is 1. The triangle inequality fails if $p < 1$ because the unit circle curves inward and the vector sum extends beyond it. At $p = 1$ the sum lands precisely on the circle, giving the edge case of the triangle inequality becoming an equality. As p grows larger than 1, we have the proper picture of the triangle inequality.

What specifically happens at $p = 1$ is that the unit ball[1] becomes *convex*, which means any line segment connecting two points in the set remains in the set. The three cases can be seen in Figure 18.2 by considering the segment connecting $(0, 1)$ to $(1, 0)$: the segment escapes the unit ball if $p < 1$, coincides with the circle if $p = 1$, and is contained within the ball if $p < 1$. We now show that convexity is precisely what makes norms work, not just in the case of the p-norm.



(a) $p = 2/3$ (b) $p = 1$ (c) $p = 4/3$

Figure 18.2 The sum of two vectors with three p-circles, showing the triangle inequality is satisfied if and only if $p \geq 1$

[1] We use the term *unit ball* to mean the set of vectors whose distance from the origin is less than or equal to one. The word *disk* would be more intuitive for the plane, but this argument concerns general vector spaces.

Definition 2

A set of vectors S is convex *if $\boldsymbol{u}, \boldsymbol{v} \in S$ implies $\lambda\boldsymbol{u}+(1-\lambda)\boldsymbol{v} \in S$ for all $0 \leq \lambda \leq 1$.*

In other words, the line segment between two points in a convex set lies entirely within the set.

Theorem 18.1

Suppose $\|\cdot\|_v : V \to \mathbb{R}$ is absolutely homogeneous and positive definite. Then v satisfies the triangle inequality (i.e., $\|\cdot\|_v$ is a norm) if and only if the unit ball $B = \{\boldsymbol{v} \in W \mid \|\boldsymbol{v}\|_v \leq 1\}$ is convex.

Proof. Suppose $\|\cdot\|_v$ satisfies the triangle inequality and let $\boldsymbol{u}, \boldsymbol{v} \in B$ and $0 \leq \lambda \leq 1$. Then

$$\begin{aligned}\|\lambda\boldsymbol{u} + (1-\lambda)\boldsymbol{v}\|_v &\leq \|\lambda\boldsymbol{u}\|_v + \|(1-\lambda)\boldsymbol{v}\|_v \\ &= \lambda\|\boldsymbol{u}\|_v + (1-\lambda)\|\boldsymbol{v}\|_v \\ &\leq \lambda + (1-\lambda) = 1\end{aligned}$$

and thus any point on the segment connecting $\boldsymbol{u}$ to $\boldsymbol{v}$ lies in B.

Conversely, suppose B is convex and assume $\boldsymbol{u}, \boldsymbol{v} \in B$, not both the zero vector. Then

$$\frac{\boldsymbol{u}+\boldsymbol{v}}{\|\boldsymbol{u}\|_v + \|\boldsymbol{v}\|_v} = \underbrace{\frac{\|\boldsymbol{u}\|_v}{\|\boldsymbol{u}\|_v + \|\boldsymbol{v}\|_v}}_{0\leq\lambda\leq 1} \cdot \underbrace{\frac{\boldsymbol{u}}{\|\boldsymbol{u}\|_v}}_{\in B} + \underbrace{\frac{\|\boldsymbol{v}\|_v}{\|\boldsymbol{u}\|_v + \|\boldsymbol{v}\|_v}}_{0\leq 1-\lambda\leq 1} \cdot \underbrace{\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_v}}_{\in B}.$$

We have thus expressed $\frac{\boldsymbol{u}+\boldsymbol{v}}{\|\boldsymbol{u}\|_v+\|\boldsymbol{v}\|_v}$ as a point lying on a line segment between two points in B and therefore by convexity must also lie in B. Thus $\frac{\|\boldsymbol{u}+\boldsymbol{v}\|_v}{\|\boldsymbol{u}\|_v+\|\boldsymbol{v}\|_v} \leq 1$, or $\|\boldsymbol{u}+\boldsymbol{v}\|_v \leq \|\boldsymbol{u}\|_v + \|\boldsymbol{v}\|_v$. □

This invites a broader view. We have demonstrated in this chapter that norms can be entirely understood from the shapes of their unit circles, and any convex simple closed curve encircling the origin defines a norm. Geometries like this are called *Minkowski geometries*, introduced in 1896 by Hermann Minkowski [75].[2]

We have been looking specifically at p-norms and the functions that parameterize their circles. More generally, we could look at any periodic functions that parameterize a curve like this as the trigonometric functions of *some* geometry whose unit circle is the chosen curve. Ellipse geometry, which we saw in some exercises, is another example of this.

If we view p-metrics as a way to explore general geometric ideas with the safety of Euclidean geometry gone but still in sight, we might view Minkowski geometry

[2] This is not to be confused with the geometry of spacetime, which unfortunately is also called Minkowski geometry.

as where that journey might lead.[3] See Exercise 19.7 and Project 19.1 for further explorations.

As another application of convexity, we resolve an outstanding claim from Chapter 4.

Theorem 18.2

If $1 < p < \infty$, then two p-circles intersect at no more than two points.

Proof. Suppose A, B, and Z are distinct points lying on the intersection of circles $\mathcal{C}$ and $\mathcal{C}'$ with centers O, O' and r, r' respectively. Assume the points are labeled so that the line OZ intersects the segment AB at some point X and that Z lies on the same side of the segment AB as O. See Figure 18.3 (left). Since $\mathcal{C}$ is convex, $r + |ZX| = |OZ| + |ZX| = |OX| \leq r$. Thus $|ZX| = 0$ and Z lies on the segment AB. The symmetric argument for Z on the opposite side of AB gives that A, B, and Z are collinear.

But then the entire segment AB must be on $\mathcal{C}$ (and $\mathcal{C}'$). To see this, suppose there were some point J on AZ such that the distance from O to J is less than r (it cannot be greater than r by convexity). Extend OJ to a point $\tilde{J}$ so that $|O\tilde{J}| = r$, and thus lies on $\mathcal{C}$. But then by construction the line $\tilde{J}B$ is not AB, and in particular it does not contain Z. Extend OZ to intersect JB at $\tilde{Z}$. But then $|O\tilde{Z}| > |OZ| = r$ and $\tilde{Z}$ does not lie in the disk bounded by $\mathcal{C}$, a contradiction to convexity since it lies on a segment connecting two of its points. See Figure 18.3 (right).

Then $\mathcal{C}$ must contain a line segment, which can be easily proved impossible with elementary calculus on $(\cos_p(t), \sin_p(t))$ (or any other parameterization) by observing the derivative is never constant on an interval (hence the exclusion of $p = 1$, where the theorem is not true). □

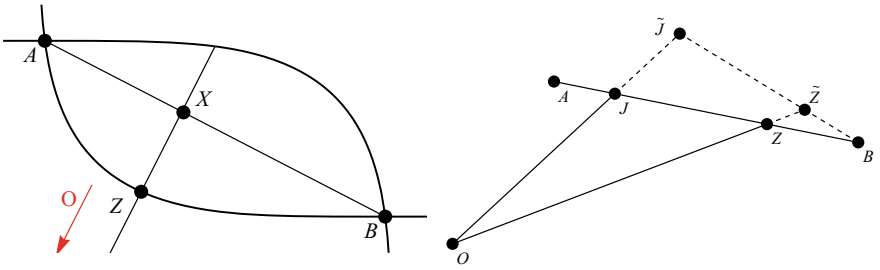


Figure 18.3 Proof of Theorem 18.2

[3] One direction to continue the journey is toward *Finsler geometry* which adapts differential geometry to accommodate Minkowski metrics.

Note that we actually proved something a bit stronger since the only properties of the p-circle that we used are convexity and that it contains no segments. For example, the argument holds if $\mathcal{C}$ is a p-circle and $\mathcal{C}'$ is a q-circle.

Exercises

▶ **Exercise 18.1** *Show that the Paris metric of Exercise 4.2 is not defined by a norm.*

▶ **Exercise 18.2** *Show that for any non-zero vector $\boldsymbol{v} \in \mathbb{R}^2$, $\|\boldsymbol{v}\|_p$ is a decreasing function of p.*

▶ **Exercise 18.3** *Show that norm equivalence is an equivalence relation. Is 2-equivalence an equivalence relation?*

▶ **Exercise 18.4** *Find a function $C(p, q)$ that returns the smallest value of C such that the p- and q-norms are equivalent.*

▶ **Exercise 18.5** *Find all 2×2 real matrices that preserve the p-norm, i.e. such that $\|A\boldsymbol{v}\|_p = \|\boldsymbol{v}\|_p$. These are the* linear isometries *of the vector space. Conclude that all p-norms for $p \neq 2$ have the same set of isometries, which is a subset of the set of Euclidean linear isometries.*[4]

▶ **Exercise 18.6** *The p-circles share common points only on the coordinate axes. What is the analog of this phenomenon in ℓ^p?*

Projects

Project 18.1 (p-non-norms)
The discussion in this chapter invites explorations of the non-metric cases where $p < 1$. We've offered $p = 1/2$ as a recurring example. What else goes wrong if $p < 0$?

But even though these non-metric cases are troublesome geometrically, they still make at least some sense analytically. We can still define and parameterize their "circles" with squigonometric functions. We can still define geometric curves like conics, pedal curves, and duals. What are they? Do they have any geometric meaning? Can the geometry enlighten analytic properties of their

[4] The upshot of Exercise 18.5 is that the transformational view of geometry has little to offer us. We will thus bypass a full discussion of isometries (e.g., [97]) and note that the isometry group of the general p-norm sits in the Euclidean group just as a dihedral group (the symmetries of the square) sits in the group of orthogonal matrices (the symmetries of the circle).

parameterizations, or vice versa? For example, consider the "-1-squircle" of Figure 4.7.

$p = 0$ is an interesting case to consider. Can you define the 0-"norm" in a way that is consistent with the limiting shape of p-circles as $p \to 0^+$? (The definition of ℓ_0 we are suggesting here is called the *Hamming distance*. It is not compelling in the plane, but the general idea is useful in computer science.)

The truly bold might want to examine complex p, but here be dragons.

Project 18.2 (Euclid and *p*-metrics)

Euclid's *Elements* [50] develop plane geometry from first principles. Could we do this for p-norm geometry? That is, can we find a consistent and complete set of axioms that model (any or all) p-norm geometries?

Expect much of the challenge to come in navigating the common notions like "line" and "angle," which we have seen can be tricky to define. More modern efforts to axiomatize geometry (e.g., Hilbert [51] and Birkhoff [11]) will provide more stable formal ground, but there is no avoiding these fundamental ambiguities in the p-metric. (See [74] for an introduction to this approach.)

It may be more practical to start farther down the road and look at how the core theorems are proved. For example, Euclid's proof that two distinct circles intersect in at most two points is as follows. Suppose two circles intersect at A, B, and C. Then the perpendicular bisector ℓ of AB is the set of points equidistant from A and B, so the centers of both circles must lie on ℓ. But by the same reasoning, the centers must also lie on the perpendicular bisector (the midset) m of BC. But ℓ and m are distinct lines and can only meet at one point. Thus the circles have the same center, providing a contradiction to the circles being distinct.

The key step in this proof was that the midsets of two points is a line, and distinct lines can only intersect once. To adapt this result to the p-metric (which we proved in Theorem 18.2 with a convexity argument), we need only introduce the assumption that the midsets of two points can only intersect once.



Duality

19

We defined p-metrics to be conjugate if $\frac{1}{p} + \frac{1}{q} = 1$. With normed vector spaces in hand we are now able to explore the geometric connections between conjugate metrics. We begin with a formal definition and geometric construction of duality that wil bring us back to our squigonometric functions, along with a promised proof that $\pi_p^{\text{arc}} = \pi_q^{\text{arc}}$ and a geometric examination of Hölder's inequality in the plane.

19.1 The dual norm and pedal curves

We are now equipped to formally explore dual norms, which are established by the following

Theorem 19.1

If $\|\cdot\|$ is a norm on the plane, then

$$\|\boldsymbol{v}\|_* = \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u} \cdot \boldsymbol{v}| \tag{19.1}$$

also defines a norm on the plane.

The right-hand side of (19.1) is read as the *supremum*, or *least upper bound*, of the set of possible sizes of the dot product $\boldsymbol{u} \cdot \boldsymbol{v}$, where $\boldsymbol{u}$ is any unit vector in the plane.

Proof. Positivity is trivial. If $\|\boldsymbol{v}\|_* = 0$ then $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ for any vector $\boldsymbol{u}$ in the unit circle for the norm, implying $\boldsymbol{v} = 0$ and thus definiteness. Absolute homogeneity is immediate from elementary properties of the dot product:

$$\|\lambda \boldsymbol{v}\|_* = \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u} \cdot \lambda \boldsymbol{v}| = |\lambda| \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u} \cdot \boldsymbol{v}| = \lambda \|\boldsymbol{v}\|_*.$$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_19

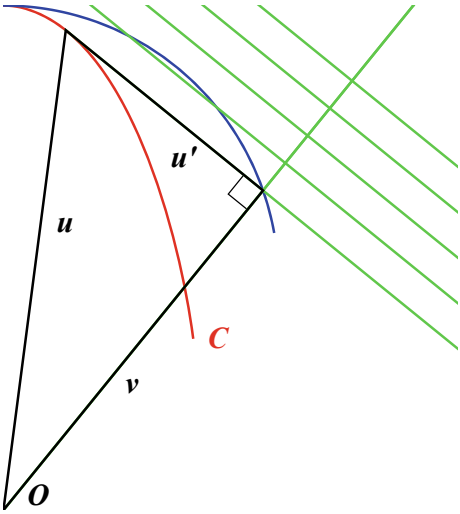


Figure 19.1 The pedal curve of a curve C

Finally, the triangle inequality comes from properties of the supremum.

$$\begin{aligned}\|\boldsymbol{v}+\boldsymbol{w}\|_* &= \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u}\cdot(\boldsymbol{v}+\boldsymbol{w})| = \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u}\cdot\boldsymbol{v}+\boldsymbol{u}\cdot\boldsymbol{w}| \\ &\leq \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u}\cdot\boldsymbol{v}| + \sup_{\|\boldsymbol{u}\|=1} |\boldsymbol{u}\cdot\boldsymbol{w}| = \|\boldsymbol{v}\|_* + \|\boldsymbol{w}\|_*.\end{aligned}$$

□

We will see that dual spaces encode much about one another, and our goal will be to describe the dual of the p-norm. We will take that up in the next section after we develop a geometric procedure to construct the dual for a general planar norm.

Our geometric development of the dual circle makes use of a construction called the *pedal curve*. We begin with a curve C and a point O called the pedal point. By assuming O is the origin (allowing us to refer to "the" pedal curve of C) we may consider the points on C to be parameterized by position vectors $\boldsymbol{u}(t)$ that also represent the segment to the pedal point.

A point on the pedal curve P is obtained by extending a tangent line at $\boldsymbol{u}$ to an orthogonal vector $\boldsymbol{p}$. The set of all such $\boldsymbol{v}$ forms the pedal curve. From Figure 19.1, we see that $\boldsymbol{v}$ is the orthogonal projection of $\boldsymbol{u}$ onto $\boldsymbol{u}'$.

This definition leads directly to a parameterization of pedal curves if the original curve has a differentiable parameterization. Suppose $\boldsymbol{u}(t) = (f(t), g(t))$. Then $\boldsymbol{u}'(t) = (f'(t), g'(t))$ with normal vector $\boldsymbol{u}'_\perp(t) = (-g'(t), f(t))$. The orthogonal projection can be obtained from the usual dot product by

$$\begin{aligned}\boldsymbol{v} = \frac{\boldsymbol{u}\cdot\boldsymbol{u}'}{\|\boldsymbol{u}'_\perp\|^2}\boldsymbol{u}'_\perp &= \frac{(f(t),g(t))\cdot(-g'(t),f'(t))}{f'(t)^2+g'(t)^2}(-g'(t),f'(t)) \\ &= \frac{f'(t)g(t)-g'(t)f(t)}{f'(t)^2+g'(t)^2}(-g'(t),f'(t))\end{aligned} \tag{19.2}$$

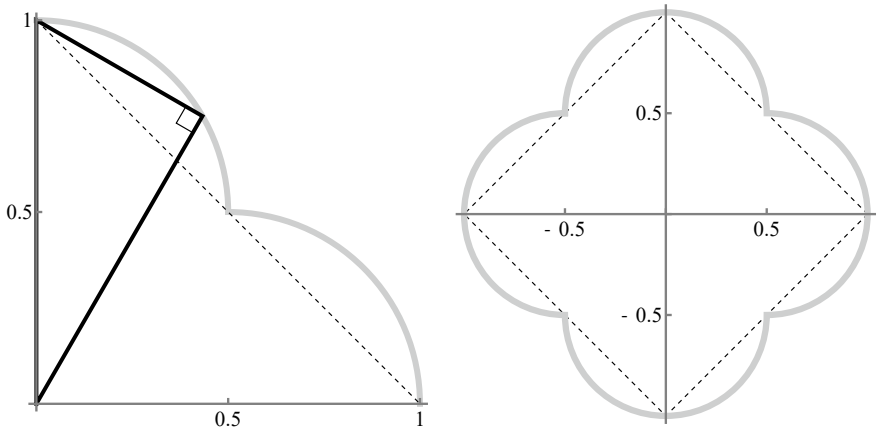


Figure 19.2 The 1-circle (dashed) and its pedal curve (gray), formed from right triangles with hypotenuses on the coordinate axes

In particular, the squigonometric functions that parameterize p-circles can be used to parameterize their pedal curves. Note that the extremal cases $p = 1$ and $p = \infty$ are not differentiable, but the construction still works. Referring to Figure 19.1, we can think of $\boldsymbol{u}'$ as "sliding" the perpendiculars down until they hit the curve. If $p = 1$, instead of a true tangent the line we will hit a vertex. Thus the right triangle in Figure 19.1 always has as its hypoteneuse the segment from the origin to a vertex, and so the feet of these perpendiculars must trace out circles (Figure 19.2). The $p = \infty$ case is the same argument rotated by 45°.

We now illustrate how to connect pedal curves to dual circles. Suppose we have a norm $\|\cdot\|$ on the plane and let $S = \{\boldsymbol{u} \in \mathbb{R}^2 \mid \|\boldsymbol{u}\| = 1\}$ be the unit circle in this norm, and denote by $\|\cdot\|_*$ and S^* the dual norm and unit circle, respectively. Choose a unit (in the Euclidean norm) direction vector $\hat{\boldsymbol{v}}$. We want to find a positive real number c such that $c\hat{\boldsymbol{v}}$ is on the unit dual circle S^*. By the definition of dual norm (19.1) and compactness of the circle, there exists a $\boldsymbol{u}$ in S such that $\boldsymbol{u} \cdot c\,\hat{\boldsymbol{v}} = \|c\,\hat{\boldsymbol{v}}\|_* = 1$. Express $\boldsymbol{u}$ in components relative to $\boldsymbol{v}$ so that $\boldsymbol{u} = a\,\hat{\boldsymbol{v}} + b\,\boldsymbol{v}_\perp$, where $\boldsymbol{v}_\perp$ is orthogonal to $\boldsymbol{v}$. Then

$$1 = \boldsymbol{u} \cdot c\,\hat{\boldsymbol{v}} = a\,\hat{\boldsymbol{v}} \cdot c\,\hat{\boldsymbol{v}} = ac.$$

Thus $c = \frac{1}{a}$ where a is the coefficient of the component of $\boldsymbol{u}$ parallel to $\boldsymbol{v}$. Since $\boldsymbol{u}$ is defined to be the point on S that maximizes this product, we can find it by sliding perpendiculars back from infinity along the ray defined by $\hat{\boldsymbol{v}}$ until it firsts meets S at a tangent. That will locate $\boldsymbol{u}$, the point on S with the largest component parallel to $\hat{\boldsymbol{v}}$. But by definition, the point $\boldsymbol{p} = a\,\hat{\boldsymbol{v}}$ on the ray that intersects this tangent is a point on the pedal curve. If we reciprocate this length we get $c\hat{\boldsymbol{v}} = \frac{1}{a}\hat{\boldsymbol{v}}$, the point on S^* parallel to $\hat{\boldsymbol{v}}$ that we were after.

The operation of reciprocating the length of a vector is called *inversion* through the Euclidean unit circle. We have thus proved the following

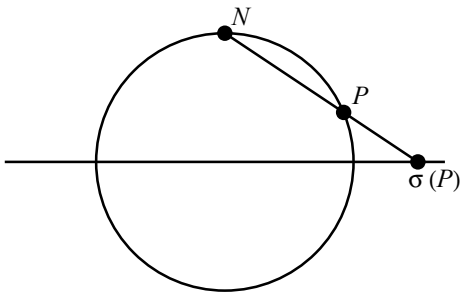


Figure 19.3 Stereographic projection of a circle to a line

Theorem 19.2

Let S be the unit circle of a norm on the plane. Then the unit circle S^* of the dual norm is the inversion of the pedal curve of S through the Euclidean unit circle.

This theorem gives us a geometric way to construct the dual, which can be better visualized if we move from the plane to the sphere $\mathbb{S}^2$ via *stereographic projection*, which is defined as follows.

We begin with the projection of a circle onto a line. Let $\mathbb{S}^1 = \{(x, y) \mid x^2 + y^2 = 1\}$ and let $N = (0, 1)$. For any point $P \neq N$ in $\mathbb{S}^1$, draw the line from N through P. The point at which this line intersects the axis $y = 0$ is the stereographic projection of the circle onto the line and we denote the x-coordinate of this point by $\sigma_1(P)$. See Figure 19.3. For a point (u, v) on $\mathbb{S}^1$, we can find an explicit formula by parameterizing the line as $(0, 1)(1 - t) + (u, v)t$. Setting the second coordinate to 0 gives $t = \frac{1}{1-v}$ and thus the line passes through $(\frac{u}{1-v}, 0)$.

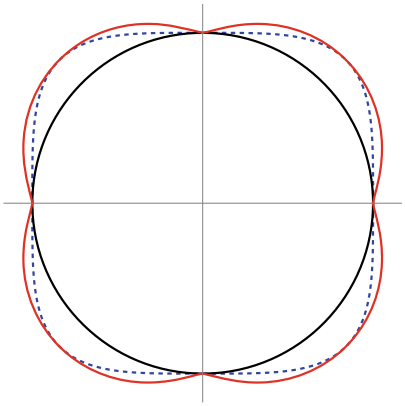
This procedure naturally generalizes to $\mathbb{S}^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$ (and higher dimensions) by taking the line through $N = (0, 0, 1)$ and any point on the sphere. A similar calculation gives the formula

$$\sigma_2(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z}\right) = \frac{1}{1-z}(x, y). \tag{19.3}$$

The reader may verify the following properties of the map $\sigma_2 : \mathbb{S}^2 - \{N\} \to \mathbb{R}^2$:

- σ_2 is bijective.
- Great circles through N map to lines through the origin.
- The equator $z = 0$ maps to the unit circle, the lower hemisphere $z < 0$ maps to the unit disk, and the upper hemisphere maps to the exterior of the unit disk.

The last point will help us with inversion. Consider a point $P(x, y, z)$ on the sphere along with its reflection through the xy-plane $P'(x, y, -z)$. Then $\sigma(P') = \frac{1}{1+z}(x, y)$



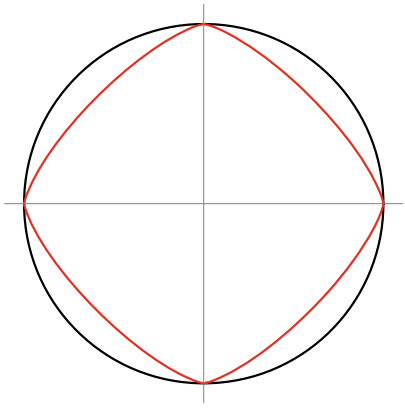
(a) The 4-circle (red) and its pedal curve (blue)



(b) Stereographic projection of the pedal curve



(c) Reflection through the equator



(d) Stereographically project back to get the dual circle

Figure 19.4 Finding the dual circle by projecting and reflecting the pedal curve

is parallel to $\sigma(P)$ and

$$\begin{aligned}\|\sigma(P)\| \cdot \|\sigma(P')\| &= \frac{1}{(1+z)(1-z)}\|(x,y)\|^2 \\ &= \frac{1}{1-z^2}(x^2+y^2) = \frac{x^2+y^2}{x^2+y^2} = 1.\end{aligned}$$

Therefore P and P' are inversions of one another through the unit circle and we can visualize the dual by calculating the pedal curve, projecting to the sphere, reflecting through the equator, and projecting back. See Figure 19.4.

We are now equipped to provide a squigonometric proof that conjugate p-norms really are dual in the sense that we have defined.

Theorem 19.3

The p-circle is the dual of the q-circle, where $\frac{1}{p} + \frac{1}{q} = 1$.

Proof. Parameterize the p-circle with $(\cos_p(t), \sin_p(t))$ and apply equation (19.2) to obtain a parameterization of the pedal curve. We restrict our calculation to the first quadrant where the p-sine and p-cosine are positive, our usual strategy for avoiding problems with potentially fractional exponents.

$$\begin{aligned} P &= \frac{f'(t)g(t) - g'(t)f(t)}{f'(t)^2 + g'(t)^2}\,(-g'(t),\, f'(t)) \\ &= \frac{-\sin_p(t)^{p-1}\sin_p(t) - \cos_p(t)^{p-1}\cos_p(t)}{\sin_p(t)^{2(p-1)} + \cos_p(t)^{2(p-1)}}\,\left(-\cos_p(t)^{p-1}, -\sin_p(t)^{p-1}\right) \\ &= \frac{1}{\sin_p(t)^{2(p-1)} + \cos_p(t)^{2(p-1)}}\,\left(\cos_p(t)^{p-1}, \sin_p(t)^{p-1}\right) \end{aligned}$$

We calculate the magnitude of this point as

$$\begin{aligned} \|P\|^2 &= \frac{1}{(\sin_p(t)^{2(p-1)} + \cos_p(t)^{2(p-1)})^2}\,\left(\cos_p(t)^{2(p-1)} + \sin_p(t)^{2(p-1)}\right) \\ &= \frac{1}{\sin_p(t)^{2(p-1)} + \cos_p(t)^{2(p-1)}}. \end{aligned}$$

By Theorem 19.2 the dual circle is the inversion of this curve and is parameterized by

$$\frac{P}{\|P\|^2} = \left(\cos_p(t)^{p-1}, \sin_p(t)^{p-1}\right).$$

We know this parameterizes the conjugate q-circle by Theorem 4.2.

Finally, we extend beyond the first quadrant by observing that the pedal curve construction and inversion respect the symmetry of the p-circle and the corresponding symmetries in the functions in our calculation.

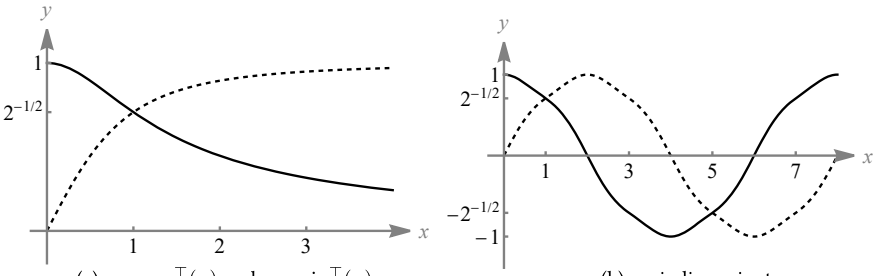
The extension to $p = 1$ and $p = \infty$ requires only technical attention to the non-differentiable points. □

A corollary is that the p-norm and q-norm are duals of each other, and thus that these spaces are *reflexive*, meaning they are isometric to the duals of their duals.

19.2 Tangential parameterization

We now define a geometrically-inspired parameterization that is particularly well-suited to exploring duality.

Let $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$ and define



(a) $y = \cos_2^\top(x)$ and $y = \sin_2^\top(x)$ (b) periodic variant

Figure 19.5 Graphs of the $p = 2$ tangential parameterization and periodic variant obtained by reflection. Cosine graphs are solid and sine graphs are dashed

$$\cos_p^\top(t) = (t^q + 1)^{-\frac{1}{p}}$$

$$\sin_p^\top(t) = (t^{-q} + 1)^{-\frac{1}{p}}$$

for $t > 0$. The reader is invited to verify the following

Theorem 19.4

The functions $\cos_p^\top(t)$ *and* $\sin_p^\top(t)$ *satisfy the following for* $t > 0$*:*

1. $\sin_p^\top(t) = \cos_p^\top(\frac{1}{t}) > 0$
2. $\sin_p^\top(t)^p + \cos_p^\top(t)^p = 1$
3. $-\frac{(\cos_p^\top)'(t)}{(\sin_p^\top)'(t)} = \left(\frac{\sin_p^\top(t)}{\cos_p^\top(t)}\right)^{p-1} = t$
4. $\lim_{t\to 0^+} \sin_p^\top(t) = \lim_{t\to\infty} \cos_p^\top(t) = 0$ *and* $\lim_{t\to\infty} \sin_p^\top(t) = \cos_p^\top(0) = 1$
5. $(\sin_p^\top)^{-1}(t) = (t^{-p} - 1)^{-\frac{1}{q}}$ *and* $(\cos_p^\top)^{-1}(t) = (t^{-p} - 1)^{\frac{1}{q}}$

We extend the definition to $t = 0$ in the obvious way and call this the *tangential parameterization* because the parameter t is the negative reciprocal of the slope of the tangent line. However, this definition requires all of $t \geq 0$ to cover the first quadrant of the p-circle and thus these functions cannot naturally be extended to a periodic function. (We can somewhat unnaturally manage this with a variant that exploits the eight-fold symmetry of the squircle, using the tangential parameterization for $0 \leq t \leq 1$ to get the first eighth and reflecting accordingly. In this case, the tangent will be encoded in the parameter modulo 1. See Figure 19.5.)

We will now explore how Keller and Vakil used this paramaterization in [54] to obtain one of our outstanding results.

Theorem 19.5

If $p, q \geq 1$ and $\frac{1}{p} + \frac{1}{q} = 1$ then $\pi_p^{arc} = \pi_q^{arc}$.

Proof. The reader may confirm the $p = 1$ and $p = \infty$ cases, allowing us to assume our parameterizations are differentiable.

Let $\boldsymbol{u}(t) = \begin{pmatrix} \cos_p^\top(t) \\ \sin_p^\top(t) \end{pmatrix}$, $\boldsymbol{v}(t) = \begin{pmatrix} \cos_q^\top(t) \\ \sin_q^\top(t) \end{pmatrix}$, and its perpendicular $\boldsymbol{v}_\perp(t) = \begin{pmatrix} -\sin_p^\top(t) \\ \cos_p^\top(t) \end{pmatrix}$. Define $f(t) = \boldsymbol{u}(t) \cdot \boldsymbol{v}_\perp(t)$ so that $f'(t) = \boldsymbol{u}'(t) \cdot \boldsymbol{v}_\perp(t) + \boldsymbol{u}(t) \cdot \boldsymbol{v}'_\perp(t)$. The key step is the following calculation which is left for the reader to confirm (or consult [54]):

$$\|\boldsymbol{u}'(t)\|_p = \boldsymbol{u}'(t) \cdot \boldsymbol{v}_\perp(t) = \frac{q}{p}(t^q + 1)^{-\frac{1}{p}-1}(t^p + 1)^{\frac{1}{p}} t^{\frac{1}{p-1}-1} \tag{19.4}$$

and similarly

$$\|\boldsymbol{v}'_\perp(t)\|_q = -\boldsymbol{v}'_\perp(t) \cdot \boldsymbol{u}(t) = \frac{p}{q}(t^p + 1)^{-\frac{1}{q}-1}(t^q + 1)^{\frac{1}{q}} t^{\frac{1}{q-1}-1}. \tag{19.5}$$

Thus

$$f'(t) = \boldsymbol{v}'_\perp(t) \cdot \boldsymbol{u}(t) + \boldsymbol{u}'(t) \cdot \boldsymbol{v}_\perp(t) = \|\boldsymbol{u}'(t)\|_p - \|\boldsymbol{v}'_\perp(t)\|_q. \tag{19.6}$$

We now integrate (19.6) from 0 to ∞. Recall that $\boldsymbol{u}$ parameterizes the first quadrant of the unit p-circle over this interval and thus $\int_0^\infty \|\boldsymbol{u}'(t)\|_p \, dt$ is the p-arclength of this curve, or $\frac{1}{4}\pi_p^{arc}$, and similarly $\int_0^\infty \|\boldsymbol{v}'_\perp(t)\|_q \, dt = \frac{1}{4}\pi_q^{arc}$.

We also note that $f(0) = \boldsymbol{u}(0) \cdot \boldsymbol{v}_\perp(0) = 0$ because the parameterizations $\boldsymbol{u}$ and $\boldsymbol{v}$ both start at the horizontal axis and are therefore parallel. They also both end parallel to the vertical axis so the same is true at ∞. Altogether we have

$$\int_0^\infty f'(t) \, dt = f(\infty) - f(0) = 0 = \frac{1}{4}\pi_p^{arc} - \frac{1}{4}\pi_q^{arc},$$

proving the theorem. □

That proof was mostly computational, so let's look at the geometry. Figure 19.6 illustrates the case $q = 4$, $p = \frac{4}{3}$. The vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are positioned with $t = 1/2$, so they are the points on the two squircles whose tangents have slope -2. As t increases, $\boldsymbol{v}(t)$ will be "ahead" of $\boldsymbol{u}(t)$, which will catch up and then pass at $t = 1$. The situation then precisely reverses for $t > 1$ (Theorem 19.4, item 1) and then line up again at $t = \infty$.

We now interpret $f(t)$ two ways. As $f(t) = \int_0^t \|\boldsymbol{u}'(u)\|_p - \|\boldsymbol{v}'(u)\|_q \, du$ we may interpret it as the cumulative signed difference in the p- and q-lengths of the arcs respective circles subtended at the points ($\boldsymbol{u}$ and $\boldsymbol{v}$) whose tangent has slope $-1/t$. But we also have $f(t) = \boldsymbol{u} \cdot \boldsymbol{v}_\perp$, which is twice the signed area of the shown triangle with vertices at $\boldsymbol{u}$,$\boldsymbol{v}$, and the origin.[1] Thus the arclengths are the same when this triangle degenerates at infinity, but these arclengths are not in lockstep.

[1] This is an elementary result of linear algebra, when we recall that "twice the area of the triangle" is the same as the area of the parallelogram they span. The determinant of the matrix with columns $\boldsymbol{u}$ and $\boldsymbol{v}$ reports just such an area, or equivalently we may use the cross-product to get the same formula.

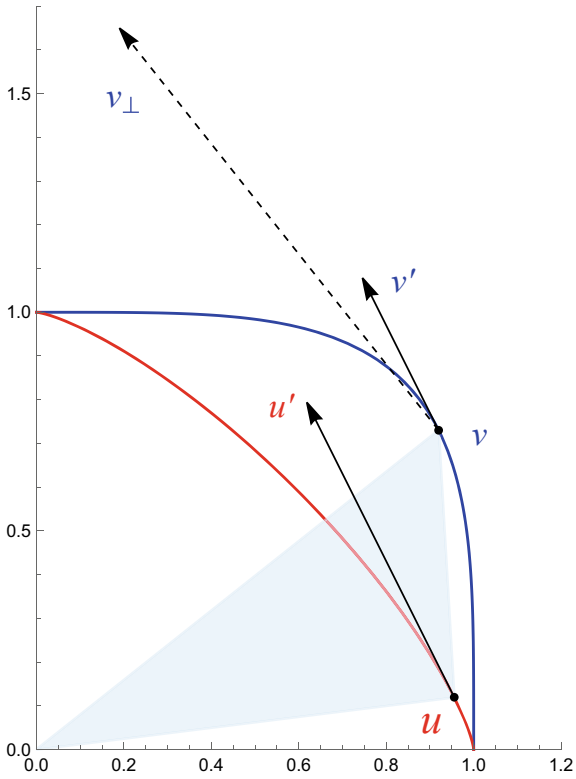


Figure 19.6 Theorem 19.5 illustrated for $p = 4/3$, $q = 4$, $t = 1/2$

Note that if $p = 2$, $\boldsymbol{v}_\perp = \boldsymbol{v}'$ and the arclengths are in lockstep because the triangle is always degenerate. This is because the radius of a Eucildean circle is perpendicular to its tangent. Recalling the discussion from Section 3.2, this relationship is off by a power of $p - 1$ in general. Observe that we could easily use the reciprocal symmetry of Theorem 19.4 to parameterize by the slope of the tangent but we would need to abandon our convention of starting our parameterizations by moving counterclockwise from the horizontal axis.

19.3 Hölder's inequality and friends

This section will survey some of the core theorems of analysis through our particular lens of planar geometry. These results generalize far beyond the plane and are key tools in the infinite dimensional vector spaces of functional analysis. By narrowing

our scope to the plane, however, we can offer a more hands-on development that reveals more of the underlying geometry. In particular, we will see what may be the most important example of a Euclidean result that "splits" across dual spaces when generalized to the p-norm. The full-powered theorems and proofs can be found in numerous standard texts.

The Euclidean result of interest is the *Cauchy-Schwarz inequality*:

Theorem 19.6

(Cauchy-Schwarz inequality).
For any planar vectors $\boldsymbol{u}, \boldsymbol{v}$, $|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|$ *with equality if and only if the vectors are parallel.*

Proof. $\|\boldsymbol{u}\|^2\|\boldsymbol{v}\|^2 - |\boldsymbol{u} \cdot \boldsymbol{v}|^2 = (u_1^2 + u_2^2)(v_1^2 + v_2^2) - (u_1v_1 + u_2v_2)^2 = u_1^2v_1^2 - 2u_1v_1u_2v_2 + u_2^2v_1^2 = (u_1v_1 + u_2v_2)^2 \geq 0$. Equality holds if and only if $(u_1, v_1) = k(u_2, v_2)$ for some k. □

Before proceeding to our analytic inequalities, we must formalize a connection between calculus and geometry.

Theorem 19.7

Let $f(x)$ *be twice differentiable on an interval* $[a, b]$. *If* $f''(x) < 0$ *for all* x *in* $[a, b]$, *then the secant line of the graph* $y = f(x)$ *between* a *and* b *lies below the graph. If* $f''(x) > 0$ *on the interval then it lies above the graph.*

This theorem formally articulates the geometric interpretation of a monotone second derivative. This property is a graph version of convexity, defined in Section 18.2, and a function whose second derivative is positive is indeed called *convex*. A negative second derivative, shown in in Figure 19.7, is *concave*. The proof is a mean value theorem argument and is deferred to Appendix D.

Theorem 19.8

(Young's inequality).
Let $a, b > 0$ *and* $\frac{1}{p} + \frac{1}{q} = 1$. *Then* $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.

Proof. First observe that if $f(x) = \ln x$, then $f''(x) = -\frac{1}{x^2} < 0$ for all $x > 0$. Let $t = \frac{1}{p}$, so $1 - t = \frac{1}{q}$. Then by Theorem 19.7,

$$\begin{aligned} \ln(ta^p + (1-t)b^q) &\geq t\ln(a^p) + (1-t)\ln(b^q) \\ &= pt\ln(a) + q(1-t)\ln(b) \\ &= \ln(a) + \ln(b) = \ln(ab). \end{aligned}$$

The theorem follows by exponentiating both sides. □

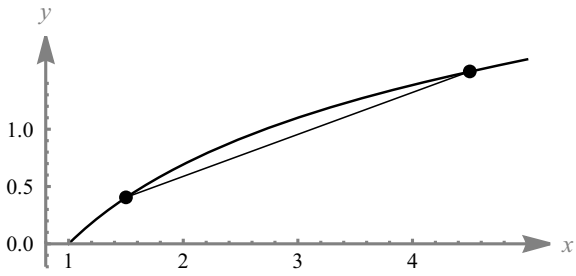


Figure 19.7 The logarithm function is concave

We are now equipped to prove Hölder's inequality. We notice that it concerns what in the Euclidean norm would be the dot product, and indeed it reduces to the Cauchy-Schwarz inequality (Theorem 19.6) there. But for general $p \neq 2$, where we have no dot product, the inequality relates the p-norm to its dual and we can see the dot product and the Cauchy-Schwarz inequality, one of the most useful tools in analysis [91], as a consequence of the self-duality of the 2-norm.

Theorem 19.9

(Hölder's inequality).
Let $\boldsymbol{x} = (x_1, x_2)$, $\boldsymbol{y} = (y_1, y_2)$, *and* $\frac{1}{p} + \frac{1}{q} = 1$, $1 \leq p \leq \infty$. *Then* $|x_1 y_1| + |x_2 y_2| \leq \|\boldsymbol{x}\|_p \|\boldsymbol{y}\|_q$.

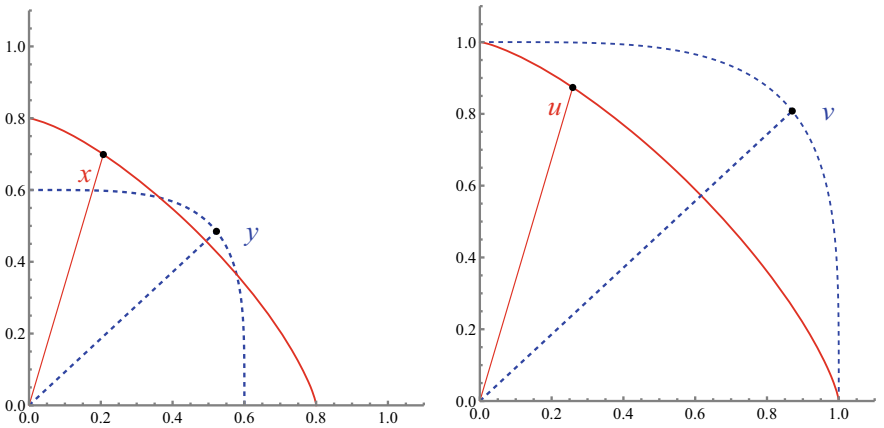
Proof. We leave the extremal case $p = 1$, $q = \infty$ to Exercise 19.8, allowing us to assume $1 < p < \infty$.

Let $\boldsymbol{u} = \frac{1}{\|\boldsymbol{x}\|_p}\boldsymbol{x}$ and $\boldsymbol{v} = \frac{1}{\|\boldsymbol{y}\|_p}\boldsymbol{y}$. Note that $\|\boldsymbol{u}\|_p = \|\boldsymbol{v}\|_q = 1$. By Young's Inequality (Theorem 19.8), $|u_1 v_1| + |u_2 v_2| \leq \frac{|u_1|^p}{p} + \frac{|v_1|^q}{q} + \frac{|u_2|^p}{p} + \frac{|v_2|^q}{q} = \frac{\|\boldsymbol{u}\|^p}{p} + \frac{\|\boldsymbol{v}\|^q}{q} = \frac{1}{p} + \frac{1}{q} = 1$. Multiply through by $\|\boldsymbol{x}\|_p \|\boldsymbol{y}\|_q$ for the desired result. □

Let's look at this proof geometrically. The vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ each lie on their own circles in the p- and q-norms. To get them in one picture, we re-scale to new vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ that lie on unit circles. We may interpret dot product as half the area of the triangle spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$ (as we did in Figure 19.6, but remember that the vectors there were linked by a common parameter and were not independent). Now we see what Young's inequality was for: it provided the bound on the area of this triangle, and it was the conjugate relationship that allowed a bound that was independent of the vectors. We recover Hölder's inequality when we scale back up (Figure 19.8).

Getting back to squigonometry and applying Hölder's inequality to our standard parameterization of the circle, $\boldsymbol{x} = (\cos_p(a), \sin_p(a))$ and $\boldsymbol{y} = (\cos_q(b), \sin_q(b))$ gives the inequality

$$|\cos_p(a)\cos_q(b)| + |\sin_p(a)\sin_q(b)| \leq 1. \tag{19.7}$$



(a) $\boldsymbol{x}$ and $\boldsymbol{y}$ lie on different circles in conjugate norms

(b) $\boldsymbol{u}$ and $\boldsymbol{v}$ re-scale the vectors so they each lie on conjugate unit circles

Figure 19.8 The proof of Hölder's inequality amounts to bounding the area of the triangle spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$

If we take $a = b$ (which really does bring us back to Figure 19.6), then we have the single-parameter version

$$|\cos_p(a)\cos_q(a)| + |\sin_p(a)\sin_q(a)| \leq 1. \tag{19.8}$$

Notice that if we consider $p = 2$ with some care with signs, we are basically looking at a different way to generalize the cosine sum formula.

Thus Hölder's inequality and its related theorems provide a source for new inequalities across conjugate squigonometric functions. Also note that they will hold for any of our various parameterizations.

Triangle inequality redux

We close by deducing the triangle inequality for the planar p-norm from Hölder's inequality. This is a planar version of a proof that generalizes to the full power of Hölder's inequality. It feels quite different from the convexity argument in Section 4.4, but remember that convexity played an essential role in proving Hölder's inequality. In this context, the triangle inequality is often called *Minkowski's inequality*.

Theorem 19.10

(Minkowski's inequality).
$\|\boldsymbol{u} + \boldsymbol{v}\|_p \leq \|\boldsymbol{u}\|_p + \|\boldsymbol{v}\|_p$

Proof.

$$\begin{aligned}
\|\boldsymbol{u}+\boldsymbol{v}\|_p^p &= |u_1+v_1|^p + |u_2+v_2|^p \\
&= |u_1+v_1|\cdot|u_1+v_1|^{p-1} + |u_2+v_2|\cdot|u_2+v_2|^{p-1} \\
&\le (|u_1|+|v_1|)\cdot|u_1+v_1|^{p-1} + (|u_2|+|v_2|)\cdot|u_2+v_2|^{p-1} \text{ (Euclidean } \triangle \text{ inequality)} \\
&\le (|u_1^p+u_2^p|^{1/p} + |v_1^p+v_2^p|^{1/p})\left(|u_1+v_1|^{q(p-1)} + |u_2+v_2|^{q(p-1)}\right)^{1/q} \text{ (Hölder's inequality)} \\
&= (|u_1^p+u_2^p|^{1/p} + |v_1^p+v_2^p|^{1/p})\left(|u_1+v_1|^{p} + |u_2+v_2|^{p}\right)^{1/q} \\
&= (\|\boldsymbol{u}\|_p + \|\boldsymbol{v}\|_p)(\|\boldsymbol{u}+\boldsymbol{v}\|_p^p)^{1/q} \\
&= (\|\boldsymbol{u}\|_p + \|\boldsymbol{v}\|_p)\|\boldsymbol{u}+\boldsymbol{v}\|_p^{p-1}
\end{aligned}$$

Divide both sides by $\|\boldsymbol{u}+\boldsymbol{v}\|_p^{p-1}$ to complete the proof. □

Exercises

▶ **Exercise 19.1** *Some authors prefer to define stereographic projection with the unit circle centered at* $(0,1)$ *and* N *at* $(0,2)$*. The image of a point* P *on the circle is still the intersection of the line through* N *and* P *with the* x*-axis. How does this change the function?*

▶ **Exercise 19.2** *Describe the unit circle in the dual to the* $p=2$ *ellipse geometry of Exercise 4.11. Use the pedal curve and inversion construction to formally calculate the parameterization. Sketch the duals for more general* p*-ellipse geometries.*

▶ **Exercise 19.3** *Prove Theorem 19.4.*

▶ **Exercise 19.4** *Prove equations (19.4) and (19.5) in the proof of Theorem 19.5.*

▶ **Exercise 19.5** *Can the tangential parameterizations be extended to* $p=1,\infty$*? What about* $0<p<1$*?*

▶ **Exercise 19.6** *Show that the pedal curve of the rectangular hyperbola* $(\cosh t, \tanh t)$ *is a lemniscate. Show that this lemniscate is also the inversion of this hyperbola.*

▶ **Exercise 19.7** *Let* ℓ *be a line through the origin. Construct a point* P *that is tangent to the unit* p*-circle and has the same slope as* ℓ*. We define the line through* P *and the origin to be* p-Minkowski orthogonal *to* ℓ*. Show that if* $\frac{1}{p}+\frac{1}{q}=1$ *and* ℓ*,* ℓ^**, and* ℓ^{**} *are lines through the origin such that* ℓ^* *is* p*-orthogonal to* ℓ *and* ℓ^{**} *is* q*-orthogonal to* ℓ^**, then* $\ell=\ell^{**}$*.*

▶ **Exercise 19.8** *Prove Hölder's inequality (Theorem 19.9) for the case* $p=1$*,* $q=\infty$*.*

▶ **Exercise 19.9** *Derive the Euclidean triangle inequality directly from the Cauchy-Schwarz inequality.*

▶ **Exercise 19.10** *Find an example that violates the expression in Hölder's inequality if we drop the requirement that* $\frac{1}{p} + \frac{1}{q} = 1$.

▶ **Exercise 19.11** *Find the variants of equations (19.7) and (19.8) if we take* $\boldsymbol{y} = (\sin_p(a), \cos_q(b))$.

Projects

Project 19.1 (Minkowski geometry)
We have often used the Minkowski perspective (Section 18.2), if not the tools, in our development of squigonometry. Because the p-metric is Minkowskian, everything about the geometry can be inferred from our study of the functions that describe its unit circle.

Many of our results can be viewed and generalized through that lens at the expense of additional machinery. Theorem 19.5, which says $\pi_p^{arc} = \pi_q^{arc}$, is one such result proved using Minkowskian techniques in [88]. But Keller and Vakil's proof in [54], which is essentially what we presented, is more elementary. Besides being more accessible, this type of argument can help ground an understanding of an idea for further generalization. There are surely other interesting stops on this journey.

Orthogonality is one place to start. We have seen how the notion of angle is limited in the p-metric, but Exercise 19.7 offers one (and there are others; also see the discussion in Project 18.2) generalization of orthogonality and a connection to duality. Explore this definition further, note the connection to the proof of Theorem 19.5, and see what other features of orthogonality can be generalized in some way. Or consider other generalizations of Euclidean ideas to the p-norm as a first step toward general Minkowski geometries. For further reading we recommend the surveys [69,70] or [31] for a more elementary invitation.

Project 19.2 (Projective and inversive geometry)
Duality is an important concept in geometry and its historical development [43]. The construction in Section 19.1 is just one approach that's particularly well-suited to our setting. More broadly, the dual arises naturally in the language of

projective geometry, wherein the lines in a space correspond to points its dual space, and vice versa. We construct the dual of parameterized curve by taking its tangent lines as points in the dual space and we could derive a formula for the parameterization of the dual curve using our generalized trigonometric functions to derive an alternate (but really the same) proof of Theorem 19.3.

Projective geometry provides general and powerful theorems that apply to conic sections and preservation of cross-ratios in Euclidean geometry. Does the projective view provide any insight into the p-metric? Certainly the Euclidean theorems depend heavily on the self-duality of the 2-norm, but perhaps there are some (presumably much less nice) results available for the p-norm.

A related concept is inversive geometry. The operation of inversion, introduced in Section 19.1 as a step in our dual construction, essentially exchanges the interior and exterior of a circle in a way that preserves angles and circles (with lines considered a special case of a circle). Thus inversive geometry can be used to elegantly prove theorems about circle configurations and angular relationships [78] particularly useful in the study of Möbius transformations and conformal geometry.[2] Is there a useful notion of inversion in a p-circle? Although the lack of a meaningful definition of angle in the p-norm is troubling, perhaps the right construction could lead to some interesting results for p-norm curves or relationships with their duals.

Project 19.3 (Hölder inequalities in squigonometry)
Hölder's inequality has myriad consequences. Start with the inequalities developed in equations (19.7) and (19.8) and exercise 19.11 and see what other squigonometric inequalities you can develop. You might also note some interesting specific cases and their geometric interpretations.

[2] Tristan Needham manages to develop most of complex analysis largely from the perspective of inversive geometry in his excellent text *Visual Complex Analysis* [79].



Analytic parameterizations

20

The generalized trigonometric functions we have examined were developed with particular purposes in mind, whether it be to give us particular information about the p-circle $|x|^p + |y|^p = 1$, or about the lemniscate or a particular ellipse. But in many applications of generalized trigonometry, these geometric motivations are not as important as their analytic properties. For example, one property of sine and cosine that we sacrificed early on was their derivative relationships. We conclude by introducing some parameterizations common in contemporary studies and viewing them with some of the tools we have developed.

20.1 Generalizing the arcsine

We recall from Chapter 1 that

$$\pi = 2\int_0^1 \frac{1}{\sqrt{1-u^2}}\,du$$

and

$$\arcsin(x) = \int_0^x \frac{1}{\sqrt{1-u^2}}\,du$$

for $x \in [-1, 1]$. We generalize the arcsine (see [57]) by simply switching the 2's in the expression to p's. We define

$$F_p(x) = \int_0^x \frac{1}{\sqrt[p]{1-u^p}}\,du,$$

where we will restrict x to $[0, 1]$ for the present. In particular, we can define a version of π from this:

$$\widetilde{\pi_p} = 2\int_0^1 \frac{1}{\sqrt[p]{1-u^p}}\,du. \tag{20.1}$$

This generalization of π has quite a different flavor than any of our previous ones. (See Figure 20.1.)

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*,
SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_20

p	π_p
1.1	20.27436239
1.2	10.47197459
1.5	4.836798061
2	3.141592654
3	2.418399152
4	2.221441469
5	2.137918664
6	2.094395102
7	2.068752111
8	2.052344306
9	2.041200539
10	2.033281477

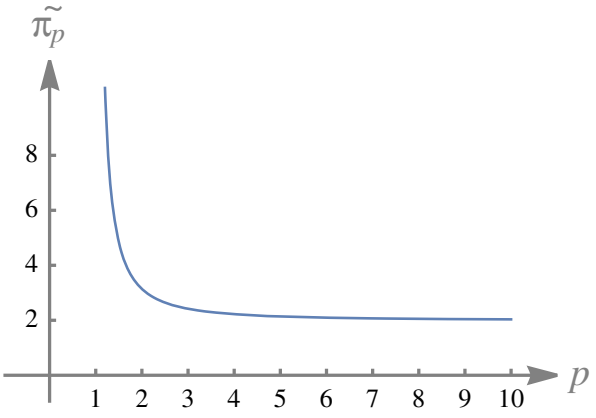


Figure 20.1 $\widetilde{\pi_p}$ values via numerical integration

As with the other generalizations of π, we have that $\widetilde{\pi_2} = \pi$. However, we see that $\pi_1 = \int_0^1 \frac{1}{1-u}\,du = \infty$.

As with the integral (6.1) for π_p, we can express $\widetilde{\pi_p}$ in terms of the beta and gamma functions by substituting $t = u^p$. Then $u = t^{1/p}$, $du = \frac{1}{p}t^{1/p-1}\,dt$ and we obtain

$$\begin{aligned}
\widetilde{\pi_p} &= 2\int_0^1 (1-u^p)^{-\frac{1}{p}}\,du \\
&= \frac{2}{p}\int_0^1 (1-t)^{-\frac{1}{p}}\,t^{\frac{1}{p}-1}\,dt \\
&= \frac{2}{p}\int_0^1 t^{\frac{1}{p}-1}(1-t)^{\frac{p-1}{p}-1}\,dt \\
&= \frac{2}{p}\,\mathrm{B}\left(\tfrac{1}{p}, \tfrac{p-1}{p}\right) \\
&= \frac{2}{p}\,\frac{\Gamma(\frac{1}{p})\Gamma(\frac{p-1}{p})}{\Gamma(\frac{1}{p}+\frac{p-1}{p})} = \frac{2\Gamma(\frac{1}{p})\Gamma(1-\frac{1}{p})}{p\,\Gamma(1)}.
\end{aligned}$$

Since $\Gamma(1) = 1$, we can use the Euler reflection formula (6.6) to write

$$\widetilde{\pi_p} = \frac{2\pi}{p\sin(\pi/p)}. \tag{20.2}$$

This gives us that

$$\lim_{p\to\infty} \widetilde{\pi_p} = \lim_{p\to\infty} \frac{2\pi}{\frac{\sin(\pi/p)}{1/p}} = \frac{2\pi}{\pi} = 2.$$

We define $\widetilde{\sin_p}(t)$ as the inverse function for $F_p(x)$; it has domain $[0, \widetilde{\pi_p}/2]$ and range $[0, 1]$. We can extend this new function to $[0, \widetilde{\pi_p}]$ in the usual manner by setting

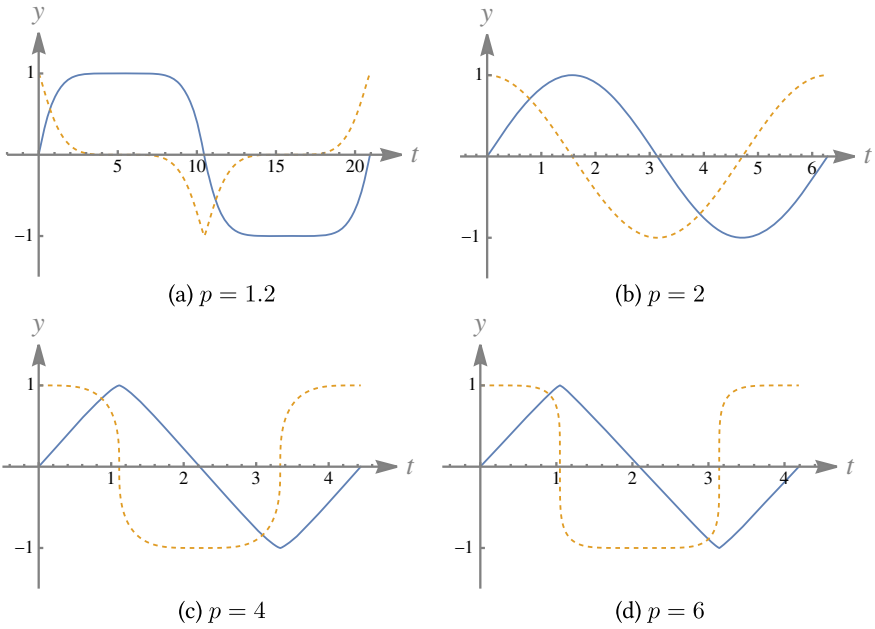


Figure 20.2 Graphs of $y = \widetilde{\sin}_p(t)$ (solid) and $y = \widetilde{\cos}_p(t)$ (dashed) for various p

$\widetilde{\sin}_p(t) = \widetilde{\sin}_p(\widetilde{\pi}_p - t)$. We can then extend further to $[-\widetilde{\pi}_p, \widetilde{\pi}_p]$ via oddness, and then to the entire real line by enacting a period of $2\widetilde{\pi}_p$. (At this point, we can go ahead and rename the function $F_p(x)$ as $\widetilde{\arcsin}_p(x)$.)

We define the partner function $\widetilde{\cos}_p$ as the derivative of $\widetilde{\sin}_p$:

$$\widetilde{\cos}_p(t) = \frac{d}{dt}\widetilde{\sin}_p(t).$$

This new cosine has even symmetry about the origin, and has period $2\widetilde{\pi}_p$. In the first quadrant $[0, \widetilde{\pi}_p/2)$, because $F_p'(x) = \frac{1}{(1-x^p)^{1/p}}$, we have

$$\widetilde{\cos}_p(t) = \left(1 - \widetilde{\sin}_p(t)^p\right)^{1/p}.$$

The symmetry and period lead to the relation

$$|\widetilde{\sin}_p(t)|^p + \left|\widetilde{\cos}_p(t)\right|^p = 1, \quad t \in \mathbb{R}. \tag{20.3}$$

Thus our new cosine and sine parameterize the unit p-circle (Figure 20.2).

Similar to our π_p^{arc}'s in Chapter 8, our new $\widetilde{\pi_p}$'s have a duality relationship. If $q = \frac{p}{p-1}$, then

$$\begin{aligned} p\,\widetilde{\pi_p} &= 2\Gamma(\tfrac{1}{p})\Gamma(1-\tfrac{1}{p}) \\ &= 2\Gamma(\tfrac{1}{p})\Gamma(\tfrac{p-1}{p}) \\ &= 2\Gamma(\tfrac{1}{p})\Gamma(\tfrac{1}{q}) \\ &= 2\Gamma(\tfrac{1}{q})\Gamma(\tfrac{q-1}{q}) \\ &= 2\Gamma(\tfrac{1}{q})\Gamma(1-\tfrac{1}{q}) = q\widetilde{\pi_q}. \end{aligned}$$

As with the previous p-trigonometric functions, we can define the tangent, secant, cotangent, and cosecant in the usual way. If we define

$$\widetilde{\tan_p}(t) = \frac{\widetilde{\sin_p}(t)}{\widetilde{\cos_p}(t)},$$

then, for $t \in [0, \widetilde{\pi_p}/2)$, we have

$$\begin{aligned} \frac{d}{dt}\widetilde{\tan_p}(t) &= \frac{d}{dt}\frac{\widetilde{\sin_p}(t)}{\widetilde{\cos_p}(t)} \\ &= \frac{\widetilde{\cos_p}(t)\frac{d}{dt}\widetilde{\sin_p}(t) - \widetilde{\sin_p}(t)\frac{d}{dt}\widetilde{\cos_p}(t)}{\widetilde{\cos_p}(t)^2} \\ &= \frac{\widetilde{\cos_p}(t)\widetilde{\cos_p}(t) + \widetilde{\sin_p}(t)\left(1-\widetilde{\sin_p}(t)^p\right)^{\frac{2}{p}-1}\widetilde{\sin_p}(t)^{p-1}}{\widetilde{\cos_p}(t)^2} \\ &= \frac{\widetilde{\cos_p}(t)^2 + \widetilde{\sin_p}(t)\left(\widetilde{\cos_p}(t)^p\right)^{\frac{2}{p}-1}\widetilde{\sin_p}(t)^{p-1}}{\widetilde{\cos_p}(t)^2} \\ &= 1 + \frac{\widetilde{\sin_p}(t)^p\widetilde{\cos_p}(t)^{2-p}}{\widetilde{\cos_p}(t)^2} \\ &= 1 + \frac{\widetilde{\sin_p}(t)^p}{\widetilde{\cos_p}(t)^p} = 1 + \widetilde{\tan_p}(t)^p. \end{aligned}$$

Employing symmetry and our knowledge of inverses, we then have

$$\widetilde{\arctan_p}(x) = \int_0^x \frac{1}{1+|u|^p}\,du, \quad x \in \mathbb{R},$$

which has range $(-\widetilde{\pi_p}/2, \widetilde{\pi_p}/2)$.

Lang and Edmunds demonstrated in [57] that the parameter of $\widetilde{\sin_p}(t)$ can be interpreted as an arclength, but not of the squircle.

The choice to force the cosine to be the derivative of sine gives these versions of sine and cosine a familiar property from classical trigonometry. What if we go the other way? Define $\widetilde{\text{Cos}_p}(t) = \int_t^{\frac{\pi_p}{2}} \widetilde{\sin_p}(u)\,du$. Then $\widetilde{\text{Cos}_p}'(t) = -\widetilde{\sin_p}(t)$ and $\widetilde{\text{Cos}_p}(\widetilde{\pi_p}/2) = 0$ and so this is an alternative derivative-defined variant of cosine.

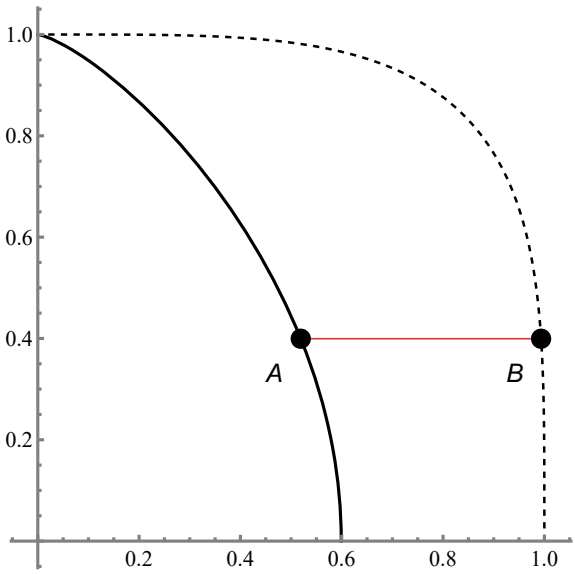


Figure 20.3 The curve Ξ_4 (solid) with 4-circle (dashed). Measuring counterclockwise from the x-axis, the p-arclength of the arc to A on Ξ is equal to the parameter t for the point $B(\widetilde{\cos}(t), \widetilde{\sin}(t))$. A and B lie on the same horizontal line

It seems natural to consider the curve Ξ parameterized by $\boldsymbol{\xi}(t) = (\widetilde{\mathrm{Cos}_p}(t), \widetilde{\sin_p}(t))$. Then the arc of this curve from the x-axis to $\boldsymbol{\xi}(t)$ has p-length

$$\int_0^t \left(\mathrm{Cos}_p'(u)^p + (\widetilde{\sin_p}'(u))^p\right)^{1/p} du = \int_0^t \left(\widetilde{\sin_p}(u)^p + \widetilde{\cos_p}(u)^p\right)^{1/p} du = t.$$

Noting that the squircle and the curve Ξ have the same second coordinate, we have shown that the parameter t of the point $(\widetilde{\cos_p}(t), \widetilde{\sin_p}(t))$ on the squircle is the p-length of the arc of Ξ subtended at the point at the same height and in the same quadrant. See Figure 20.3.

Thus Ξ is a curve whose arclength tracks the parameter of the corresponding squircle. Only in the ideal world where $p = 2$ does the squircle do this for itself.

20.2 Two other generalizations

In [63], Lindqvist, Peetre and Borwein define the inverse p-sine, inverse p-cosine, and inverse p-tangent as in Chapter 5, but define their version of π_p as

$$\widehat{\pi_p} = p \int_0^1 \frac{1}{(1-u^p)^{(p-1)/p}}\, du. \tag{20.4}$$

That is,

$$\widehat{\pi_p} = \frac{p}{2}\pi_p = \frac{\Gamma(1/p)^2}{\Gamma(2/p)} = \mathrm{B}\left(\tfrac{1}{p}, \tfrac{1}{p}\right).$$

We can notate the inverses of these new functions as $\widehat{\sin_p}$, $\widehat{\cos_p}$, and so on, but it can be shown that these new functions are, in fact, our previous areal p-trigonometric functions. (See Exercises.)

A very different generalization [62] comes from a solution to a particular nonlinear eigenvalue problem.[1] New sine, cosine, and tangent functions – $\widehat{S_p}$, $\widehat{C_p}$ and $\widehat{T_p}$, respectively – for $p > 1$ are defined by the equations

$$t = \int_0^{\widehat{S_p}(t)} \frac{1}{\left(1 - \frac{u^p}{p-1}\right)^{1/p}}\, du, \quad t = \int_{\widehat{C_p}(t)}^{\sqrt[p]{p-1}} \frac{1}{\left(1 - \frac{u^p}{p-1}\right)^{1/p}}\, du, \qquad (20.5)$$

$$t = \int_0^{\widehat{T_p}(t)} \frac{1}{1 + \frac{u^p}{p-1}}\, du.$$

Note the curious upper bound on the second integral – it's not 1, although $\sqrt[p]{p-1} \to 1$ as $p \to \infty$. (In particular, this sine and cosine do not parameterize a unit p-circle.) We can then define the generalization of π for these functions.

$$\widehat{\Pi_p} = 2\widehat{S_p^{-1}}(\sqrt[p]{p-1}) = 2\widehat{C_p^{-1}}(0) = 2\int_0^{\sqrt[p]{p-1}} \frac{1}{\left(1 - \frac{u^p}{p-1}\right)^{1/p}}\, du$$

Set $v = \frac{u^p}{p-1}$. Then $u = \sqrt[p]{p-1}\, v^{1/p}$, $du = \sqrt[p]{p-1} \cdot \frac{1}{p} v^{1/p-1}\, dv$, and

$$\begin{aligned}
\widehat{\Pi_p} &= 2\int_0^1 \frac{1}{(1-v)^{1/p}} \cdot \sqrt[p]{p-1} \cdot \frac{1}{p} v^{1/p-1}\, dv \\
&= \frac{2\sqrt[p]{p-1}}{p} \int_0^1 (1-v)^{-1/p} v^{1/p-1}\, dv \\
&= \frac{2\sqrt[p]{p-1}}{p} \mathrm{B}\left(\tfrac{1}{p}, \tfrac{p-1}{p}\right) \\
&= \frac{2\sqrt[p]{p-1}}{p} \Gamma\left(\tfrac{1}{p}\right) \Gamma\left(1 - \tfrac{1}{p}\right) \\
&= \frac{2\pi\sqrt[p]{p-1}}{p \sin(\pi/p)},
\end{aligned}$$

[1] Much contemporary interest in generalized trigonometry comes from the eigenvalue problems they solve. Briefly, define the classical Laplacian by $\Delta u = \nabla \cdot \nabla u$. This is a fundamental operator in analysis and physics, and the solutions to the eigenvalue problem $\Delta u = -\lambda u$ give rise to the classical trigonometric functions and the construction of Fourier series. A generalized Laplacian takes the form $\Delta_p u = \nabla \cdot (|\nabla u|^{p-2} \nabla u)$ (note that $\Delta_2 = \Delta$) and the squigonometric functions emerge as solutions to a corresponding eigenproblem – you might try calculating p-Laplacian on the conjugate sine $u = \sin_q$. As with much of what we have seen with these functions, they can play similar roles as their classical counterparts but the story is much more complicated. A full exposition is beyond our scope and we refer the reader to the literature (e.g., [9,27,57,60]) for further details.

by (6.6). Note that $\widehat{\Pi}_p = \sqrt[p]{p-1}\,\widetilde{\pi}_p$. Thus $\widehat{S}_p$, $\widehat{C}_p$, etc. are scaled versions of the functions defined in Section 20.1 with generally longer periods and higher amplitudes (at least for $p > 2$).

There are some major differences though. If q is such that $\frac{1}{p} + \frac{1}{q} = 1$, then $\widehat{\Pi}_p = \widehat{\Pi}_q$ and $\widehat{\Pi}_2 = \pi$ is the *maximum* value of $\widehat{\Pi}_p$. (The proof is left as an exercise.) In particular, since $\widehat{\Pi}_p \to 2$ as $p \to \infty$, then $\widehat{\Pi}_p \to 2$ as $p \to 1^+$.

The generalization of the Pythagorean relationship for $\widehat{S}_p$ and $\widehat{C}_p$ can be established from the first integral in (20.5). Let q be conjugate to p once again and assume for the moment that $t \in [0, \widehat{\Pi}_p/2)$. Then set

$$\frac{v^q}{q-1} = 1 - \frac{u^p}{p-1}.$$

This gives us $pv^{q-1}\,dv = -qu^{p-1}\,du$ and thus

$$\begin{aligned} t &= \int_0^{\widehat{S}_p(t)} \frac{1}{\left(1 - \frac{u^p}{p-1}\right)^{1/p}}\,du \\ &= \int_{\sqrt[q]{q-1}}^{v^*} \frac{1}{\left(\frac{v^q}{q-1}\right)^{1/p}} \left(-\frac{pv^{q-1}}{qu^{p-1}}\right) dv, \end{aligned}$$

where

$$v^* = \left[(q-1)\left(1 - \frac{\widehat{S}_p(t)^p}{p-1}\right)\right]^{1/q}.$$

We then obtain

$$\begin{aligned} t &= \int_{v^*}^{\sqrt[q]{q-1}} \frac{1}{\left(\frac{v^q}{q-1}\right)^{1/p}} \left\{ \frac{pv^{q-1}}{q\left[(p-1)\left(1 - \frac{v^q}{q-1}\right)\right]^{(p-1)/p}} \right\} dv \\ &= \int_{v^*}^{\sqrt[q]{q-1}} \frac{(q-1)^{1/p}}{v^{1/(p-1)}} \left\{ \frac{pv^{q-1}}{q(p-1)^{1/q}\left(1 - \frac{v^q}{q-1}\right)^{1/q}} \right\} dv \end{aligned}$$

Writing everything in terms of q leads to

$$\begin{aligned} t &= \int_{v^*}^{\sqrt[q]{q-1}} \frac{(q-1)^{(q-1)/q}}{v^{q-1}} \left\{ \frac{v^{q-1}}{(q-1)\left(\frac{1}{q-1}\right)^{1/q}\left(1 - \frac{v^q}{q-1}\right)^{1/q}} \right\} dv \\ &= \int_{v^*}^{\sqrt[q]{q-1}} (q-1)^{-1/q} \left\{ \frac{(q-1)^{1/q}}{\left(1 - \frac{v^q}{q-1}\right)^{1/q}} \right\} dv \\ &= \int_{v^*}^{\sqrt[q]{q-1}} \frac{1}{\left(1 - \frac{v^q}{q-1}\right)^{1/q}}\,dv. \end{aligned}$$

This is the second integral in (20.5) if $v^* = \widehat{C_q}(t)$; that is, if

$$\widehat{C_q}(t) = \left[(q-1)\left(1 - \frac{\widehat{S_p}(t)^p}{p-1}\right)\right]^{1/q}.$$

Rearranging this equation and extending the result to the entire real line in the usual manner gives us the Pythagorean-style relationship we were seeking:

$$\frac{|\widehat{S_p}(t)|^p}{p-1} + \frac{|\widehat{C_q}(t)|^q}{q-1} = 1, \tag{20.6}$$

for all $t \in \mathbb{R}$, where $\frac{1}{p} + \frac{1}{q} = 1$. We can use similar techniques to find formulas for the $\widehat{T_p}$ function in terms of the $\widehat{S_p}$ and $\widehat{C_p}$ functions. We leave this as an exercise for the reader.

20.3 Two-parameter functions

A common theme in developing each set of p-trigonometric functions has been first to define an arcsine function through an integral of the form

$$\arcsin^*_{p,q}(x) = \int_0^x (1-u^q)^{-1/p}\, du$$

and then define $\sin^*_{p,q}$ as the inverse function of $\arcsin_{p,q}$. We can then define

$$\pi^*_{p,q} = 2\int_0^1 (1-u^q)^{-1/p}\, du.$$

These are generalizations of some of our other squigonometric functions and variants of π as $\sin^*_{p,q}(t) = \sin_p(t)$ and $\pi^*_{p,q} = \pi_p$, where p and q are conjugate. Similarly, $\sin^*_{p,p}(t) = \widetilde{\sin_p}$. (We leave verification of this an exercise.) However, in [27,57], the companion cosine function $\cos^*_{p,q}(t)$ is defined as the derivative of $\sin^*_{p,q}$, so the definition of cosine lines up with $\widetilde{\cos_p}(t)$ but not $\cos_p(t)$.

When $p \neq 2$, the possible trigonometric functions sprawl before us like a wall-to-wall carpet that almost, but not quite, fits a room: some parts are smooth, while some bumps emerge. If we step on one bump to smooth it out, a different bump will make itself known somewhere else. It is mostly a matter of preference but one must take care to keep straight which versions have which properties. Unless $p = 2$, we have multiple reasonable options.

There are further journeys one can take in continuing exploration of generalized trigonometric functions. The functions $\widetilde{\sin_p}(\cdot)$ and $\widetilde{\cos_p}(\cdot)$ defined earlier in the chapter offer a generalization of Fourier analysis, the foundations of which are laid out in [57]. (See also the project below.) The geometries we have studied are special cases of Minkowski geometry, which offers a more general lens through which to see these problems. Duality offers another set of tools. Studies of higher dimensions may also open new territory.

Overall, this is a subject with a long but fractured history (cf. [2,30,42,68,83, 105] and the references therein) and there are many connections among the special functions associated with p-norm geometry yet to explore.

Exercises

▶ **Exercise 20.1** *Calculate* $\lim_{p\to 1^+} (p-1)\widetilde{\pi_p}$.

▶ **Exercise 20.2** *Find* $\frac{d}{dt}\widetilde{\sec_p}(t)$.

▶ **Exercise 20.3**

1. *Prove that* $\widehat{\pi_p} \to \infty$ *as* $p \to \infty$.
2. *Show that* $\widehat{\pi_p} \approx 2p$ *for large* p.

▶ **Exercise 20.4** *Show that*

$$\widehat{\sin_p}\left(\frac{\widehat{\pi_p}}{2p}\right) = \sqrt[p]{\frac{1}{2}} = \widehat{\cos_p}\left(\frac{\widehat{\pi_p}}{2p}\right).$$

▶ **Exercise 20.5** *Show that* $|\widehat{\sin_p}(t)|^p + |\widehat{\cos_p}(t)|^p = 1$ *for all* $t \in \mathbb{R}$.

▶ **Exercise 20.6** *Show that the p-trigonometric functions arising from the definition (20.4) of $\widehat{\pi_p}$ are exactly the same functions as the areal p-trigonometric functions defined in Chapter 3.*

▶ **Exercise 20.7** *Prove that if* $\frac{1}{p} + \frac{1}{q} = 1$, *then* $\widehat{\Pi_p} = \widehat{\Pi_q}$ *and that* $\widehat{\Pi_2} = \pi$ *is the maximum value.*

▶ **Exercise 20.8** *Use a similar technique to that used to prove (20.6) to show that*

$$\widehat{T_p}(t) = \frac{\widehat{S_p}(t)}{\sqrt[p]{p-1}\,|\widehat{C_q}(t)|^{q-2}\,\widehat{C_q}(t)},$$

where p and q are conjugate. (Hint: Assume first that $0 \le t \le \widehat{\Pi_p}/2$. *Use the substitution*

$$u^p = \frac{v^p}{1+v^p}$$

in the integral defining $\widehat{S_p}$.)

▶ **Exercise 20.9** *Verify that* $\sin_p^*(t) = \widetilde{\sin_{p,p}}^*(t)$ *and* $\sin_{p,q}^*(t) = \sin_p(t)$ *where p and q are conjugate.*

▶ **Exercise 20.10** *Show that*

$$\pi^*_{p,q} = \frac{2}{q}\,\mathrm{B}(\tfrac{1}{p}, 1 - \tfrac{1}{q}).$$

▶ **Exercise 20.11** *Prove that for* $p \in (1, \infty)$, $2 \le \pi^*_{p,q} \le 4$ *where* p *and* q *are conjugate.*

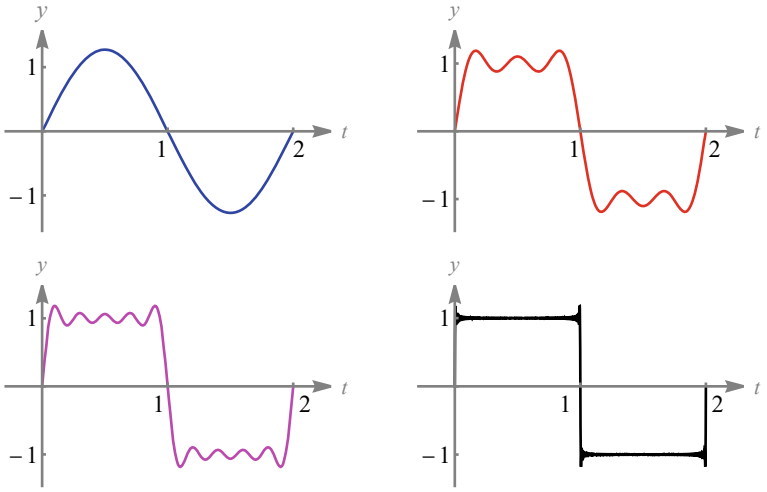
Projects

Project 20.1 (Auxilliary curves)
The squircle parameterization $(\widetilde{\cos}_p(t), \widetilde{\sin}_p(t))$ from Section 20.1 was developed a little differently from our other parameterizations in that it was motivated by a differential (cosine is the derivative of sine) rather than a geometric property. We then saw some of the geometry captured by the auxilliary curve Ξ, which was parameterized by what would have been the other differential property to have chosen (sine is the negative derivative of cosine). And of course, everything beautifully aligns when $p = 2$.

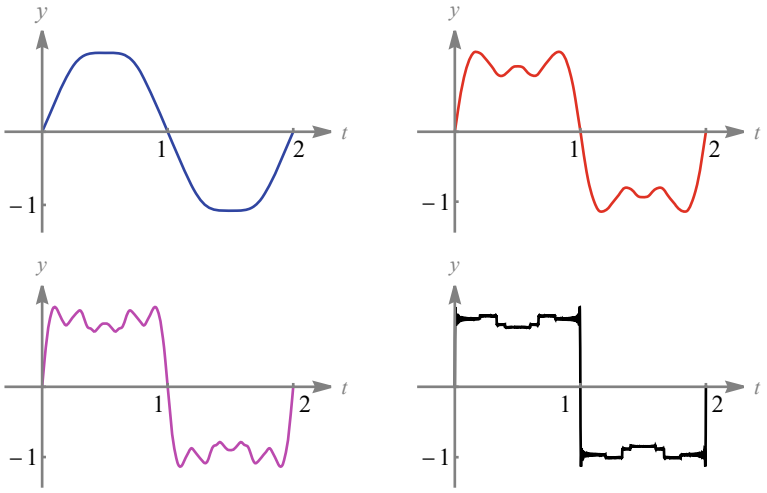
What are these auxiliary curves? Do they have any interesting geometric properties of their own? This idea of one curve capturing the geometry of another was used in our study of duality. Are there other ways to apply this approach?

Project 20.2 (Sum formulas)
We saw back in Chapter 16 that one can find generalizations of the "angle" sum formulas in the elliptic cases $p = 3$ and $p = 4$. More is known if we extend to the two-parameter definitions.

Takeuchi [95] studies the particularly interesting case of $p = 6$ and finds a double-angle formula for the 6-sine (see Appendix B), demonstrating how these techniques can be applied to non-elliptic cases. This case was studied by Ramanujan [10] as a generalization of the lemniscate function. [76] finds formulas for non-conjugate pairs (3/2, 2), (3/2, 6), (3, 6), and (6/5, 3) and notes other known sum formulas: (2, 3) in [20], (2, 4) (i.e., $\tanh_4$ of formula (16.4)), (2, 6) in [90], and (6/5, 2) in [95]. Much remains unknown.



(a) Fourier Series



(b) "Four"-ier series

Figure 20.4 Plots of $\sum_{k=1,3,5\ldots}^{2n-1} \frac{4}{k\pi_p} \sin_p(k\pi_p t)$ for $n = 1, 3, 5, 100$ and $p = 2, 4$. This approximates a square wave for $p = 2$

Project 20.3 (Fourier series)
Most functions we deal with everyday tend to be nice – continuous, differentiable, and integrable. If we seek to analyze a function, it can be useful to express it in terms of nice so-called *basis functions*. A (possibly infinite) set of functions $\{x_n\}$ is a *(Schauder) basis* for another set of functions X if every $x \in X$ can be written in the form $x = \sum_{n=0}^{\infty} \alpha_n x_n$ for some complex numbers α_n.

In the case of odd periodic functions on, say, the interval $[-1, 1]$, we can use the basis functions

$$\{\sin(\pi t), \sin(2\pi t), \sin(3\pi t), \cdots\}$$

as building blocks. We can then represent a periodic function $f(t)$ by the series

$$f(t) = b_1 \sin(\pi t) + b_2 \sin(2\pi t) + b_3 \sin(3\pi t) + \cdots = \sum_{n=1}^{\infty} b_n \sin(n\pi t). \tag{20.7}$$

This is an example of a *Fourier sine series*. (See [94, Chapter 4] for an excellent introduction to Fourier series. Also see Figures 20.4 and A.17.)

We know several things about this series. We define the space of functions $L^q[a, b]$ to be the set of functions $g(t)$ such that

$$\int_a^b |g(t)|^q \, dt$$

is finite. (See also Section 18.1). We can impose a norm (see Chapter 19) on functions g in $L^q[a, b]$:

$$\|g\|_q = \left(\int_a^b |g(t)|^q \, dt \right)^{\frac{1}{q}}$$

We also know that for any function $f(t)$ in $L^q[0, 1], q \in (1, \infty)$, if we extend f to an odd function on $[-1, 1]$, $f(t)$ has a unique Fourier sine series representing it. That is,

$$\left\| f(t) - \sum_{n=1}^{N} b_n \sin(n\pi t) \right\|_q \to 0$$

as $N \to \infty$.

This means that $\{\sin(n\pi t)\}_{n=1}^{\infty}$ is a basis for $L^q[0, 1]$ for every $q \in (1, \infty)$. (See [28, 12.10.1].)

The coefficients b_n in (20.7), called *Fourier sine coefficients*, are given by

$$b_n = 2 \int_0^1 f(t) \sin(n\pi t) \, dt.$$

Two questions naturally arise from these definitions:

1. What are the Fourier sine coefficients for $f(t) = \sin_p(n\pi_p t)$?
2. Do the functions $\{\sin_p(n\pi_p t)\}_{n=1}^{\infty}$ (p fixed) themselves form a basis for $L^q[0, 1]$? In other words, can we represent odd functions on $[-1, 1]$ with Fourier p-sine series?

A great starting place to explore these two problems is [16], where the questions are answered for the functions $\{\widetilde{\sin_p}(n\widetilde{\pi_p} t)\}_{n=1}^{\infty}$.

There are also Fourier cosine series (to represent even periodic functions) and complete Fourier series, which use both sines and cosines. Our questions could be applied to these series and basis functions as well.

Project 20.4 (Unifying the theory)
Although our focus has been primarily on the areal parameterization, we have explored several different parameterizations of the p-circle, each with its own geometric and analytic character. We have also developed some tools to relate them in specific instances. Is there a way to unify these tools? That could mean a lot of different things. For example, perhaps we might classify these parameterizations by the ζ in the CIVP (7.1) and use these to tease out families of identities.

Curve menagerie

A

We offer here a collection of curves and diagrams that illustrate various squigonometric phenomena in different p-metrics. We typically illustrate the cases $p = 1/2$ (the semimetric case), $p = 1, \infty$ (the dual extreme cases), $p = 4/3, 4$ (dual), and $p = 2$ (Euclidean). Limiting examples are used in some cases where a figure is ill-defined or not instructive. We include references to the relevant section of the text where appropriate. All figures in this appendix were drawn in Mathematica (Figures A.1, A.2, A.3, A.4, A.5, A.6, A.7, A.8, A.9, A.10, A.11, A.12, A.13, A.14, A.15, A.16, A.17 and A.18).

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_A

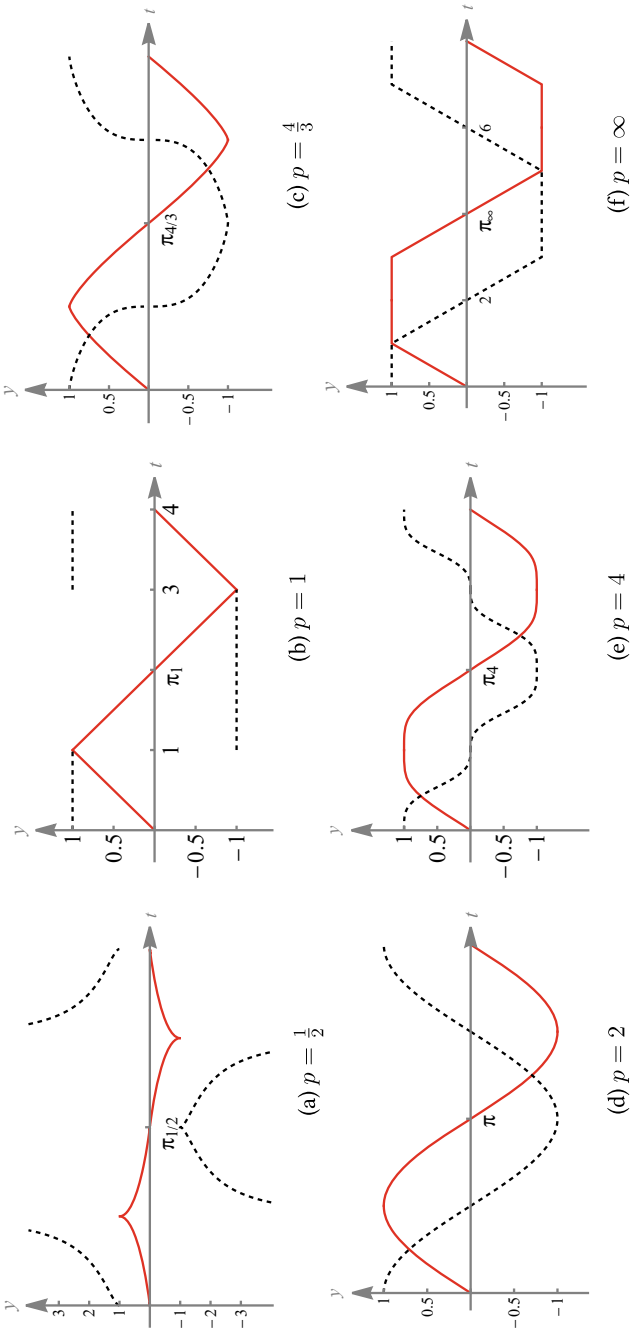


Fig. A.1 One period of the curve $y = \sin_p(t)$ (solid) and its derivative (dashed). (Section 3.1)

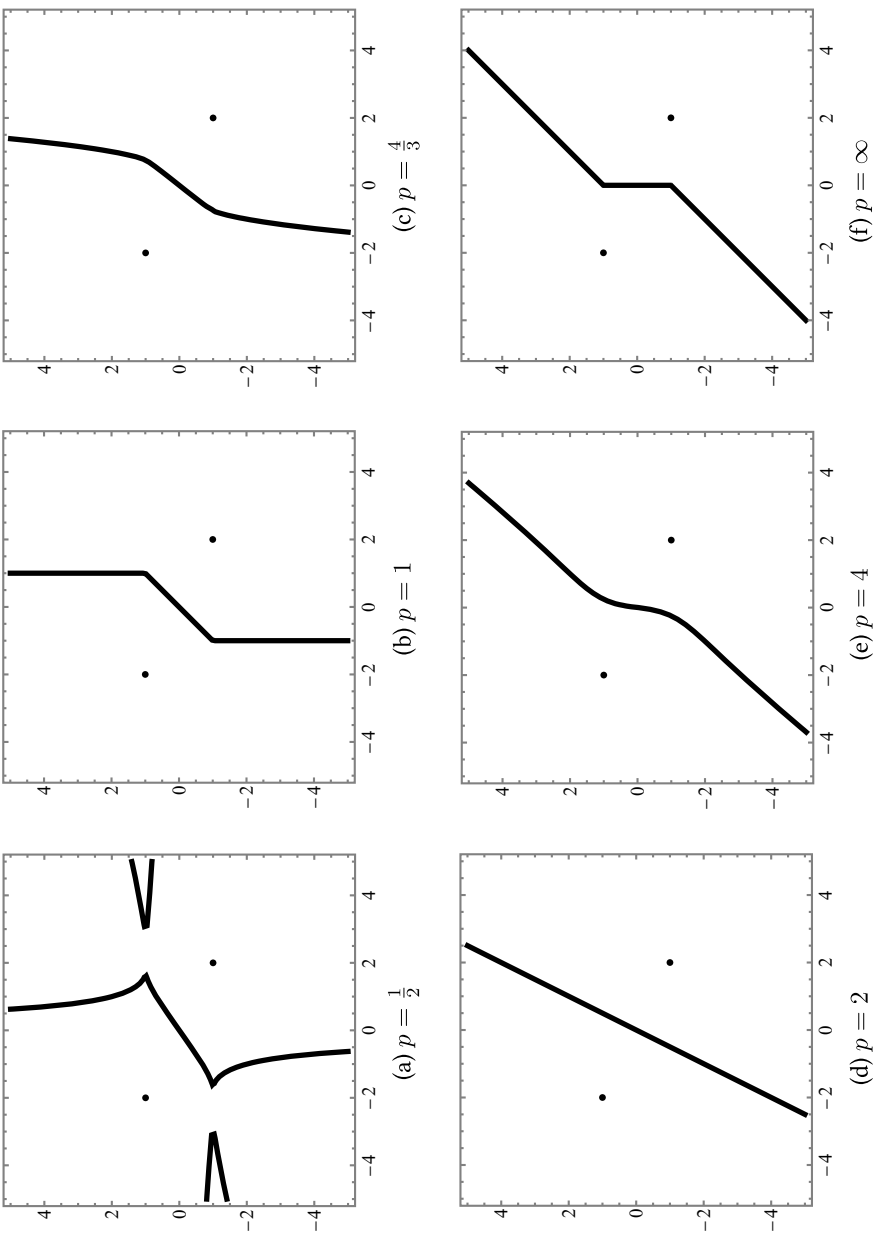


Fig. A.2 The p-metric midset of the points $A(2, -1)$ and $B(-2, 1)$ (Section 4.2)

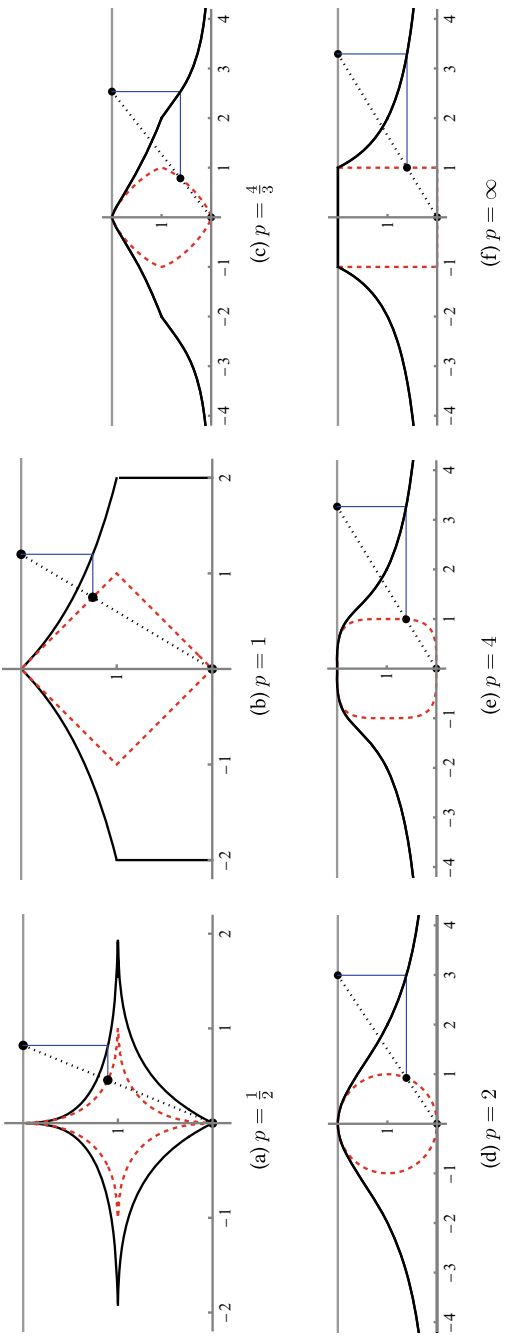


Fig. A.3 Witches of Agnesi (Exercise 4.12)

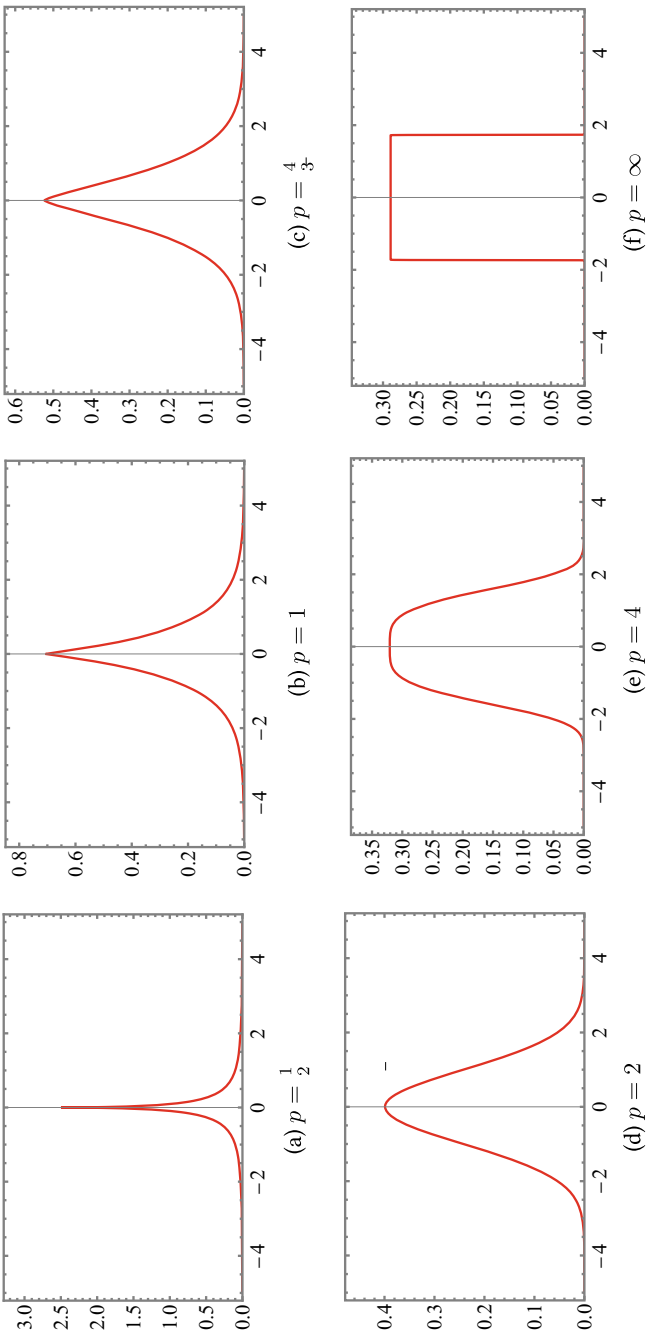


Fig. A.4 Generalized normal distributions (Exercise 6.12) with mean 0 and variance 1

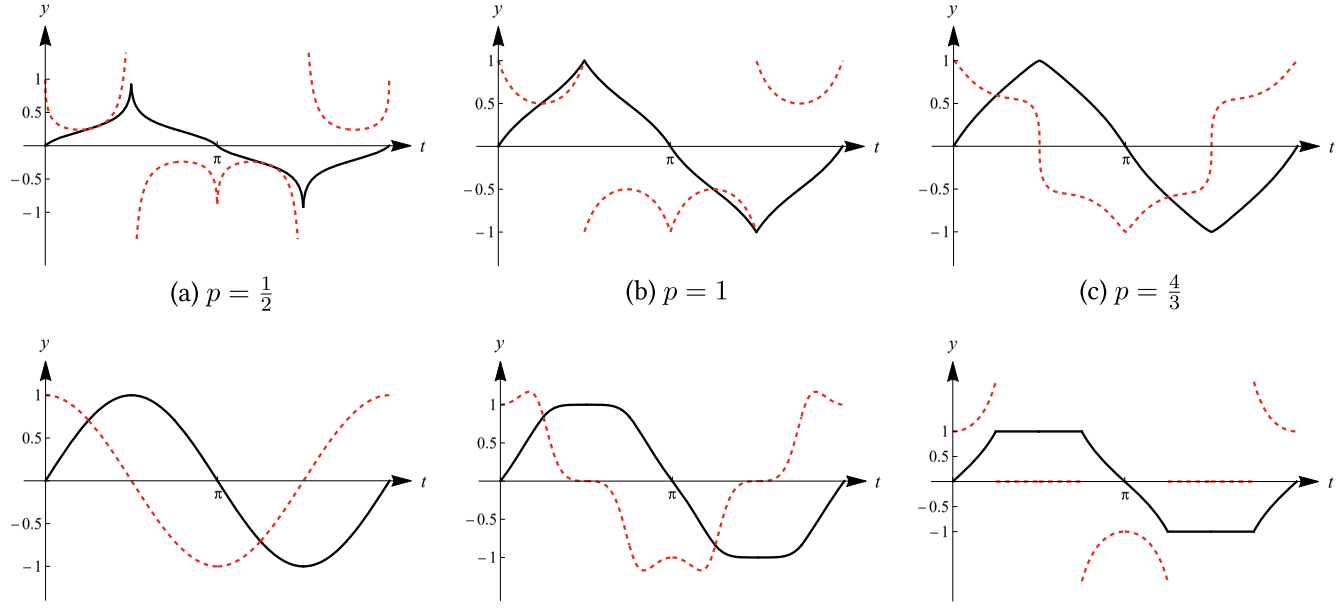


Fig. A.5 Graphs of $y = \sin_p^{\text{ang}}(t)$ (black) and its derivative (red, dashed). (Section 7.3)

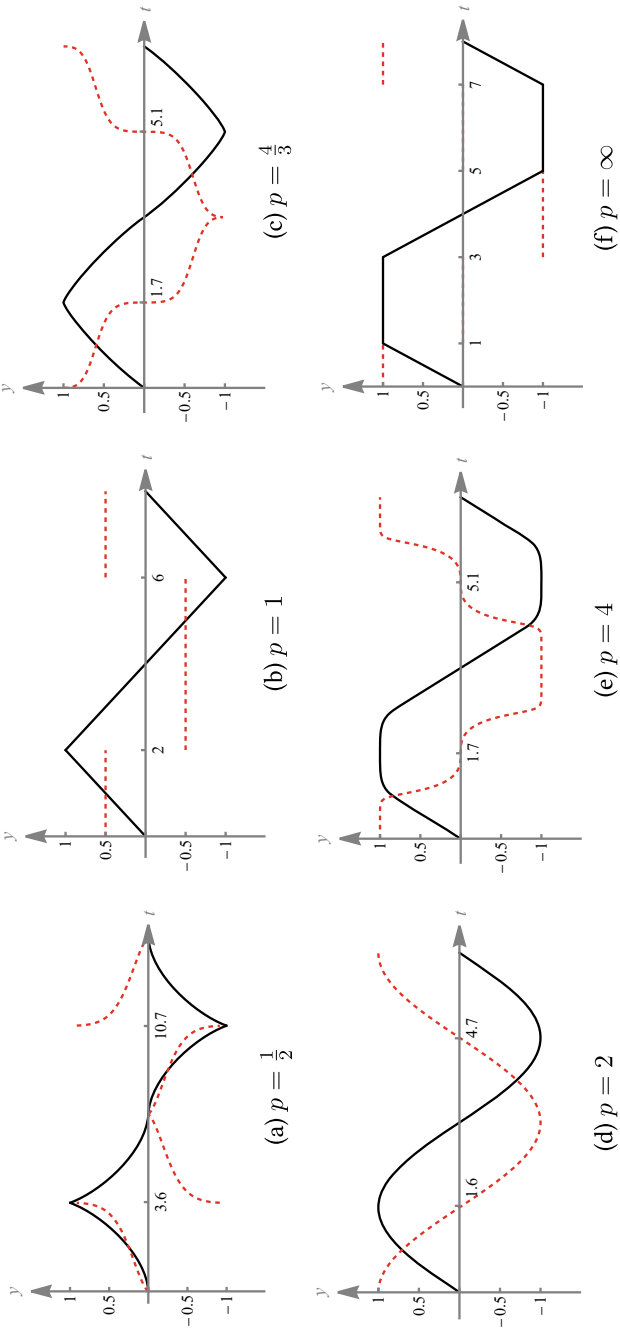


Fig. A.6 Graphs of $y = \sin_p^{\text{arc}}(t)$ (black) and its derivative (red, dashed). (Section 8.3)

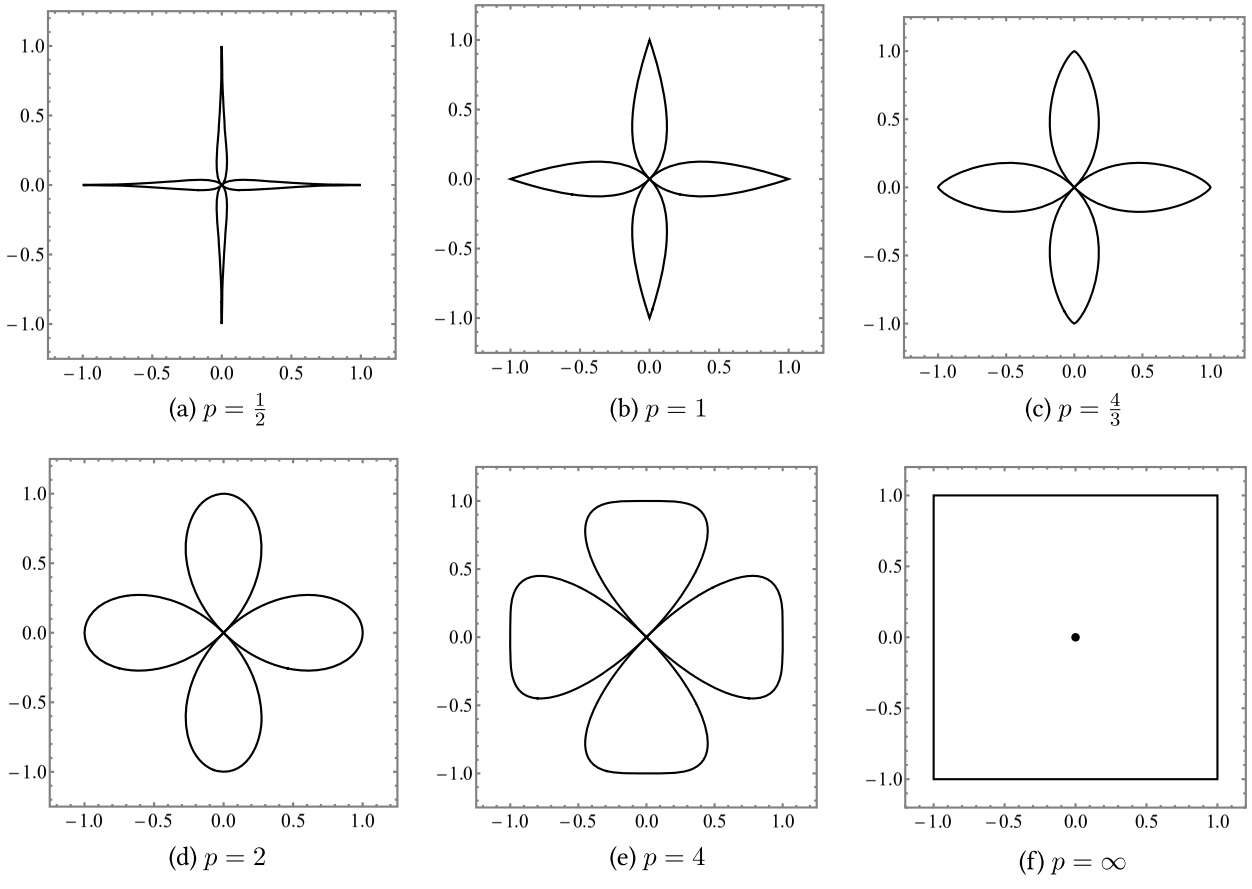


Fig. A.7 Four-leaf roses $r = \cos_p^p(t) - \sin_p^p(t)$, r measured in the p-metric. (Section 13.2)

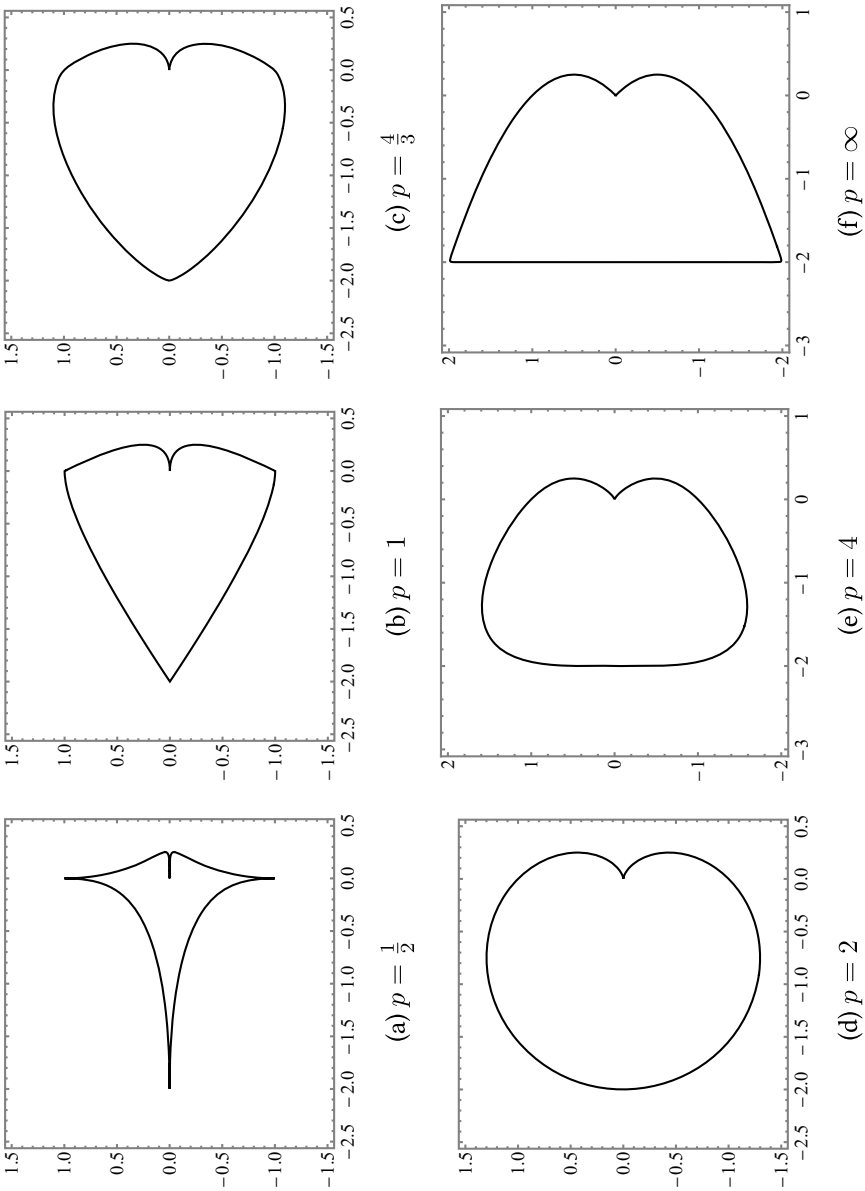


Fig. A.8 Cardioids, $r = 1 - \cos_p(t)$, r measured in the p-metric. (Section 13.2)

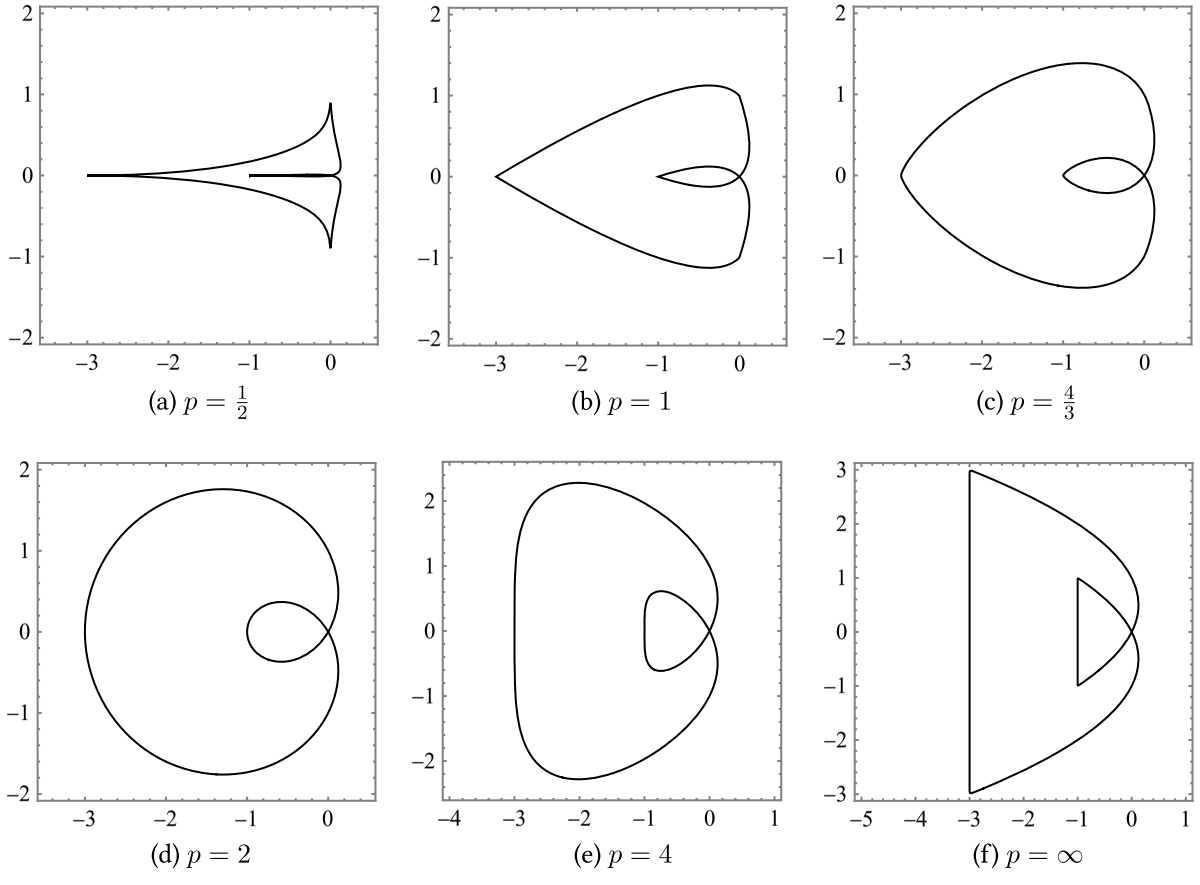
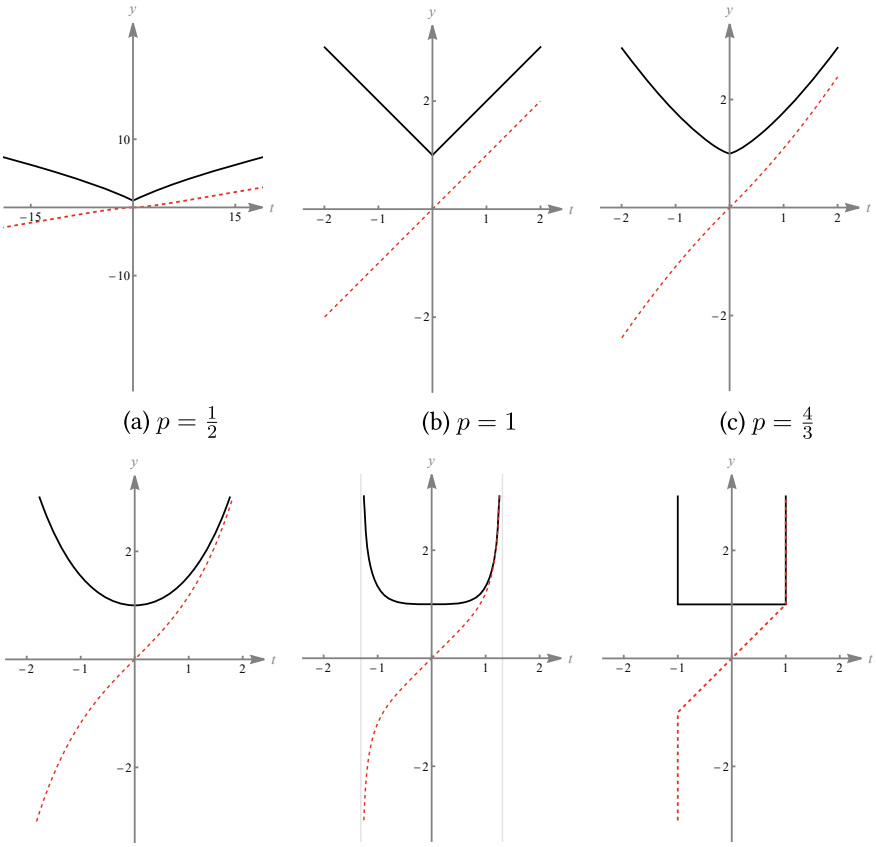


Fig. A.9 Limaçons, $r = 1 - 2\cos_p(t)$, r measured in the p-metric. (Section 13.2)



(a) $p = \frac{1}{2}$ (b) $p = 1$ (c) $p = \frac{4}{3}$

(d) $p = 2$ (e) $p = 4$ (f) $p = \infty$

Fig. A.10 $y = \cosh_p(t)$ (solid) and $y = \sinh_p(t)$ (dashed). (Section 14.1)

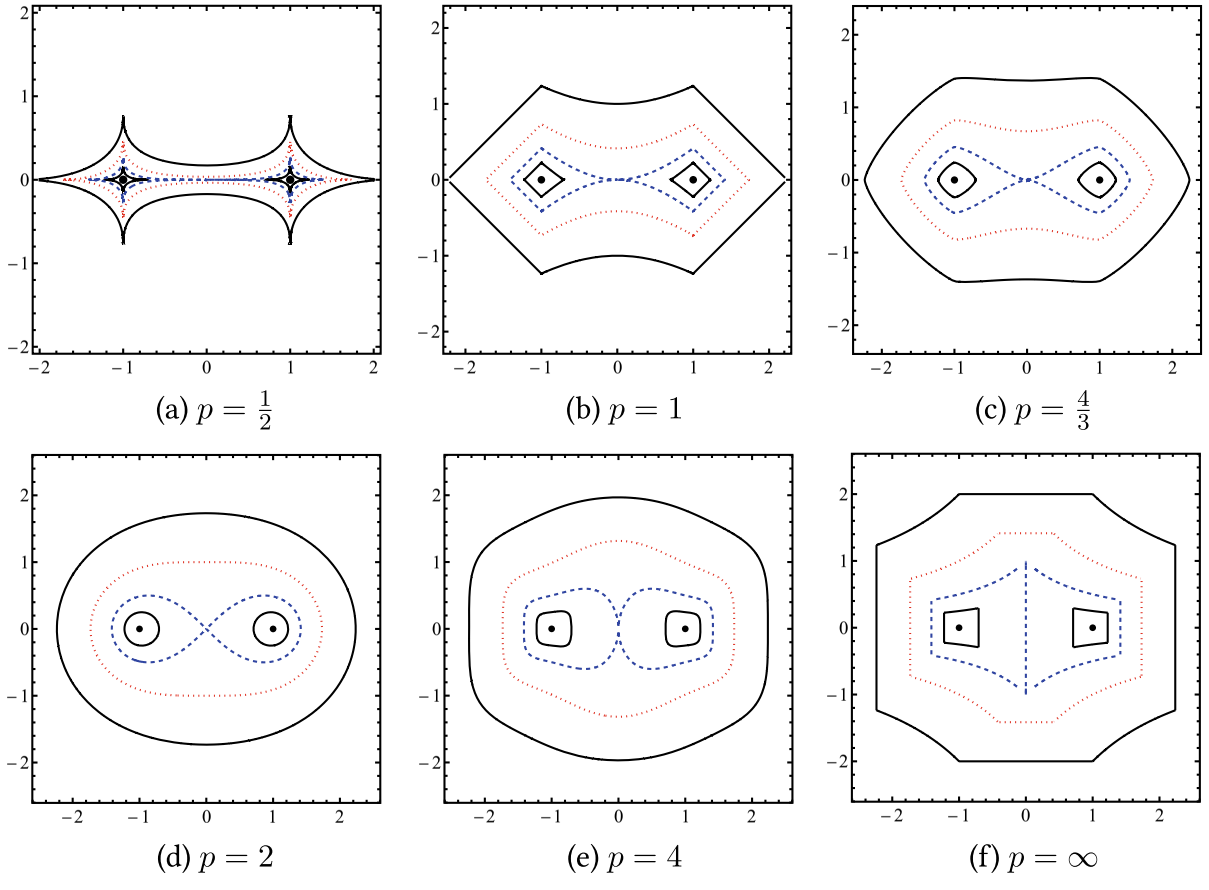


Fig. A.11 Generalized lemniscates $\{P \mid d(P, A) \cdot d(P, B) = k\}$ for $k = 4, 2, 1, \frac{1}{2}$ (black, dotted red, dashed blue, black, resp.) with points $A(-1, 0)$ and $B(1, 0)$. (Exercise 16.1)

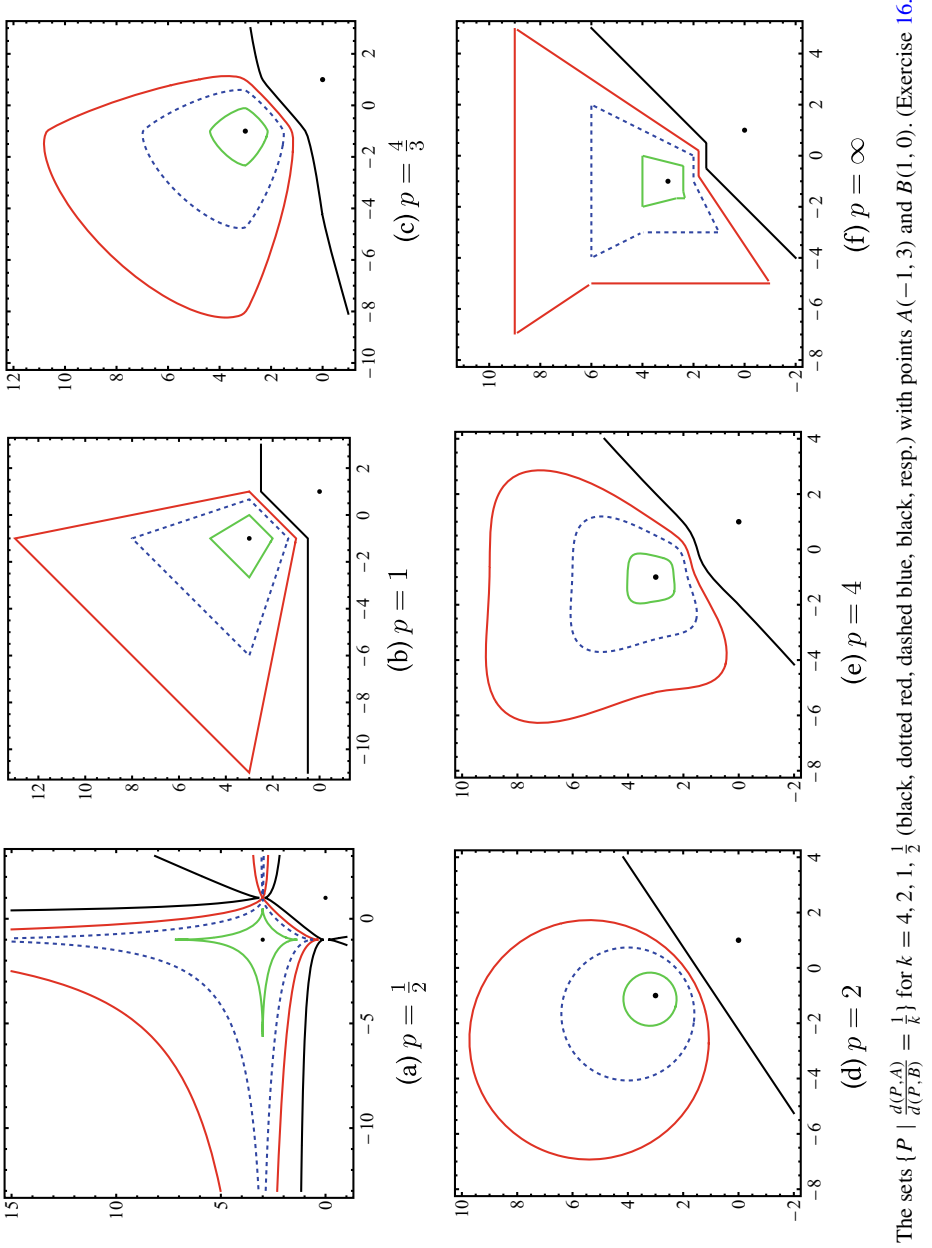


Fig. A.12 The sets $\{P \mid \frac{d(P,A)}{d(P,B)} = \frac{1}{k}\}$ for $k = 4, 2, 1, \frac{1}{2}$ (black, dotted red, dashed blue, black, resp.) with points $A(-1, 3)$ and $B(1, 0)$. (Exercise 16.2)

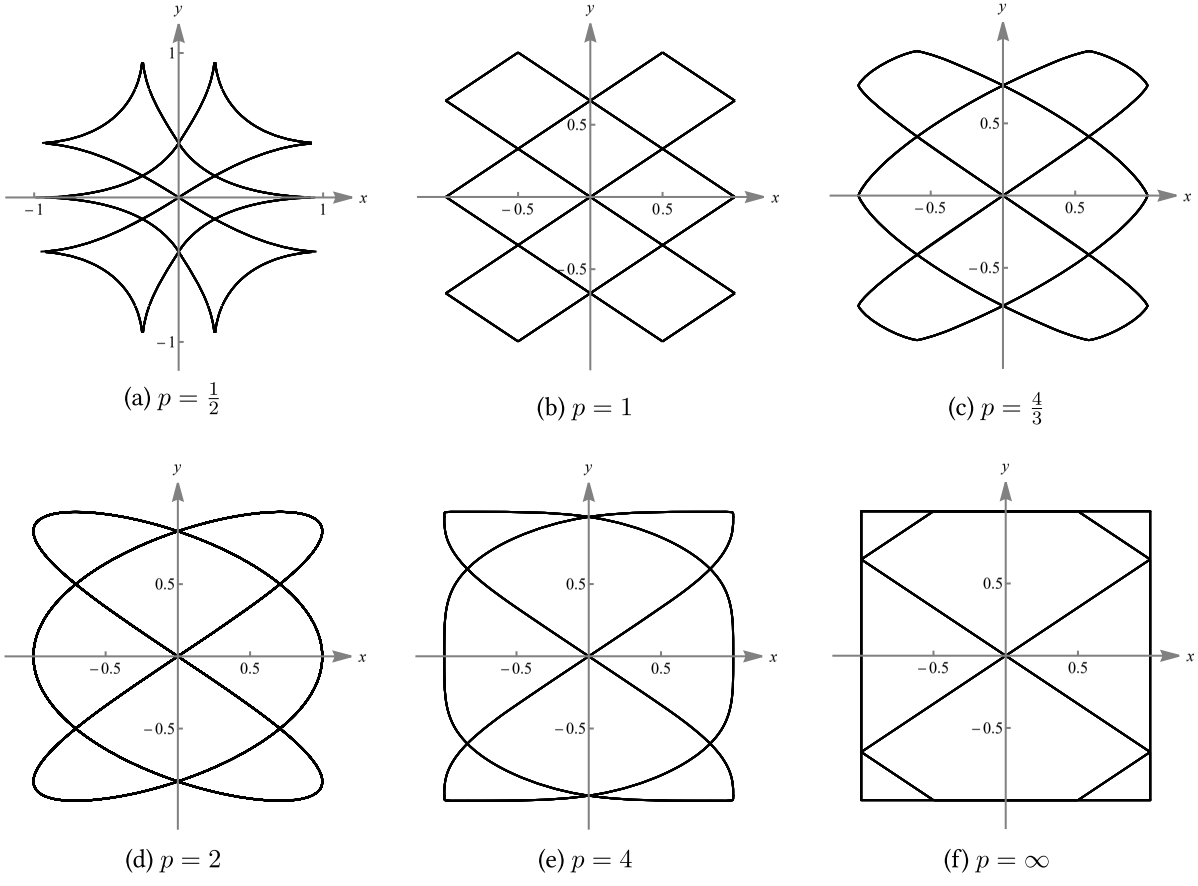
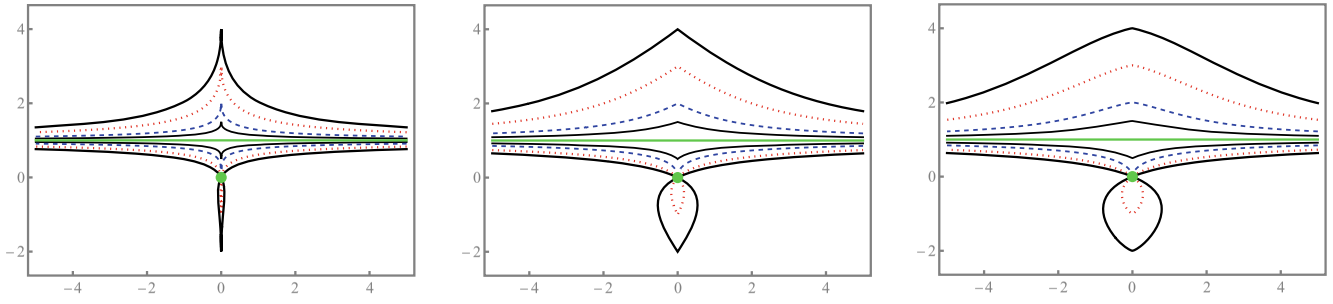
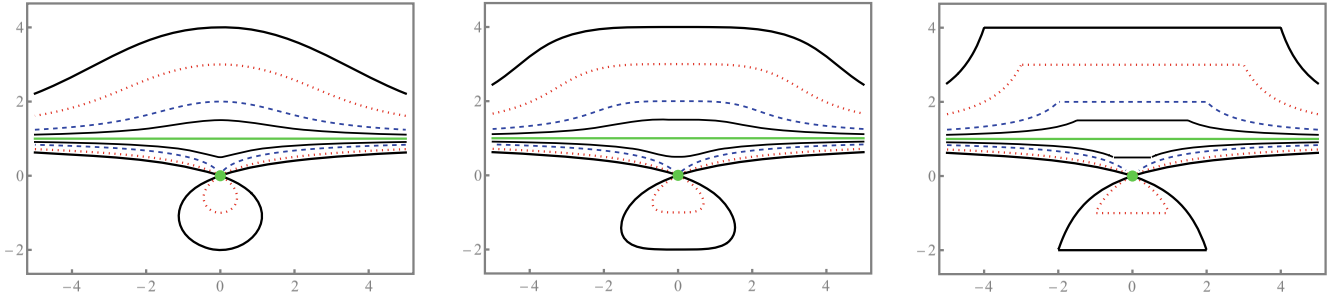


Fig. A.13 Lissajous-like curves $(\cos_p(3t), \sin_p(5t))$ (Project 16.2)



(a) $p = \frac{1}{2}$ (b) $p = 1$ (c) $p = \frac{4}{3}$



(d) $p = 2$ (e) $p = 4$ (f) $p = \infty$

Fig. A.14 A conchoid of a curve $\mathcal{C}$ with respect to a point P and parameter d is the locus of points X such that if M is the intersection of line OX and $\mathcal{C}$, then $|MX|_p = d$. Here in green, $\mathcal{C}$ is the line $y = 1$ and O is the origin. Viewing from top to bottom, these are generalized conchoids for $d = 3, 2, 1$, and $1/2$ (resp., black, dotted red, dashed blue, black) with distances measured in the p-metric. (Project 16.2)

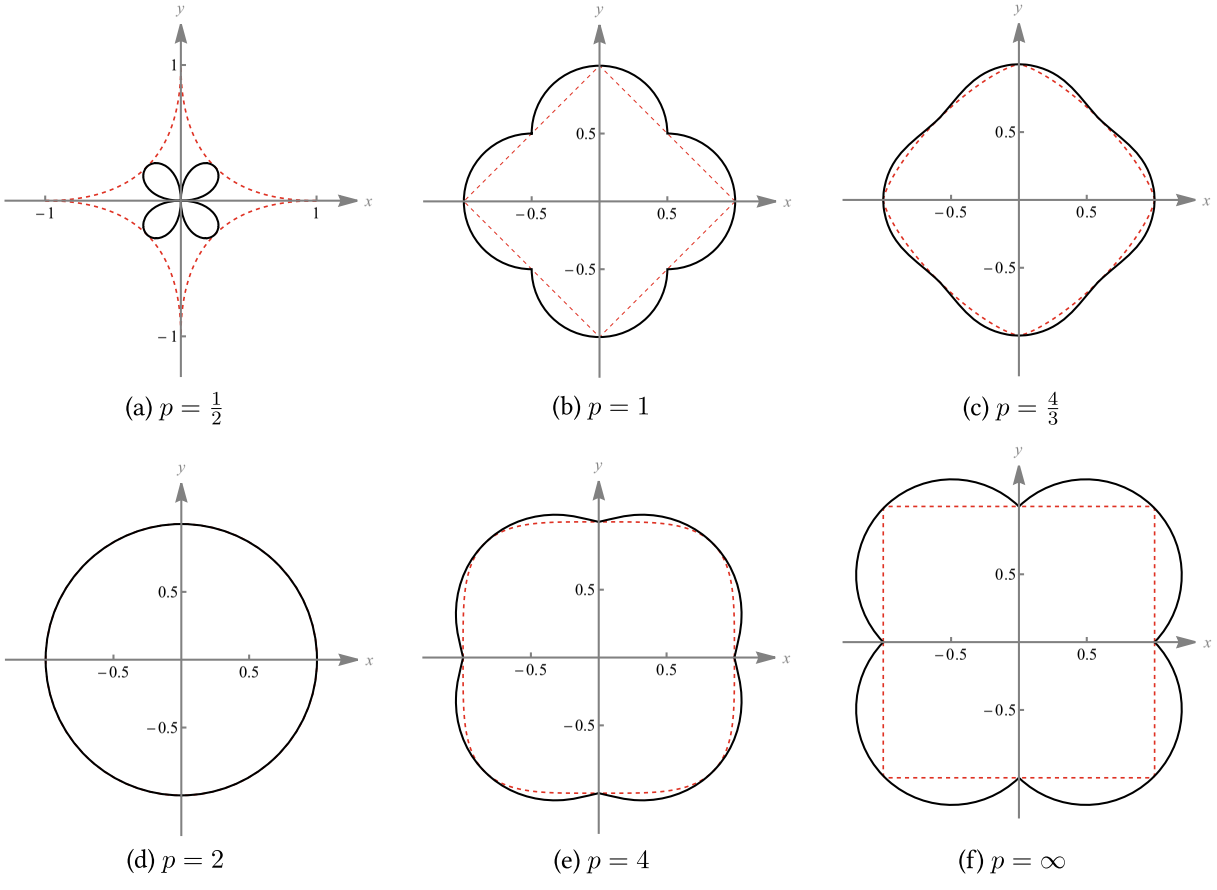


Fig. A.15 p-circles (dashed) and their pedal curves (solid). The Euclidean circle's pedal curve is itself. (Section 19.1)

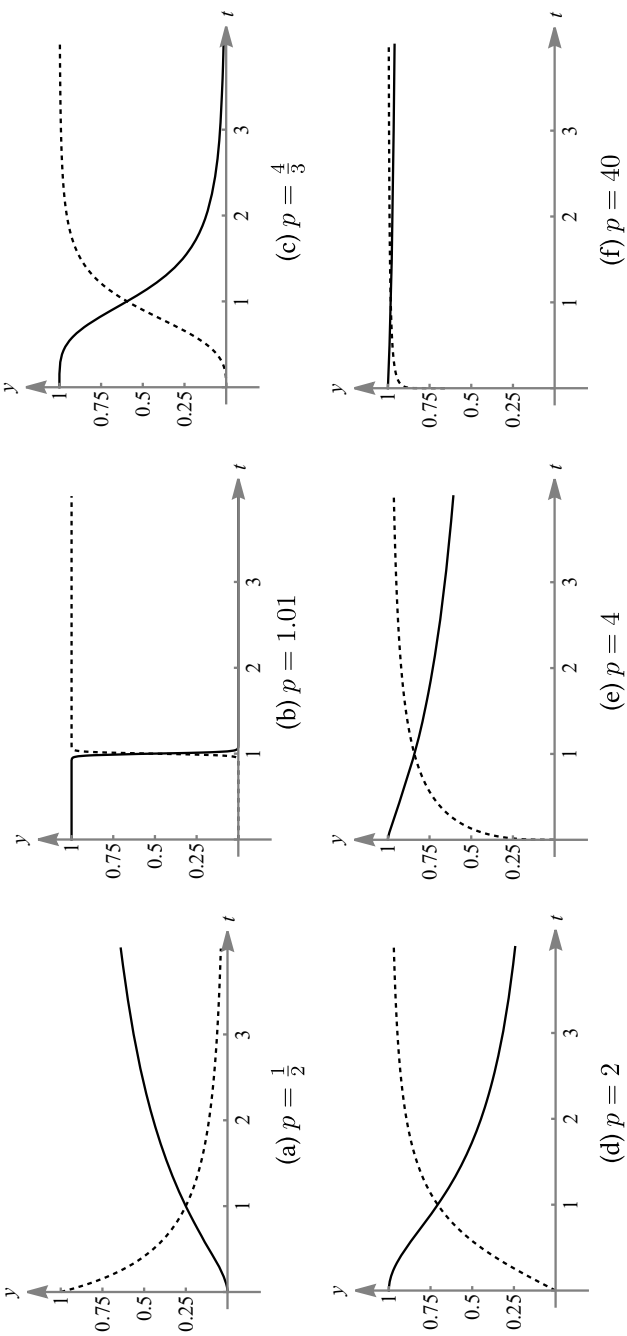


Fig. A.16 Graphs of the tangential squigonometric functions $\cos_p^{\top}(t)$ (solid) and $\sin_p^{\top}(t)$ (dashed). Ill-defined cases $p = 1, \infty$ are illustrated with limiting examples. (Section 19.2)

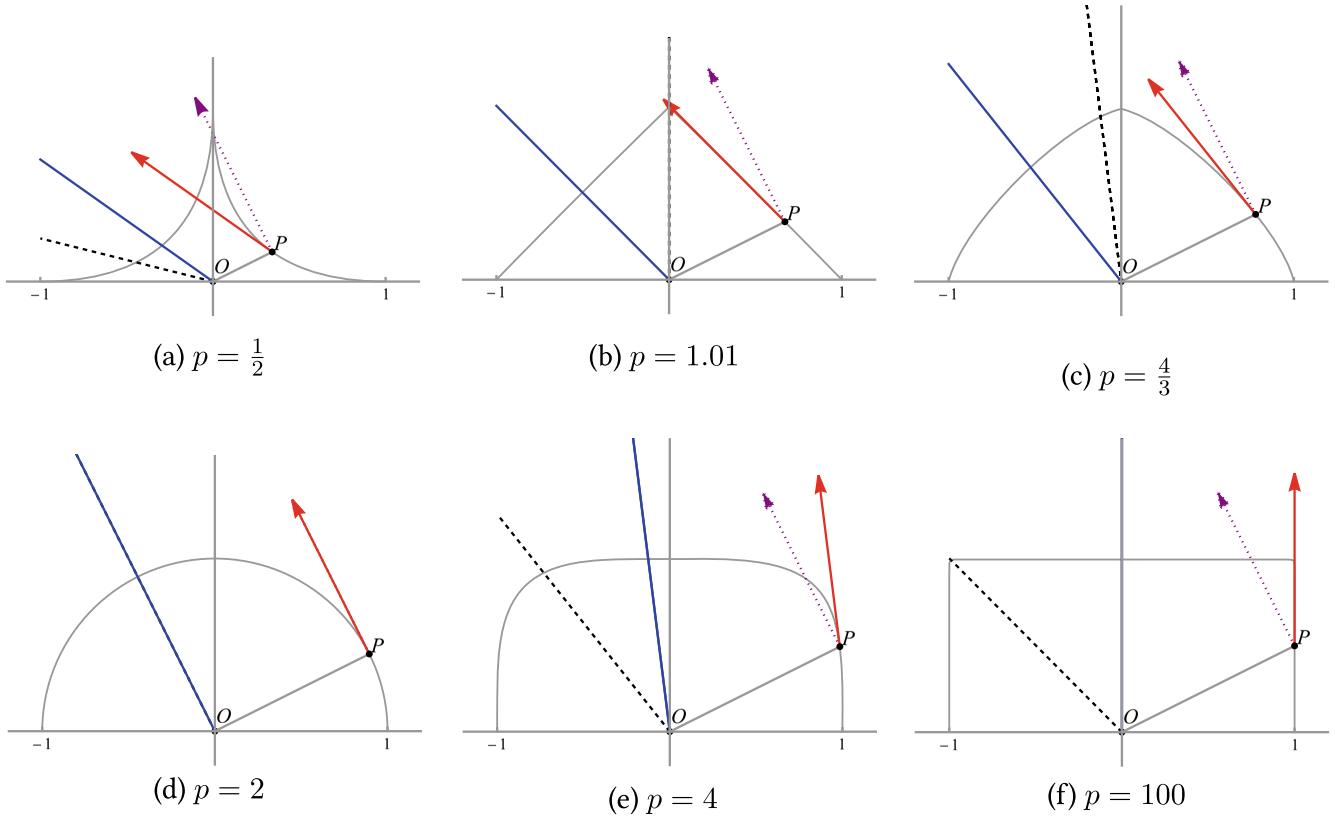


Fig. A.17 Line segment from the origin O of slope $\frac{1}{2}$ meeting the p-squircle at P, along with a unit perpendicular to $\overline{OP}$ at P (dotted arrow), a unit tangent at P (red arrow), and p- and q-Minkowski orthogonals to $\overline{OP}$, where $\frac{1}{p} + \frac{1}{q} = 1$ (blue and dashed, resp.; see Exercise 19.7). Ill-defined cases $p = 1, \infty$ are illustrated with limiting examples

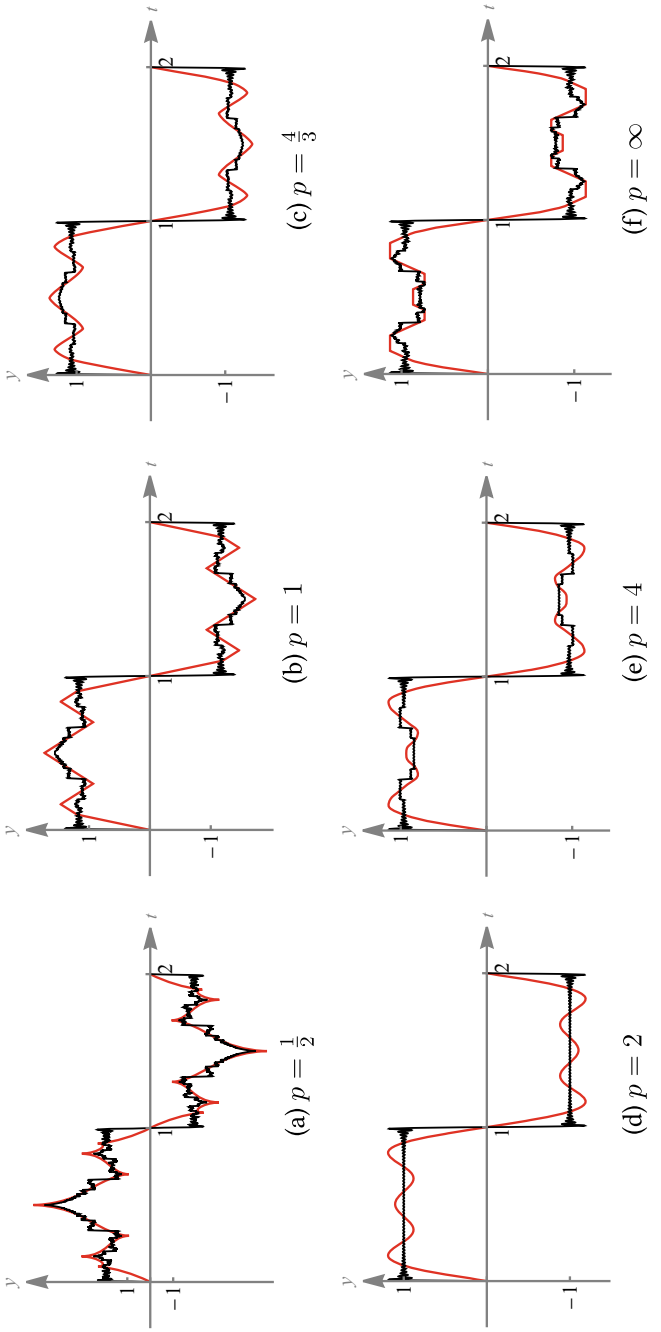


Fig. A.18 Plots of $\sum_{k=1}^{2n-1} \frac{4}{k\pi_p} \sin_p(k\pi_p t)$ for $n = 3$ and $n = 50$. (Project 20.3)

Formulas and integrals

B

B.1 Addition and doubling formulas

1. $\sin_6(2a) = \dfrac{2^{1/6}\sin_6(a)\cos_6(a)^{1/5}\sqrt{3+\sqrt{1+32\sin_6(a)^6\cos_6(a)^{6/5}}}}{(1+32\sin_6(a)^6\cos_6(a)^{6/5})^{1/4}\left(1+\sqrt{1+32\sin_6(a)^6\cos_6(a)^{6/5}}\right)^{1/6}}$

2. $\cos_4(a+b) = \dfrac{(\cos_4(a)\cos_4(b))^2-(\sin_4(a)\sin_4(b))^2}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2-(\sin_4(a)\sin_4(b))^2\right)^4+(\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b))^4}}$

3. $\sin_4(a+b) = \dfrac{\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b)}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2-(\sin_4(a)\sin_4(b))^2\right)^4+(\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b))^4}}$

4. $\tan_4(a+b) = \dfrac{\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b)}{(\cos_4(a)\cos_4(b))^2-(\sin_4(a)\sin_4(b))^2}$

5. $\cosh_4(a+b) = \dfrac{(\cos_4(a)\cos_4(b))^2+(\sin_4(a)\sin_4(b))^2}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2+(\sin_4(a)\sin_4(b))^2\right)^4+(\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b))^4}}$

6. $\sinh_4(a+b) = \dfrac{\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b)}{\sqrt[4]{\left((\cos_4(a)\cos_4(b))^2+(\sin_4(a)\sin_4(b))^2\right)^4+(\sin_4(a)\cos_4(a)+\sin_4(b)\cos_4(b))^4}}$

7. $\tanh_4(a+b) = \dfrac{\sinh_4(a)\cosh_4(a)+\sinh_4(b)\cosh_4(b)}{(\cosh_4(a)\cosh_4(b))^2+(\sinh_4(a)\sinh_4(b))^2}$

8. $\cos_4(2a) = \dfrac{\cos_4(a)^2-\sin_4(a)^2}{\sqrt[4]{\left(\cos_4(a)^4-\sin_4(a)^4\right)^4+(2\sin_4(a)\cos_4(a))^4}}$

9. $\sin_4(2a) = \dfrac{2\sin_4(a)\cos_4(a)}{\sqrt[4]{\left(\cos_4(a)^4-\sin_4(a)^4\right)^4+(2\sin_4(a)\cos_4(a))^4}}$

10. $\tan_4(2a) = \dfrac{2\tan_4(a)}{1-\tan_4(a)^2}\cdot\dfrac{1}{\cos_4(a)^2}$

11. $\cosh_4(2a) = \dfrac{\cosh_4(a)^2+\sinh_4(a)^2}{\sqrt[4]{\left(\cosh_4(a)^2+\sinh_4(a)^2\right)^4+(2\sinh_4(a)\cosh_4(a))^4}}$

12. $\sinh_4(2a) = \dfrac{2\sinh_4(a)\cosh_4(a)}{\sqrt[4]{\left(\cosh_4(a)^2+\sinh_4(a)^2\right)^4+(2\sinh_4(a)\cosh_4(a))^4}}$

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R. D. Poodiack and W. E. Wood, *Squigonometry: The Study of Imperfect Circles*, SUMS Readings, https://doi.org/10.1007/978-3-031-13783-9_B

13. $\tanh_4(2a) = \frac{2\tanh_4(a)}{1+\tanh_4(a)^2} \cdot \frac{1}{\cosh_4(a)^2}$
14. $\sin_3(2a) = \frac{\sin_3(a)(1+\cos_3(a)^3)}{\cos_3(a)(1+\sin_3(a)^3)}$
15. $\cos_3(2a) = \frac{\cos_3(a)^3-\sin_3(a)^3}{\cos_3(a)(1+\sin_3(a)^3)}$
16. $\mathrm{sn}(a+b) = \frac{\mathrm{sn}(b)\,\mathrm{cn}(a)\,\mathrm{dn}(a)+\mathrm{sn}(a)\,\mathrm{cn}(b)\,\mathrm{dn}(b)}{1-k^2\,\mathrm{sn}(a)^2\,\mathrm{sn}(b)^2}$
17. $\mathrm{cn}(a+b) = \frac{\mathrm{cn}(a)\,\mathrm{cn}(b)-\mathrm{sn}(a)\,\mathrm{dn}(a)\,\mathrm{sn}(b)\,\mathrm{dn}(b)}{1-k^2\,\mathrm{sn}(a)^2\,\mathrm{sn}(b)^2}$
18. $\mathrm{dn}(a+b) = \frac{\mathrm{dn}(a)\,\mathrm{dn}(b)-k^2\,\mathrm{sn}(a)\,\mathrm{cn}(a)\,\mathrm{sn}(b)\,\mathrm{cn}(b)}{1-k^2\,\mathrm{sn}(a)^2\,\mathrm{sn}(b)^2}$
19. $\mathrm{sl}(a+b) = \frac{\mathrm{sl}(a)\,\mathrm{cl}(b)+\mathrm{sl}(b)\,\mathrm{cl}(a)}{1-\mathrm{sl}(a)\,\mathrm{sl}(b)\,\mathrm{cl}(a)\,\mathrm{cl}(b)}$
20. $\mathrm{cl}(a+b) = \frac{\mathrm{cl}(a)\,\mathrm{cl}(b)-\mathrm{sl}(a)\,\mathrm{sl}(b)}{1+\mathrm{sl}(a)\,\mathrm{sl}(b)\,\mathrm{cl}(a)\,\mathrm{cl}(b)}$
21. $\mathrm{sl}(2a) = \frac{2\,\mathrm{sl}(a)\,\mathrm{cl}(a)}{1-\mathrm{sl}(a)^2\,\mathrm{cl}(a)^2}$
22. $\mathrm{cl}(2a) = \frac{\mathrm{cl}(a)^2-\mathrm{sl}(a)^2}{1+\mathrm{sl}(a)^2\,\mathrm{cl}(a)^2}$

B.2 Integral table

Based on the integral table in [42].

Basic forms

1. $\int \cos_p(u)^{p-1}\,du = \sin_p(u) + C$
2. $\int \sin_p(u)^{p-1}\,du = -\cos_p(u) + C$
3. $\int \cos_p(u)^p\,du = \frac{1}{2}[u + \cos_p(u)\sin_p(u)] + C$
4. $\int \sin_p(u)^p\,du = \frac{1}{2}[u - \cos_p(u)\sin_p(u)] + C$
5. $\int \sec_p(u)^2\,du = \tan_p(u) + C$
6. $\int \csc_p(u)^2\,du = -\cot_p(u) + C$
7. $\int \sec_p(u)^2 \sin_p(u)^{p-1}\,du = \int \frac{\sin_p(u)^{p-1}}{\cos_p(u)^2}\,du = \sec_p(u) + C$
8. $\int \csc_p(u)^2 \cos_p(u)^{p-1}\,dt = \int \frac{\cos_p(u)^{p-1}}{\sin_p(u)^2}\,du = -\csc_p(u) + C$

9. $\displaystyle\int \frac{1}{(a^p-u^p)^{\frac{p-1}{p}}}\,du = \frac{1}{a^{p-2}}\operatorname{arcsin}_p\left(\frac{u}{a}\right)+C$

10. $\displaystyle\int \frac{1}{(a^p+u^p)^{\frac{2}{p}}}\,du = \frac{1}{a}\operatorname{arctan}_p\left(\frac{u}{a}\right)+C$

11. $\displaystyle\int \frac{u^{p-3}}{(u^p-a^p)^{\frac{p-1}{p}}}\,du = \frac{1}{a}\operatorname{arcsec}_p\left(\frac{u}{a}\right)+C$

12. $\displaystyle\int \sqrt[p]{a^p-u^p}\,du = \frac{1}{2}\left[a^2\operatorname{arcsin}_p\left(\frac{u}{a}\right)+u\sqrt[p]{a^p-u^p}\right]+C,\quad (a>0)$

13. $\displaystyle\int \sqrt[p]{a^p+u^p}\,du = \frac{u}{2}\sqrt[p]{a^p+u^p}+\frac{a^2}{2}\operatorname{gd}_p^{-1}\left(\arctan_p\left(\frac{u}{a}\right)\right)+C,\quad (a>0)$

14. $\displaystyle\int \sqrt[p]{u^p-a^p}\,du = \frac{u}{2}\sqrt[p]{u^p-a^p}-\frac{a^2}{2}\operatorname{gd}_p^{-1}\left(\operatorname{arcsec}_p\left(\frac{u}{a}\right)\right)+C,\quad (a>0)$

15. $\displaystyle\int \frac{1}{(a^p-u^p)^{\frac{2}{p}}}\,du = \frac{1}{a}\operatorname{gd}_p^{-1}\left(\operatorname{arcsin}_p\left(\frac{u}{a}\right)\right)+C,\quad (a>0)$

16. $\displaystyle\int \frac{1}{(u^p-a^p)^{\frac{2}{p}}}\,du = -\frac{1}{a}\operatorname{gd}_p^{-1}\left(\operatorname{arcsin}_p\left(\frac{a}{u}\right)\right)+C,\quad (a>0)$

p-Trigonometric forms

17. $\displaystyle\int \sin_p(u)\,du = \frac{1}{2}\sin_p(u)^2\cdot{}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};\sin_p(u)^p\right)+C$

18. $\displaystyle\int \tan_p(u)\,du = \frac{1}{2}\tan_p(u)^2\cdot{}_2F_1\left(\begin{matrix}\frac{2}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};-\tan_p(u)^p\right)+C$

19. $\displaystyle\int \sec_p(u)^p\,du = \int\frac{1}{\cos_p(u)^p}\,du = \frac{p-2}{p-1}u+\frac{1}{p-1}\frac{\sin_p(u)}{\cos_p(u)^{p-1}}+C$

20. $\displaystyle\int \csc_p(u)^p\,du = \int\frac{1}{\sin_p(u)^p}\,du = \frac{p-2}{p-1}u-\frac{1}{p-1}\frac{\cos_p(u)}{\sin_p(u)^{p-1}}+C$

21. $\displaystyle\int \sin_p(u)^m\cos_p(u)^{p-1}\,du = \frac{1}{m+1}\sin_p(u)^{m+1}+C\quad (m\neq -1)$

22. $\displaystyle\int \cos_p(u)^m\sin_p(u)^{p-1}\,du = -\frac{1}{m+1}\cos_p(u)^{m+1}+C\quad (m\neq -1)$

23. $\displaystyle\int \frac{\cos_p(u)^{p-1}}{\sin_p(u)}\,du = \ln|\sin_p(u)|+C$

24. $\displaystyle\int \frac{\sin_p(u)^{p-1}}{\cos_p(u)}\,du = -\ln|\cos_p(u)|+C = \ln|\sec_p(u)|+C$

25. $$\int \frac{1}{\cos_p(u)\sin_p(u)}\,du = \ln|\tan_p(u)| + C = -\ln|\cot_p(u)| + C$$

26. $$\int (\cos_p(u)^p - \sin_p(t)^p)\,du = \cos_p(u)\sin_p(u) + C$$

27. $$\begin{aligned}\int (\sec_p(u)^p - \csc_p(u)^p)\,du &= \int \left(\frac{1}{\cos_p(u)^p} - \frac{1}{\sin_p(u)^p}\right) du \\ &= \frac{1}{p-1}\sec_p(u)^{p-1}\csc_p(u)^{p-1} + C \\ &= \frac{1}{p-1}\,\frac{1}{\cos_p(u)^{p-1}\sin_p(u)^{p-1}} + C\end{aligned}$$

28. $$\int (\cos_p(u)^p - \sin_p(u)^p)\cos_p(u)^m \sin_p(u)^m\,du = \frac{1}{m+1}\cos_p(u)^{m+1}\sin_p(u)^{m+1} + C, \quad (m \neq -1)$$

29. $$\int \frac{\cos_p(u)^p - \sin_p(u)^p}{\cos_p(u)\sin_p(u)}\,du = \ln|\cos_p(u)\sin_p(u)| + C$$

30. $$\int \cos_p(u)^m\,du = \frac{1}{m-p+2}\cos_p(u)^{m-p+1}\sin_p(u) + \frac{m-p+1}{m-p+2}\int \cos_p(u)^{m-p}\,du, \quad (m \neq p-2)$$

31. $$\int \sin_p(u)^m\,du = -\frac{1}{m-p+2}\sin_p(u)^{m-p+1}\cos_p(u) + \frac{m-p+1}{m-p+2}\int \sin_p(u)^{m-p}\,du, \quad (m \neq p-2)$$

32. $$\begin{aligned}\int \sec_p(u)^m\,du &= \int \frac{1}{\cos_p(u)^m}\,du \\ &= \frac{1}{m-1}\,\frac{\sin_p(u)}{\cos_p(u)^{m-1}} + \frac{m-2}{m-1}\int \frac{1}{\cos_p(u)^{m-p}}\,du, \\ &= \frac{1}{m-1}\sec_p(u)^{m-2}\tan_p(u) + \frac{m-2}{m-1}\int \sec_p(u)^{m-p}\,du, \quad (m \neq 1)\end{aligned}$$

33. $$\begin{aligned}\int \csc_p(u)^m\,du &= \int \frac{1}{\sin_p(u)^m}\,du \\ &= -\frac{1}{m-1}\,\frac{\cos_p(u)}{\sin_p(u)^{m-1}} + \frac{m-2}{m-1}\int \frac{1}{\sin_p(u)^{m-p}}\,du, \\ &= -\frac{1}{m-1}\csc_p(u)^{m-2}\cot_p(u) + \frac{m-2}{m-1}\int \csc_p(u)^{m-p}\,du, \quad (m \neq 1)\end{aligned}$$

34. $$\int \tan_p(u)^{p-1}\,du = \frac{1}{p-2}\sec_p(u)^{p-2} + C, \quad (p \neq 2)$$

35. $$\int \cot_p(u)^{p-1}\,du = -\frac{1}{p-2}\csc_p(u)^{p-2} + C, \quad (p \neq 2)$$

36. $$\int \tan_p(u)^m\,du = \frac{1}{m-1}\,\frac{\sin_p(u)^{m-p+1}}{\cos_p(u)^{m-1}} - \frac{m-p+1}{m-1}\int \tan_p(u)^{m-p}\,du, \quad (m \neq 1)$$

37. $\displaystyle\int \cot_p(u)^m\,du = -\frac{1}{m-1}\,\frac{\cos_p(u)^{m-p+1}}{\sin_p(u)^{m-1}} - \frac{m-p+1}{m-1}\int \cot_p(u)^{m-p}\,du,$

$(m \neq 1)$

Inverse p-trigonometric forms

38. $\displaystyle\int \operatorname{arcsin}_p(u)\,du = u\operatorname{arcsin}_p(u) - \tfrac{1}{2}u^2\cdot{}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};u^p\right) + C$

39. $\displaystyle\int \operatorname{arctan}_p(u)\,du = u\operatorname{arctan}_p(u) - \tfrac{1}{2}u^2\cdot{}_2F_1\left(\begin{matrix}\frac{2}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};-u^p\right) + C$

40. $\displaystyle\int u^{p-2}\operatorname{arcsin}_p(u)\,du = \frac{1}{p-1}\left(u^{p-1}\operatorname{arcsin}_p(u) + \sqrt[p]{1-u^p}\right) + C$

41. $\displaystyle\int u^{p-2}\operatorname{arctan}_p(u)\,du = \frac{1}{p-1}\left(u^{p-1}\operatorname{arctan}_p(u) - \frac{1}{p-2}(1+u^p)^{\frac{p-2}{p}}\right) + C$

42. $\displaystyle\int u^{p-1}\operatorname{arcsin}_p(u)\,du = \frac{1}{p+1}\left(u^{p+1}\cdot{}_2F_1\left(\begin{matrix}\frac{1}{p},\ \frac{p-1}{p}\\ \frac{2p+1}{p}\end{matrix};u^p\right)\right) + C$

43. $\displaystyle\int u^{p-1}\operatorname{arctan}_p(u)\,du = \frac{1}{p+1}\left(u^{p+1}\cdot{}_2F_1\left(\begin{matrix}\frac{1}{p},\ \frac{p-1}{p}\\ \frac{2p+1}{p}\end{matrix};-u^p\right)\right) + C$

44. $\displaystyle\int u^n\operatorname{arcsin}_p(u)\,du = \frac{1}{n+1}\left[u^{n+1}\operatorname{arcsin}_p(u) - \int \frac{u^{n+1}}{(1-u^p)^{\frac{p-1}{p}}}\,du\right] \quad (n \neq -1)$

45. $\displaystyle\int u^n\operatorname{arctan}_p(u)\,du = \frac{1}{n+1}\left[u^{n+1}\operatorname{arctan}_p(u) - \int \frac{u^{n+1}}{(1+u^p)^{\frac{2}{p}}}\,du\right] \quad (n \neq -1)$

p-Hyperbolic, p-exponential and p-logarithmic forms

46. $\displaystyle\int \sinh_p(u)\,du = \frac{1}{2}\sinh_p(u)^2\cdot{}_2F_1\left(\begin{matrix}\frac{p-1}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};-\sinh_p(u)^p\right) + C$

47. $\displaystyle\int \tanh_p(u)\,du = \frac{1}{2}\tanh_p(u)^2\cdot{}_2F_1\left(\begin{matrix}\frac{2}{p},\ \frac{2}{p}\\ \frac{p+2}{p}\end{matrix};\tanh_p(u)^p\right) + C$

48. $\displaystyle\int \cosh_p(u)^{p-1}\,du = \sinh_p(u) + C$

49. $\int \sinh_p(u)^{p-1}\,du = \cosh_p(u) + C$

50. $\int (\cosh_p(u)^{p-1} + \sinh_p(u)^{p-1})\,du = \exp_p(u) + C$

51. $\int \sinh_p(u)^p\,du = \frac{1}{2}[\cosh_p(u)\sinh_p(u) - u] + C$

52. $\int \cosh_p(u)^p\,du = \frac{1}{2}[\cosh_p(u)\sinh_p(u) + u] + C$

53. $\int \operatorname{sech}_p(t)^2\,du = \int \frac{1}{\cosh_p(u)^2}\,du = \tanh_p(u) + C$

54. $\int \operatorname{csch}_p(u)^2\,dt = \int \frac{1}{\sinh_p(u)^2}\,du = -\coth_p(u) + C$

55. $\int \operatorname{sech}_p(u)^p\,du = \int \frac{1}{\cosh_p(u)^p}\,du = \frac{p-2}{p-1}\,u + \frac{1}{p-1}\,\frac{\sinh_p(u)}{\cosh_p(u)^{p-1}} + C$

56. $\int \operatorname{csch}_p(u)^p\,du = \int \frac{1}{\sinh_p(u)^p}\,du = -\frac{p-2}{p-1}\,u - \frac{1}{p-1}\,\frac{\cosh_p(u)}{\sinh_p(u)^{p-1}} + C$

57. $\int \cosh_p(u)^{p-1}\sinh_p(u)^m\,du = \frac{1}{m+1}\sinh_p(u)^{m+1} + C \quad (m \neq -1)$

58. $\int \sinh_p(u)^{p-1}\cosh_p(u)^m\,du = \frac{1}{m+1}\cosh_p(u)^{m+1} + C \quad (m \neq -1)$

59. $\int \frac{\cosh_p(u)^{p-1}}{\sinh_p(u)}\,du = \ln|\sinh_p(u)| + C$

60. $\int \frac{\sinh_p(u)^{p-1}}{\cosh_p(u)}\,du = \ln\cosh_p(u) + C$

61. $\int \frac{1}{\cosh_p(u)\sinh_p(u)}\,du = \ln|\tanh_p(u)| + C = -\ln|\coth_p(u)| + C$

62. $\int (\cosh_p(u)^p + \sinh_p(u)^p)\,du = \cosh_p(u)\sinh_p(u) + C$

63. $$\begin{aligned}\int (\operatorname{sech}_p(u)^p + \operatorname{csch}_p(u)^p)\,du &= \int \left(\frac{1}{\cosh_p(u)^p} + \frac{1}{\sinh_p(u)^p}\right) du \\ &= -\frac{1}{p-1}\operatorname{sech}_p(u)^{p-1}\operatorname{csch}_p(u)^{p-1} + C \\ &= -\frac{1}{p-1}\,\frac{1}{\cosh_p(u)^{p-1}\sinh_p(u)^{p-1}} + C\end{aligned}$$

64. $\int (\cosh_p(u)^p + \sinh_p(u)^p)\cosh_p(u)^m\sinh_p(u)^m\,du = \frac{1}{m+1}\cosh_p(u)^{m+1}\sinh_p(u)^{m+1} + C, \quad (m \neq -1)$

65. $\int \frac{\cosh_p(u)^p + \sinh_p(u)^p}{\cosh_p(u)\sinh_p(u)}\,du = \ln|\cosh_p(u)\sinh_p(u)| + C$

66. $$\int \cosh_p(u)^m\,du = \frac{1}{m-p+2}\cosh_p(u)^{m-p+1}\sinh_p(u) + \frac{m-p+1}{m-p+2}\int \cosh_p(u)^{m-p}\,du, \quad (m \neq p-2)$$

67. $$\int \sinh_p(u)^m\,du = \frac{1}{m-p+2}\sinh_p(u)^{m-p+1}\cosh_p(u) + \frac{m-p+1}{m-p+2}\int \sinh_p(u)^{m-p}\,du, \quad (m \neq p-2)$$

68. $$\begin{aligned}\int \operatorname{sech}_p(u)^m\,du &= \int \frac{1}{\cosh_p(u)^m}\,du \\ &= \frac{1}{m-1}\frac{\sinh_p(u)}{\cosh_p(u)^{m-1}} + \frac{m-2}{m-1}\int \frac{1}{\cosh_p(u)^{m-p}}\,du, \\ &= \frac{1}{m-1}\operatorname{sech}_p(u)^{m-2}\tanh_p(u) + \frac{m-2}{m-1}\int \operatorname{sech}_p(u)^{m-p}\,du, \quad (m \neq 1)\end{aligned}$$

69. $$\begin{aligned}\int \operatorname{csch}_p(u)^m\,du &= \int \frac{1}{\sinh_p(u)^m}\,du \\ &= -\frac{1}{m-1}\frac{\cosh_p(u)}{\sinh_p(u)^{m-1}} - \frac{m-2}{m-1}\int \frac{1}{\sinh_p(u)^{m-p}}\,du, \\ &= -\frac{1}{m-1}\operatorname{csch}_p(u)^{m-2}\coth_p(u) - \frac{m-2}{m-1}\int \operatorname{csch}_p(u)^{m-p}\,du, \quad (m \neq 1)\end{aligned}$$

70. $$\int \tanh_p(u)^{p-1}\,du = -\frac{1}{p-2}\operatorname{sech}_p(u)^{p-2} + C, \quad (p \neq 2)$$

71. $$\int \coth_p(u)^{p-1}\,du = -\frac{1}{p-2}\operatorname{csch}_p(u)^{p-2} + C, \quad (p \neq 2)$$

72. $$\int \tanh_p(u)^m\,du = -\frac{1}{m-1}\frac{\sinh_p(u)^{m-p+1}}{\cosh_p(u)^{m-1}} + \frac{m-p+1}{m-1}\int \tanh_p(u)^{m-p}\,du, \quad (m \neq 1)$$

73. $$\int \coth_p(u)^m\,du = -\frac{1}{m-1}\frac{\cosh_p(u)^{m-p+1}}{\sinh_p(u)^{m-1}} + \frac{m-p+1}{m-1}\int \coth_p(u)^{m-p}\,du, \quad (m \neq 1)$$

74. $$\int \frac{1}{\cosh_p(\ln_p(u))^{p-1} + \sinh_p(\ln_p(u))^{p-1}}\,du = \ln_p(u) + C$$

75. $$\int \frac{1}{\ln_p(u)[\cosh_p(\ln_p(u))^{p-1} + \sinh_p(\ln_p(u))^{p-1}]}\,du = \ln_p(\ln_p(u)) + C$$

Integral expressions for π_p

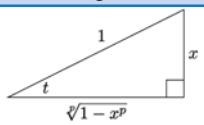
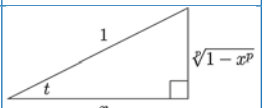
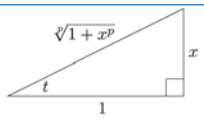
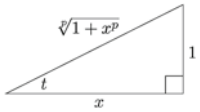
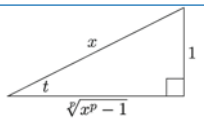
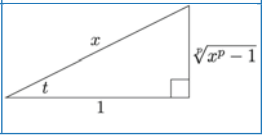
76. $$\pi_p = 2\int_0^1 \frac{1}{(1-u^p)^{\frac{p-1}{p}}}\,du$$

77. $$\pi_p = 2\int_0^\infty \frac{1}{(1+u^p)^{\frac{2}{p}}}\,du$$

78. $\pi_p = 2\displaystyle\int_1^{\infty} \frac{u^{p-3}}{(u^p-1)^{\frac{p-1}{p}}}\,du$

79. $\pi_p = 4\displaystyle\int_0^{1} \sqrt[p]{1-u^p}\,du$

B.3 Relationships between squigonometric functions and inverse squigonometric functions

t	$\sin_p(t)$	$\cos_p(t)$	$\tan_p(t)$	Diagram
$\arcsin_p(x)$	x	$\sqrt[p]{1-x^p}$	$\dfrac{x}{\sqrt[p]{1-x^p}}$	
$\arccos_p(x)$	$\sqrt[p]{1-x^p}$	x	$\dfrac{\sqrt[p]{1-x^p}}{x}$	
$\arctan_p(x)$	$\dfrac{x}{\sqrt[p]{1+x^p}}$	$\dfrac{1}{\sqrt[p]{1+x^p}}$	x	
$\arccot_p(x)$	$\dfrac{1}{\sqrt[p]{1+x^p}}$	$\dfrac{x}{\sqrt[p]{1+x^p}}$	$\dfrac{1}{x}$	
$\arccsc_p(x)$	$\dfrac{1}{x}$	$\dfrac{\sqrt[p]{x^p-1}}{x}$	$\dfrac{1}{\sqrt[p]{x^p-1}}$	
$\arcsec_p(x)$	$\dfrac{\sqrt[p]{x^p-1}}{x}$	$\dfrac{1}{x}$	$\sqrt[p]{x^p-1}$	

B.4 Formulas for p = 1/2

1. $\arcsin_{\frac{1}{2}}(x) = x - \frac{2}{3}x^{3/2},\ 0 \le x \le 1$
2. $$\sin_{\frac{1}{2}}(t) = \frac{\left(1 - i\sqrt{3}\right)(216t - 81)}{72\sqrt[3]{3}\sqrt[3]{24t^2 + 8\sqrt{3}\sqrt{3t^4 - t^3} - 36t + 9}} - \frac{1}{8}\sqrt[3]{3}\left(1 + i\sqrt{3}\right)\sqrt[3]{24t^2 + 8\sqrt{3}\sqrt{3t^4 - t^3} - 36t + 9} + \frac{3}{4},$$ $$0 \le t \le \tfrac{1}{3} = \tfrac{1}{2}\pi_{\frac{1}{2}}$$
3. $\arctan_{\frac{1}{2}}(x) = 2\sqrt{x} - 2\ln\left(\sqrt{x} + 1\right)$
4. $\tan_{\frac{1}{2}}(t) = \left(W\left(-e^{-\frac{t}{2}-1}\right) + 1\right)^2$
   $W(t)$ is the Lambert-W function, the inverse of $f(w) = we^w$.

B.5 Gudermannian identities

1. $$\begin{aligned}\mathrm{gd}_p(t) &= \int_0^t \cosh_p(u)^{p-3}\,dt \\ &= \arccos_p(\mathrm{sech}_p(t)) \cdot \mathrm{sgn}(t) \\ &= \arctan_p(\sinh_p(t)) \\ &= \mathrm{arcsec}_p(\cosh(t)) \cdot \mathrm{sgn}(t) \\ &= \arcsin_p(\tanh_p(t))\end{aligned}$$
2. $\mathrm{gd}_p^{-1}(t) = \int_0^t \cos_p(u)^{p-3}\,du$
3. $\mathrm{gd}_p^{-1}(t) = -\omega\ \mathrm{gd}(\omega^{p-1}t), \quad \omega^p = -1.$
4. $\mathrm{gd}_1(t) = \ln(1 + |t|)$
5. $\mathrm{gd}_2^{-1}(t) = \ln|\sec(t) + \tan(t)|$
6. $\mathrm{gd}_3(t) = \mathrm{gd}_3^{-1}(t) = t$

Parameterization primer

C

Our story begins with the assertion that $(\cos(t), \sin(t))$ is the natural parameterization of the circle. As the fundamental objects of this book are parameterized curves, we offer here an overview of the key tools and terms associated to parameterizations. The material in this appendix is common to most standard calculus texts, e.g. [17], to which we refer the reader for further background and exercises.

A parameterization of a curve in the plane is a function $f(t) = (u(t), v(t))$. For each value t in the domain of f we get a point in the plane. The parameter t may have some geometric or physical meaning. A common interpretation is time, and we think of a particle as being at the point $f(t)$ at time t. The range of $f(t)$, often called the *trace*, is the path that the particle will follow as time varies.

A simple example is a line, such as the set of points (x, y) satisfying $y = 2x + 1$. With no additional effort, this expression gives a parameterization $(t, 2t + 1)$, and indeed graphs of functions are the special case of parameterizations in which the parameter is simply the x-coordinate, i.e. $(t, f(t))$.

But there are others. This line is also parameterized by $f(t) = (5 - 4t, 11 - 8t)$. Notice this satisfies our functional relationship $y = 11-8t = 1+2(5-4t) = 1+2x$. This suggests a trick to parameterize a line for which you know two points. If we knew only that the line went through the points $P(1, 3)$ and $Q(5, 11)$, then we may think of the points as vectors and define $f(t) = (1, 3)t + (5, 11)(1 - t)$ so that $f(0) = Q$ and $f(1) = P$. Expand this out to see that it really is the same $f(t)$.

Beware of depending too heavily on the Cartesian relationship. $(t^2, 2t^2 + 1)$ satisfies $v(t) = 2u(t) + 1$, but since $u(t)$ is never negative the trace is a ray from the origin and not the whole line. Or consider $(\cos t, 2\cos t + 1)$ which models a particle oscillating along a segment of the line. Satisfying $y = 2x + 1$ means our curve is a subset of the line, but not necessarily all of it.

We are concerned with more interesting curves, like circles. Often the curve will be expressed in terms of a parameter that tells us something about the curve. Arclength is a particularly useful parameterization, wherein we can think of the particle moving so that it will trace a path of length t in t units of time. Or t may indicate the angle

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of the ray from the origin to $f(t)$. Or it may say something about area. In the case of the circle, the classical parameterization $(\cos(t), \sin(t))$, $0 \le t \le 2\pi$, does all three.

But there are other parameterizations. $(\sin(t), \cos(t))$ traces out a circle, but starts at $(0, 1)$. So does $(-\sin(t), \cos(t))$, but the particle moves clockwise. $(\sin(2t), \cos(2t))$ goes twice as fast, so it only requires a domain of $0 \le t \le \pi$. Notice that all of these satisfy $u(t)^2 + v(t)^2 = 1$.

The trigonometric functions are not necessary to trace out circular arcs. The parameterization $(\frac{1}{\sqrt{1+t^2}}, \frac{t}{\sqrt{1+t^2}})$ satisfies our functional relationship, but observe that the first coordinate is always positive. We only get a semicircle as t ranges over all reals, and we never quite reach $x = 0$. Nonetheless, we can still think of these algebraic functions as variants of sine and cosine.

Since a parameterized curve is essentially just a list of coordinates, we can learn much about them by combining our familiar analysis of the coordinate plane with a little calculus.

A parameterization $f(t) = (u(t), v(t))$ is differentiable if the component functions are. Our curves will usually be differentiable except possibly at a few points. Its derivative $f'(t) = (u'(t), v'(t))$ reports the velocity of the particle, which is best considered as a vector. The distance from $f'(t)$ to the origin is the speed, and the location of the point is the direction. For example, if $f(t) = (\sin(2t), \cos(2t))$, then $f'(t) = (2\cos(2t), -2\sin(2t))$. Notice that $\|f'(t)\| = 2$, which we may consider the speed, and the direction of motion is perpendicular to the radius. This is a key property of Euclidean circles.

We must be careful not to conflate our geometric understanding of single-variable functions with parameterizations. For example, let $f(t) = (t^3, 2t^3 + 1)$. This is our line again, but notice $f'(0) = (0, 0)$. We may be accustomed to associating a zero derivative with a feature of the curve, but there is nothing particularly special about this point on the line. What this is capturing is that our particle moves to the right along the line from $-\infty$ but slows down as it approaches the origin, where it actually stops. Then it immediately starts accelerating as it scoots off to the right along the line at increasing speed. The zero derivative was capturing something about the particle's motion, not its trace.

We can recover that familiar information by returning to Cartesian thinking. If we want the slope of the line, we still want $\frac{dy}{dx}$. We can use the chain rule to express it in terms of the parameterization:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{v'(t)}{u'(t)}.$$

Thus the points where $v'(t) = 0$ and $u'(t) = 0$ locate the horizontal and vertical tangents, respectively.

Proofs of formulas and theorems

D

Proof of (6.3)

We first write

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt = 2\int_0^\infty u^{2x-1} e^{-u^2}\, du$$

via the substitution $t = u^2$, $dt = 2u\, du$. This gives us

$$\begin{aligned}\Gamma(x)\Gamma(y) &= 2\int_0^\infty u^{2x-1} e^{-u^2}\, du \cdot 2\int_0^\infty v^{2y-1} e^{-v^2}\, dv \\ &= 4\int_0^\infty \int_0^\infty e^{-(u^2+v^2)} u^{2x-1} v^{2y-1}\, du\, dv\end{aligned}$$

We switch to polar coordinates with $u = r\cos\theta$ and $v = r\sin\theta$ and get

$$\begin{aligned}\Gamma(x)\Gamma(y) &= 4\int_0^\infty \int_0^{\pi/2} e^{-r^2} r^{2(x+y)-2} \cos^{2x-1}\theta \sin^{2y-1}\theta\, r\, dr\, d\theta \\ &= 2\int_0^\infty e^{-r^2} r^{2(x+y)-1}\, dr \cdot 2\int_0^{\pi/2} \cos^{2x-1}\theta \sin^{2y-1}\theta\, d\theta\end{aligned}$$

In the first integral, if we set $s = r^2$, we have that $ds = 2r\, dr$ and $dr = \frac{1}{2}s^{-1/2}\, ds$. For the last integral, we set $w = \cos^2\theta$. Then $\sin\theta = (1-w)^{1/2}$, and $dw = -2\cos\theta\sin\theta\, d\theta$, $d\theta = -\frac{1}{2}w^{-1/2}(1-w)^{-1/2}\, dw$. Thus

$$\begin{aligned}\Gamma(x)\Gamma(y) &= 2\int_0^\infty e^{-s} s^{(x+y)-1/2} \cdot \tfrac{1}{2}s^{-1/2}\, ds \\ &\qquad \cdot 2\int_0^1 w^{x-1/2} \cdot (1-w)^{y-1/2} \cdot \tfrac{1}{2}w^{-1/2}(1-w)^{-1/2}\, dw \\ &= \int_0^\infty s^{(x+y)-1} e^{-s}\, ds \int_0^1 w^{x-1}(1-w)^{y-1}\, dw \\ &= \Gamma(x+y)\, \mathrm{B}(x,y).\end{aligned}$$

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Therefore

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

Proof of (6.4)

Recall that

$$\mathrm{B}(x, y) = \int_0^1 u^{x-1}(1-u)^{y-1}\, du.$$

Multiplying and dividing the integrand by $(1-u)^{x-1}$ yields

$$\mathrm{B}(x, y) = \int_0^1 \left(\frac{u}{1-u}\right)^{x-1} (1-u)^{x+y-2}\, du.$$

Let $t = \frac{u}{1-u}$. When $u = 0$, we have $t = 0$. As $u \to 1^-$, we have $t \to \infty$. Since $u = \frac{t}{1+t}$ and $du = \frac{1}{(1+t)^2}\, dt$, we obtain

$$\begin{aligned}
\mathrm{B}(x, y) &= \int_0^\infty t^{x-1} \left(1 - \frac{t}{1+t}\right)^{x+y-2} \frac{1}{(1+t)^2}\, dt \\
&= \int_0^\infty t^{x-1} \left(\frac{1}{1+t}\right)^{x+y-2} (1+t)^{-2}\, dt \\
&= \int_0^\infty t^{x-1} (1+t)^{-(x+y)+2} (1+t)^{-2}\, dt \\
&= \int_0^\infty t^{x-1} (1+t)^{-(x+y)}\, dt,
\end{aligned}$$

which is (6.4).

Proof of (6.7)

The proof is due to Weisstein [102]. Recall the definition of the beta function:

$$\mathrm{B}(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 u^{x-1}(1-u)^{y-1}\, du.$$

We thus have

$$\mathrm{B}(z, z) = \frac{\Gamma^2(z)}{\Gamma(2z)} = \int_0^1 u^{z-1}(1-u)^{z-1}\, du.$$

We now set $u = (1+t)/2$. Then $du = dt/2$ and

$$\begin{aligned}\frac{\Gamma^2(z)}{\Gamma(2z)} &= \int_{-1}^{1} \left(\frac{1+t}{2}\right)^{z-1} \left(1 - \frac{1+t}{2}\right)^{z-1} \frac{dt}{2} \\ &= \frac{1}{2}\int_{-1}^{1} \left(\frac{1+t}{2}\right)^{z-1} \left(\frac{1-t}{2}\right)^{z-1} dt \\ &= \frac{1}{2^{1+2(z-1)}} \int_{-1}^{1} (1-t^2)^{z-1}\, dt \\ &= 2^{1-2z} \int_{-1}^{1} (1-t^2)^{z-1}\, dt \\ &= 2^{1-2z} \left[2\int_{0}^{1} (1-t^2)^{z-1}\, dt\right], \end{aligned} \tag{D.1}$$

by symmetry. We then note that if we substitute $u = t^2$ into the definition of the beta function, then $du = 2t\, dt$ and we get

$$\begin{aligned}\mathrm{B}(x,y) &= \int_0^1 u^{x-1}(1-u)^{y-1}\, du \\ &= 2\int_0^1 t^{2(x-1)}(1-t^2)^{y-1}\, t\, dt \\ &= 2\int_0^1 t^{2x-1}(1-t^2)^{y-1}\, dt. \end{aligned}$$

Comparing the last expression to (D.1), we see that

$$\frac{\Gamma^2(z)}{\Gamma(2z)} = 2^{1-2z}\,\mathrm{B}(\tfrac{1}{2}, z) = 2^{1-2z}\frac{\Gamma(\frac{1}{2})\Gamma(z)}{\Gamma(z+\frac{1}{2})}.$$

Solving for $\Gamma(2z)$ and recalling that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, we get

$$\begin{aligned}\Gamma(2z) &= \frac{\Gamma(z)\Gamma(z+\frac{1}{2})}{2^{1-2z}\Gamma(\frac{1}{2})} \\ &= \frac{\Gamma(z)\Gamma(z+\frac{1}{2})}{2^{1-2z}\sqrt{\pi}}, \end{aligned}$$

which is the Legendre doubling formula (6.7).

Proof of Theorem 19.7

Let x be in the interval $[a, b]$. Then the point on the secant line between a and b that lies above x is $f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$. We want to determine whether this point lies above or below the graph, so we are interested in the sign of the difference between the y-coordinates of this point and $f(x)$, i.e. the following expression:

$$\Delta = f(a) + \frac{f(b)-f(a)}{b-a}(x-a) - f(x) \tag{D.2}$$

We need to get from $f(x)$ to $f''(x)$, so the rest of the argument is algebraic manipulation to invoke the Mean Value Theorem twice.

$$\begin{aligned}
\Delta &= f(a) + \frac{f(b) - f(a)}{b - a}(x - a) - f(x) \\
&= \frac{b - a}{b - a} f(a) + \frac{f(b) - f(a)}{b - a}(x - a) - \frac{b - a}{b - a} f(x) \\
&= \frac{b - a}{b - a} f(a) + \frac{x - a}{b - a} f(b) - \frac{x - a}{b - a} f(a) - \frac{b - x + x - a}{b - a} f(x) \\
&= \frac{b - x}{b - a} f(a) + \frac{x - a}{b - a} f(b) - \frac{b - x}{b - a} f(x) - \frac{x - a}{b - a} f(x) \\
&= \frac{b - x}{b - a}(f(a) - f(x)) + \frac{x - a}{b - a}(f(b) - f(x))
\end{aligned}$$

We are ready for our first application of the Mean Value Theorem, which we apply separately to each summand to produce c_1, c_2 such that $a < c_1 < x$ and $x < c_2 < b$ and $f'(c_1) = \frac{f(x)-f(a)}{x-a}$ and $f'(c_2) = \frac{f(x)-f(b)}{x-b}$. The algebra proceeds:

$$\begin{aligned}
\Delta &= \frac{b - x}{b - a}(a - x) f'(c_1) + \frac{x - a}{b - a}(b - x) f'(c_2) \\
&= \frac{(b - x)(x - a)}{b - a}(f'(c_2) - f'(c_1))
\end{aligned}$$

We invoke the Mean Value Theorem again to obtain a d such that $c_1 < d < c_2$ and $f''(d) = \frac{f'(c_2)-f'(c_1)}{c_2-c_1}$. We then have $\Delta = \frac{(b-x)(x-a)(c_2-c_1)}{b-a} f''(d)$.

Recall that $a < c_1 < x < c_2 < b$, so all of the terms in front of $f''(d)$ are positive. We conclude that Δ has the same sign as $f''(d)$, which is the same at every point in the interval. Thus $\Delta > 0$ if $f'' > 0$, meaning the line lies above the function, and similarlly $\Delta < 0$ if $f'' < 0$. This completes the proof.

Alternate Pi Days

E

"Pi Day" is a celebration of the number π celebrated on March 14, which is written 3/14 in the U.S. date format [1]. Since π_p varies from 2 to 4, we offer alternative "pi days" throughout February and March. The dates and the corresponding p are listed below.

February				March			
Day	p	Day	p	Day	p	Day	p
01	1.00502	16	1.08445	01	1.80476	16	2.03082
02	1.01007	17	1.09005	02	1.81823	17	2.04798
03	1.01515	18	1.09569	03	1.83191	18	2.06545
04	1.02026	19	1.10137	04	1.8458	19	2.08324
05	1.02541	20	1.10709	05	1.8599	20	2.10135
06	1.0306	21	1.11286	06	1.87422	21	2.1198
07	1.03582	22	1.11867	07	1.88876	22	2.13859
08	1.04107	23	1.12452	08	1.90353	23	2.15774
09	1.04636	24	1.13042	09	1.91854	24	2.17726
10	1.05169	25	1.13636	10	1.9338	25	2.19716
11	1.05705	26	1.14235	11	1.9493	26	2.21746
12	1.06246	27	1.14839	12	1.96506	27	2.23817
13	1.0679	28	1.15448	13	1.98108	28	2.2593
14	1.07338	29	1.16061	14	1.99738	29	2.28087
15	1.07889			15	2.01396	30	2.30288
						31	2.32537

Casual Pi Days

"Casual Pi Day" (or "Pi Approximation Day") is celebrated on July 22nd, which is usually written 22/7 outside the United States. In case this does not fit with your academic year, we offer a few other opportunities to eat pie throughout the year. (To

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be fully proper, you should endeavor to make your pie in the shape of the p-circle for the date you choose!)

We have derived these from the first continued fraction convergents for π_p. (See Chapter 12.) Note that beyond $p = 6$, the first non-integer convergent denominator is generally bigger than 12, so for these, feel free to enjoy some pie on January 3rd or 4th.

p	Dates
4/3	Feb. 5, Nov. 28
2	Jul. 22
5/2	Feb. 7, Mar. 10, May 17, Aug. 27
3	Feb. 7
4	Mar. 11, Jul. 26
5	May 19
6	Jun. 23, Jul. 27
> 6	Jan. 3, Jan. 4

We get different options with other parameterizations, of course.

We might also mention "tau day," celebrated on June 28th (6/28 in the US date format). This recognizes a view [48] that the "standard" π would have been better defined as $\tau = 2\pi$, *i.e.* the ratio of the circumference of a circle to its radius rather than its diameter. This is an unresolvable semantic debate on which the authors are officially neutral, but we encourage any argument that results in a deeper understanding of π (or τ). Anyone looking for a math holiday later in the year might explore τ_p.

Selected exercise hints and solutions

F

1.1. $u' = -(u^2 + 1)$, $u(\frac{\pi}{2}) = 0$.

1.2. $x(t) = \cot(t)$, $y(t) = \csc(t)$ solves the CIVP

$$\begin{cases} x'(t) = -y(t)^2 \\ y'(t) = -x(t)y(t) \\ x\left(\dfrac{\pi}{2}\right) = 0 \\ y\left(\dfrac{\pi}{2}\right) = 1 \end{cases}$$

and $(x(t), y(t))$ parameterizes a hyperbola.

1.3.

$$\begin{cases} x'(t) = -2y(t) \\ y'(t) = 2x(t) \\ x(0) = 1 \\ y(0) = 0, \end{cases}$$

2.2. One possibility is

$$x(t) = \begin{cases} 1 & 0 \le t \le 1, \\ 2 - t & 1 \le t \le 3, \\ -1 & 3 \le t \le 5, \\ t - 6 & 5 \le t \le 7, \\ 1 & 7 \le t \le 8. \end{cases}$$

$$y(t) = \begin{cases} t & 0 \le t \le 1, \\ 1 & 1 \le t \le 3, \\ 4 - t & 3 \le t \le 5, \\ -1 & 5 \le t \le 7, \\ t - 8 & 7 \le t \le 8. \end{cases}$$

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3.1. Show that $x = \cos_p(-t)$ and $y = -\sin_p(-t)$ satisfy CIVP (3.2).

3.2. Start with the equation $\cos_p(t)^p + \sin_p(t)^p = 1$ and divide through by, respectively, $\cos_p(t)^p$ and $\sin_p(t)^p$.

3.6.

1. This parameterization satisfies $x^p + ((1 - x^p)^{1/p})^p = 1$, so it lies on the p-circle. It is continuous from $(0, 1)$ to $(1, 0)$ so it must be the complete curve for $0 \le x \le 1$. In this case, $x' = 1$ and clearly $y' \neq 0$ so (3.2) won't hold.
2. This is the parameterization obtained by thinking of the circle as the graph of a function. Thus the area of the circle is four times the area under this curve, or $4\int_0^1 (1 - x^p)^{(1/p)}\, dx$.
3. The argument remains unchanged if we consider x to be a continuous function of t (monotone was included mainly to restrict to the first quadrant).
4. Without continuity, we could get something like $x(t) = 0$ if $1/2 \le t \le 1$, $x(t) = 1$ if $0 \le t < 1/2$, which maps entirely to two points $(0, 1)$ and $(1, 0)$ (but still on the circle!).

3.7. Use a theorem whose proof cannot be contained in the margin of this book.

4.1.

1	7	7.00
$\frac{4}{3}$	$\left(4\,2^{2/3} + 3\sqrt[3]{3}\right)^{3/4}$	5.91
2	5	5.00
4	$\sqrt[4]{337}$	4.28
10	$5^{3/10}\,\sqrt[10]{8861}$	4.02
∞	4	4.00

4.2. Checking the cases in the definition separately for each property will reveal that they all hold because they hold in the Euclidean norm, and similarly for the inequality. We can save some work by observing that $\|\boldsymbol{u} - \boldsymbol{v}\| \le \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$. For example, if $\boldsymbol{u} \neq \lambda\boldsymbol{v}$ then $d_P(\boldsymbol{u}, \boldsymbol{v}) = \|\boldsymbol{u}\| + \|\boldsymbol{v}\| = \|\boldsymbol{u} - \boldsymbol{w}\| + \|\boldsymbol{v} - \boldsymbol{w}\| \le d_P(\boldsymbol{u}, \boldsymbol{w}) + d_P(\boldsymbol{v}, \boldsymbol{w})$. The Paris distance and Euclidean distance between two points are equal if and only if they both lie on the same line through the origin.

4.4. $\left(\sqrt[4]{\frac{1}{2} \pm \frac{\sqrt{2}}{3}}, \sqrt[4]{\frac{1}{2} \mp \frac{\sqrt{2}}{3}}\right)$ in the first quadrant, then negate coordinates (x, y, and both) to reflect to the others for a total of eight points.

4.6. The midset is the line $y = x$. The expression simplifies to $y^3 - x^3 + x - y$. This is obviously satisfied when $y = x$, and factoring the cubic will reveal that there are no other solutions.

4.8. The slopes of the two lines are $\frac{2^{\frac{1}{p}-1}}{1+2^{\frac{1}{p}-1}}$ and $\frac{2^{\frac{1}{p}-1}}{-1+2^{\frac{1}{p}-1}}$ for $p > 1$. Notice the appearance of the conjugate.

4.9. $\frac{4e}{1-e^2}$

4.11. $\cos^{\mathcal{E}}(t) = a\cos(t)$, $\sin^{\mathcal{E}}(t) = b\sin(t)$

5.1. A glance at Figure 5.3 should convince us that

$$\text{Area} = \frac{1}{2}y(1-y^p)^{\frac{1}{p}} + \int_{(1-y^p)^{1/p}}^{1} (1-t^p)^{\frac{1}{p}}\,dt.$$

Differentiating this with respect to y yields

$$\begin{aligned}(\text{Area})' &= \frac{1}{2}(1-y^p)^{\frac{1}{p}} + \frac{1}{2}y(1-y^p)^{\frac{1}{p}-1}(-y^{p-1}) + y(1-y^p)^{\frac{1}{p}-1}(y^{p-1}) \\ &= \frac{1}{2}(1-y^p)^{\frac{1}{p}} + \frac{1}{2}y^p(1-y^p)^{\frac{1}{p}-1} \\ &= \frac{1}{2}(1-y^p)^{\frac{1}{p}-1}.\end{aligned}$$

We can then integrate this expression to get

$$\text{Area} = \frac{1}{2}\int_0^y (1-u^p)^{\frac{1}{p}-1}\,du = \frac{\arcsin_p(y)}{2}.$$

5.2. Redo the solution for Exercise 5.1 with respect to x.

5.3. $\arcsin_1(x) = \int_0^x \frac{1}{(1-u)^0}\,du = \int_0^x du$. Then consider the limit as $p \to \infty$ of $(1-u^p)^{\frac{p-1}{p}}$.

5.4.

1. $\frac{12}{\sqrt[4]{21,361}} = \frac{12(21,361)^{\frac{3}{4}}}{21,361} \approx 0.992604.$
2. $\frac{4}{\sqrt[3]{407}} = \frac{4(407)^{\frac{2}{3}}}{407} \approx 0.944568$
3. $\frac{\sqrt[5]{243-x^5}}{3}$

5.5. The original triangle has perimeter $2+2^{1/p}$. The rotated triangle has perimeter $2^{\frac{1}{2}} + 2^{\frac{1}{2}+\frac{1}{p}}$. They are equal at $p = 2$, of course, and nowhere else because the function defined as the difference of the two perimeters is monotone.

6.1. They weren't too far off (0.6% relative error). $\pi_{\text{Egypt}} = \left(\frac{16}{9}\right)^2 \approx 3.1605 \approx \pi_{2.0316}$.

6.2. Define $f(t) = \sin(t+\pi_p/2)$ and $g(t) = \cos(t+\pi_p/2)$. Then $f'(t) = g(t)^{p-1}$, $g'(t) = -f(t)^{p-1}$, $g(0) = \cos(\pi_p/2) = 0$, and $f(0) = (1-\cos(\pi_p/2)^p)^{1/p} = 1$. Thus f and g satisfy the same CIVP (3.2) as cosine and sine. The others are similar, or they can be deduced from this one.

6.4. Use the result of Exercise 5.1 with $y = 1$ to obtain the area of the first quadrant portion of the p-circle, then use symmetry.

6.5. Look at the definition of the p-arcsecant in Chapter 4 and think of the location of a vertical asymptote for the p-secant.

6.7. $\pi_{1/2} = 2/3$.

6.8. Verify that the derivatives agree at the multiples of $\frac{\pi_p}{2}$, but for non-integer p you will eventually get negative exponents.

6.9. $\sin(\frac{\pi}{p}) = \sin(\pi(1-\frac{1}{q})) = \sin(\pi - \frac{\pi}{q}) = \sin(\frac{\pi}{q})$. For the second part, observe that $\Gamma(1/p)\Gamma(1/q) = \Gamma(1/p)\Gamma(1-1/p)$ and apply the Euler reflection formula (6.6). Then use (6.5). Exercise 6.9 is relevant because the expression $\pi_p\pi_q$ is symmetric in p and q but the formula seemingly is not, until we observe that $\sin(\frac{\pi}{p}) = \sin(\frac{\pi}{q})$. Or we could take the second part as a convoluted solution to the first part.

6.10. Let $I = \int_{-\infty}^{\infty} e^{-u^2}\,du$, and express I^2 as a double integral over the plane, then switch to polar coordinates.

6.11. Use symmetry to write the integral as $2\int_0^{\infty} e^{-u^p}\,du$, then substitute $w = u^p$ and use the definition of Γ.

6.12. For simplicity, assume $\mu = 0$ and $k = 1$ and calculate $\int_{-\infty}^{\infty} f(x)\,dx = 2\int_0^{\infty} f(x)\,dx$ with the substitution $w = u^p$ and apply the definition of $\Gamma(x)$. The variance is $\frac{\Gamma(\frac{3}{p})}{\Gamma(\frac{1}{p})}k^2$, obtained from a similar integration. We can find an exponential distribution at $p = 1$, normal at $p = 2$, and uniform at $p = \infty$.

7.2. If we check the signs, we will see the solution functions can never be negative, i.e. we will only get a quarter-squircle.

7.4. $\sin_{\infty}^{\text{ang}}(t) = \tan t$ for $0 \le t \le \pi/4$, 1 for $\pi/4 \le t \le \pi/2$, and then extend by symmetry.

7.5. When $p = 1$, the integral in Equation (7.8) becomes

$$I_1 = 2\int_0^1 \frac{1}{(1-u)^0((1-u)^{\frac{2}{1}} + u^2)}\,du = 2\int_0^1 \frac{1}{(1-u)^2 + u^2}\,du$$

Simplification and completing the square leads to

$$I_1 = \int_0^1 \frac{1}{u^2 - u + \frac{1}{2}}\,du = \int_0^1 \frac{1}{(u-\frac{1}{2})^2 + \frac{1}{4}}\,du = 4\int_0^1 \frac{1}{(2u-1)^2 + 1}\,du.$$

Substituting $w = 2u - 1$ gives

$$I_1 = 2\int_{-1}^1 \frac{1}{w^2+1}\,dw = 2\arctan(w)\Big]_{-1}^1 = \pi.$$

For $p = 2$, the integral in (7.8) becomes

$$I_2 = 2\int_0^1 \frac{1}{(1-u^2)^{\frac{1}{2}}((1-u^2)^{\frac{2}{2}} + u^2)}\,du = 2\int_0^1 \frac{1}{\sqrt{1-u^2}}\,du = 2\arcsin(u)\Big]_0^1 = \pi.$$

8.1. $\pi_1^{\text{arc}} = 4\int_0^{1/2}\left(1 + (\frac{1}{u} - 1)^0\right)^1\,du = 4\int_0^{1/2} 2\,du = 4$

8.2. $\pi_2^{\text{arc}} = \int_0^1 (\frac{1}{u} + \frac{1}{1-u})^{1/2}\,du) = \int_0^1 \frac{du}{\sqrt{u(1-u)}}$. Substituting $w = (1-u^2)$ gives $\pi_2^{arc} = \int_1^0 \frac{1}{w}\frac{1}{\sqrt{1-w^2}} \cdot (-2w)\,dw = 2\int_0^1 \frac{dw}{\sqrt{1-w^2}} = 2\arcsin(1) = \pi$.

8.3. If we let w represent the arclength tangent and z represent the arclength secant, a solution is

$$\begin{cases} w'(t) &= \frac{z(t)^{p+1}}{(1+w(t)^{p(p-1)})^{1/p}} \\ z'(t) &= \frac{z(t)^2 w(t)^{p-1}}{(1+w(t)^{p(p-1)})^{1/p}} \\ w(0) &= 0 \\ z(0) &= 1. \end{cases}$$

8.4.

$$\text{arcsec}_p^{\text{arc}}(x) = \int_1^x \frac{(1-(u^p-1)^{p-1})^{\frac{1}{p}}}{u^2(u^p-1)^{\frac{p-1}{p}}}\,du, \quad x \geq 1.$$

8.6. $2\pi_4$

8.7. It is always 4.

8.8. Express the domain of the parameterization as a union of intervals on which the curve can be expressed as the graph of a monotone function.

9.1. $-\frac{(x^{10}+1024)^{9/10}}{9216x^9} + C$

9.2. $\frac{45}{256\sqrt[4]{2}} - \frac{9}{64}$

9.3. $\frac{11}{12\sqrt[5]{2}}$

9.4. $x\arctan_4 x - \frac{1}{2}\operatorname{arcsinh} x + C$

9.6. Use formula (9.7) with either $m = 0$ and $n = k$ or vice versa.

10.1 **and** 10.2. Use formulas (6.8) and (6.5).

10.3 Use Equation (6.2) a lot along with the fact that $\Gamma(1) = 1$.

10.5 Write

$$\sec_p(t)^p = 1 + \tan_p(t)^p = 1 + \frac{\sin_p(t)^p}{\cos_p(t)^p}$$

and integrate the last term by parts, remembering that

$$\sin_p(t)^p = \sin_p(t) \cdot \sin_p(t)^{p-1}.$$

11.1. -4, implying the power series has a non-zero coefficient on an even power.

11.2. Use the Binomial Theorem to produce a series for $(1+x^p)^{-\frac{2}{p}}$, integrate it term by term, then use the fact that $\arctan_p(0) = 0$.

12.1. $\pi_{12} \approx 3.9594365$. Since the series is alternating, we can bound the error by the next term in the partial sum. We get an error bound $< 10^{-6}$ around $n = 190$ terms.

12.3. Apply (11.3) and (11.4).

13.6. This is a Euclidean octahedron with vertices at the unit coordinate vectors. Classical formulas then give a volume of $\frac{4}{3}$.

13.8. Note that the unit ∞-sphere is a cube with common side length 2.

14.1. $\operatorname{arctanh}_{3/2} x + C$ and $3 \operatorname{arcsinh}_6(x^{1/3}) + C$.

14.7. Use the definitions of Υ_p and π_p to write

$$\frac{\Upsilon_p}{\pi_p} = \frac{1}{2} \cdot \frac{\Gamma(1 - \frac{2}{p})}{\Gamma(1 - \frac{1}{p})} \cdot \frac{\Gamma(\frac{2}{p})}{\Gamma(\frac{1}{p})}.$$

Now employ the Euler reflection formula (6.6) and the well known double-angle formula for sine.

14.8. Use the last identity in Theorem 14.3, part 1.

16.1. See Figure A.11.

16.2. $k = 1$ is the p-midset of the points. If $p = 2$ and $k \neq 1$ then the curve is a circle. See Figure A.12.

16.3. Use Theorem 14.2 and the fact that $\omega^4 = -1$.

17.7. Set $\alpha = \frac{L}{6}$ and note that $\operatorname{sl}(3\alpha) = \operatorname{sl}(2\alpha+\alpha) = 1$ and $\operatorname{cl}(3\alpha) = \operatorname{cl}(2\alpha+\alpha) = 0$. Also notice that $\operatorname{sl}(\alpha) = \operatorname{sl}(2\alpha-\alpha)$ and $\operatorname{cl}(\alpha) = \operatorname{cl}(2\alpha-\alpha)$. If we set $u = \operatorname{sl}(2\alpha)$, $w = \operatorname{sl}(\alpha)$, $v = \operatorname{cl}(2\alpha)$, $z = \operatorname{cl}(\alpha)$, and $P = uvwz$, we can set up four equations in these four unknowns. We can then introduce κ by noting that $u = \kappa v$ and $z = \kappa w$.

18.1. Show that it is not translation-invariant.

18.2. Consider the p-circle centered at the origin on which $\boldsymbol{v}$ lies.

18.3. Reflexivity: any norm is 1-equivalent to itself. Symmetry: if norm X is C-equivalent to norm Y then norm Y is C-equivalent to norm X. Transitivity: if norm X is C-equivalent to norm Y and norm Y is D-equivalent to Z, then norm X is CD-equivalent to norm Z. Transitivity may fail if we restrict to a specific constant.

18.6. Consider sequences for which all terms are equal to 0 except for a single term that is equal to 1.

19.1. The revised definition effectively re-scales the image, with the bottom hemisphere now mapping to a disk of radius $2\sqrt{2}$ instead of 1.

19.4. See [54].

19.5. Analytically, we get $\cos_1^\top(t) = \begin{cases} 1 & \text{if } 0 < t < 1 \\ \frac{1}{2} & \text{if } t = 1 \\ 0 & \text{if } t > 1 \end{cases}$ and $\sin_1^\top(t) = 1 - \cos_1^\top(t)$.

But there was no hope to parameterize the 1-circle with the slope of the tangent line since that slope is constant. Similar problems occur with $p = \infty$, but they go away if $0 < p < 1$. See Figure A.16.

19.6. This can be done computationally since we have formulas for pedal curves and inversion for a given parameterization, but note that these formulas may not parameterize the lemniscate exactly the same way. Parameterizing the hyperbola with $(\sec_p(t), \tan_p(t))$ may offer more comfortable territory in working with identities.

20.1. $\lim_{p \to 1^+} (p-1)\widetilde{\pi_p} = 2.$

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